OPERATOR-NORM DISCREPANCY OF RANDOM BANDED TOEPLITZ CONTRACTIONS

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ABSTRACT. For complex matrices $A_1, \dots, A_n$ write $\operatorname{disc}_{\mathrm{op}}(A_1, \dots, A_n) = \min_{\varepsilon \in \{-1,1\}^n} \left\| \sum_i \varepsilon_i A_i \right\|_{\mathrm{op}}$. We study this quantity when the A_i are independent random banded Toeplitz or Hankel contractions. The deterministic theory already settles two important subregimes. When $d \geq 4w$ and the half-bandwidth w is at most the number of matrices n, the banded theorem of the companion paper [6] gives $\operatorname{disc}_{\mathrm{op}}(A_1, \dots, A_n) \leq C\sqrt{n \log(e + w/n)}$ on *every* realization, so $\operatorname{disc}_{\mathrm{op}} = O(\sqrt{n})$ almost surely, with a constant free of the matrix dimension d; no probabilistic input is used, and the multilevel version replaces w by w^r. The full-bandwidth theorem of the same companion gives the pathwise bound $C\sqrt{n \log(e + d/n)}$, so the random full-bandwidth question is already true deterministically whenever d/n is bounded. There is also a second full-bandwidth endpoint: with the expected-norm normalization $A = T_d(\xi)/\mathbb{E}\|T_d(\xi)\|_{\mathrm{op}}$, every fixed sign pattern already gives $O(\sqrt{n})$ with high probability, by a direct net bound for the random Toeplitz symbol; every polynomial-sized deterministic list of sign patterns is harmless by the same union bound, and the same holds for independent random lists after conditioning on the list, even with only a polynomial expected size bound. More generally, a finite sign family of size M is uniformly controlled at the entropy-sensitive scale $\sqrt{n(1 + \log M/\log d)}$. In the logarithmic sample window $n = O(\log d)$, all sign patterns satisfy the square-root bound simultaneously. Thus the expected-normalized random full-bandwidth model is a theorem, not a conjecture. Its sampled robust edge certificates also become actual full Toeplitz/Hankel operator-norm certificates in the same unbounded-aspect range, after the fixed-sign theorem supplies a uniform $O(1)$ bound on the normalized summands. What remains out of reach for the deterministic contraction theory is the genuinely realization-normalized contraction model in the unbounded-aspect regime $d/n \to \infty$, where the available pathwise tools still certify only the growing $\sqrt{n \log(e + d/n)}$ scale. We explain why the intrinsic-freeness theory of Bandeira, Boedihardjo and van Handel does not by itself certify $O(\sqrt{n})$ for this contraction problem: Toeplitz matrices are highly delocalized in Hilbert–Schmidt norm, so the free-deviation correction is not visibly lower order. The remaining sign-optimized contraction question is reduced to a concrete edge-variance profile: if the normalized Fourier samples have average square mass $O(1/\log d)$ at every edge node, then independent random signs already give $O(\sqrt{n})$, and this edge profile is verified in the expected-normalized model when $n \gtrsim \log d$. A profile-normalized variant also gives fixed-sign, sign-family entropy, coefficient-family entropy, almost-sure triangular array, and explicit dimension-schedule bounds for independent non-identical lags with arbitrary deterministic variance envelopes. A normalizer-tail criterion shows that, for any sub-Gaussian law, the same conclusion follows once the one-matrix lower tail $\mathbb{P}\{\|T_d(\xi)\|_{\mathrm{op}} < a\,\mathbb{E}\|T_d(\xi)\|_{\mathrm{op}}\}$, multiplied by $\min\{n, \log^2(e + d)\}$, tends to zero. On the same edge-profile event all scalar edge-grid sign moments have the corresponding sub-Gaussian size, the sign-averaged edge maximum is $O(\sqrt{n})$, short random search amplifies the signing success probability with binomial success-count control, and good signings are Hamming-stable on balls of radius $\Theta(\log d)$, with the same binomial control for sampled robust centers; in particular $O_B(1)$ sampled signs give a robust certificate with failure at most d^{-B}. A deterministic Borel transfer template upgrades every eventual edge-grid robust-center count in the full-bandwidth range $d_m/n_m \to \infty$, $\log d_m = o(n_m)$, to actual realization-normalized Toeplitz/Hankel operator-norm robust certificates, with the same centers and Hamming radii. For the convex-concentrated coefficient classes this gives an explicit high-probability operator-norm certificate theorem: $O_B(1)$ sampled signs contain a logarithmic-radius Hamming ball that is robust for both full Toeplitz and full Hankel contractions; fixed-size sampled lists even contain a $1 - \rho$ robust-center fraction, and every sign in a polynomial-size sampled batch is robust with high

1. Introduction

For self-adjoint matrices $A_1, \dots, A_n$ the matrix discrepancy is

$$\operatorname{disc}(A_1, \dots, A_n) = \min_{\varepsilon \in \{-1,1\}^n} \left\| \sum_{i=1}^{n} \varepsilon_i A_i \right\|_{\mathrm{op}},$$

and for arbitrary complex matrices we use the same formula and write $\operatorname{disc}_{\mathrm{op}}$. The Matrix Spencer conjecture asks whether self-adjoint $d \times d$ contractions $A_1, \dots, A_n$ with $d \le n$ always satisfy $\operatorname{disc} = O(\sqrt{n})$, with a constant independent of d; the general case is open, and the best dimension-uniform bounds carry a logarithmic loss from noncommutative Khintchine inequalities. We refer to the companion paper [6] for the deterministic theory and to the surveys cited there for the background.

This paper concerns the *random* version of the problem for one structured family: independent random banded Toeplitz, or Hankel, contractions. The question is whether the discrepancy of a typical realization is governed by the favorable $\sqrt{n}$ scale or by the unfavorable $\sqrt{n \log d}$ scale that a naive union bound would suggest. We separate the answer into two regimes by the half-bandwidth.

For bounded bandwidth the answer is already settled, and it requires no randomness. The deterministic banded theorem of [6] bounds the discrepancy of *any* banded Toeplitz contractions, on a pathwise basis, by a quantity that depends on the bandwidth but not on the dimension. Applying it on each realization gives an almost-sure $O(\sqrt{n})$ bound for half-bandwidth $w \le n$ in dimensions $d \ge 4w$, dimension-free in d (Theorem 3.2). So for banded inputs the existential "do signs make the operator norm $O(\sqrt{n})$" is already answered in the affirmative, deterministically. We state this cleanly because it frames precisely what is left.

There is also a deterministic full-bandwidth endpoint that should be separated from the genuinely random question. The scalar Toeplitz theorem of [6] gives, for arbitrary full Toeplitz contractions,

$$\operatorname{disc}_{\mathrm{op}}(A_1, \dots, A_n) \le C\sqrt{n \log(e + d/n)}.$$

Consequently the full-bandwidth random question is already true on every realization in any bounded-aspect-ratio regime $d \le \Lambda n$, with a constant depending only on Λ. What remains is the full-bandwidth regime $w = d - 1$ with $d/n \to \infty$. In that regime the deterministic pathwise method of [6] reverts to the growing $\sqrt{n \log(e + d/n)}$ scale. The expected-normalized model $A = T_d(\xi)/N_d$, $N_d = \mathbb{E}\|T_d(\xi)\|_{\mathrm{op}}$, is easier than this pathwise contraction problem: Theorem 4.1 below shows that any fixed sign pattern already gives $O(\sqrt{n})$ with high probability, and Corollary 4.9 gives the entropy-sensitive uniform version for finite sign families. Thus the expected-normalized full random Toeplitz model is settled. What remains out of reach is a high-probability,

realization-wise signing theorem for honest full-bandwidth random contractions in the unbounded-aspect regime. The generic matrix Khintchine bound, using only $\|A_i\|_{\mathrm{op}} \le 1$, still gives the safe upper bound $\sqrt{n \log d}$.

One might hope that the sharp matrix-concentration theory of Bandeira, Boedihardjo and van Handel [1], which removes the Khintchine logarithm whenever the summands behave as though freely independent, would certify $O(\sqrt{n})$ for the contraction problem. The direct route is not currently justified, for a structural reason that we record in Remark 4.2: an expected-normalized Toeplitz matrix is *maximally delocalized*, with Hilbert–Schmidt norm of order $\sqrt{d/\log d}$, so the free-deviation parameter $v(X)$ that controls the correction term in [1] sits at the same order as the standard deviation $\sigma(X)$, rather than below it. In that regime the $(\log d)^{3/4}$ error of the intrinsic-freeness bound is at least the size of the main term, and the bound collapses back to the Khintchine scale. We confirmed numerically that the ratio $v(X)/\sigma(X)$ for signed Toeplitz superpositions stays bounded away from zero across a range of dimensions, in the band 2.5 to 5.3, which is consistent with the parameter count and rules out the free-probability shortcut.

Despite these obstacles we believe the realization-normalized full-bandwidth contraction class is nonetheless well behaved. The worst-case factor $\sqrt{\log(d/n)}$ is forced by adversarial Toeplitz matrices whose symbols conspire to peak together; for independent random symbols there is no such conspiracy. The remaining question is the following.

Question 1.1 (Random full Toeplitz contractions)**.** *Let $B_i = T_d(\xi^{(i)})$, $1 \le i \le n$, be independent full Toeplitz matrices drawn from a fixed mean-zero sub-Gaussian unit-variance law, and set*

$$A_i := B_i / \|B_i\|_{\mathrm{op}}.$$

Suppose $n \le d \le e^{o(n)}$ and $d/n \to \infty$. Is

$$\mathrm{disc}_{\mathrm{op}}(A_1, \dots, A_n) = O(\sqrt{n})$$

with probability tending to one as $n \to \infty$, with a constant independent of d? The same question can be asked for the corresponding random Hankel model.

The evidence and the reduction are the content of Section 5. The deterministic theory of [6] already removes all d-dependence once $w \le n$, while Theorem 4.1 removes the expected-normalized full-bandwidth case, Corollary 4.8 shows that every polynomial-sized prescribed deterministic sign list, or independent random sign list with polynomial expected size, is harmless, and Corollary 4.9 gives the sharper interpolation according to the sign-family entropy $\log |\mathcal{S}|$. In particular, Corollary 4.10 shows that in the logarithmic sample window all sign patterns are harmless simultaneously. Thus Question 1.1 asks only whether realization-normalized typical full-bandwidth contractions behave like moderately banded ones. After the standard circulant approximation, signing Toeplitz matrices is signing the d Fourier samples

of their symbols, and the discrepancy is the vector discrepancy of those samples in ℓ^∞. Proposition 5.5 below shows that it is enough to prove a uniform empirical variance bound for those samples, and Proposition 5.13 proves that this profile has exactly the expected scale in the easier expected-normalized model. Corollary 5.2 records the deterministic robust form of the reduction: any edge-grid Hamming ball at the $O(\sqrt{n})$ scale is an actual $O(\sqrt{n})$ operator-norm robust certificate for full Toeplitz and Hankel contractions in the unbounded-aspect asymptotic range. Corollary 5.4 packages the same observation for all almost-sure sampled-count branches: eventual edge-grid robust batches become actual Toeplitz/Hankel operator-norm robust batches with the same centers and Hamming radii. Corollary 5.3 extends the transfer to arbitrary uniformly operator-bounded Toeplitz normalizations, and Corollary 6.10 applies it to the expected-normalized sampled robust centers. Theorem 5.14 gives the precise remaining one-matrix input: a lower tail whose product with $\min\{n, \log^2(e+d)\}$ tends to zero transfers the expected profile to realization normalization. Corollary 5.15 separates the deterministic data-dependent input: with M_a low-normalizer samples, the profile is $O(n/\log(e+d) + M_a \log(e+d))$ and the sign scale is $O(\sqrt{n + M_a \log^2(e+d)})$. The dyadic refinement Corollary 5.16 replaces this single low-normalizer count by a weighted budget $W_a = \sum_r 4^r M_{a,r} + \ell M_{a,\infty}$, $\ell = \log(e+d)$, so moderately low normalizers are charged by their actual dyadic depth and the $O(\sqrt{n})$ scale follows from $W_a = O(n/\ell)$. Corollary 5.17 converts this realized condition into a one-matrix dyadic-tail input: if $\mu^{\mathrm{dy}}_{d,a} = \mathbb{E}W_a/n$ is $o(1/\ell)$, Markov already gives the square-root package with high probability, and if $\mu^{\mathrm{dy}}_{d,a} \le \theta/\ell$ then the excess budget has Bernstein cost $\exp\{-cn/\ell^2\}$. Corollary 5.18 gives an equivalent practical check through the truncated inverse-square normalizer mass, and through inverse p-moments with the explicit factor $\ell^{(2-p)_+/2}$. Corollary 5.19 pushes the same inverse-moment input through finite random search, success-count, Hamming stability, and polynomial-confidence robust certificates. Corollary 5.20 gives the corresponding abundance, robust-fraction, and polynomial-batch versions under the same inverse-moment matrix exception. Corollary 5.21 records the finite arbitrary-batch version: the discarded fraction may be as small as a power of d with no upper bound on the sampled batch size. Corollary 5.23 promotes these count and robust-batch conclusions to almost-sure triangular-array statements under the matching summable sign tails, with the same Chernoff-scale replacement as the dyadic-tail criterion. Corollary 5.24 then lets the discarded robust-center fraction tend to zero for arbitrary sampled batches, provided ρ_m stays above the natural binomial summability scale. Corollary 5.25 transfers this inverse-moment sampled-batch package from edge-grid robust centers to actual Toeplitz/Hankel operator-norm robust centers under the full-bandwidth aspect assumptions. The schedule form then extracts all-integer and polynomial-dimension consequences from polynomial decay of this inverse mass, sparse stretched-exponential consequences from

logarithmic decay, and the critical $\log^3 d$ integer window from an $O(1/\ell)$ inverse-mass bound. Corollary 5.27 gives a checkable sufficient condition: a power lower-tail curve $F_{d,a}(t) \lesssim \ell^{-\gamma} t^{\beta}$, $\beta > 2$, forces $\mu^{\mathrm{dy}}_{d,a} \lesssim \ell^{-\gamma}$. In the critical $\gamma = 1$ case the mass at the top cutoff aN_d may be only order $1/\ell$, rather than $1/\ell^2$, provided the deeper tail has the quadratic-plus decay needed to make the dyadic weights summable. Corollary 5.35 pushes this power-curve input through the sampled random-search and Hamming-stable robust-center machinery, so the same weaker critical-density hypothesis controls not only existence of a good signing but also finite randomized certificate search. Corollaries 5.36, 5.37, and 5.38 record the corresponding finite abundance, robust-fraction, and polynomial-batch all-robust forms under the same power-curve matrix exception, with the Bernstein replacement in the critical density. Corollary 5.39 is the corresponding almost-sure triangular-array form: summability of the power-curve matrix exceptions gives eventual randomized certificates, and in the critical $\gamma = 1$ density the Bernstein cost only requires $\sum_m \exp\{-cn_m/\ell_m^2\} < \infty$. Corollary 5.40 adds the actual sampled algorithmic conclusion: after the same power-curve matrix summability, adding $\sum_m d_m^{-\alpha} < \infty$ makes $O_\alpha(1)$ sampled signs per dimension eventually contain robust certificates almost surely. Corollary 5.41 gives the corresponding growing-batch sampled-count theorem: summability of the sampled-list tail and the binomial count tail makes the whole sampled list controlled and all but a prescribed number of sampled signs good robust centers, eventually almost surely. Corollaries 5.42 and 5.43 promote the finite power-curve abundance package to almost-sure robust-fraction and all-robust polynomial-batch forms under the same matrix summability, with only the explicit sampled-sign summability terms added. Corollary 5.44 extracts the full integer-dimension critical-density consequence: the $\log^3 d$ sample window gives not only eventual one-certificate searches, but also eventual constant-size robust-fraction tests and all-robust polynomial batches. Corollary 5.45 uses the same all-integer critical window with the sampled-count theorem: for arbitrary sampled batch sizes the discarded robust-center fraction may vary with d and tend to zero, provided $\rho_d \geq ed^{-\tau}$ eventually for some $\tau < B - 1$. Corollary 5.46 extracts the sparse dimension schedules explicitly: if $d_m \geq \exp(cm^\kappa)$, then $\gamma > 1 + 1/\kappa$ is enough in the Markov form, and the critical $\gamma = 1$ Bernstein form needs only $n_m \geq C\ell_m^{2+\rho}$ for any $\rho > 0$. Corollary 5.47 is the actual sampled-list sparse version: under the same sparse power-curve windows, any fixed positive confidence exponent gives a constant number of sampled signs per dimension which eventually contain logarithmically robust certificates almost surely. Corollary 5.48 gives the corresponding sparse growing-batch count theorem: after only the sampled-list and binomial count summability terms, all but a prescribed number of sampled signs are eventually good robust centers, and polynomial-in-d_m or subcritical stretched-exponential batches are eventually entirely robust. Corollary 5.49 uses the same count tail with $r_m = \lceil \rho M_m \rceil$: on these sparse

schedules, an eventual $1-\rho$ robust fraction holds for arbitrary sampled batch sizes, with only the entropy-sensitive maximum remembering $\log M_m$. Corollary 5.50 lets this discarded fraction vary with m, so $\rho_m \to 0$ is allowed whenever $\rho_m \geq e d_m^{-\tau}$ eventually for some $\tau < B$. Corollary 5.28 is the fixed-budget consequence $M_a = O(n/\log^2(e+d))$, and carries the random-sign tail, sign-averaged maximum, deterministic certificate, and scalar edge-grid moment bounds; the same profile also gives a sampled-sign-list entropy bound, gives binomial success counts for independent random search, and makes the certificate robust under $\Theta(\log d)$ sign flips with sampled robust-center tails. Corollary 5.31 packages this as a joint matrix-and-search statement: $O_B(1)$ sampled signs find such a robust certificate with additional failure d^{-B}. In the larger sample window $n/\log^2(e+d) \to \infty$, Corollary 5.76 gives the tunable Chernoff tradeoff: $p_d(a) \leq \theta/\log^2(e+d)$ and a fixed empirical budget $R \geq 1$, $R > \theta$, cost $\exp\{-I(R,\theta)n/\log^2(e+d)\}$. Corollary 5.63 makes the summable polynomial regime concrete: if $p_d(a) \leq Cd^{-B}$ with $B > 1$, no dimension sparsification is needed, and the almost-sure profile holds eventually along the full sequence of integer dimensions. Corollary 5.52 gives the same explicit polylogarithmic, logarithmic-window, and polynomial failure rates for the full random-search and robust-center certificate package. Corollary 5.32 packages the corresponding finite abundance form: after the same non-Chernoff matrix exception, the whole sampled list, the $1-\rho$ good-sign fraction, and the $1-\rho$ robust-center fraction hold simultaneously outside an explicit binomial sign tail. Corollary 5.33 turns this into a constant-size randomized robust-fraction test with an added d^{-S} confidence term, and Corollary 5.34 gives the finite polynomial-batch all-robust consequence $M \leq d^{\beta}$. Corollary 5.54 records that the same Borel–Cantelli transfer also preserves the sampled random-search bounds and logarithmic Hamming-stable robust-center certificates along the triangular array. Corollary 5.55 upgrades this to actual growing sampled batches: after the sampled-list maximum tail and the binomial failure-count tail are summable, all but the prescribed number of sampled signs are eventually good and robust. In the polynomial lower-tail regime, batches $M_m \leq d_m^{\beta}$ are eventually entirely robust along polynomial dimension schedules whenever $(A-\beta)\kappa > 1$. Corollary 5.56 isolates the logarithmic-tail version in its sharp summability form: in logarithmic windows it is enough that $\sum_m (\log(e+d_m))^{\min\{\gamma,2\}-B} < \infty$, together with the sampled sign count tails. The later stretched, log-corrected, and iterated-log schedule corollaries are explicit integral-test checks of this single condition. Corollary 5.57 recasts this as a dimension-schedule design rule: it is enough to choose any lower logarithmic growth profile $h_m \leq \log(e+d_m)$ with $\sum_m h_m^{\min\{\gamma,2\}-B} < \infty$. Corollary 5.58 then lets the discarded robust-center fraction vary with m: arbitrary sampled batch sizes have a $1-\rho_m$ robust fraction whenever ρ_m stays above the natural exponential binomial-tail threshold, so $\rho_m \to 0$ is allowed. Corollary 5.59 takes $r_d = \lceil \rho M_d \rceil$ in the full integer polynomial-tail regime: arbitrary sampled batch sizes have an eventual $1-\rho$ robust-center

fraction once the count exponent satisfies $A > 1$. Corollary 5.60 lets this full-integer discarded fraction vary with d: if $\rho_d \geq ed^{-\tau}$ for some $\tau < A - 1$, then $\rho_d \to 0$ is allowed and the robust-center fraction tends to one with no upper bound on the batch size. Corollary 5.61 extracts the corresponding constant-size robust-fraction theorem: once the matrix transfer is summable and $\sum_m d_m^{-S} < \infty$, $O_{\rho,S}(1)$ sampled signs eventually contain a $1 - \rho$ fraction of robust centers. Corollary 5.62 upgrades this from a conditional sign-search statement to actual sampled-list success: if $\sum_m d_m^{-\alpha} < \infty$, then $O_\alpha(1)$ sampled signs find robust certificates for all but finitely many m almost surely over both the matrices and the sign samples. Corollary 5.64 then records the sparser logarithmic-tail regime: polynomial lower-normalizer tails, and logarithmic tails with exponent greater than 3, give the almost-sure profile along exponentially sparse dimension schedules. Corollary 5.65 upgrades this sparse regime to actual sampled robust certificates: on exponentially sparse schedules the sampled-list summand $d_m^{-\alpha}$ is summable for every $\alpha > 0$. Corollary 5.66 gives the polynomial dimension-schedule counterpart: if $p_d(a) \leq Cd^{-B}$ and $d_m \gtrsim m^\kappa$, then the same almost-sure profile follows whenever $B\kappa > 1$, after choosing the auxiliary d^{-A} event with $A\kappa > 1$. Corollary 5.67 adds the sampled-list version on the same schedules, with the additional summability condition $\alpha\kappa > 1$ for the finite random-sign list; the robust-fraction form only needs $S\kappa > 1$ for the fraction confidence. In logarithmic sample windows $n_m \lesssim \log^\gamma d_m$ along stretched-exponential schedules $d_m \geq \exp(cm^\kappa)$, Corollary 5.68 improves this logarithmic threshold to $B > \min\{\gamma, 2\} + 1/\kappa$. Corollary 5.69 records the log-corrected endpoint: for $d_m \geq \exp(cm^\kappa(\log m)^\lambda)$, the same conclusion holds at $\kappa(B - \min\{\gamma, 2\}) = 1$ provided $\lambda(B - \min\{\gamma, 2\}) > 1$, with the sampled-certificate and robust-fraction consequences on the same summable schedule. Corollary 5.70 adds the matching sampled-count theorem at this endpoint, including all-robust polynomial-in-d_m batches and subcritical log-corrected stretched-exponential batches. Corollary 5.71 pushes the same sampled-count statement through one more integral-test layer: if the first log-corrected endpoint is itself critical, an additional $\log(e + \log(e + m))^\mu$ factor with $\mu(B - \min\{\gamma, 2\}) > 1$ again gives eventual sampled-count certificates, with the corresponding lexicographic batch-growth rule. Corollary 5.57 is the schedule design version of the same mechanism: any deterministic lower profile $h_m \leq \log(e + d_m)$ with $\sum_m h_m^{\min\{\gamma,2\}-B} < \infty$ suffices, including the same all-robust polynomial-batch and arbitrary-batch fraction consequences. Corollary 5.58 sharpens the arbitrary-batch fraction conclusion: the discarded fraction may be a sequence $\rho_m \to 0$, as long as $\rho_m \geq ed_m^{-\tau}$ eventually for some $\tau < A$. Corollary 5.72 adds the actual sampled-list version of this logarithmic-window transfer; on these stretched-exponential schedules, every $\alpha > 0$ gives a constant-size random sign list which eventually contains robust certificates. Corollary 5.73 gives the matching robust-fraction version: for every $S > 0$ and $0 < \rho \leq 1$, a constant-size sampled list eventually has

$O(\sqrt{n_m})$ maximum and a $1-\rho$ fraction of logarithmically robust centers. Corollary 5.74 adds the corresponding growing-batch count form: on the same logarithmic-window sparse schedules, summable sampled-list and binomial count tails force all but the prescribed number of sampled signs to be good and robust eventually. Corollary 5.75 takes $r_m = \lceil \rho M_m \rceil$ in that count form, giving an eventual $1-\rho$ robust-center fraction for arbitrary sampled batch sizes. Corollary 5.98 is the specialization $R = 2L$, $\theta = L$, while Corollary 5.97 records the relative excess form $R = (1+\eta)L$. Corollary 5.77 records the summable almost-sure profile/signing/moment version. Corollary 5.78 carries these Chernoff and relative-excess rates to the sampled random-search and robust-center certificate package. Corollary 5.79 records the finite joint matrix-and-sign form of that package: sampled-list maxima, good-sign counts, and robust-center counts all inherit the Chernoff matrix exception plus their deterministic edge-profile sign tails. Corollary 5.80 packages the corresponding finite abundance form: with the sampled-list maximum controlled, all but a chosen ρ-fraction of sampled signs are good and robust outside the matching binomial fraction tail. Corollary 5.81 turns this into a constant-size randomized test: for prescribed ρ and polynomial confidence exponent S, $O_{\rho,S}(1)$ sampled signs already certify a $1-\rho$ robust fraction. Corollary 5.82 extracts the finite polynomial-batch consequence: if $M \le d^{\beta}$, then every sampled sign in the batch is a robust center with only polynomial additional failure. Corollary 5.83 extracts the finite actual sampled-list consequence: $O_B(1)$ sampled signs find a robust certificate with only the additional d^{-B} sign-search failure. Corollary 5.84 records the summable almost-sure search version; together they include an all-dimensions consequence at $n(d) \gtrsim \log^3 d$. Corollary 5.85 upgrades the summable Chernoff search transfer to actual sampled-list robust certificates after adding the single summable sign-search term $d_m^{-\alpha}$. Corollary 5.86 upgrades the finite joint tails to a Borel count theorem for growing sampled-sign batches: after a summable sampled-list maximum tail and a summable binomial failure-count tail, all but a prescribed number of sampled signs are eventually good and robust. Corollary 5.87 specializes this count theorem to the full integer sequence in the critical $\log^3 d$ Chernoff window: arbitrary sampled batch sizes have robust-center fractions tending to one whenever the discarded fraction satisfies $\rho_d \ge e d^{-\tau}$ eventually for some $\tau < B-1$. Corollary 5.88 extracts the constant-size robust-fraction consequence: if the d_m^{-S} confidence series is summable, then $O_{\rho,S}(1)$ sampled signs eventually contain a $1-\rho$ fraction of robust centers almost surely. Corollary 5.89 makes this explicit on polynomial dimension schedules $d_m \gtrsim m^{\kappa}$, where the confidence requirement becomes $S\kappa > 1$. Corollary 5.90 extracts the general critical-log schedule criterion. Corollary 5.91 makes the sparse large-sample consequence explicit: along stretched-exponential schedules, any sample window above the quadratic logarithmic scale is already summable. Corollary 5.92 adds the actual sampled-list version on the same sparse schedules, with no lower bound on α beyond positivity. Corollary 5.93 adds the matching robust-fraction

version: on the same sparse schedules every $S > 0$ gives a constant-size sampled list with an eventual $1 - \rho$ robust fraction. Corollary 5.94 records the corresponding growing-batch count form: on stretched-exponential schedules, polynomial-in-d_m batches are eventually entirely robust as soon as the sign-tail exponent beats the batch exponent. Corollary 5.95 again takes $r_m = \lceil \rho M_m \rceil$, so arbitrary sampled batch sizes have an eventual $1 - \rho$ robust-center fraction under the same sparse Chernoff windows. Corollary 5.96 allows the Chernoff discarded fraction to vary with m and tend to zero, provided $\rho_m \geq e d_m^{-\tau}$ eventually for some $\tau < B$. Theorem 5.99 verifies this input under convex concentration, covering real or complex Gaussian and bounded coefficient laws as well as Rademacher coefficients, and Corollary 5.100 adds the unbounded strongly log-concave real and complex product laws. Corollary 5.101 further closes the class under fixed finite independent linear combinations, giving examples such as Gaussian plus Rademacher noise. Corollary 5.102 adds centered bounded Gaussian scale mixtures whose volatility is uniformly bounded above and below; this covers variance mixtures by conditioning on the volatility field and using Gaussian comparison plus concentration. Theorem 5.103 then removes identical distribution from the conclusion: the n matrices may be drawn from different laws in any uniform lower-normalizer library, including the convex, strongly-log-concave, finite-convolution, and bounded scale-mixture classes just listed. Corollary 5.104 records the pathwise invariance under arbitrary positive matrix-level radial scalings, so covered realization-normalized models may be multiplied by heavy-tailed amplitudes without changing the normalized array. Theorem 5.105 further replaces the uniform per-law lower-tail requirement by an aggregate lower-tail budget $\sum_i p_{d,i}(a) \min\{1, \log^2(e+d)/n\} = o(1)$, allowing sparse bad-law contamination or nonuniform law quality. Theorem 5.106 replaces the cutoff budget by an aggregate dyadic depth budget $\log(e+d) n^{-1} \sum_i \mathbb{E} Y_{d,i,a}^{\mathrm{dy}} = o(1)$, which distinguishes moderately low normalizers from genuinely tiny ones. Corollary 5.107 makes this criterion checkable from aggregate truncated inverse-normalizer moments. Corollary 5.108 upgrades these heterogeneous conclusions to sampled Toeplitz/Hankel operator certificates, robust-fraction tests, polynomial batches, and almost-sure sampled-batch schedules. Corollary 5.109 records the corresponding almost-sure sampled operator schedules under aggregate inverse-moment summability. Corollary 5.110 extracts the fixed-fraction, vanishing-fraction, and polynomial-batch consequences for arbitrary sampled batch sizes. Corollary 5.111 adds the entropy-sensitive whole-list edge and operator-norm maximum to the same heterogeneous inverse-moment sampled-batch conclusion. Corollary 5.112 extracts the concrete all-integer, polynomial-schedule, stretched-exponential, and Bernstein-window thresholds for these batch-maximal operator certificates. Corollary 5.113 records the corresponding concrete aggregate inverse-p thresholds. Corollary 5.114 extends those thresholds to random law environments whose empirical inverse-p budget failures are summable. Corollary 5.115 gives expectation and random-catalog criteria

which imply that summability. Corollary 5.116 records the sharper independent bounded-catalog Bernstein check. Corollary 5.117 gives the corresponding predictable bounded-catalog Freedman check for adaptive environment selection. Corollary 5.118 removes boundedness by adding a summable tail-excess budget. Corollary 5.119 verifies that excess from conditional moment envelopes. Corollary 5.120 chooses the truncation level automatically from a target tail exponent. Corollary 5.121 makes the same automatic truncation simultaneous over finite active menus of predictable catalog environments. Corollary 5.122 extracts concrete active-prefix growth thresholds from that menu criterion. Corollary 5.123 allows the exposed active-prefix length itself to be random, after adding only a summable overflow-tail condition. Corollary 5.124 verifies that condition from sub-Poisson proposal-count and occupancy envelopes. Corollary 5.125 derives the same envelope from adaptive one-step activation intensities. Corollary 5.126 allows those predictable intensity budgets to fail on a summable exceptional set. Corollary 5.127 gives direct moment tests for that random intensity-envelope failure. Corollary 5.128 instead certifies the random intensity envelope by scalar Freedman tails, and Corollary 5.129 adds a truncation-and-excess variant for unbounded predictable loads. Corollary 5.130 chooses the truncation automatically from a conditional moment envelope and records the resulting growth-rate tests. Corollary 5.131 then chooses the deterministic active prefix from the same intensity envelope. Corollary 5.132 combines this automatic prefix with active-menu catalog budgets to give the sampled certificates in one check. Corollary 5.133 extracts the corresponding concrete rate schedules. Corollary 5.134 turns their catalog moment series into explicit sample-size lower bounds under aggregate catalog-growth envelopes. Corollary 5.135 then extracts compatible sample-size windows from real catalog-growth exponents. Corollary 5.136 turns those aggregate exponents into per-stream catalog moment budgets. Corollary 5.137 also tunes the mean-budget exponent, leaving only per-stream first and s-moment catalog rates. Corollary 5.138 turns the same input into constant-size sampled operator-norm robust certificates with logarithmic Hamming radius, while Corollary 5.139 gives the matching operator-norm robust-fraction and polynomial-batch abundance forms. Corollary 5.140 records the almost-sure sampled-batch schedule version. Finally Corollary 5.142 records the Gaussian specialization. Thus the remaining edge-profile gap is a normalizer-tail problem for sub-Gaussian laws beyond these concentration, finite-convolution, and bounded scale-mixture classes. Equivalently, in the residual model one must control $\max_\omega |\sum_i \varepsilon_i P_i(\omega)|$ over the $O(d)$ edge nodes ω, where P_i is the normalized random symbol of A_i. The fluctuation of a single such edge maximum is the near-circle edge problem of the companion paper [7], whose local-limit problem for the unit-disk zero count remains open and is driven by the near-circle shell. Question 1.1 is the point at which that edge analysis and the deterministic banded discrepancy theory meet.

We are explicit throughout about what is proved and what is not. Theorem 3.2 (the pathwise banded bound), Theorem 3.4 (the bounded-aspect full-bandwidth endpoint), Theorem 4.1 and Corollaries 4.8–4.10 (the expected-normalized full random model and its deterministic/random polynomial-list, sign-entropy, and logarithmic-window uniform-in-sign strengthenings), Theorem 4.2 and Corollaries 4.3, 4.4, 4.5, and 4.6–4.7 (profile-normalized independent non-identical lag arrays and their finite and almost-sure sign/coefficient-family entropy bounds), and Corollary 6.10 (expected-normalized sampled operator certificates), Theorem 5.14 and its quantitative and almost-sure refinements, including Corollary 5.2, Corollary 5.3, Corollary 5.4, Corollaries 5.29, 5.30, 5.52, 5.77, 5.78, 5.79, 5.80, 5.81, 5.82, 5.83, 5.18, 5.19, 5.20, 5.21, 5.22, 5.23, 5.24, 5.25, 5.26, 5.36, 5.37, 5.38, 5.41, 5.42, 5.43, 5.44, 5.45, 5.46, 5.47, 5.48, 5.49, 5.50, 5.84, 5.85, 5.86, 5.87, 5.88, 5.89, 5.90, 5.91, 5.92, 5.93, 5.94, 5.95, 5.96, 5.54, 5.32, 5.33, 5.34, 5.55, 5.56, 5.57, 5.58, 5.59, 5.60, 5.61, 5.63, 5.66, 5.68, 5.73, 5.74, 5.75, 5.70, 5.71, and Theorem 5.99 (the convex-concentrated realization-normalized case), together with Corollary 5.100 (strongly log-concave real and complex coefficient laws), Corollary 5.101 (finite independent linear combinations of convex-concentrated laws), Corollary 5.102 (bounded Gaussian scale-mixture coefficient laws), Theorem 5.103 (heterogeneous non-identically distributed lower-normalizer libraries), Corollary 5.104 (matrix-level radial scalings), Theorem 5.105 (heterogeneous average lower-tail budgets), Theorem 5.106 (heterogeneous dyadic lower-tail budgets), Corollary 5.107 (heterogeneous inverse-moment lower-normalizer budgets), Corollary 5.108 (heterogeneous sampled operator certificates and batches), Corollary 5.109 (heterogeneous inverse-moment sampled operator schedules), Corollary 5.110 (heterogeneous inverse-moment sampled operator robust fractions), Corollary 5.111 (heterogeneous inverse-moment sampled-list operator maxima), Corollary 5.112 (concrete heterogeneous inverse-moment sampled-list operator schedules), Corollary 5.113 (concrete heterogeneous aggregate inverse-p sampled-list operator schedules), Corollary 5.114 (random-environment aggregate inverse-p sampled-list operator schedules), Corollary 5.115 (expectation and random-catalog criteria for those schedules), Corollary 5.116 (independent bounded-catalog Bernstein criteria for those schedules), Corollary 5.117 (predictable bounded-catalog Freedman criteria for those schedules), Corollary 5.118 (truncated Freedman criteria for unbounded predictable catalogs), Corollary 5.119 (conditional-moment criteria for the truncated predictable-catalog schedules), Corollary 5.120 (automatic truncation schedules for those moment criteria), Corollary 5.121 (active-menu automatic truncation for predictable catalogs), Corollary 5.122 (concrete active-prefix schedules for those menus), Corollary 5.123 (random active-prefix counts with summable overflow), Corollary 5.124 (sub-Poisson active-count overflow tests), Corollary 5.125 (predictable active-intensity overflow

tests), Corollary 5.126 (random predictable-intensity overflow tests), Corollary 5.127 (moment tests for random predictable-intensity envelopes), Corollary 5.128 (Freedman tests for random predictable-intensity envelopes), Corollary 5.129 (truncated Freedman tests for unbounded predictable-intensity envelopes), Corollary 5.130 (automatic moment truncation for predictable active intensities), Corollary 5.131 (automatic deterministic prefixes from predictable active-intensity moments), Corollary 5.132 (end-to-end automatic active-prefix sampled certificates), Corollary 5.133 (concrete end-to-end automatic active-prefix schedules), Corollary 5.134 (sample-size forms of those schedules), Corollary 5.135 (compatible sample-size windows for catalog-growth exponents), Corollary 5.136 (per-stream catalog budgets for those compatible windows), Corollary 5.137 (first-moment-tuned per-stream compatible windows), Corollary 5.138 (constant-size sampled operator-norm robust certificates in the same lower-tail classes) and Corollary 5.139 (robust-fraction and polynomial-batch operator-certificate abundance), and Corollary 5.140 (the corresponding almost-sure sampled-batch schedules), are theorems. Together they imply Corollary 5.142 and the same realization-normalized discrepancy conclusion for Rademacher, bounded real or complex, strongly log-concave real or complex coefficient laws, bounded Gaussian scale mixtures, and fixed finite independent linear combinations of those convex-concentrated inputs, even when the matrices are sampled heterogeneously from such a uniform library; the heterogeneous sampled-certificate corollary gives the same randomized search and robust-batch forms for those libraries, and the average-budget theorem allows nonuniform law quality whenever the aggregate low-normalizer budget is small. Question 1.1 remains open at the full arbitrary sub-Gaussian level; Section 5 gives the remaining lower-tail reduction and supporting heuristics. Section 6 records proved fixed-edge fourth- and sixth-moment identities, uniform phase-square cancellation and scalar even-moment estimates in the expected-normalized model, packages these scalar moments into an expected-normalized edge-maximum moment/signing and sampled robust-certificate theorem, and gives a moment-method template for the still-open all-frequency higher-moment problem.

Companions. This note sits between two companions. The deterministic discrepancy engine, including the banded Toeplitz theorem used in Section 3 and the circulant reduction used in Section 5, is [6]. The fine behavior of the random-Toeplitz spectral edge, including the local limit problem that controls the obstruction in Section 5, is [7].

2. Setup

We use the conventions of [6]. For a finite lag sequence $(a_k)_{|k|<d}$ the Toeplitz matrix is $T_d(a) = (a_{r-s})_{0\le r,s<d}$, and if $\varphi \in L^\infty(\mathbb{T})$ has $\widehat{\varphi}(k) = a_k$ for $|k| < d$ then φ induces $T_d(a)$. A Toeplitz matrix has *half-bandwidth* w if $a_k = 0$ for $|k| > w$, equivalently if its symbol is a trigonometric polynomial

of degree at most w. The full-bandwidth case is $w = d - 1$. The reversal matrix J_d, $(J_d x)_r = x_{d-1-r}$, turns a Hankel matrix into a Toeplitz matrix of equal operator norm, so every statement below transfers from Toeplitz to Hankel by the Toeplitz–Hankel reversal argument of [6]; we phrase the results for Toeplitz matrices.

A *random Toeplitz matrix* is $T_d(\xi)$ with i.i.d. entries $\xi_{-(d-1)}, \dots, \xi_{d-1}$, mean zero and unit variance. Its operator norm is of order $\sqrt{d \log d}$ [2]; we write

$$N_d := \mathbb{E}\,\|T_d(\xi)\|_{\mathrm{op}} \asymp \sqrt{d \log d},$$

and call $A = T_d(\xi)/N_d$ an *expected-normalized random Toeplitz matrix*; it satisfies $\mathbb{E}\,\|A\|_{\mathrm{op}} \asymp 1$, but it is not a contraction almost surely. When a statement needs honest contractions we instead assume the realized bound $\|A\|_{\mathrm{op}} \le 1$. A *random banded Toeplitz contraction of half-bandwidth w* means a banded Toeplitz random matrix normalized so that this realized operator-norm bound holds; the pathwise results below use only that bound and make no use of the law.

Throughout, C, c are absolute constants that may change from line to line, and ε ranges over $\{-1, 1\}^n$.

3. Pathwise deterministic bounds

The banded and bounded-aspect full-bandwidth regimes need no probability. We restate the deterministic inputs from the companion paper and read off the random corollaries.

Theorem 3.1 (Banded Toeplitz signing; [6])**.** *There is an absolute constant C such that the following holds. Let $A_i = T_d(p_i)$ be arbitrary complex Toeplitz matrices whose symbols have degree at most w, with $\|A_i\|_{\mathrm{op}} \le 1$, and suppose $d \ge 4w$ and $w \ge 1$. Then*

$$\mathrm{disc}_{\mathrm{op}}(A_1, \dots, A_n) \le C\sqrt{n \log\left(e + \frac{w}{n}\right)}.$$

The multilevel version from [6] replaces w by w^r for r-level banded block Toeplitz or block Hankel contractions with per-coordinate half-bandwidth w.

This is a statement about every fixed tuple of matrices satisfying the hypotheses. Since a random banded Toeplitz contraction satisfies them on every realization, the conclusion holds almost surely with no probabilistic argument.

Theorem 3.2 (Pathwise random banded discrepancy)**.** *Let $A_1, \dots, A_n$ be random banded Toeplitz contractions of half-bandwidth w, drawn from any joint law whatsoever, normalized so that $\|A_i\|_{\mathrm{op}} \le 1$ almost surely, on a dimension $d \ge 4w$. Then on every realization*

$$\mathrm{disc}_{\mathrm{op}}(A_1, \dots, A_n) \le C\sqrt{n \log\left(e + \frac{w}{n}\right)},$$

with C absolute. In particular, for $w \le n$,

$$\operatorname{disc}_{\mathrm{op}}(A_1, \dots, A_n) = O(\sqrt{n}) \qquad \textit{almost surely,}$$

with a constant independent of the dimension d. For r-level banded block Toeplitz or block Hankel contractions of per-coordinate half-bandwidth w the same holds with w replaced by w^r, so the bound is $O_r(\sqrt{n})$ almost surely whenever $w^r \le n$, again uniformly in the ambient dimension.

Proof. Fix a realization. The matrices A_i are banded Toeplitz contractions of half-bandwidth w with $d \ge 4w$, so Theorem 3.1 applies and yields the displayed bound; the bound is the same on every realization, hence holds surely. For $w \le n$ the logarithm is at most $\log(e+1)$, a constant, so the right side is $O(\sqrt{n})$, with no dependence on d because none appears in the bound. The multilevel statement is identical, using the multilevel banded theorem of [6] and $w^r \le n$. □

So for bounded bandwidth the random discrepancy problem is fully resolved, and the resolution is deterministic and dimension-free. Randomness contributes nothing because the deterministic bound already covers the worst case in the band. This is the baseline against which the full-bandwidth question is the real content.

Remark 3.3. The dimension-freeness is the substantive point. A random banded Toeplitz contraction lives in $\mathrm{M}_d(\mathbb{C})$ with d possibly enormous, yet its signed superposition has operator norm $O(\sqrt{n})$ as soon as $w \le n$, with no penalty for the size of the ambient matrices. This is the random shadow of the entropy mechanism of [6]: a banded symbol carries $O(w)$ scalar degrees of freedom regardless of d, so the relevant Fourier norming set has size $O(w)$, not $O(d)$.

The full-bandwidth scalar theorem of [6] gives a second pathwise endpoint. It is not dimension-free for $d/n \to \infty$, but it settles the random question whenever the aspect ratio is bounded.

Theorem 3.4 (Pathwise full bandwidth at bounded aspect ratio)**.** *There is an absolute constant C such that every collection of arbitrary complex $d \times d$ Toeplitz contractions satisfies*

$$\operatorname{disc}_{\mathrm{op}}(A_1, \dots, A_n) \le C\sqrt{n \log\left(e + \frac{d}{n}\right)}.$$

Consequently, for every fixed $\Lambda < \infty$, every random full Toeplitz contraction model with $d \le \Lambda n$ satisfies

$$\operatorname{disc}_{\mathrm{op}}(A_1, \dots, A_n) \le C_\Lambda \sqrt{n}$$

on every realization. The same statements hold for Hankel contractions.

Proof. The first display is the scalar Toeplitz theorem of [6]. If $d \le \Lambda n$, then $\log(e + d/n) \le \log(e + \Lambda)$, giving the stated constant $C_\Lambda = C\sqrt{\log(e+\Lambda)}$.

The Hankel case follows from the Toeplitz–Hankel reversal argument in the same companion paper. □

4. The full-bandwidth split

We now take $w = d-1$ and $d/n \to \infty$, and we ask when random signs or optimized signs improve on the deterministic $\sqrt{n \log(e + d/n)}$ bound. Expected normalization and realization normalization lead to different questions.

4.1. **Expected normalization is already solved by fixed signs.** We use two normalizations and keep them apart. The *entry* normalization takes $B = T_d(\xi)$ with unit-variance entries, so $\|B\|_{\mathrm{op}} \asymp \sqrt{d \log d}$ by Meckes [2] and $\|B\|_{\mathrm{HS}} \asymp d$. The *expected* normalization divides by the operator-norm scale, $A = T_d(\xi)/N_d$ with $N_d = \mathbb{E}\,\|T_d(\xi)\|_{\mathrm{op}} \asymp \sqrt{d \log d}$, so that $\mathbb{E}\,\|A\|_{\mathrm{op}} \asymp 1$. This normalization preserves the exact Toeplitz superposition structure under any fixed choice of signs, and that observation already gives the square-root bound.

Theorem 4.1 (Expected-normalized full Toeplitz is fixed-sign)**.** *Let $d \geq 2$, and let $B_i = T_d(\xi^{(i)})$, $1 \leq i \leq n$, be independent full Toeplitz matrices whose lag variables are independent, mean zero, unit variance, and have sub-Gaussian norm at most K. Put*

$$N_d := \mathbb{E}\,\|T_d(\xi)\|_{\mathrm{op}} \asymp_K \sqrt{d \log d}, \qquad A_i := B_i/N_d.$$

Then for every $L > 0$ and every deterministic sign vector $\varepsilon \in \{-1, 1\}^n$,

$$\mathbb{P}\left\{ \left\| \sum_{i=1}^{n} \varepsilon_i A_i \right\|_{\mathrm{op}} > C_{L,K} \sqrt{n} \right\} \leq d^{-L}.$$

Consequently, with probability at least $1 - d^{-L}$,

$$\mathrm{disc}_{\mathrm{op}}(A_1, \ldots, A_n) \leq C_{L,K} \sqrt{n}$$

in the expected-normalized full Toeplitz model, without optimizing the signs. The same conclusion holds for the corresponding expected-normalized Hankel model.

Proof. Fix the signs. The signed sum of the unnormalized matrices is again Toeplitz:

$$\sum_i \varepsilon_i B_i = T_d(\eta), \qquad \eta_k := \sum_{i=1}^{n} \varepsilon_i \xi_k^{(i)}.$$

The variables η_k, $|k| < d$, are independent, centered, and sub-Gaussian with norm at most $CK\sqrt{n}$. Let

$$Q(e^{i\theta}) = \sum_{|k|<d} \eta_k e^{ik\theta}$$

be the associated trigonometric symbol. Since $T_d(\eta)$ is a compression of Laurent multiplication by Q,

$$\|T_d(\eta)\|_{\mathrm{op}} \le \|Q\|_{L^\infty(\mathbb{T})}.$$

For each fixed θ, $Q(e^{i\theta})$ has sub-Gaussian norm at most $CK\sqrt{nd}$. Choose a uniform grid $\Gamma \subset \mathbb{T}$ of $8d$ points. The Bernstein inequality for trigonometric polynomials gives $\|Q'\|_\infty \le d\,\|Q\|_\infty$, hence $\|Q\|_\infty \le 2\max_\Gamma |Q|$. Therefore, for a large enough constant $C_{L,K}$,

$$\mathbb{P}\{\|Q\|_\infty > C_{L,K}\sqrt{nd\log d}\} \le 8d \cdot 2\exp(-cC_{L,K}^2 \log d/K^2) \le d^{-L}.$$

Meckes' two-sided estimate [2] gives $N_d \ge c_K\sqrt{d\log d}$. Dividing by N_d yields the displayed $C_{L,K}\sqrt{n}$ bound. The discrepancy bound follows by taking this one fixed sign vector in the minimum. Hankel matrices are reduced to Toeplitz matrices by the reversal J_d, which preserves operator norm. □

Theorem 4.2 (Profile-normalized independent lags are fixed-sign)**.** *Let $d \ge 2$. For $1 \le i \le n$ and $|k| < d$, let $\zeta_{i,k}$ be independent centered real or complex random variables. Let $\sigma_{i,k} \ge 0$ be deterministic numbers such that $\zeta_{i,k}$ has sub-Gaussian norm at most $K\sigma_{i,k}$, and put*

$$S_i := \left(\sum_{|k|<d} \sigma_{i,k}^2\right)^{1/2}.$$

Let $B_i = T_d((\zeta_{i,k})_{|k|<d})$, and define

$$A_i := \begin{cases} B_i/(S_i\sqrt{\log(e+d)}), & S_i > 0, \\ 0, & S_i = 0. \end{cases}$$

Then for every $L > 0$ and every deterministic sign vector $\varepsilon \in \{-1,1\}^n$,

$$\mathbb{P}\left\{\left\|\sum_{i=1}^n \varepsilon_i A_i\right\|_{\mathrm{op}} > C_{L,K}\sqrt{n}\right\} \le d^{-L}.$$

More generally, if $\mathcal{S} \subset \{-1,1\}^n$ is deterministic and $|\mathcal{S}| \le d^\alpha$, then with probability at least $1 - d^{-L}$,

$$\max_{\varepsilon\in\mathcal{S}} \left\|\sum_{i=1}^n \varepsilon_i A_i\right\|_{\mathrm{op}} \le C_{L,\alpha,K}\sqrt{n}.$$

The same conclusions hold for the corresponding profile-normalized full Hankel matrices.

Proof. It is enough to prove the sign-list assertion; the first display is the case $|\mathcal{S}| = 1$. Fix a sign vector ε, let $I = \{i : S_i > 0\}$, and write the Toeplitz symbol of the signed normalized sum as

$$Q_\varepsilon(e^{i\theta}) = \sum_{i\in I} \varepsilon_i \frac{1}{S_i\sqrt{\ell}} \sum_{|k|<d} \zeta_{i,k} e^{ik\theta}, \qquad \ell = \log(e+d).$$

For each fixed θ, the i-th inner normalized summand has sub-Gaussian norm at most

$$C_K \frac{(\sum_{|k|<d} \sigma_{i,k}^2)^{1/2}}{S_i \sqrt{\ell}} = \frac{C_K}{\sqrt{\ell}}.$$

Independence over i therefore gives $Q_\varepsilon(e^{i\theta})$ sub-Gaussian norm at most $C_K\sqrt{n/\ell}$, uniformly in θ and ε.

Let $\Gamma \subset \mathbb{T}$ be a uniform grid of $8d$ points. Since Q_ε is a trigonometric polynomial of degree at most $d-1$, Bernstein's inequality gives $\|Q_\varepsilon\|_{L^\infty(\mathbb{T})} \le 2\max_{\theta\in\Gamma} \left|Q_\varepsilon(e^{i\theta})\right|$. The Toeplitz operator norm is bounded by $\|Q_\varepsilon\|_{L^\infty(\mathbb{T})}$. Hence, after increasing $C_{L,\alpha,K}$,

$$\begin{aligned}\mathbb{P}\left\{\max_{\varepsilon\in\mathcal{S}} \left\|\sum_i \varepsilon_i A_i\right\|_{\mathrm{op}} > C_{L,\alpha,K}\sqrt{n}\right\} &\le 16d\,|\mathcal{S}|\,\exp(-c_K C_{L,\alpha,K}^2 \ell) \\ &\le d^{-L}.\end{aligned}$$

The Hankel case follows by multiplying the corresponding Toeplitz matrix by the reversal J_d, which preserves operator norm. □

Corollary 4.3 (Profile-normalized sign entropy bound)**.** *In the setting of Theorem 4.2, fix $L > 0$. Let $\mathcal{S} \subset \{-1,1\}^n$ be a deterministic sign family with $|\mathcal{S}| \le M$, where $M \ge 1$. Then with probability at least $1 - d^{-L}$,*

$$\max_{\varepsilon\in\mathcal{S}} \left\|\sum_{i=1}^n \varepsilon_i A_i\right\|_{\mathrm{op}} \le C_K \sqrt{n\left(L + 1 + \frac{\log(eM)}{\log(e+d)}\right)}.$$

The same statement holds if $\mathcal{S}$ is random but independent of the matrices and satisfies $\mathbb{E}\,|\mathcal{S}| \le M$. In particular, with probability at least $1 - d^{-L}$,

$$\max_{\varepsilon\in\{-1,1\}^n} \left\|\sum_{i=1}^n \varepsilon_i A_i\right\|_{\mathrm{op}} \le C_K \sqrt{n\left(L + 1 + \frac{n}{\log(e+d)}\right)},$$

and if $n \le \lambda \log(e+d)$, all sign patterns are simultaneously $O_{L,K,\lambda}(\sqrt{n})$. The same conclusions hold for the corresponding profile-normalized full Hankel matrices.

Proof. The proof of Theorem 4.2, before the final constant is chosen, gives the fixed-sign tail

$$\mathbb{P}\left\{\left\|\sum_i \varepsilon_i A_i\right\|_{\mathrm{op}} > C_K u\sqrt{n}\right\} \le (e+d)^{-u^2} \qquad (u \ge 1)$$

for every deterministic ε. Take

$$u^2 = L + 1 + \frac{\log(eM)}{\log(e+d)}.$$

A union bound over deterministic $\mathcal{S}$ gives the first assertion. For a random independent $\mathcal{S}$, condition on the realized family and then average:

$$\mathbb{P}\left\{\max_{\varepsilon\in\mathcal{S}}\left\|\sum_i \varepsilon_i A_i\right\|_{\mathrm{op}} > C_K u\sqrt{n}\right\} \le \mathbb{E}\,|\mathcal{S}|\;(e+d)^{-u^2} \le d^{-L}.$$

Taking $M = 2^n$ gives the whole-cube estimate. The Hankel statement follows again by Toeplitz–Hankel reversal. □

Corollary 4.4 (Profile-normalized Borel sign entropy)**.** *For each m, let $d_m \ge 2$, $n_m \ge 1$, and let $(A_{m,i})_{i\le n_m}$ be a profile-normalized full Toeplitz array satisfying the hypotheses of Theorem 4.2 with the same sub-Gaussian constant K. Let $\mathcal{S}_m \subset \{-1,1\}^{n_m}$ be nonempty and deterministic, or nonempty almost surely, random, and independent of the m-th matrix array. Assume $\mathbb{E}\,|\mathcal{S}_m| \le M_m$, $M_m \ge 1$. If $L_m \ge 1$ and*

$$\sum_m d_m^{-L_m} < \infty,$$

then, almost surely, for all sufficiently large m,

$$\max_{\varepsilon\in\mathcal{S}_m}\left\|\sum_{i=1}^{n_m}\varepsilon_i A_{m,i}\right\|_{\mathrm{op}} \le C_K\sqrt{n_m\left(L_m+1+\frac{\log(eM_m)}{\log(e+d_m)}\right)}.$$

In particular, if $M_m = 2^{n_m}$, then the same event gives the simultaneous whole-cube bound

$$\max_{\varepsilon\in\{-1,1\}^{n_m}}\left\|\sum_{i=1}^{n_m}\varepsilon_i A_{m,i}\right\|_{\mathrm{op}} \le C_K\sqrt{n_m\left(L_m+1+\frac{n_m}{\log(e+d_m)}\right)}$$

eventually almost surely. If additionally $n_m \le \lambda\log(e+d_m)$ and $\sum_m d_m^{-L} < \infty$ for some fixed L, every sign pattern is eventually $O_{K,L,\lambda}(\sqrt{n_m})$. The same conclusions hold for the corresponding profile-normalized full Hankel arrays.

Proof. Apply Corollary 4.3 to the m-th array with $L = L_m$. The exceptional probabilities are bounded by $d_m^{-L_m}$, which is summable. The first Borel–Cantelli lemma gives the eventual Toeplitz conclusion; independence across m is not needed. The whole-cube and logarithmic-window claims are the special cases $M_m = 2^{n_m}$ and $n_m \le \lambda\log(e+d_m)$. The Hankel statement follows by applying the same argument after Toeplitz–Hankel reversal. □

Corollary 4.5 (Profile-normalized sign-entropy schedules)**.** *In the setting of Corollary 4.4, the following concrete schedule consequences hold.*

(i) *If $d_m \ge m^\kappa$ eventually for some $\kappa > 0$ and $\mathbb{E}\,|\mathcal{S}_m| \le d_m^\alpha$ for some $\alpha < \infty$, then almost surely, for all sufficiently large m,*

$$\max_{\varepsilon\in\mathcal{S}_m}\left\|\sum_{i=1}^{n_m}\varepsilon_i A_{m,i}\right\|_{\mathrm{op}} \le C_{K,\alpha,\kappa}\sqrt{n_m}.$$

(ii) *If $d_m \ge \exp(cm^\sigma)$ eventually for some $c,\sigma > 0$ and $\mathbb{E}\,|\mathcal{S}_m| \le d_m^\alpha$ for some $\alpha < \infty$, then the same conclusion holds with a constant $C_{K,\alpha}$.*

(iii) *In either dimension-growth schedule, if* $\mathcal{S}_m = \{-1,1\}^{n_m}$ *and* $n_m \le \lambda \log(e + d_m)$ *eventually, then almost surely every sign pattern is eventually* $O_{K,\lambda,\kappa}(\sqrt{n_m})$ *in the polynomial schedule and* $O_{K,\lambda}(\sqrt{n_m})$ *in the stretched-exponential schedule.*

The same conclusions hold for the corresponding profile-normalized full Hankel arrays.

Proof. For the polynomial schedule choose $L > \max\{1, \kappa^{-1}\}$. Then

$$\sum_m d_m^{-L} \le \sum_m m^{-\kappa L} < \infty.$$

For the stretched-exponential schedule take $L = 1$, since

$$\sum_m d_m^{-1} \le \sum_m \exp(-cm^\sigma) < \infty.$$

In both cases, Corollary 4.4 applies with a fixed summable tail exponent. If $\mathbb{E}\,|\mathcal{S}_m| \le d_m^\alpha$, then

$$\frac{\log(eM_m)}{\log(e + d_m)} \le C_\alpha,$$

uniformly for $d_m \ge 2$, giving the first two bounds. For the whole cube, take $M_m = 2^{n_m}$; the assumption $n_m \le \lambda \log(e + d_m)$ gives

$$\frac{\log(eM_m)}{\log(e + d_m)} \le C_\lambda.$$

The Hankel statement is again the Toeplitz statement after reversal. □

Corollary 4.6 (Profile-normalized coefficient-family entropy)**.** *In the setting of Theorem 4.2, fix* $L > 0$. *Let* $\mathcal{V} \subset \mathbb{C}^n$ *be a finite nonempty deterministic coefficient family with* $|\mathcal{V}| \le M$, $M \ge 1$, *and*

$$\|v\|_2 \le R \qquad (v \in \mathcal{V}).$$

Then with probability at least $1 - d^{-L}$,

$$\max_{v\in\mathcal{V}} \left\| \sum_{i=1}^n v_i A_i \right\|_{\mathrm{op}} \le C_K R \sqrt{L + 1 + \frac{\log(eM)}{\log(e + d)}}.$$

The same statement holds if $\mathcal{V}$ *is finite and nonempty almost surely, random but independent of the matrices, satisfies* $\mathbb{E}\,|\mathcal{V}| \le M$, *and obeys* $\|v\|_2 \le R$ *for all* $v \in \mathcal{V}$ *almost surely. The same conclusions hold for the corresponding profile-normalized full Hankel matrices.*

Proof. If $R = 0$, every displayed sum vanishes, so assume $R > 0$. Fix a deterministic $v \in \mathbb{C}^n$, let $I = \{i : S_i > 0\}$, and write

$$Q_v(e^{i\theta}) = \sum_{i\in I} v_i \frac{1}{S_i\sqrt{\ell}} \sum_{|k|<d} \zeta_{i,k} e^{ik\theta}, \qquad \ell = \log(e + d).$$

For each fixed θ, the i-th normalized summand has sub-Gaussian norm at most $C_K/\sqrt{\ell}$. Independence over i gives

$$\left\|Q_v(e^{i\theta})\right\|_{\psi_2} \le C_K \frac{\|v\|_2}{\sqrt{\ell}} \le C_K \frac{R}{\sqrt{\ell}},$$

uniformly in θ and v. Using the same $8d$-point grid and Bernstein inequality as in the proof of Theorem 4.2, we obtain the fixed-vector tail

$$\mathbb{P}\left\{\left\|\sum_i v_i A_i\right\|_{\mathrm{op}} > C_K u R\right\} \le (e+d)^{-u^2} \qquad (u \ge 1).$$

Choose

$$u^2 = L + 1 + \frac{\log(eM)}{\log(e+d)}.$$

The union bound over deterministic $\mathcal{V}$ gives the asserted probability. For a random independent $\mathcal{V}$, condition on the realized family and average, using $\mathbb{E}\,|\mathcal{V}| \le M$. The Hankel claim follows from Toeplitz–Hankel reversal. □

Corollary 4.7 (Profile-normalized Borel coefficient entropy)**.** *For each m, let $d_m \ge 2$, $n_m \ge 1$, and let $(A_{m,i})_{i \le n_m}$ be a profile-normalized full Toeplitz array satisfying the hypotheses of Theorem 4.2 with the same sub-Gaussian constant K. Let $\mathcal{V}_m \subset \mathbb{C}^{n_m}$ be finite and nonempty deterministic, or finite and nonempty almost surely, random, and independent of the m-th matrix array. Assume*

$$\mathbb{E}\,|\mathcal{V}_m| \le M_m, \qquad \sup_{v \in \mathcal{V}_m} \|v\|_2 \le R_m$$

almost surely, where $M_m \ge 1$. If $L_m \ge 1$ and

$$\sum_m d_m^{-L_m} < \infty,$$

then, almost surely, for all sufficiently large m,

$$\max_{v \in \mathcal{V}_m} \left\|\sum_{i=1}^{n_m} v_i A_{m,i}\right\|_{\mathrm{op}} \le C_K R_m \sqrt{L_m + 1 + \frac{\log(eM_m)}{\log(e+d_m)}}.$$

In particular, if $d_m \ge m^\kappa$ eventually and $M_m \le d_m^\alpha$ for some $\kappa > 0$, $\alpha < \infty$, then the right side is $O_{K,\alpha,\kappa}(R_m)$ eventually almost surely. If $d_m \ge \exp(cm^\sigma)$ eventually and $M_m \le d_m^\alpha$, then it is eventually $O_{K,\alpha}(R_m)$. The same conclusions hold for the corresponding profile-normalized full Hankel arrays.

Proof. Apply Corollary 4.6 to the m-th array with $L = L_m$. The exceptional probabilities are bounded by $d_m^{-L_m}$, so Borel–Cantelli gives the eventual Toeplitz conclusion. In the polynomial schedule choose $L > \max\{1, \kappa^{-1}\}$, and in the stretched-exponential schedule choose $L = 1$. Then $\sum_m d_m^{-L} < \infty$ in the corresponding schedule, while $M_m \le d_m^\alpha$ gives

$$\frac{\log(eM_m)}{\log(e+d_m)} \le C_\alpha.$$

The Hankel statement again follows by Toeplitz–Hankel reversal. □

Corollary 4.8 (Expected-normalized polynomial sign lists are harmless)**.** *In the setting of Theorem 4.1, fix $L > 0$ and $\alpha < \infty$. Let $\mathcal{S} \subset \{-1,1\}^n$ be a deterministic sign list with $|\mathcal{S}| \le d^\alpha$. Then with probability at least $1 - d^{-L}$,*

$$\max_{\varepsilon \in \mathcal{S}} \left\| \sum_{i=1}^n \varepsilon_i A_i \right\|_{\mathrm{op}} \le C_{L,\alpha,K} \sqrt{n}.$$

The same simultaneous statement holds if $\mathcal{S}$ is random but independent of the matrices and either satisfies $|\mathcal{S}| \le d^\alpha$ almost surely or only satisfies $\mathbb{E}\,|\mathcal{S}| \le d^\alpha$. It also holds for the corresponding expected-normalized Hankel model.

Proof. Apply Theorem 4.1 with exponent $L + \alpha + 1$. For each fixed sign vector the exceptional probability is at most $d^{-L-\alpha-1}$. A union bound over deterministic $\mathcal{S}$ gives exceptional probability at most d^{-L-1}, hence at most d^{-L}. The Hankel case again follows by Toeplitz–Hankel reversal.

If $\mathcal{S}$ is random and independent of the matrices, condition on the realized list and use the fixed-sign union bound:

$$\mathbb{P}\left\{ \max_{\varepsilon \in \mathcal{S}} \left\| \sum_{i=1}^n \varepsilon_i A_i \right\|_{\mathrm{op}} > C_{L,\alpha,K} \sqrt{n} \;\middle|\; \mathcal{S} \right\} \le |\mathcal{S}|\; d^{-L-\alpha-1}.$$

Taking expectations gives an exceptional probability at most $\mathbb{E}\,|\mathcal{S}|\; d^{-L-\alpha-1} \le d^{-L-1} \le d^{-L}$. The almost-sure size assumption is the special case $\mathbb{E}\,|\mathcal{S}| \le d^\alpha$. □

Corollary 4.9 (Expected-normalized sign entropy bound)**.** *In the setting of Theorem 4.1, fix $L > 0$. Let $\mathcal{S} \subset \{-1,1\}^n$ be a deterministic sign family with $|\mathcal{S}| \le M$, where $M \ge 1$. Then with probability at least $1 - d^{-L}$,*

$$\max_{\varepsilon \in \mathcal{S}} \left\| \sum_{i=1}^n \varepsilon_i A_i \right\|_{\mathrm{op}} \le C_K \sqrt{n \left(L + 1 + \frac{\log(eM)}{\log(e+d)} \right)}.$$

The same statement holds if $\mathcal{S}$ is random but independent of the matrices and satisfies $\mathbb{E}\,|\mathcal{S}| \le M$. In particular, with probability at least $1 - d^{-L}$,

$$\max_{\varepsilon \in \{-1,1\}^n} \left\| \sum_{i=1}^n \varepsilon_i A_i \right\|_{\mathrm{op}} \le C_K \sqrt{n \left(L + 1 + \frac{n}{\log(e+d)} \right)}$$

in the expected-normalized Toeplitz and Hankel models.

Proof. The proof of Theorem 4.1, before the exponent is fixed, gives the following tail estimate: after enlarging C_K, for every fixed sign vector ε and every $u \ge 1$,

$$\mathbb{P}\left\{ \left\| \sum_{i=1}^n \varepsilon_i A_i \right\|_{\mathrm{op}} > C_K u \sqrt{n} \right\} \le (e+d)^{-u^2}.$$

Indeed, for the signed symbol Q, the $8d$-point net gives

$$\mathbb{P}\{\|Q\|_\infty > C_K u\sqrt{nd\log(e+d)}\} \le (e+d)^{-u^2}.$$

Dividing by $N_d \ge c_K\sqrt{d\log(e+d)}$ gives the displayed tail.

Set

$$u^2 = L + 2 + \frac{\log(eM)}{\log(e+d)}.$$

For deterministic $\mathcal{S}$, the union bound gives

$$\mathbb{P}\left\{\max_{\varepsilon\in\mathcal{S}}\left\|\sum_{i=1}^n \varepsilon_i A_i\right\|_{\mathrm{op}} > C_K u\sqrt{n}\right\} \le M(e+d)^{-L-2}(eM)^{-1} \le d^{-L},$$

after adjusting constants for $d \ge 2$. Absorbing the harmless $+2$ into the displayed $+1$ changes only C_K.

If $\mathcal{S}$ is random and independent of the matrices, condition on $\mathcal{S}$ and take expectations in the same inequality; the factor $|\mathcal{S}|$ is replaced by $\mathbb{E}\,|\mathcal{S}| \le M$. Finally take $M = 2^n$. Since $\log(e2^n) \le C(1+n)$, the stated all-sign bound follows, with another absolute change of C_K. The Hankel case is again obtained by reversal. □

Corollary 4.10 (Expected-normalized signs are uniformly harmless at logarithmic size)**.** *In the setting of Theorem 4.1, fix $L > 0$ and $\lambda < \infty$. If $n \le \lambda \log d$, then with probability at least $1 - d^{-L}$,*

$$\max_{\varepsilon\in\{-1,1\}^n}\left\|\sum_{i=1}^n \varepsilon_i A_i\right\|_{\mathrm{op}} \le C_{L,\lambda,K}\sqrt{n}.$$

The same simultaneous-in-sign statement holds for the corresponding expected-normalized Hankel model.

Proof. This is the all-sign specialization in Corollary 4.9. If $n \le \lambda\log d$, then $n/\log(e+d) \le C\lambda$, so the entropy factor is bounded in terms of L, λ only. □

Remark 4.11. Theorem 4.1 is why the expected-normalized full random model should not be listed as open. It does not contradict the deterministic worst-case lower bound $\sqrt{n\log(e+d/n)}$: the latter is attained by adversarial Toeplitz symbols arranged to peak together, whereas the random expected-normalized sum is just another random Toeplitz matrix whose symbol has variance n. What the theorem does not provide is a pathwise statement for every tuple of full Toeplitz contractions, nor does it by itself settle the realization-normalized model of Question 1.1, where each matrix is rescaled by its own random operator norm before the signing problem is posed.

4.2. **Intrinsic freeness does not certify the square-root bound.** The sharp matrix-concentration theory of Bandeira, Boedihardjo and van Handel [1] improves on the Khintchine logarithm precisely when the summands behave as though freely independent. For a self-adjoint Gaussian model

$X = \sum_i g_i A_i$ with i.i.d. standard Gaussian g_i, their bound has the schematic form

$$\mathbb{E}\,\|X\|_{\mathrm{op}} \le \|X_{\mathrm{free}}\| + C\, v(X)^{1/2}\, \sigma(X)^{1/2}\, (\log d)^{3/4}, \tag{4.1}$$

where $\sigma(X)^2 = \|\mathbb{E}X^2\|_{\mathrm{op}}$ is the variance parameter, $\|X_{\mathrm{free}}\|$ is the operator norm of the free (or matrix-semicircular) model with the same covariance, and $v(X)$ is the free-deviation parameter, a covariance norm that measures how far the summands are from being free. The improvement over Khintchine occurs when $v(X) \ll \sigma(X)$, so that the $(\log d)^{3/4}$ term is lower order; the free model then carries the main term with no logarithm.

Proposition 4.12 (Toeplitz matrices are maximally delocalized)**.** *Let $A = T_d(\xi)/N_d$ be a random full Toeplitz matrix with unit-variance entries, $N_d \asymp \sqrt{d \log d}$. Then*

$$\|A\|_{\mathrm{HS}} \asymp \frac{\sqrt{d}}{\sqrt{\log d}} \gg 1 = \|A\|_{\mathrm{op}} \ \ (\textit{order}).$$

Consequently, for the Gaussian Toeplitz model $X = \sum_i g_i A_i$ on $n \le d$ such expected-normalized matrices, the free-deviation parameter and the variance parameter are of the same order,

$$v(X) \asymp \sigma(X),$$

rather than $v(X) \ll \sigma(X)$. The intrinsic-freeness bound (4.1) therefore has its $(\log d)^{3/4}$ correction term at least as large as its main term, and reduces to the Khintchine scale; it does not certify $\mathbb{E}\,\|X\|_{\mathrm{op}} = O(\sqrt{n})$.

Proof. The Hilbert–Schmidt norm of an unnormalized $d \times d$ Toeplitz matrix $T_d(\xi)$ is

$$\|T_d(\xi)\|_{\mathrm{HS}}^2 = \sum_{0 \le r,s < d} |\xi_{r-s}|^2 = \sum_{|k|<d} (d - |k|)\, |\xi_k|^2 \asymp d^2$$

in expectation, since each lag k appears $d - |k|$ times and the ξ_k have unit variance; hence $\|T_d(\xi)\|_{\mathrm{HS}} \asymp d$. Dividing by $N_d \asymp \sqrt{d \log d}$,

$$\|A\|_{\mathrm{HS}} = \frac{\|T_d(\xi)\|_{\mathrm{HS}}}{N_d} \asymp \frac{d}{\sqrt{d \log d}} = \frac{\sqrt{d}}{\sqrt{\log d}}.$$

An expected-normalized Toeplitz matrix with operator norm of order 1 and Hilbert–Schmidt norm of order $\sqrt{d}/\sqrt{\log d}$ has stable rank $\|A\|_{\mathrm{HS}}^2 / \|A\|_{\mathrm{op}}^2 \asymp d/\log d$, i.e. it is delocalized over almost the full dimension. This is the opposite of a low-rank perturbation, for which $v \ll \sigma$.

The deviation parameter $v(X)$ in [1] is, up to the normalization there, controlled from below by the operator norm of the matrix of Hilbert–Schmidt overlaps of the A_i, and for the variance parameter one has $\sigma(X)^2 = \|\sum_i A_i^2\|_{\mathrm{op}} \asymp n$ since the A_i have operator norm of order one and lie in nearly orthogonal Toeplitz directions. The overlap matrix has diagonal entries $\|A_i\|_{\mathrm{HS}}^2 \asymp d/\log d$, which are far larger than 1; tracking these through the

definition of v in [1] gives $v(X)^2 \gtrsim \sigma(X)^2$, so $v(X) \asymp \sigma(X)$ rather than $v(X) \ll \sigma(X)$. With $v \asymp \sigma \asymp \sqrt{n}$ the correction term in (4.1) is

$$v(X)^{1/2}\sigma(X)^{1/2}(\log d)^{3/4} \asymp \sqrt{n}\,(\log d)^{3/4},$$

which exceeds the main term $\|X_{\mathrm{free}}\| \asymp \sqrt{n}$ once $\log d \to \infty$. The bound then reads $\mathbb{E}\,\|X\|_{\mathrm{op}} \lesssim \sqrt{n}\,(\log d)^{3/4}$, and since $(\log d)^{3/4} > (\log d)^{1/2}$ this is in fact weaker than the plain Khintchine bound $\sqrt{n \log d}$; in particular it does not give $O(\sqrt{n})$. □

Remark 4.13 (Numerical check). The ratio $v(X)/\sigma(X)$ was evaluated directly for signed superpositions of independent random Toeplitz contractions across a range of dimensions and sample sizes. It remained bounded away from zero throughout, taking values in the band 2.5 to 5.3, with no decay as d grew. This is consistent with Proposition 4.12: the parameter count forces v to stay at the scale of σ, so the intrinsic-freeness route cannot certify the square-root bound for this class, and the question is genuinely about typical realizations rather than about a soft concentration estimate.

5. The residual question and its reduction

The expected-normalized theorem of Section 4 removes the easiest random model. We now isolate the harder realization-normalized contraction question and identify the precise analytic obstacle.

5.1. **Why the worst case should not be typical.** The deterministic bound $\sqrt{n \log(e + d/n)}$ is sharp in the worst case: the matching lower bound of [6, Thm. 5.30] is realized by circulants whose symbols are arranged, adversarially, so that for every sign pattern some common frequency carries a large signed sum. Such a configuration is a measure-zero conspiracy. For independent random symbols the values $P_i(\omega)$ at a fixed frequency ω are independent mean-zero scalars of bounded variance, and the obstruction to $O(\sqrt{n})$ is whether, across the $O(d)$ frequencies simultaneously, the best sign pattern can keep every $|\sum_i \varepsilon_i P_i(\omega)|$ at the $\sqrt{n}$ scale. The deterministic theory already does this for every band up to $w \le n$, i.e. for every symbol supported on $O(n)$ Fourier modes; the residual question is whether the typical realization-normalized full symbol, although supported on all d modes, has no worse a signed edge than a banded one.

5.2. **The circulant reduction.** We make the reduction explicit. Recall from [6, §5] that signing $d \times d$ circulants is, on the nose, complex vector discrepancy of their d Fourier symbols: if C_i is the circulant with symbol vector $v(C_i) = (P_i(\omega))_\omega \in \mathbb{C}^d$, where ω ranges over the dth roots of unity, then

$$\left\|\sum_{i=1}^n \varepsilon_i C_i\right\|_{\mathrm{op}} = \max_{\omega} \left|\sum_{i=1}^n \varepsilon_i P_i(\omega)\right| = \left\|\sum_{i=1}^n \varepsilon_i v(C_i)\right\|_{\infty}.$$

A full Toeplitz contraction is approximated by a circulant of one larger size up to a controlled boundary error, the de la Vallée Poussin construction

of [6, Rem. 5.12], so up to constants the discrepancy of random Toeplitz contractions is the ℓ^∞ vector discrepancy of the random sample vectors $v(C_i) \in \mathbb{C}^d$.

Proposition 5.1 (Reduction to edge control)**.** *Question 1.1 has an affirmative answer if the following is true. Let $v_1, \dots, v_n \in \mathbb{C}^d$ be the random symbol-sample vectors of independent random realization-normalized Toeplitz contractions, $n \le d \le e^{o(n)}$ and $d/n \to \infty$. Then with probability tending to one there exist signs ε with*

$$\max_{\omega} \left| \sum_{i=1}^{n} \varepsilon_i v_i(\omega) \right| \le C\sqrt{n}.$$

Proof. By the circulant approximation,

$$\operatorname{disc}_{\mathrm{op}}(A_1, \dots, A_n) \le C \max_{\omega} \left| \sum_{i} \varepsilon_i v_i(\omega) \right| + E,$$

where E is the boundary error of the de la Vallée Poussin step, of order $o(\sqrt{n})$ for the relevant parameters by [6, Rem. 5.12]. The displayed hypothesis bounds the first term by $C\sqrt{n}$, hence $\operatorname{disc}_{\mathrm{op}} = O(\sqrt{n})$ on the high-probability event. □

Corollary 5.2 (Robust edge-to-operator transfer)**.** *Let (d_m, n_m) be a sequence with*

$$n_m \le d_m, \qquad d_m/n_m \to \infty, \qquad \log d_m = o(n_m).$$

For each m, let $B_{m,i} = T_{d_m}(\xi^{(m,i)})$, $1 \le i \le n_m$, be full Toeplitz matrices with symbols $P_{m,i}$, and put

$$A_{m,i} := \frac{B_{m,i}}{\|B_{m,i}\|_{\mathrm{op}}}, \qquad v_{m,i}(\omega) := \frac{P_{m,i}(\omega)}{\|B_{m,i}\|_{\mathrm{op}}}, \qquad \omega^{d_m} = 1.$$

Let $\mathcal{F}_m \subset \{-1, 1\}^{n_m}$ be any sign family. If, for some constant L, all sufficiently large m satisfy

$$\max_{\varepsilon \in \mathcal{F}_m} \max_{\omega^{d_m}=1} \left| \sum_{i=1}^{n_m} \varepsilon_i v_{m,i}(\omega) \right| \le L\sqrt{n_m},$$

then there is $L' < \infty$, depending only on L and the deterministic Toeplitz–circulant reduction constants, such that all sufficiently large m satisfy

$$\max_{\varepsilon \in \mathcal{F}_m} \left\| \sum_{i=1}^{n_m} \varepsilon_i A_{m,i} \right\|_{\mathrm{op}} \le L'\sqrt{n_m}.$$

The same conclusion holds for the corresponding full Hankel contractions $\widetilde{A}_{m,i} = J_{d_m} A_{m,i}$. In particular, any Hamming ball of signs with $O(\sqrt{n_m})$ edge maximum is an $O(\sqrt{n_m})$ operator-norm robust certificate for both the realization-normalized Toeplitz and Hankel families.

Proof. The proof of Proposition 5.1 gives, uniformly over sign vectors,

$$\left\|\sum_{i=1}^{n_m}\varepsilon_i A_{m,i}\right\|_{\mathrm{op}} \le C_{\mathrm{red}} \max_{\omega^{d_m}=1}\left|\sum_{i=1}^{n_m}\varepsilon_i v_{m,i}(\omega)\right| + E_m,$$

where the de la Vallée Poussin boundary error satisfies $E_m = o(\sqrt{n_m})$ under $d_m/n_m \to \infty$ and $\log d_m = o(n_m)$. Taking the maximum over $\mathcal{F}_m$ and absorbing E_m into the constant gives the Toeplitz claim. Multiplication by the reversal J_{d_m} preserves operator norm and turns full Toeplitz into full Hankel matrices, so the Hankel claim follows identically. □

Corollary 5.3 (Bounded-normalization robust edge-to-operator transfer)**.** *Let (d_m, n_m) be a sequence with*

$$n_m \le d_m, \qquad d_m/n_m \to \infty, \qquad \log d_m = o(n_m).$$

For each m, let $A_{m,i}$, $1 \le i \le n_m$, be full Toeplitz matrices with symbols $Q_{m,i}$, and assume

$$\max_{1\le i\le n_m} \|A_{m,i}\|_{\mathrm{op}} \le K_0$$

for all sufficiently large m. Let $\mathcal{F}_m \subset \{-1,1\}^{n_m}$ be any sign family. If, for some constant L, all sufficiently large m satisfy

$$\max_{\varepsilon\in\mathcal{F}_m}\max_{\omega^{d_m}=1}\left|\sum_{i=1}^{n_m}\varepsilon_i Q_{m,i}(\omega)\right| \le L\sqrt{n_m},$$

then there is $L' < \infty$, depending only on L, K_0, and the deterministic Toeplitz–circulant reduction constants, such that all sufficiently large m satisfy

$$\max_{\varepsilon\in\mathcal{F}_m}\left\|\sum_{i=1}^{n_m}\varepsilon_i A_{m,i}\right\|_{\mathrm{op}} \le L'\sqrt{n_m}.$$

The same conclusion holds for the corresponding full Hankel matrices $\widetilde{A}_{m,i} = J_{d_m}A_{m,i}$. Thus every logarithmic Hamming ball whose edge-grid maximum is $O(\sqrt{n_m})$ is an actual operator-norm robust certificate for any uniformly operator-bounded Toeplitz normalization.

Proof. The proof of Proposition 5.1 is deterministic and uses only the edge-grid bound and a uniform bound on the individual Toeplitz summands in the de la Vallée Poussin boundary term. With $\max_i \|A_{m,i}\|_{\mathrm{op}} \le K_0$, it gives, uniformly over all sign vectors,

$$\left\|\sum_{i=1}^{n_m}\varepsilon_i A_{m,i}\right\|_{\mathrm{op}} \le C_{\mathrm{red}} \max_{\omega^{d_m}=1}\left|\sum_{i=1}^{n_m}\varepsilon_i Q_{m,i}(\omega)\right| + E_m(K_0),$$

where $E_m(K_0) = o_{K_0}(\sqrt{n_m})$ under $d_m/n_m \to \infty$ and $\log d_m = o(n_m)$. Taking the maximum over $\mathcal{F}_m$ and absorbing the boundary term proves the Toeplitz claim. Multiplication by J_{d_m} preserves operator norm and turns full Toeplitz matrices into full Hankel matrices, so the Hankel claim is identical. □

Corollary 5.4 (Borel robust batches transfer to operator norm)**.** *Let (d_m, n_m) satisfy*

$$n_m \le d_m, \qquad d_m/n_m \to \infty, \qquad \log d_m = o(n_m).$$

For each m, let $B_{m,i} = T_{d_m}(\xi^{(m,i)})$, $1 \le i \le n_m$, be full Toeplitz matrices with symbols $P_{m,i}$, and put

$$A_{m,i} := \frac{B_{m,i}}{\|B_{m,i}\|_{\mathrm{op}}}, \qquad \widetilde{A}_{m,i} := J_{d_m} A_{m,i}, \qquad v_{m,i}(\omega) := \frac{P_{m,i}(\omega)}{\|B_{m,i}\|_{\mathrm{op}}}.$$

For each m, let $\delta_m^{(1)}, \dots, \delta_m^{(M_m)}$ be sampled signs, and let $1 \le r_m \le M_m$. Suppose that on an event Ω, all but finitely many m have a set $S_m \subset \{1, \dots, M_m\}$ with $|S_m| \ge M_m - r_m + 1$ such that, for some constants $L < \infty$ and radii q_m,

$$\max_{\omega^{d_m}=1} \left| \sum_{i=1}^{n_m} \varepsilon_i' v_{m,i}(\omega) \right| \le L\sqrt{n_m}$$

for every $s \in S_m$ and every sign vector ε' with

$$\#\{i : \varepsilon_i' \ne \delta_{m,i}^{(s)}\} \le q_m.$$

Then, on the same event Ω, after increasing L by a deterministic constant depending only on the Toeplitz–circulant reduction, all but finitely many m have the same set S_m and the same radii q_m satisfying

$$\left\| \sum_{i=1}^{n_m} \varepsilon_i' A_{m,i} \right\|_{\mathrm{op}} \le L'\sqrt{n_m}, \qquad \left\| \sum_{i=1}^{n_m} \varepsilon_i' \widetilde{A}_{m,i} \right\|_{\mathrm{op}} \le L'\sqrt{n_m}$$

for every $s \in S_m$ and every ε' in the same Hamming ball. Consequently, every almost-sure sampled-count or robust-fraction theorem in this paper whose conclusion supplies logarithmic-radius $O(\sqrt{n_m})$ edge-grid robust centers also supplies actual realization-normalized full Toeplitz and full Hankel operator-norm robust centers in the above full-bandwidth range.

Proof. For each sufficiently large m, let $\mathcal{F}_m$ be the union of the Hamming balls centered at $\{\delta_m^{(s)} : s \in S_m\}$ with radius q_m. The hypothesis gives

$$\max_{\varepsilon' \in \mathcal{F}_m} \max_{\omega^{d_m}=1} \left| \sum_{i=1}^{n_m} \varepsilon_i' v_{m,i}(\omega) \right| \le L\sqrt{n_m}.$$

Apply Corollary 5.2 to the sign family $\mathcal{F}_m$. The Toeplitz estimate follows immediately, and the Hankel estimate follows from the same reversal argument used there. The final sentence is just this deterministic implication applied on the almost-sure event supplied by any of the sampled-count Borel corollaries. □

General deterministic vector-discrepancy theorems for d coordinates still allow the growing $\sqrt{n \log(e + d/n)}$ scale in this range. Thus Proposition 5.1 transfers the remaining problem to the joint behavior of the n random edge vectors: one must use randomness of the symbols to rule out the adversarial common-frequency peaking that makes the deterministic bound sharp.

Proposition 5.5 (Edge-variance profile criterion)**.** *Let $v_1, \dots, v_n \in \mathbb{C}^d$ be arbitrary edge-sample vectors and set*

$$V^2 := \max_{1\le j\le d} \sum_{i=1}^{n} |v_i(j)|^2 .$$

If

$$V^2 \le C_0 \frac{n}{\log(e+d)},$$

then for every $A > 0$, independent Rademacher signs satisfy

$$\mathbb{P}_\varepsilon \left\{ \max_{1\le j\le d} \left| \sum_{i=1}^{n} \varepsilon_i v_i(j) \right| > C_{A,C_0}\sqrt{n} \right\} \le d^{-A}.$$

Consequently, under this deterministic variance-profile bound, the edge-control hypothesis in Proposition 5.1 holds with high probability over the signs.

Proof. For a fixed coordinate j, the complex Hoeffding inequality gives

$$\mathbb{P}_\varepsilon \left\{ \left| \sum_i \varepsilon_i v_i(j) \right| > t \right\} \le 4\exp\left(-c\, \frac{t^2}{\sum_i |v_i(j)|^2} \right) \le 4\exp\left(-c\, \frac{t^2}{V^2} \right),$$

by applying the real inequality to the real and imaginary parts. A union bound over the d coordinates yields

$$\mathbb{P}_\varepsilon \left\{ \max_j \left| \sum_i \varepsilon_i v_i(j) \right| > t \right\} \le 4d\exp\left(-c\, \frac{t^2}{V^2} \right).$$

With $t = C_{A,C_0}\sqrt{n}$ and $V^2 \le C_0 n/\log(e+d)$, the right-hand side is at most d^{-A} after increasing C_{A,C_0}. □

Corollary 5.6 (Random search under an edge profile)**.** *With the notation of Proposition 5.5, let $\delta^{(1)}, \dots, \delta^{(M)}$ be independent uniform sign vectors in $\{-1,1\}^n$, independent of everything else, where $M \ge 1$ is an integer. For every $u \ge 0$,*

$$\mathbb{P}_\delta \left\{ \max_{1\le s\le M} \max_{1\le j\le d} \left| \sum_{i=1}^{n} \delta_i^{(s)} v_i(j) \right| > CV\sqrt{\log(e+d) + \log(eM) + u} \right\} \le e^{-u}.$$

Consequently, if $V^2 \le C_0 n/\log(e+d)$, then with δ-probability at least $1-e^{-u}$,

$$\max_{1\le s\le M} \max_{1\le j\le d} \left| \sum_{i=1}^{n} \delta_i^{(s)} v_i(j) \right| \le C_{C_0}\sqrt{n\left(1 + \frac{\log(eM)+u}{\log(e+d)}\right)}.$$

Moreover, under the same variance-profile hypothesis, for every $A > 0$ and every integer $M \ge 1$,

$$\mathbb{P}_\delta \left\{ \min_{1\le s\le M} \max_{1\le j\le d} \left| \sum_{i=1}^{n} \delta_i^{(s)} v_i(j) \right| > C_{A,C_0}\sqrt{n} \right\} \le d^{-AM}.$$

More generally, if

$$G_M := \#\left\{1 \le s \le M : \max_{1\le j\le d}\left|\sum_{i=1}^{n}\delta_i^{(s)}v_i(j)\right| \le C_{A,C_0}\sqrt{n}\right\},$$

then for every integer $1 \le r \le M$*,*

$$\mathbb{P}_\delta\{G_M \le M - r\} \le \binom{M}{r} d^{-Ar}.$$

Thus, once the matrix realization has the edge profile, M *independent random-sign trials amplify the failure probability from* d^{-A} *to* d^{-AM}*, and the number of sampled certificates has the corresponding binomial upper tail for failures.*

Proof. The proof of Proposition 5.5 gives, for one sign vector δ,

$$\mathbb{P}_\delta\left\{\max_j\left|\sum_i \delta_i v_i(j)\right| > t\right\} \le 4d\exp(-ct^2/V^2).$$

Taking a union bound over $s = 1, \dots, M$ gives

$$\mathbb{P}_\delta\left\{\max_s\max_j\left|\sum_i \delta_i^{(s)} v_i(j)\right| > t\right\} \le 4dM\exp(-ct^2/V^2).$$

With

$$t = CV\sqrt{\log(e+d) + \log(eM) + u},$$

the last expression is at most e^{-u} after increasing the absolute constant C. Substituting $V^2 \le C_0 n/\log(e+d)$ gives the displayed entropy-sensitive sampled-list bound.

For the final assertions, Proposition 5.5 says that a single random sign vector fails the $C_{A,C_0}\sqrt{n}$ threshold with probability at most d^{-A}. The M trials are independent, so the probability that all M fail is at most d^{-AM}. If F_s is the failure event for trial s, then for every fixed set $R \subset \{1, \dots, M\}$ of size r,

$$\mathbb{P}_\delta\left(\bigcap_{s\in R} F_s\right) \le d^{-Ar}.$$

The event $G_M \le M - r$ means that at least r of the failure events occur. A union bound over the r-subsets R gives $\binom{M}{r}d^{-Ar}$. □

Proposition 5.7 (Edge-profile scalar moment criterion)**.** *Let* $v_1, \dots, v_n \in \mathbb{C}^d$ *and set*

$$V^2 := \max_{1\le j\le d}\sum_{i=1}^{n}|v_i(j)|^2.$$

Then, simultaneously for every integer $k \ge 1$*,*

$$\max_{1\le j\le d}\mathbb{E}_\varepsilon\left|\sum_{i=1}^{n}\varepsilon_i v_i(j)\right|^{2k} \le (CkV^2)^k.$$

In particular, if $V^2 \le C_0 n/\log(e+d)$, *then*

$$\max_{1\le j\le d} \mathbb{E}_\varepsilon \left|\sum_{i=1}^n \varepsilon_i v_i(j)\right|^{2k} \le \left(C_{C_0} k \,\frac{n}{\log(e+d)}\right)^k \qquad (k\ge 1).$$

Proof. For deterministic complex scalars a_i, the standard Rademacher moment bound gives

$$\left(\mathbb{E}_\varepsilon \left|\sum_i \varepsilon_i a_i\right|^{2k}\right)^{1/(2k)} \le C\sqrt{k}\left(\sum_i |a_i|^2\right)^{1/2}.$$

Indeed, apply the real sub-Gaussian moment bound to the real and imaginary parts and use $|z| \le |\operatorname{Re} z| + |\operatorname{Im} z|$. Taking $a_i = v_i(j)$, maximizing over j, and raising to the power $2k$ proves the first display; the second follows from the displayed bound on V^2. □

Corollary 5.8 (Edge-profile moment bound for the edge maximum)**.** *With the notation of Proposition 5.7, put*

$$M_\varepsilon := \max_{1\le j\le d}\left|\sum_{i=1}^n \varepsilon_i v_i(j)\right|.$$

Then, for every integer $k \ge 1$,

$$\mathbb{E}_\varepsilon M_\varepsilon^{2k} \le d\,(CkV^2)^k,$$

and hence

$$\left(\mathbb{E}_\varepsilon M_\varepsilon^{2k}\right)^{1/(2k)} \le C\, d^{1/(2k)}\sqrt{k}\, V.$$

In particular, if $V^2 \le C_0 n/\log(e+d)$ *and* $k \ge \log(e+d)$, *then*

$$\left(\mathbb{E}_\varepsilon M_\varepsilon^{2k}\right)^{1/(2k)} \le C_{C_0}\sqrt{n}\,\sqrt{\frac{k}{\log(e+d)}}.$$

Taking $k = \lceil (A+2)\log(e+d)\rceil$ *and applying Markov's inequality recovers the high-probability edge-control conclusion of Proposition 5.5.*

Proof. Since $M_\varepsilon^{2k} \le \sum_{j=1}^d |\sum_i \varepsilon_i v_i(j)|^{2k}$, the first display follows from Proposition 5.7. Taking $2k$-th roots gives the second display. If $k \ge \log(e+d)$, then $d^{1/(2k)} \le e^{1/2}$, and the displayed bound on V^2 gives the third display. For $k = \lceil (A+2)\log(e+d)\rceil$, the third display is at most $C_{A,C_0}\sqrt{n}$; increasing the constant in the threshold and using Markov's inequality gives a d^{-A} tail bound. □

Corollary 5.9 (Edge-profile signing certificate)**.** *With the notation of Proposition 5.7,*

$$\mathbb{E}_\varepsilon \max_{1\le j\le d}\left|\sum_{i=1}^n \varepsilon_i v_i(j)\right| \le C\,V\sqrt{\log(e+d)}.$$

Consequently,

$$\min_{\varepsilon\in\{-1,1\}^n}\max_{1\le j\le d}\left|\sum_{i=1}^n \varepsilon_i v_i(j)\right| \le C\,V\sqrt{\log(e+d)}.$$

In particular, if $V^2 \le C_0 n/\log(e+d)$*, then there is a sign vector* $\varepsilon\in\{-1,1\}^n$ *for which*

$$\max_{1\le j\le d}\left|\sum_{i=1}^n \varepsilon_i v_i(j)\right| \le C_{C_0}\sqrt{n}.$$

Proof. Let M_ε be the edge maximum from Corollary 5.8 and put $k_0 = \lceil \log(e+d)\rceil$. Since $M_\varepsilon \ge 0$,

$$\mathbb{E}_\varepsilon M_\varepsilon \le \left(\mathbb{E}_\varepsilon M_\varepsilon^{2k_0}\right)^{1/(2k_0)} \le C\,d^{1/(2k_0)}\sqrt{k_0}\,V \le C\,V\sqrt{\log(e+d)}.$$

The minimum over signs is bounded by the average over signs. The final display follows by substituting $V^2 \le C_0 n/\log(e+d)$. □

Corollary 5.10 (Hamming-stable edge-profile certificates)**.** *With the notation of Proposition 5.7, suppose a sign vector* $\varepsilon\in\{-1,1\}^n$ *satisfies*

$$\max_{1\le j\le d}\left|\sum_{i=1}^n \varepsilon_i v_i(j)\right| \le R.$$

If $\varepsilon'\in\{-1,1\}^n$ *differs from* ε *in at most* r *coordinates, then*

$$\max_{1\le j\le d}\left|\sum_{i=1}^n \varepsilon'_i v_i(j)\right| \le R + 2\sqrt{r}\,V.$$

Consequently, if $V^2 \le C_0 n/\log(e+d)$*, then for every* $A>0$*, with Rademacher probability at least* $1-d^{-A}$ *a random sign vector* ε *is the center of a Hamming ball of radius*

$$r_0 = \lfloor c_{A,C_0}\log(e+d)\rfloor$$

on which every sign vector ε' *satisfies*

$$\max_{1\le j\le d}\left|\sum_{i=1}^n \varepsilon'_i v_i(j)\right| \le C_{A,C_0}\sqrt{n}.$$

Equivalently, at least a $1-d^{-A}$ *fraction of all sign vectors are centers of such logarithmic-radius good Hamming balls. Moreover, if* $\delta^{(1)},\dots,\delta^{(M)}$ *are independent uniform sign vectors and* H_M *is the number of sampled signs that are centers of such radius-*r_0 *good Hamming balls, then for every integer* $1\le r\le M$*,*

$$\mathbb{P}_\delta\{H_M \le M-r\} \le \binom{M}{r} d^{-Ar}.$$

Proof. Let $F=\{i:\varepsilon_i\ne\varepsilon'_i\}$, so $|F|\le r$. For each coordinate j,

$$\sum_i \varepsilon'_i v_i(j) - \sum_i \varepsilon_i v_i(j) = \sum_{i\in F}(\varepsilon'_i-\varepsilon_i)v_i(j).$$

Since $|\varepsilon_i' - \varepsilon_i| \le 2$, Cauchy's inequality gives

$$\left|\sum_{i\in F}(\varepsilon_i' - \varepsilon_i)v_i(j)\right| \le 2\sum_{i\in F}|v_i(j)| \le 2\sqrt{r}\left(\sum_{i\in F}|v_i(j)|^2\right)^{1/2} \le 2\sqrt{r}\,V.$$

Taking the maximum over j proves the deterministic stability estimate.

Under $V^2 \le C_0 n/\log(e+d)$, Proposition 5.5 gives with sign probability at least $1 - d^{-A}$ a center ε satisfying

$$\max_j \left|\sum_i \varepsilon_i v_i(j)\right| \le C_{A,C_0}\sqrt{n}.$$

Choose $c_{A,C_0} > 0$ small enough that

$$2\sqrt{r_0}\,V \le C'_{A,C_0}\sqrt{n}.$$

The first part then gives the claimed bound on the whole Hamming ball after enlarging C_{A,C_0}. Since Rademacher probability is uniform measure on the sign cube, the final fraction statement is the same assertion.

For the sampled-center assertion, let E_s be the event that $\delta^{(s)}$ is not the center of such a good Hamming ball. The preceding paragraph gives $\mathbb{P}(E_s) \le d^{-A}$. For every fixed $R \subset \{1,\dots,M\}$ of size r, independence gives

$$\mathbb{P}_\delta\left(\bigcap_{s\in R} E_s\right) \le d^{-Ar}.$$

If $H_M \le M - r$, at least r of the events E_s occur. A union bound over the r-subsets R gives the displayed estimate. □

Corollary 5.11 (Polynomial-confidence robust signing search)**.** *With the notation of Proposition 5.5, assume*

$$V^2 \le C_0\frac{n}{\log(e+d)}.$$

Then there are constants C_{C_0} and $c_{C_0} > 0$ such that the following holds. For every $B > 0$, if

$$M \ge \lceil B \rceil$$

independent uniform sign vectors $\delta^{(1)},\dots,\delta^{(M)} \in \{-1,1\}^n$ are sampled, then with probability at least $1 - d^{-B}$ at least one sampled sign vector is the center of a Hamming ball of radius

$$r_0 = \lfloor c_{C_0}\log(e+d)\rfloor$$

on which every sign vector ε' satisfies

$$\max_{1\le j\le d}\left|\sum_{i=1}^n \varepsilon_i' v_i(j)\right| \le C_{C_0}\sqrt{n}.$$

Thus a constant-size random list gives a robust $O(\sqrt{n})$ signing certificate with arbitrary polynomial confidence.

Proof. Apply Corollary 5.10 with exponent $A = 1$. A single uniform sign fails to be the center of the displayed good Hamming ball with probability at most d^{-1}. The M trials are independent, so the probability that none of them is such a robust center is at most

$$d^{-M} \le d^{-B}.$$

□

Remark 5.12 (The remaining random input). For the realization-normalized Toeplitz contraction model in Question 1.1, the vectors in Proposition 5.1 have the form

$$v_i(\omega) = \frac{P_i(\omega)}{\|T_d(\xi^{(i)})\|_{\mathrm{op}}}, \qquad \omega^d = 1.$$

Thus Proposition 5.5 reduces the residual question to the empirical edge-delocalization estimate

$$\max_{\omega^d=1} \sum_{i=1}^{n} \frac{|P_i(\omega)|^2}{\|T_d(\xi^{(i)})\|_{\mathrm{op}}^2} \lesssim \frac{n}{\log d}$$

with high probability. This is stronger and more targeted than asking a generic vector-discrepancy theorem to beat the worst case: it says that, after realization normalization, no edge node should carry more than its natural average share of the n random symbols. The expected-normalized theorem above is the same heuristic with the random denominators replaced by the deterministic scale N_d. Theorem 5.141 removes the random-denominator difficulty for standard Gaussian coefficients by combining Gaussian norm concentration with a bad-normalizer truncation. The general form of that argument is Theorem 5.14: apart from standard sub-Gaussian root-value bounds and a crude Hilbert–Schmidt floor, the only missing one-matrix input is

$$\mathbb{P}\{\|T_d(\xi)\|_{\mathrm{op}} < a\,\mathbb{E}\,\|T_d(\xi)\|_{\mathrm{op}}\} \min\{n, \log^2(e+d)\} = o(1).$$

Thus the residual problem is no longer an arbitrary dependence question between edge values and normalizers; it is enough to rule out a logarithmic-density population of low-normalizer samples.

Proposition 5.13 (Expected edge profile)**.** *In the expected-normalized model of Theorem 4.1, let*

$$v_i(\omega) := \frac{P_i(\omega)}{N_d}, \qquad \omega^d = 1,$$

where P_i is the full Toeplitz symbol of B_i. For every $A > 0$, if $n \ge C_{A,K} \log(e+d)$, then

$$\mathbb{P}\left\{\max_{\omega^d=1} \sum_{i=1}^{n} |v_i(\omega)|^2 > C_{A,K} \frac{n}{\log(e+d)}\right\} \le d^{-A}.$$

Thus the edge-variance profile required in Proposition 5.5 holds automatically in the expected-normalized model, at the natural $n/\log d$ scale.

Proof. Fix a dth root of unity ω. The random variable

$$P_i(\omega) = \sum_{|k|<d} \xi_k^{(i)} \omega^k$$

is centered sub-Gaussian with norm at most $C_K\sqrt{d}$, hence $|P_i(\omega)|^2$ is sub-exponential with norm at most $C_K d$ and mean at most $C_K d$. Since $N_d \geq c_K\sqrt{d\log(e+d)}$, the variables $|v_i(\omega)|^2$ are independent nonnegative sub-exponential variables with mean and sub-exponential norm $O_K(1/\log(e+d))$. Bernstein's inequality therefore gives

$$\mathbb{P}\left\{\sum_{i=1}^n |v_i(\omega)|^2 > C_{A,K}\frac{n}{\log(e+d)}\right\} \leq \exp(-c_{A,K}n)$$

after increasing $C_{A,K}$. If $n \geq C_{A,K}\log(e+d)$, the right side is at most d^{-A-1}. A union bound over the d roots gives the claim. □

Theorem 5.14 (Normalizer-tail criterion)**.** *Let $B_i = T_d(\xi^{(i)})$, $1 \leq i \leq n$, be independent full Toeplitz matrices whose lag variables are independent, mean zero, unit variance, and have sub-Gaussian norm at most K. Let P_i be their symbols and put*

$$N_d := \mathbb{E}\,\|T_d(\xi)\|_{\mathrm{op}}\,.$$

Assume $n \leq d$, $\log d = o(n)$, and that for some $a > 0$

$$p_d(a) := \mathbb{P}\{\|T_d(\xi)\|_{\mathrm{op}} < aN_d\} \quad \textit{satisfies} \quad p_d(a)\min\{n, \log^2(e+d)\} = o(1).$$

Then

$$\max_{\omega^d=1}\sum_{i=1}^n \frac{|P_i(\omega)|^2}{\|B_i\|_{\mathrm{op}}^2} \leq C_{K,a}\,\frac{n}{\log(e+d)}$$

with probability tending to one. Consequently the realization-normalized edge control in Proposition 5.1 holds with high probability over the matrices and independent signs. On the same high-probability matrix event,

$$\mathbb{E}_\varepsilon \max_{\omega^d=1}\left|\sum_{i=1}^n \varepsilon_i \frac{P_i(\omega)}{\|B_i\|_{\mathrm{op}}}\right| \leq C_{K,a}\sqrt{n},$$

so in particular a deterministic sign vector with $O_{K,a}(\sqrt{n})$ edge maximum exists. Also on this event, simultaneously for every integer $k \geq 1$,

$$\max_{\omega^d=1}\mathbb{E}_\varepsilon\left|\sum_{i=1}^n \varepsilon_i \frac{P_i(\omega)}{\|B_i\|_{\mathrm{op}}}\right|^{2k} \leq \left(C_{K,a}k\,\frac{n}{\log(e+d)}\right)^k.$$

If also $d/n \to \infty$, then

$$\mathrm{disc}_{\mathrm{op}}(B_1/\|B_1\|_{\mathrm{op}}, \ldots, B_n/\|B_n\|_{\mathrm{op}}) = O_{K,a}(\sqrt{n})$$

with probability tending to one.

Proof. Write $\ell = \log(e+d)$, and split

$$G = \{i : \|B_i\|_{\mathrm{op}} \geq aN_d\}, \qquad G^c = \{1, \dots, n\} \setminus G.$$

On the event of Proposition 5.13,

$$\max_{\omega^d=1} \sum_{i \in G} \frac{|P_i(\omega)|^2}{\|B_i\|_{\mathrm{op}}^2} \leq a^{-2} \max_{\omega^d=1} \sum_{i=1}^{n} \frac{|P_i(\omega)|^2}{N_d^2} \leq C_{K,a} \frac{n}{\ell},$$

with probability tending to one, since $\ell = o(n)$.

The number $M = |G^c|$ of bad normalizers satisfies

$$\mathbb{P}\{M > n/\ell^2\} \leq p_d(a) \min\{n, \ell^2\} = o(1).$$

For $n \geq \ell^2$ this is Markov's inequality; for $n < \ell^2$ the event forces at least one bad normalizer and is bounded by $np_d(a)$. It remains to bound the cost of those bad samples. A union bound over $i \leq n \leq d$ and the d roots gives

$$\max_{i, \omega^d=1} |P_i(\omega)| \leq C_K \sqrt{d\ell}$$

with probability tending to one, because each $P_i(\omega)$ has sub-Gaussian norm at most $C_K\sqrt{d}$.

We also need a denominator floor which is much weaker than the desired operator-norm scale. Since $\mathbb{E}\,|\xi|^2 = 1$ and $\mathbb{E}\,|\xi|^4 \leq C_K$, Paley–Zygmund gives constants $c_K, \tau_K > 0$ such that $\mathbb{P}\{|\xi|^2 \geq \tau_K\} \geq c_K$. For $|k| \leq d/2$ the Toeplitz weight $d - |k|$ is at least $d/2$, so a Chernoff bound gives

$$\|B_i\|_{\mathrm{HS}}^2 = \sum_{|k|<d} (d - |k|) \left|\xi_k^{(i)}\right|^2 \geq c_K d^2$$

for all $i \leq n$, with probability tending to one. Therefore

$$\|B_i\|_{\mathrm{op}} \geq d^{-1/2} \|B_i\|_{\mathrm{HS}} \geq c_K \sqrt{d}$$

uniformly in $i \leq n$ on the same event.

On the intersection of these events, every bad sample contributes at most

$$\frac{|P_i(\omega)|^2}{\|B_i\|_{\mathrm{op}}^2} \leq C_K \ell$$

to each root ω. Since $M \leq n/\ell^2$, the total bad contribution is at most $C_K n/\ell$, uniformly over the roots. Combining good and bad samples proves the displayed profile.

The signed edge-control statement is Proposition 5.5 applied on this high-probability matrix event. The conditional expectation and existence statement is Corollary 5.9. The final discrepancy bound follows from Proposition 5.1. The scalar moment estimate is Proposition 5.7 applied to the same profile. □

Corollary 5.15 (Data-dependent bad-normalizer decomposition)**.** *Fix $A > 0$ and $a > 0$. Let $B_i = T_d(\xi^{(i)})$, $1 \leq i \leq n$, be independent full Toeplitz matrices*

whose lag variables are independent, mean zero, unit variance, and have sub-Gaussian norm at most K. Let P_i be their symbols, put $N_d = \mathbb{E}\,\|T_d(\xi)\|_{\mathrm{op}}$, write $\ell = \log(e+d)$, and set

$$M_a := \#\{1 \le i \le n : \|B_i\|_{\mathrm{op}} < aN_d\}.$$

There is a constant $C_{A,K,a}$ such that, whenever

$$n \le d, \qquad n \ge C_{A,K,a}\ell,$$

one has, outside an event of probability at most $C_{A,K,a}d^{-A}$,

$$\max_{\omega^d=1} \sum_{i=1}^{n} \frac{|P_i(\omega)|^2}{\|B_i\|_{\mathrm{op}}^2} \le C_{A,K,a}\left(\frac{n}{\ell} + M_a\ell\right).$$

On the same matrix event, independent signs satisfy

$$\mathbb{P}_\varepsilon\left\{\max_{\omega^d=1}\left|\sum_{i=1}^{n}\varepsilon_i\frac{P_i(\omega)}{\|B_i\|_{\mathrm{op}}}\right| > C_{A,K,a}\sqrt{n + M_a\ell^2}\right\} \le d^{-A},$$

and

$$\mathbb{E}_\varepsilon \max_{\omega^d=1}\left|\sum_{i=1}^{n}\varepsilon_i\frac{P_i(\omega)}{\|B_i\|_{\mathrm{op}}}\right| \le C_{A,K,a}\sqrt{n + M_a\ell^2}\,.$$

Thus a deterministic sign vector with edge maximum $C_{A,K,a}\sqrt{n + M_a\ell^2}$ exists. On the same event, simultaneously for every integer $k \ge 1$,

$$\max_{\omega^d=1}\mathbb{E}_\varepsilon\left|\sum_{i=1}^{n}\varepsilon_i\frac{P_i(\omega)}{\|B_i\|_{\mathrm{op}}}\right|^{2k} \le \left(C_{A,K,a}k\left(\frac{n}{\ell} + M_a\ell\right)\right)^k.$$

Proof. Let

$$G = \{i : \|B_i\|_{\mathrm{op}} \ge aN_d\}.$$

The expected-normalized edge-profile event of Proposition 5.13 gives

$$\max_{\omega^d=1}\sum_{i\in G}\frac{|P_i(\omega)|^2}{\|B_i\|_{\mathrm{op}}^2} \le C_{A,K,a}\frac{n}{\ell}$$

outside an event of probability at most d^{-A}, after increasing the constant and using $n \ge C_{A,K,a}\ell$.

The same union bound and Paley–Zygmund/Chernoff Hilbert–Schmidt floor used in the proof of Theorem 5.14 give, outside another $C_{A,K}d^{-A}$ event,

$$\max_{i,\omega^d=1}|P_i(\omega)| \le C_{A,K}\sqrt{d\ell}, \qquad \min_i \|B_i\|_{\mathrm{op}} \ge c_K\sqrt{d}.$$

Hence every index outside G contributes at most $C_{A,K}\ell$ to each edge node. Since there are exactly M_a such indices, the bad-normalizer contribution is at most $C_{A,K}M_a\ell$. Combining the good and bad contributions proves the data-dependent profile.

Condition on matrices satisfying that profile and put

$$V^2 := C_{A,K,a}\left(\frac{n}{\ell} + M_a\ell\right).$$

The Hoeffding-and-union-bound proof of Proposition 5.5 gives

$$\mathbb{P}_\varepsilon\left\{\max_{\omega^d=1}\left|\sum_i \varepsilon_i \frac{P_i(\omega)}{\|B_i\|_{\mathrm{op}}}\right| > t\right\} \le 4d\exp(-ct^2/V^2).$$

Taking $t = C_{A,K,a}\sqrt{V^2\ell}$ gives the displayed tail, since $V^2\ell \le C_{A,K,a}(n + M_a\ell^2)$. The sign-averaged maximum and deterministic certificate follow from Corollary 5.9; the scalar moment estimates follow from Proposition 5.7. □

Corollary 5.16 (Dyadic bad-normalizer decomposition)**.** *Fix $A > 0$ and $a > 0$. In the setting of Corollary 5.15, put $\ell = \log(e + d)$ and*

$$J_\ell := \left\lceil \frac{1}{2}\log_2(e+\ell)\right\rceil.$$

For $0 \le r < J_\ell$, define

$$M_{a,r} := \#\left\{1 \le i \le n : 2^{-(r+1)}aN_d \le \|B_i\|_{\mathrm{op}} < 2^{-r}aN_d\right\},$$

and put

$$M_{a,\infty} := \#\left\{1 \le i \le n : \|B_i\|_{\mathrm{op}} < 2^{-J_\ell}aN_d\right\}.$$

Set

$$W_a := \sum_{0\le r<J_\ell} 4^{r+1}M_{a,r} + \ell\, M_{a,\infty}.$$

There is a constant $C_{A,K,a}$ such that, whenever

$$n \le d, \qquad n \ge C_{A,K,a}\ell,$$

one has, outside an event of probability at most $C_{A,K,a}d^{-A}$,

$$\max_{\omega^d=1}\sum_{i=1}^n \frac{|P_i(\omega)|^2}{\|B_i\|_{\mathrm{op}}^2} \le C_{A,K,a}\left(\frac{n}{\ell} + W_a\right).$$

On the same matrix event, independent signs satisfy

$$\mathbb{P}_\varepsilon\left\{\max_{\omega^d=1}\left|\sum_{i=1}^n \varepsilon_i \frac{P_i(\omega)}{\|B_i\|_{\mathrm{op}}}\right| > C_{A,K,a}\sqrt{n + W_a\ell}\right\} \le d^{-A},$$

and

$$\mathbb{E}_\varepsilon \max_{\omega^d=1}\left|\sum_{i=1}^n \varepsilon_i \frac{P_i(\omega)}{\|B_i\|_{\mathrm{op}}}\right| \le C_{A,K,a}\sqrt{n + W_a\ell}\,.$$

Thus a deterministic sign vector with edge maximum $C_{A,K,a}\sqrt{n + W_a\ell}$ exists. On the same event, simultaneously for every integer $k \ge 1$,

$$\max_{\omega^d=1}\mathbb{E}_\varepsilon\left|\sum_{i=1}^n \varepsilon_i \frac{P_i(\omega)}{\|B_i\|_{\mathrm{op}}}\right|^{2k} \le \left(C_{A,K,a}k\left(\frac{n}{\ell} + W_a\right)\right)^k.$$

In particular, if the realized dyadic bad-normalizer budget satisfies

$$W_a \le R\frac{n}{\ell}$$

for some $R \ge 1$, then the realization-normalized edge profile, random-sign tail, sign-averaged maximum, deterministic certificate, and scalar moment package all hold at the $C_{A,K,a,R}\sqrt{n}$ scale.

Proof. Let

$$G = \{i : \|B_i\|_{\mathrm{op}} \ge aN_d\}.$$

As in Corollary 5.15, the expected-normalized profile event gives

$$\max_{\omega^d=1} \sum_{i\in G} \frac{|P_i(\omega)|^2}{\|B_i\|_{\mathrm{op}}^2} \le C_{A,K,a}\frac{n}{\ell}$$

outside a d^{-A} event. Intersect this with the standard root-value and Hilbert–Schmidt floor event from the proof of Theorem 5.14:

$$\max_{i,\omega^d=1} |P_i(\omega)| \le C_{A,K}\sqrt{d\ell}, \qquad \min_i \|B_i\|_{\mathrm{op}} \ge c_K\sqrt{d}.$$

Using $N_d \ge c_K\sqrt{d\ell}$, the first of these bounds implies

$$\max_{i,\omega^d=1} \frac{|P_i(\omega)|^2}{N_d^2} \le C_{A,K}.$$

If i lies in the r-th dyadic bin, then

$$\frac{|P_i(\omega)|^2}{\|B_i\|_{\mathrm{op}}^2} \le C_{A,K,a}\, 4^{r+1} \qquad (\omega^d = 1).$$

The final bin is controlled by the Hilbert–Schmidt floor instead:

$$\frac{|P_i(\omega)|^2}{\|B_i\|_{\mathrm{op}}^2} \le C_{A,K}\ell.$$

Summing over all dyadic bins gives the displayed profile bound with W_a.

Condition on any matrix realization satisfying this profile and put

$$V^2 := C_{A,K,a}\left(\frac{n}{\ell} + W_a\right).$$

The Hoeffding-and-union-bound proof of Proposition 5.5 gives

$$\mathbb{P}_\varepsilon\left\{\max_{\omega^d=1}\left|\sum_i \varepsilon_i \frac{P_i(\omega)}{\|B_i\|_{\mathrm{op}}}\right| > t\right\} \le 4d\exp(-ct^2/V^2).$$

Taking $t = C_{A,K,a}\sqrt{V^2\ell}$ yields the displayed sign tail, since $V^2\ell \le C_{A,K,a}(n + W_a\ell)$. The expectation and deterministic certificate follow from Corollary 5.9, and the scalar moment estimates follow from Proposition 5.7. The final assertion is the specialization $W_a \le Rn/\ell$. □

Corollary 5.17 (Probabilistic dyadic-tail criterion)**.** *Fix $A > 0$, $a > 0$, and $R \ge 1$. In the setting of Corollary 5.16, let $Y_{d,a}$ be the one-matrix dyadic*

depth variable

$$Y_{d,a} := \sum_{0\le r<J_\ell} 4^{r+1}\mathbf{1}\Big\{2^{-(r+1)}aN_d \le \|T_d(\xi)\|_{\mathrm{op}} < 2^{-r}aN_d\Big\} + \ell\,\mathbf{1}\Big\{\|T_d(\xi)\|_{\mathrm{op}} < 2^{-J_\ell}aN_d\Big\},$$

and put

$$\mu^{\mathrm{dy}}_{d,a} := \mathbb{E}Y_{d,a}.$$

Then, whenever $n \le d$ *and* $n \ge C_{A,K,a}\ell$,

$$\mathbb{P}\left\{\max_{\omega^d=1}\sum_{i=1}^{n}\frac{|P_i(\omega)|^2}{\|B_i\|^2_{\mathrm{op}}} > C_{A,K,a,R}\frac{n}{\ell}\right\} \le C_{A,K,a,R}d^{-A} + \frac{\ell\,\mu^{\mathrm{dy}}_{d,a}}{R}.$$

On the complement of this event, the random-sign tail, sign-averaged maximum, deterministic certificate, and scalar edge-grid moment package of Corollary 5.16 hold at the $C_{A,K,a,R}\sqrt{n}$ *scale. In particular, if* $\mu^{\mathrm{dy}}_{d,a} = o(1/\ell)$, *then the realization-normalized edge profile holds with probability tending to one; if also* $d/n \to \infty$, *the corresponding full Toeplitz/Hankel contractions have* $O_{K,a,R}(\sqrt{n})$ *discrepancy with probability tending to one.*

There is also a large-sample exponential form. There is an absolute $c_{R,\theta} > 0$ *such that, if* $0 < \theta < R$ *and*

$$\mu^{\mathrm{dy}}_{d,a} \le \frac{\theta}{\ell},$$

then the preceding exceptional probability can be replaced by

$$C_{A,K,a,R}d^{-A} + \exp\left(-c_{R,\theta}\frac{n}{\ell^2}\right).$$

Thus the dyadic lower-tail input gives the same Chernoff-scale profile probability as the single-cutoff theory, but with the low-normalizer mass weighted by its actual dyadic depth.

Proof. For independent matrices $B_i = T_d(\xi^{(i)})$, the realized budget in Corollary 5.16 is

$$W_a = \sum_{i=1}^{n} Y^{(i)}_{d,a}$$

with $Y^{(i)}_{d,a}$ independent copies of $Y_{d,a}$. Hence

$$\mathbb{P}\left\{W_a > R\frac{n}{\ell}\right\} \le \frac{\ell\,\mathbb{E}W_a}{Rn} = \frac{\ell\,\mu^{\mathrm{dy}}_{d,a}}{R}$$

by Markov's inequality. On the event $W_a \le Rn/\ell$, Corollary 5.16 gives the displayed profile and the signing, expectation, certificate, and scalar-moment conclusions outside only the standard $C_{A,K,a}d^{-A}$ matrix event. Proposition 5.1 and the Toeplitz–Hankel reversal give the final discrepancy consequence.

For the exponential form, first note that

$$Y_{d,a} \le C\ell.$$

Indeed $J_\ell = \lceil \frac{1}{2}\log_2(e+\ell) \rceil$, so $4^{J_\ell} \le C(e+\ell) \le C'\ell$, after increasing the absolute constant to cover bounded ℓ. If $\mu^{\mathrm{dy}}_{d,a} \le \theta/\ell$, then

$$\mathbb{E}W_a \le \theta \frac{n}{\ell}.$$

Bernstein's inequality for independent variables bounded by $C\ell$, with

$$\sum_i \mathrm{Var}(Y^{(i)}_{d,a}) \le C\ell\, \mathbb{E}W_a \le C\theta n,$$

gives

$$\mathbb{P}\left\{W_a > R\frac{n}{\ell}\right\} \le \exp\left(-c\frac{(R-\theta)^2 n^2/\ell^2}{\theta n + (R-\theta)n}\right) \le \exp\left(-c_{R,\theta}\frac{n}{\ell^2}\right).$$

Combining this with the d^{-A} matrix event proves the last assertion. □

Corollary 5.18 (Inverse-moment dyadic-tail criterion)**.** *Fix $A > 0$, $a > 0$, and $R \ge 1$. In the setting of Corollary 5.17, define the one-matrix truncated inverse-square normalizer mass*

$$\nu^{(2)}_{d,a} := \mathbb{E}\left[\mathbf{1}_{\{\|T_d(\xi)\|_{\mathrm{op}} < aN_d\}} \min\left\{\left(\frac{aN_d}{\|T_d(\xi)\|_{\mathrm{op}}}\right)^2, \ell\right\}\right], \qquad \ell = \log(e+d),$$

where the minimum is interpreted as ℓ on the event $\|T_d(\xi)\|_{\mathrm{op}} = 0$. Then

$$\mu^{\mathrm{dy}}_{d,a} \le C\,\nu^{(2)}_{d,a}$$

with an absolute constant C. Consequently, whenever $n \le d$ and $n \ge C_{A,K,a}\ell$,

$$\mathbb{P}\left\{\max_{\omega^d=1}\sum_{i=1}^n \frac{|P_i(\omega)|^2}{\|B_i\|^2_{\mathrm{op}}} > C_{A,K,a,R}\frac{n}{\ell}\right\} \le C_{A,K,a,R}d^{-A} + \frac{C\ell\,\nu^{(2)}_{d,a}}{R}.$$

On the complement, the profile, random-sign tail, sign-averaged maximum, deterministic certificate, and scalar edge-grid moment conclusions of Corollary 5.16 hold at the $C_{A,K,a,R}\sqrt{n}$ scale.

There is also a large-sample form. If $0 < \theta < R$ and

$$C\,\nu^{(2)}_{d,a} \le \frac{\theta}{\ell},$$

then the preceding exceptional probability can be replaced by

$$C_{A,K,a,R}d^{-A} + \exp\left(-c_{R,\theta}\frac{n}{\ell^2}\right).$$

Thus any estimate of the truncated inverse-square normalizer mass at order $O(1/\ell)$ is enough for the Chernoff-scale dyadic transfer.

More generally, for $p > 0$ put

$$I_{d,a}^{(p)} := \mathbb{E}\left[\mathbf{1}_{\{\|T_d(\xi)\|_{\mathrm{op}} < aN_d\}} \left(\frac{aN_d}{\|T_d(\xi)\|_{\mathrm{op}}}\right)^p\right].$$

Then

$$\nu_{d,a}^{(2)} \le \ell^{(2-p)_+/2}\, I_{d,a}^{(p)}.$$

In particular, any finite inverse p-moment with $p \ge 2$ controls the dyadic criterion directly, while $0 < p < 2$ only costs the explicit factor $\ell^{1-p/2}$.

Proof. Let $X = \|T_d(\xi)\|_{\mathrm{op}}$ and $Z = aN_d/X$ on the event $\{X < aN_d\}$. If X lies in the r-th finite dyadic bin from the definition of $Y_{d,a}$, then $Z^2 > 4^r$, and hence

$$4^{r+1} \le 4Z^2.$$

At the same time $r < J_\ell$, so

$$4^{r+1} \le 4^{J_\ell} \le C\ell.$$

Therefore the finite-bin contribution is bounded by

$$4^{r+1} \le C \min\{Z^2, \ell\}.$$

On the bottom bin $X < 2^{-J_\ell} aN_d$, one has $Z^2 > 4^{J_\ell} \ge \ell$, so the bottom weight ℓ is exactly bounded by $\min\{Z^2, \ell\}$. Thus

$$Y_{d,a} \le C\, \mathbf{1}_{\{X < aN_d\}} \min\{Z^2, \ell\}$$

pointwise, which gives $\mu_{d,a}^{\mathrm{dy}} \le C\nu_{d,a}^{(2)}$. The probability and large-sample conclusions are the corresponding Markov and Bernstein parts of Corollary 5.17.

Finally, for $u \ge 1$,

$$\min\{u^2, \ell\} \le \ell^{(2-p)_+/2} u^p.$$

Indeed, for $p \ge 2$ this is immediate from $u^2 \le u^p$, while for $0 < p < 2$ it follows by considering separately $u \le \sqrt{\ell}$ and $u > \sqrt{\ell}$. Applying this with $u = Z$ proves the final inverse-moment bound. □

Corollary 5.19 (Inverse-moment random-search and robust certificates)**.** *Fix $A > 0$, $a > 0$, and $R \ge 1$. In the setting of Corollary 5.18, write*

$$v_i(\omega) := \frac{P_i(\omega)}{\|B_i\|_{\mathrm{op}}}, \qquad \omega^d = 1, \qquad \ell = \log(e + d).$$

Then, whenever $n \le d$ and $n \ge C_{A,K,a,R}\ell$, outside a matrix event of probability at most

$$C_{A,K,a,R} d^{-A} + \frac{C\ell\, \nu_{d,a}^{(2)}}{R},$$

the following all hold conditionally on the matrices.

First, for every integer $M \ge 1$, if $\delta^{(1)}, \dots, \delta^{(M)}$ are independent uniform sign vectors, then for every $u \ge 0$,

$$\mathbb{P}_\delta\left\{\max_{1 \le s \le M} \max_{\omega^d = 1} \left|\sum_{i=1}^n \delta_i^{(s)} v_i(\omega)\right| > C_{A,K,a,R}\sqrt{n\left(1 + \frac{\log(eM) + u}{\ell}\right)}\right\} \le e^{-u}.$$

Second, for every integer $M \geq 1$,

$$\mathbb{P}_\delta \left\{ \min_{1\leq s\leq M} \max_{\omega^d=1} \left| \sum_{i=1}^{n} \delta_i^{(s)} v_i(\omega) \right| > C_{A,K,a,R}\sqrt{n} \right\} \leq d^{-AM}.$$

More generally, if G_M *is the number of sampled sign vectors whose edge maximum is at most* $C_{A,K,a,R}\sqrt{n}$, *then for every integer* $1 \leq r \leq M$,

$$\mathbb{P}_\delta\{G_M \leq M - r\} \leq \binom{M}{r} d^{-Ar}.$$

Finally, with Rademacher probability at least $1 - d^{-A}$, *a random sign vector is the center of a Hamming ball of radius*

$$r_0 = \lfloor c_{A,K,a,R}\ell \rfloor$$

on which every signing has edge maximum at most $C_{A,K,a,R}\sqrt{n}$. *If* H_M *counts the sampled signs that are centers of such good Hamming balls, then for every integer* $1 \leq r \leq M$,

$$\mathbb{P}_\delta\{H_M \leq M - r\} \leq \binom{M}{r} d^{-Ar}.$$

Consequently, for every $B > 0$, *if*

$$M \geq \lceil B \rceil$$

independent uniform signs are sampled independently of the matrices, then the joint matrix-and-sign probability that no sampled sign is such a robust center is at most

$$C_{A,K,a,R} d^{-A} + \frac{C\ell\, \nu_{d,a}^{(2)}}{R} + d^{-B}.$$

There is also a large-sample form. If C_* *denotes the absolute constant in Corollary 5.18,* $0 < \theta < R$, *and*

$$C_* \nu_{d,a}^{(2)} \leq \frac{\theta}{\ell},$$

then in all preceding statements the matrix exceptional probability can be replaced by

$$C_{A,K,a,R} d^{-A} + \exp\left(-c_{R,\theta} \frac{n}{\ell^2}\right),$$

and the joint robust-search failure by the same expression plus d^{-B}. *For every* $p > 0$, *all occurrences of* $\nu_{d,a}^{(2)}$ *may be replaced by*

$$\ell^{(2-p)_+/2} \mathbb{E}\left[\mathbf{1}_{\{\|T_d(\xi)\|_{\mathrm{op}} < aN_d\}} \left(\frac{aN_d}{\|T_d(\xi)\|_{\mathrm{op}}} \right)^p \right].$$

Proof. Corollary 5.18 gives the edge profile

$$\max_{\omega^d=1} \sum_{i=1}^{n} |v_i(\omega)|^2 \leq C_{A,K,a,R} \frac{n}{\ell}$$

outside the displayed matrix event. Condition on any realization satisfying this profile. The sampled-list maximum, d^{-AM} amplification, and binomial good-sign count follow from Corollary 5.6, after absorbing the profile constant. The Hamming radius and robust-center binomial tail follow from Corollary 5.10. The polynomial-confidence robust search assertion follows from Corollary 5.11 with B there equal to the displayed confidence exponent, followed by a union bound with the matrix event.

The large-sample replacement is the large-sample part of Corollary 5.18. The final inverse p-moment replacement is also the final assertion of Corollary 5.18. □

Corollary 5.20 (Inverse-moment sampled abundance)**.** *Fix $A > 0$, $a > 0$, $R \geq 1$, $B > 0$, and a fraction $0 < \rho \leq 1$. In the setting of Corollary 5.18, write*

$$\ell = \log(e + d), \qquad v_i(\omega) := \frac{P_i(\omega)}{\|B_i\|_{\mathrm{op}}}, \qquad \omega^d = 1.$$

Let $\delta^{(1)}, \dots, \delta^{(M)}$ be independent uniform sign vectors, independent of the matrices, where $M \geq 1$, and put

$$r_\rho = \lceil \rho M \rceil.$$

There are constants $C_{A,B,K,a,R}$ and $c_{A,B,K,a,R} > 0$ such that, whenever

$$n \leq d, \qquad n \geq C_{A,B,K,a,R}\ell,$$

then, for every $u \geq 0$, with probability at least

$$1 - C_{A,B,K,a,R}d^{-A} - \frac{C\ell\,\nu^{(2)}_{d,a}}{R} - e^{-u} - 2\binom{M}{r_\rho}d^{-Br_\rho},$$

all three conclusions below hold simultaneously.

First, the whole sampled list obeys

$$\max_{1\leq s\leq M}\max_{\omega^d=1}\left|\sum_{i=1}^{n}\delta_i^{(s)}\frac{P_i(\omega)}{\|B_i\|_{\mathrm{op}}}\right| \leq C_{A,B,K,a,R}\sqrt{n\left(1+\frac{\log(eM)+u}{\ell}\right)}.$$

Second, at least $M - r_\rho + 1$ sampled signs have edge maximum at most

$$C_{A,B,K,a,R}\sqrt{n}.$$

Third, at least $M - r_\rho + 1$ sampled signs are centers of Hamming balls of radius

$$\lfloor c_{A,B,K,a,R}\ell \rfloor$$

on which every signing ε' has edge maximum at most

$$C_{A,B,K,a,R}\sqrt{n}.$$

In particular, at least a $1 - \rho$ fraction of the sampled signs are good, and at least a $1 - \rho$ fraction are robust centers.

The preceding display has two useful specializations. First, fix $S > 0$, and choose $B > 0$ and $M \geq 1$ so that

$$B \geq 2\log_2(e/\rho), \qquad M \geq \frac{2(S+1)}{B\rho}.$$

If $d \geq 2$, then, taking $u = S\ell$, with probability at least

$$1 - C_{A,B,K,a,R}d^{-A} - \frac{C\ell\,\nu_{d,a}^{(2)}}{R} - 2d^{-S},$$

the whole sampled list has edge maximum $C_{A,B,K,a,R,S,\rho}\sqrt{n}$, and at least a $1-\rho$ fraction of the sampled signs are robust centers of radius $\lfloor c_{A,B,K,a,R}\ell \rfloor$.

Second, fix $\beta \geq 0$, $\sigma > 0$, and suppose $M \leq d^{\beta}$. Then, after enlarging constants depending on β, σ, with probability at least

$$1 - C_{A,K,a,R,\beta,\sigma}d^{-A} - \frac{C\ell\,\nu_{d,a}^{(2)}}{R} - d^{-\sigma},$$

every sampled sign is the center of a Hamming ball of radius

$$\lfloor c_{A,K,a,R,\beta,\sigma}\ell \rfloor$$

on which every signing has edge maximum at most

$$C_{A,K,a,R,\beta,\sigma}\sqrt{n}.$$

In particular, the entire polynomial-size sampled batch is robust and good.

There is also a large-sample form. If C_* denotes the absolute constant in Corollary 5.18, $0 < \theta < R$, and

$$C_*\nu_{d,a}^{(2)} \leq \frac{\theta}{\ell},$$

then in every displayed probability bound above the matrix part

$$C_{A,K,a,R}d^{-A} + \frac{C\ell\,\nu_{d,a}^{(2)}}{R}$$

may be replaced, after changing constants, by

$$C_{A,K,a,R}d^{-A} + \exp\left(-c_{R,\theta}\frac{n}{\ell^2}\right).$$

For every $p > 0$, all occurrences of $\nu_{d,a}^{(2)}$ may again be replaced by the inverse p-moment expression from Corollary 5.19.

Proof. The general abundance statement is Corollary 5.19 combined with a union bound over the sampled-list event, the good-sign count event, and the robust-center count event, using the sign-count exponent B. The matrix failure is exactly the Markov-form matrix failure in Corollary 5.19.

For the robust-fraction specialization, use

$$\binom{M}{r_\rho} \leq (e/\rho)^{r_\rho}$$

and $B \geq 2\log_2(e/\rho)$ to get, for $d \geq 2$,

$$\binom{M}{r_\rho} d^{-Br_\rho} \leq d^{-Br_\rho/2}.$$

Since $r_\rho \geq \rho M$, the lower bound on M gives $Br_\rho/2 \geq S+1$, hence the two binomial failure terms are bounded by d^{-S}. Taking $u = S\ell$ also gives $e^{-u} \leq d^{-S}$, and the sampled-list maximum is $O_{A,B,K,a,R,S,\rho}(\sqrt{n})$ because M is fixed in terms of B, S, ρ.

For the polynomial-batch specialization, apply the robust-center count with sign exponent $B_{\mathrm{sign}} = \beta + \sigma$ and $r = 1$. The conditional sign failure is at most

$$\binom{M}{1} d^{-(\beta+\sigma)} \leq d^{-\sigma}.$$

On the complementary event all sampled signs are robust centers, and therefore all have edge maximum at the same $O(\sqrt{n})$ scale. The large-sample and inverse p-moment replacements are the corresponding final assertions of Corollary 5.19. □

Corollary 5.21 (Inverse-moment finite vanishing discarded fractions)**.** *Fix $A > 0$, $a > 0$, $R \geq 1$, a sign-count exponent $B > 0$, and $0 < \tau < B$. In the setting of Corollary 5.18, write*

$$\ell = \log(e+d), \qquad v_i(\omega) = \frac{P_i(\omega)}{\|B_i\|_{\mathrm{op}}}, \qquad \omega^d = 1.$$

Let $\delta^{(1)}, \dots, \delta^{(M)}$ be independent uniform sign vectors, independent of the matrices, where $M \geq 1$. Let $0 < \rho \leq 1$ satisfy

$$\rho \geq e\, d^{-\tau},$$

and put $r_\rho = \lceil \rho M \rceil$. There are constants $C_{A,B,K,a,R}$ and $c_{A,B,K,a,R} > 0$ such that, whenever

$$d \geq 2, \qquad n \leq d, \qquad n \geq C_{A,B,K,a,R}\ell,$$

then, for every $u \geq 0$, with probability at least

$$1 - C_{A,B,K,a,R} d^{-A} - \frac{C\ell\, \nu^{(2)}_{d,a}}{R} - e^{-u} - 2d^{-(B-\tau)},$$

all three conclusions below hold simultaneously.

First, the whole sampled list obeys

$$\max_{1\leq s\leq M} \max_{\omega^d=1} \left| \sum_{i=1}^{n} \delta_i^{(s)} v_i(\omega) \right| \leq C_{A,B,K,a,R} \sqrt{n\left(1 + \frac{\log(eM)+u}{\ell}\right)}.$$

Second, at least $M - r_\rho + 1$ sampled signs have edge maximum at most

$$C_{A,B,K,a,R}\sqrt{n}.$$

Third, at least $M - r_\rho + 1$ *sampled signs are centers of Hamming balls of radius*

$$\lfloor c_{A,B,K,a,R}\ell \rfloor$$

on which every signing has edge maximum at most

$$C_{A,B,K,a,R}\sqrt{n}.$$

In particular, at least a $1-\rho$ *fraction of the sampled signs are robust centers. No upper bound on* M *is required; only the sampled-list maximum retains the entropy factor* $\log(eM)$.

There is also a large-sample form. If C_* *denotes the absolute constant in Corollary 5.18,* $0 < \theta < R$, *and*

$$C_*\nu_{d,a}^{(2)} \le \frac{\theta}{\ell},$$

then the matrix part of the displayed probability bound may be replaced, after changing constants, by

$$C_{A,K,a,R}d^{-A} + \exp\left(-c_{R,\theta}\frac{n}{\ell^2}\right).$$

For every $p > 0$, *all occurrences of* $\nu_{d,a}^{(2)}$ *may again be replaced by the inverse* p*-moment expression from Corollary 5.19.*

Proof. Apply Corollary 5.20 with the same fraction ρ. The only point is to simplify the binomial count tail. Since

$$\binom{M}{r_\rho} \le \left(\frac{eM}{r_\rho}\right)^{r_\rho} \le \left(\frac{e}{\rho}\right)^{r_\rho},$$

the lower bound $\rho \ge ed^{-\tau}$ gives

$$\binom{M}{r_\rho} d^{-Br_\rho} \le d^{-(B-\tau)r_\rho} \le d^{-(B-\tau)}.$$

Substituting this into the finite abundance theorem gives the displayed probability and the sampled-list, good-count, and robust-center-count conclusions. Finally,

$$M - r_\rho + 1 \ge M - \rho M,$$

so the robust-center count is at least a $1-\rho$ fraction. The large-sample and inverse p-moment variants are the corresponding final assertions of Corollary 5.20. □

Corollary 5.22 (Almost-sure inverse-moment dyadic-tail transfer)**.** *Fix* $A > 0$, $a > 0$, *and* $R \ge 1$. *Let* $(d_m, n_m)_{m\ge1}$ *be a triangular-array sequence with* $n_m \le d_m$, *put* $\ell_m = \log(e + d_m)$, *and assume*

$$n_m \ge C_{A,K,a,R}\ell_m \qquad (m \ge 1).$$

For each m, let $B_{m,i} = T_{d_m}(\xi^{(m,i)})$, $1 \le i \le n_m$, be independent full Toeplitz matrices whose lag variables are independent, mean zero, unit variance, and have sub-Gaussian norm at most K, and let $P_{m,i}$ be their symbols. Define

$$\nu^{(2)}_{m,a} := \mathbb{E}\left[\mathbf{1}_{\{\|T_{d_m}(\xi)\|_{\mathrm{op}} < aN_{d_m}\}} \min\left\{\left(\frac{aN_{d_m}}{\|T_{d_m}(\xi)\|_{\mathrm{op}}}\right)^2, \ell_m\right\}\right],$$

where the minimum is interpreted as ℓ_m on the event $\|T_{d_m}(\xi)\|_{\mathrm{op}} = 0$. If

$$\sum_{m=1}^{\infty}\left(d_m^{-A} + \frac{\ell_m \nu^{(2)}_{m,a}}{R}\right) < \infty,$$

then, with probability one over the matrix triangular array, all but finitely many m satisfy

$$\max_{\omega^{d_m}=1} \sum_{i=1}^{n_m} \frac{|P_{m,i}(\omega)|^2}{\|B_{m,i}\|^2_{\mathrm{op}}} \le C_{A,K,a,R}\frac{n_m}{\ell_m}.$$

For those m, the signing, sign-averaged maximum, deterministic-certificate, and scalar edge-grid moment conclusions of Corollary 5.16 hold with (d, n, ℓ) replaced by (d_m, n_m, ℓ_m) at scale $C_{A,K,a,R}\sqrt{n_m}$.

Moreover, put

$$v_{m,i}(\omega) := \frac{P_{m,i}(\omega)}{\|B_{m,i}\|_{\mathrm{op}}}, \qquad \omega^{d_m} = 1.$$

On the same almost-sure event, all but finitely many m also have the conditional sampled-list maximum, d_m^{-AM} success amplification, binomial good-sign counts, Hamming radius $\lfloor c_{A,K,a,R}\ell_m \rfloor$, and robust-center binomial tails of Corollary 5.54, with $v_{m,i}$ as above and constants depending on A, K, a, R.

There is also a large-sample form. If, for the absolute constant C in Corollary 5.18, there is a $0 < \theta < R$ such that

$$C\nu^{(2)}_{m,a} \le \frac{\theta}{\ell_m}$$

for all sufficiently large m, and if

$$\sum_{m=1}^{\infty}\left(d_m^{-A} + \exp\left(-c_{R,\theta}\frac{n_m}{\ell_m^2}\right)\right) < \infty,$$

then all preceding almost-sure matrix and conditional sign-search conclusions hold with this Bernstein summability condition in place of the Markov summability condition.

Finally, suppose independent uniform sign lists

$$\delta^{(1)}_m, \dots, \delta^{(M_m)}_m \in \{-1, 1\}^{n_m}, \qquad M_m \ge \lceil \alpha \rceil,$$

are sampled independently of the matrices. If

$$\sum_{m=1}^{\infty}\left(d_m^{-A} + \frac{\ell_m \nu^{(2)}_{m,a}}{R} + d_m^{-\alpha}\right) < \infty,$$

then, with probability one over the matrices and sampled signs, all but finitely many m contain a sampled sign that is the center of a Hamming ball of radius $\lfloor c_{A,K,a,R}\ell_m \rfloor$ on which every signing ε' satisfies

$$\max_{\omega^{d_m}=1} \left| \sum_{i=1}^{n_m} \varepsilon'_i \frac{P_{m,i}(\omega)}{\|B_{m,i}\|_{\mathrm{op}}} \right| \le C_{A,K,a,R}\sqrt{n_m}.$$

The same sampled-certificate conclusion holds under the large-sample Bernstein summability condition after adding the term $d_m^{-\alpha}$.

For every $p > 0$, all displayed summability conditions remain valid if $\nu^{(2)}_{m,a}$ is replaced by

$$\ell_m^{(2-p)_+/2}\mathbb{E}\left[\mathbf{1}_{\{\|T_{d_m}(\xi)\|_{\mathrm{op}}<aN_{d_m}\}}\left(\frac{aN_{d_m}}{\|T_{d_m}(\xi)\|_{\mathrm{op}}}\right)^p\right].$$

Proof. Apply Corollary 5.18 at each index m. The Markov part gives a matrix exceptional probability bounded by a constant times

$$d_m^{-A} + \frac{\ell_m \nu^{(2)}_{m,a}}{R},$$

so the first displayed summability condition and Borel–Cantelli give the eventual edge profile. The signing, sign-averaged maximum, deterministic-certificate, and scalar moment statements are deterministic consequences of that same good profile event in Corollary 5.16.

Once the edge profile holds, Corollaries 5.6 and 5.10, applied with the profile constant $C_{A,K,a,R}$, give the conditional sampled-list, success-count, and robust Hamming-center conclusions. This is exactly the package displayed in Corollary 5.54, with constants depending also on R.

If $C\nu^{(2)}_{m,a} \le \theta/\ell_m$ eventually, the large-sample part of Corollary 5.18 replaces the Markov matrix exception by

$$C_{A,K,a,R}d_m^{-A} + \exp\left(-c_{R,\theta}\frac{n_m}{\ell_m^2}\right).$$

The Bernstein summability assumption therefore gives the same almost-sure eventual conclusions.

For the sampled-certificate assertion, condition on a matrix realization with the eventual edge profile and apply Corollary 5.11 with $B = \alpha$. The failure probability for the sampled sign list is at most $d_m^{-\alpha}$, so the displayed joint summability condition and Borel–Cantelli give an eventual sampled robust center. The Bernstein variant is identical, with the matrix exception replaced as above. The final inverse p-moment replacement is the last assertion of Corollary 5.18. □

Corollary 5.23 (Almost-sure inverse-moment sampled abundance)**.** *Fix $A > 0$, $a > 0$, $R \ge 1$, and a sign-count exponent $B_{\mathrm{s}} > 0$. Let $(d_m, n_m)_{m\ge1}$, ℓ_m, $B_{m,i} = T_{d_m}(\xi^{(m,i)})$, $P_{m,i}$, and $\nu^{(2)}_{m,a}$ be as in Corollary 5.22, with*

$$n_m \le d_m, \qquad n_m \ge C_{A,B_{\mathrm{s}},K,a,R}\ell_m.$$

At each m, independently of the matrices, sample independent uniform sign vectors

$$\delta_m^{(1)},\dots,\delta_m^{(M_m)}\in\{-1,1\}^{n_m},\qquad M_m\geq 1.$$

Let $u_m\geq 0$ and $1\leq r_m\leq M_m$ be deterministic parameters. If

$$\sum_{m=1}^{\infty}\left(d_m^{-A}+\frac{\ell_m\nu_{m,a}^{(2)}}{R}+e^{-u_m}+\binom{M_m}{r_m}d_m^{-B_{\mathrm{s}}r_m}\right)<\infty,$$

then, with probability one over the matrices and sampled signs, all but finitely many m satisfy all three conclusions below, with

$$v_{m,i}(\omega):=\frac{P_{m,i}(\omega)}{\|B_{m,i}\|_{\mathrm{op}}},\qquad \omega^{d_m}=1.$$

First,

$$\max_{1\leq s\leq M_m}\max_{\omega^{d_m}=1}\left|\sum_{i=1}^{n_m}\delta_{m,i}^{(s)}v_{m,i}(\omega)\right|\leq C_{A,B_{\mathrm{s}},K,a,R}\sqrt{n_m\left(1+\frac{\log(eM_m)+u_m}{\ell_m}\right)}.$$

Second, at least M_m-r_m+1 sampled signs have edge maximum at most

$$C_{A,B_{\mathrm{s}},K,a,R}\sqrt{n_m}.$$

Third, at least M_m-r_m+1 sampled signs are centers of Hamming balls of radius

$$\lfloor c_{A,B_{\mathrm{s}},K,a,R}\ell_m\rfloor$$

on which every signing has edge maximum at most

$$C_{A,B_{\mathrm{s}},K,a,R}\sqrt{n_m}.$$

There are two useful consequences. First, fix $0<\rho\leq 1$ and $S>0$. Choose B_{s} and an integer $M\geq 1$ so that

$$B_{\mathrm{s}}\geq 2\log_2(e/\rho),\qquad M\geq\frac{2(S+1)}{B_{\mathrm{s}}\rho}.$$

If $d_m\geq 2$ and

$$\sum_{m=1}^{\infty}\left(d_m^{-A}+\frac{\ell_m\nu_{m,a}^{(2)}}{R}+d_m^{-S}\right)<\infty,$$

then any fixed sampled list of this size M has, eventually almost surely, edge maximum $C_{A,B_{\mathrm{s}},K,a,R,S,\rho}\sqrt{n_m}$ on the whole list and a $1-\rho$ fraction of logarithmic Hamming-robust centers.

Second, fix $\beta\geq 0$ and $\sigma>0$. If $M_m\leq d_m^{\beta}$ and

$$\sum_{m=1}^{\infty}\left(d_m^{-A}+\frac{\ell_m\nu_{m,a}^{(2)}}{R}+d_m^{-\sigma}\right)<\infty,$$

then, eventually almost surely, every sampled sign in the polynomial-size batch is the center of a Hamming ball of radius

$$\lfloor c_{A,K,a,R,\beta,\sigma}\ell_m\rfloor$$

on which every signing has edge maximum at most

$$C_{A,K,a,R,\beta,\sigma}\sqrt{n_m}.$$

In particular, the whole sampled batch is robust and good.

There is also a large-sample form. If C_* denotes the absolute constant in Corollary 5.18, $0 < \theta < R$, and

$$C_*\nu_{m,a}^{(2)} \le \frac{\theta}{\ell_m}$$

for all sufficiently large m, then each displayed summability condition above remains valid with the matrix part

$$d_m^{-A} + \frac{\ell_m \nu_{m,a}^{(2)}}{R}$$

replaced by

$$d_m^{-A} + \exp\left(-c_{R,\theta}\frac{n_m}{\ell_m^2}\right).$$

For every $p > 0$, all occurrences of $\nu_{m,a}^{(2)}$ may again be replaced by the inverse p-moment expression from Corollary 5.22.

Proof. For the general count statement, apply Corollary 5.20 at index m, with

$$\rho_m = \frac{r_m}{M_m}.$$

Then $r_{\rho_m} = r_m$, and the constants in the finite abundance statement do not depend on ρ_m. The displayed summability condition dominates the finite matrix-and-sign failure probabilities, so Borel–Cantelli gives the eventual sampled-list maximum, good-sign count, and robust-center count.

For the fixed robust-fraction consequence, set

$$r_m = \lceil \rho M \rceil, \qquad u_m = S\ell_m.$$

As in the proof of Corollary 5.20,

$$\binom{M}{r_m} d_m^{-B_s r_m} \le d_m^{-S} \qquad (d_m \ge 2),$$

after increasing constants, and $e^{-u_m} \le d_m^{-S}$. The whole sampled list is $O_{A,B_s,K,a,R,S,\rho}(\sqrt{n_m})$ because M is fixed, and

$$M - r_m + 1 \ge M - \rho M.$$

Thus at least a $1-\rho$ fraction of the sampled list consists of robust centers.

For the polynomial-batch consequence, apply the polynomial-batch specialization of Corollary 5.20 at each m. Its additional sign failure is at most $d_m^{-\sigma}$, so the displayed summability condition again gives the eventual all-robust batch conclusion by Borel–Cantelli. The large-sample and inverse p-moment replacements are the corresponding final assertions of Corollary 5.20. □

Corollary 5.24 (Inverse-moment vanishing discarded fractions)**.** *Fix $A > 0$, $a > 0$, $R \geq 1$, a sign-count exponent $B_{\mathrm{s}} > 0$, and $0 < \tau < B_{\mathrm{s}}$. In the setting of Corollary 5.23, assume $d_m \geq 2$ eventually, let $M_m \geq 1$ be arbitrary deterministic sampled-batch sizes, let $u_m \geq 0$, and let $(\rho_m)_{m\geq 1} \subset (0,1]$ satisfy*

$$\rho_m \geq e\, d_m^{-\tau} \qquad \text{for all sufficiently large } m.$$

Put

$$r_m = \lceil \rho_m M_m \rceil.$$

If

$$\sum_{m=1}^{\infty} \left(d_m^{-A} + \frac{\ell_m \nu_{m,a}^{(2)}}{R} + e^{-u_m} + d_m^{-(B_{\mathrm{s}}-\tau)} \right) < \infty,$$

then, with probability one over the matrices and sampled signs, all but finitely many m satisfy the sampled-list maximum bound from Corollary 5.23 and at least a $1-\rho_m$ fraction of the M_m sampled signs have edge maximum

$$C_{A,B_{\mathrm{s}},K,a,R}\sqrt{n_m}$$

and are centers of Hamming balls of radius

$$\lfloor c_{A,B_{\mathrm{s}},K,a,R}\ell_m \rfloor$$

on which every signing has edge maximum at most

$$C_{A,B_{\mathrm{s}},K,a,R}\sqrt{n_m}.$$

No upper bound on M_m is required. In particular, if $\rho_m \to 0$, then the robust-center fraction tends to one along the sampled batches, while the whole-list maximum retains only the entropy factor $\log(eM_m)$.

If the large-sample hypothesis

$$C_* \nu_{m,a}^{(2)} \leq \frac{\theta}{\ell_m}$$

holds eventually for some $0 < \theta < R$, then the same conclusion holds with the matrix summand

$$d_m^{-A} + \frac{\ell_m \nu_{m,a}^{(2)}}{R}$$

replaced by

$$d_m^{-A} + \exp\left(-c_{R,\theta} \frac{n_m}{\ell_m^2} \right).$$

For every $p > 0$, the same inverse p-moment replacement as in Corollary 5.23 is valid.

Proof. Apply Corollary 5.23 with the displayed r_m. It remains only to check its binomial count-tail summability. Since $r_m = \lceil \rho_m M_m \rceil$,

$$\binom{M_m}{r_m} \leq \left(\frac{eM_m}{r_m} \right)^{r_m} \leq \left(\frac{e}{\rho_m} \right)^{r_m}.$$

The lower bound on ρ_m gives, for all large m,

$$\binom{M_m}{r_m} d_m^{-B_{\mathrm{s}} r_m} \le d_m^{-(B_{\mathrm{s}}-\tau) r_m} \le d_m^{-(B_{\mathrm{s}}-\tau)}.$$

The displayed summability condition therefore implies the general sampled-count hypothesis of Corollary 5.23. Finally,

$$M_m - r_m + 1 \ge M_m - \rho_m M_m,$$

so the robust-center count is at least a $1-\rho_m$ fraction. The large-sample and inverse p-moment variants are the corresponding final assertions of Corollary 5.23. □

Corollary 5.25 (Inverse-moment operator-norm robust batches)**.** *In the setting of Corollary 5.23, assume in addition that*

$$d_m/n_m \to \infty, \qquad \log d_m = o(n_m).$$

Put

$$A_{m,i} := \frac{B_{m,i}}{\|B_{m,i}\|_{\mathrm{op}}}, \qquad \widetilde{A}_{m,i} := J_{d_m} A_{m,i}.$$

If the summability hypothesis of Corollary 5.23 holds for parameters u_m and r_m, then, with probability one over the matrices and sampled signs, all but finitely many m satisfy the sampled-list edge maximum bound from that corollary and at least $M_m - r_m + 1$ sampled signs are centers of Hamming balls of radius

$$\lfloor c_{A,B_{\mathrm{s}},K,a,R} \ell_m \rfloor$$

on which every signing ε' obeys both operator-norm estimates

$$\left\| \sum_{i=1}^{n_m} \varepsilon'_i A_{m,i} \right\|_{\mathrm{op}} \le C_{A,B_{\mathrm{s}},K,a,R} \sqrt{n_m}, \qquad \left\| \sum_{i=1}^{n_m} \varepsilon'_i \widetilde{A}_{m,i} \right\|_{\mathrm{op}} \le C_{A,B_{\mathrm{s}},K,a,R} \sqrt{n_m},$$

after increasing the constant.

Consequently, under the hypotheses of Corollary 5.24, the same almost-sure event has the following stronger conclusion: all but finitely many m have at least a $1-\rho_m$ fraction of the sampled signs as logarithmic-radius robust centers for the realization-normalized full Toeplitz and full Hankel operator-norm discrepancy problems. No upper bound on M_m is required; only the sampled-list edge maximum keeps the entropy factor $\log(eM_m)$.

The large-sample Bernstein and inverse p-moment replacements from Corollaries 5.23 and 5.24 remain valid.

Proof. The abundance corollary gives, eventually almost surely, at least $M_m - r_m + 1$ sampled signs whose Hamming balls have edge maximum at most $C_{A,B_{\mathrm{s}},K,a,R}\sqrt{n_m}$. Apply Corollary 5.2 to each of these Hamming balls, or equivalently to their union, and absorb the deterministic reduction constant. This gives the Toeplitz and Hankel operator-norm estimates on the same almost-sure event.

For the vanishing-fraction statement, use Corollary 5.24; its count estimate gives

$$M_m - r_m + 1 \geq M_m - \rho_m M_m.$$

The large-sample and inverse p-moment variants are inherited from the same two inverse-moment Borel corollaries, since the edge-to-operator transfer is deterministic. □

Corollary 5.26 (Concrete inverse-moment Borel schedules)**.** *Fix $a > 0$, $R \geq 1$, and $C_0 < \infty$. For each d, let $\ell_d = \log(e + d)$ and let $\nu_{d,a}^{(2)}$ be the truncated inverse-square normalizer mass from Corollary 5.18. Let $(d_m, n_m)_{m\geq 1}$ be a triangular-array sequence with $n_m \leq d_m$, put $\ell_m = \log(e + d_m)$, and let $B_{m,i} = T_{d_m}(\xi^{(m,i)})$, $1 \leq i \leq n_m$, be independent full Toeplitz matrices whose lag variables are independent, mean zero, unit variance, and have sub-Gaussian norm at most K, with symbols $P_{m,i}$.*

In each of the following Markov-form situations, with constants depending on the displayed matrix-event exponent A, the almost-sure matrix and conditional sign-search conclusions of Corollary 5.22 hold:

(i) *$d_m = m$, $A > 1$,*

$$C_{A,K,a,R}\ell_m \leq n_m \leq d_m, \qquad \nu_{d,a}^{(2)} \leq C_0 d^{-B} \quad (d \geq 1)$$

for some $B > 1$.

(ii) *$d_m \geq c_0 m^{\kappa}$ for some $c_0, \kappa > 0$ and $A > 0$,*

$$C_{A,K,a,R}\ell_m \leq n_m \leq d_m, \qquad \nu_{d,a}^{(2)} \leq C_0 d^{-B} \quad (d \geq 1),$$

and $A\kappa > 1$, $B\kappa > 1$.

(iii) *$d_m \geq \exp(c_0 m^{\kappa})$ for some $c_0, \kappa > 0$ and $A > 0$,*

$$C_{A,K,a,R}\ell_m \leq n_m \leq d_m, \qquad \nu_{d,a}^{(2)} \leq C_0 \ell_d^{-B} \quad (d \geq 1),$$

and $B > 1 + 1/\kappa$.

Thus, in each case, with probability one over the matrices, all but finitely many m satisfy

$$\max_{\omega^{d_m}=1} \sum_{i=1}^{n_m} \frac{|P_{m,i}(\omega)|^2}{\|B_{m,i}\|_{\mathrm{op}}^2} \leq C_{A,K,a,R} \frac{n_m}{\ell_m},$$

together with the dyadic signing, sign-averaged maximum, deterministic-certificate, scalar moment, sampled-list, success-count, Hamming-ball, and robust-center conditional conclusions of Corollary 5.22.

If independent sign lists

$$\delta_m^{(1)}, \ldots, \delta_m^{(M_m)} \in \{-1, 1\}^{n_m}, \qquad M_m \geq \lceil \alpha \rceil,$$

are also sampled, then the actual sampled robust-certificate conclusion of Corollary 5.22 also holds whenever $\sum_m d_m^{-\alpha} < \infty$. In particular, this requires $\alpha > 1$ in the all-integer case, $\alpha\kappa > 1$ in the polynomial schedule case, and no lower bound beyond $\alpha > 0$ in the stretched-exponential case.

The robust-fraction and polynomial-batch conclusions of Corollary 5.23 also hold on the same Markov-form schedules after adding the corresponding sign summability condition. Thus a fixed $O_{\rho,S}(1)$ sampled list has an eventual $1-\rho$ fraction of logarithmically robust centers whenever $\sum_m d_m^{-S} < \infty$, which means $S > 1$ in the all-integer case, $S\kappa > 1$ in the polynomial schedule case, and any $S > 0$ in the stretched-exponential case. If $M_m \le d_m^{\beta}$, then the whole sampled batch is eventually robust whenever $\sum_m d_m^{-\sigma} < \infty$, which means $\sigma > 1$, $\sigma\kappa > 1$, and any $\sigma > 0$, respectively. The vanishing discarded-fraction conclusion of Corollary 5.24 also holds with no upper bound on M_m: for any sign-count exponent B_{s} and any $0 < \tau < B_{\mathrm{s}}$, arbitrary sampled batches have an eventual $1-\rho_m$ robust-center fraction whenever $\rho_m \ge e d_m^{-\tau}$ eventually, $\sum_m e^{-u_m} < \infty$, and $\sum_m d_m^{-(B_{\mathrm{s}}-\tau)} < \infty$. Thus the all-integer case requires $B_{\mathrm{s}} - \tau > 1$, the polynomial schedule case requires $(B_{\mathrm{s}} - \tau)\kappa > 1$, and the stretched-exponential case needs only $\tau < B_{\mathrm{s}}$.

There is also a critical Bernstein form. Let C_* be the absolute constant in Corollary 5.18. Suppose

$$\nu_{d,a}^{(2)} \le \frac{\eta}{\ell_d} \qquad (d \ge 1), \qquad C_*\eta < R.$$

Choose $C_*\eta < \theta < R$. Then the same eventual matrix and conditional sign-search conclusions hold in each triangular array for which

$$\sum_{m=1}^{\infty} \left(d_m^{-A} + \exp\left(-c_{R,\theta} \frac{n_m}{\ell_m^2} \right) \right) < \infty.$$

Consequently, along all integer dimensions, if $A > 1$ and

$$n(d) \le d, \qquad n(d) \ge q \log^3(e+d) \quad \text{for all large } d, \qquad q > \frac{1}{c_{R,\theta}},$$

then the conclusions hold almost surely for all sufficiently large d. On stretched-exponential schedules $d_m \ge \exp(c_0 m^{\kappa})$, they hold if

$$n_m \ge q\, \ell_m^2 \log(e+m) \quad \text{eventually with} \quad q > \frac{1}{c_{R,\theta}},$$

and also under the simpler superquadratic logarithmic window

$$n_m \ge Q\, \ell_m^{2+\tau} \qquad \text{eventually}$$

for any $Q, \tau > 0$. The corresponding actual sampled robust-certificate claims again follow after adding any sign-list exponent α with $\sum_m d_m^{-\alpha} < \infty$. The same critical Bernstein schedules inherit the fixed robust-fraction conclusion after adding $\sum_m d_m^{-S} < \infty$, and the all-robust polynomial-batch conclusion after adding $\sum_m d_m^{-\sigma} < \infty$. Thus the all-integer $\log^3 d$ window requires $S > 1$ and $\sigma > 1$, while the stretched-exponential critical windows allow every $S, \sigma > 0$. They also inherit the vanishing discarded-fraction conclusion under the same extra sign-series condition $\sum_m d_m^{-(B_{\mathrm{s}}-\tau)} < \infty$: hence $B_{\mathrm{s}} - \tau > 1$ in the all-integer critical $\log^3 d$ window and only $\tau < B_{\mathrm{s}}$ on the stretched-exponential critical windows.

Proof. It is enough to verify the summability hypotheses in Corollary 5.22. In the all-integer polynomial-decay case,

$$\sum_m m^{-A} < \infty, \qquad \sum_m \ell_m \nu^{(2)}_{d_m,a} \le C_0 \sum_m \log(e+m) m^{-B} < \infty$$

because $A, B > 1$. In the polynomial schedule case,

$$d_m^{-A} \le C m^{-A\kappa}, \qquad \ell_m \nu^{(2)}_{d_m,a} \le C_{B,C_0,c_0}(\log(e+m)) m^{-B\kappa},$$

and both series converge under $A\kappa > 1$ and $B\kappa > 1$. In the stretched-exponential logarithmic-decay case, $\sum_m d_m^{-A} < \infty$ and

$$\ell_m \nu^{(2)}_{d_m,a} \le C_0 \ell_m^{1-B} \le C_{B,C_0,c_0} m^{\kappa(1-B)},$$

which is summable precisely when $B > 1 + 1/\kappa$. This proves the Markov-form matrix and conditional sign-search conclusions. The sampled-list assertion is the sampled-certificate part of Corollary 5.22, since the additional summability condition is exactly $\sum_m d_m^{-\alpha} < \infty$. The robust-fraction and polynomial-batch addenda follow in the same way from Corollary 5.23; the displayed conditions $S > 1$, $S\kappa > 1$, $\sigma > 1$, and $\sigma\kappa > 1$ are exactly the convergence tests for the extra series $\sum_m d_m^{-S}$ and $\sum_m d_m^{-\sigma}$, while stretched-exponential growth makes both series converge for every positive exponent. The vanishing discarded-fraction addendum follows from Corollary 5.24; the conditions $B_{\mathrm{s}} - \tau > 1$ and $(B_{\mathrm{s}} - \tau)\kappa > 1$ are exactly the convergence tests for $\sum_m d_m^{-(B_{\mathrm{s}}-\tau)}$, while stretched-exponential growth makes this series converge for every $B_{\mathrm{s}} - \tau > 0$.

For the critical branch, the hypothesis on η gives

$$C_* \nu^{(2)}_{d_m,a} \le \frac{\theta}{\ell_m}$$

for every m, so the large-sample part of Corollary 5.22 applies under the displayed Bernstein summability condition. Along integer dimensions,

$$\exp\left(-c_{R,\theta} \frac{n(d)}{\log^2(e+d)}\right) \le (e+d)^{-c_{R,\theta} q},$$

which is summable by the lower bound on q; $\sum_d d^{-A}$ is summable because $A > 1$. On a stretched-exponential schedule,

$$\exp\left(-c_{R,\theta} \frac{n_m}{\ell_m^2}\right) \le (e+m)^{-c_{R,\theta} q}$$

under the first displayed lower bound on n_m, and the exponent assumption again gives summability. Under $n_m \ge Q \ell_m^{2+\tau}$, the same term is bounded by $\exp(-cQ\ell_m^{\tau}) \le \exp(-c' m^{\kappa\tau})$, which is summable. The sampled robust-certificate addendum is again the sampled part of Corollary 5.22. The robust-fraction and polynomial-batch critical addenda are the corresponding parts of Corollary 5.23, with the same sign-series convergence tests as above. The critical vanishing discarded-fraction addendum is the corresponding

large-sample form of Corollary 5.24, again with only the extra sign-series convergence test. □

Corollary 5.27 (Power-curve dyadic-tail criterion)**.** *Fix $A > 0$, $a > 0$, $L > 0$, $\gamma \geq 1$, and $\beta > 2$. In the setting of Corollary 5.17, put*

$$F_{d,a}(t) := \mathbb{P}\left\{ \|T_d(\xi)\|_{\mathrm{op}} < t\, a N_d \right\}, \qquad 0 < t \leq 1.$$

Assume that, for the dimension d,

$$F_{d,a}(t) \leq L\, \ell^{-\gamma} t^{\beta} \qquad (0 < t \leq 1), \qquad \ell = \log(e + d).$$

Then

$$\mu_{d,a}^{\mathrm{dy}} \leq C_\beta L\, \ell^{-\gamma}.$$

Consequently, whenever $n \leq d$, $n \geq C_{A,K,a}\ell$, and $R \geq 1$,

$$\mathbb{P}\left\{ \max_{\omega^d=1} \sum_{i=1}^{n} \frac{|P_i(\omega)|^2}{\|B_i\|_{\mathrm{op}}^2} > C_{A,K,a,R} \frac{n}{\ell} \right\} \leq C_{A,K,a,R} d^{-A} + \frac{C_\beta L}{R\, \ell^{\gamma-1}}.$$

On the complement, the profile, random-sign tail, sign-averaged maximum, deterministic certificate, and scalar edge-grid moment conclusions of Corollary 5.16 hold at the $C_{A,K,a,R}\sqrt{n}$ scale.

In the critical-density case $\gamma = 1$, if $R > C_\beta L$, the exceptional probability can instead be bounded by

$$C_{A,K,a,R,\beta,L} d^{-A} + \exp\left(-c_{R,\beta,L} \frac{n}{\ell^2} \right).$$

Thus a power lower-tail curve with exponent strictly larger than 2 permits top-scale low-normalizer mass of order $1/\log d$, rather than the single-cutoff order $1/\log^2 d$, as long as the deeper tail is charged by the dyadic profile.

Proof. For each $0 \leq r < J_\ell$, the r-th dyadic event in the definition of $Y_{d,a}$ has probability at most $F_{d,a}(2^{-r})$. Hence

$$\begin{aligned} \mu_{d,a}^{\mathrm{dy}} &\leq \sum_{0 \leq r < J_\ell} 4^{r+1} F_{d,a}(2^{-r}) + \ell F_{d,a}(2^{-J_\ell}) \\ &\leq \frac{4L}{\ell^\gamma} \sum_{r \geq 0} 2^{(2-\beta)r} + L\, \ell^{1-\gamma} 2^{-\beta J_\ell}. \end{aligned}$$

Since $\beta > 2$, the geometric sum is finite. Also $2^{J_\ell} \geq (e + \ell)^{1/2}$, and therefore

$$\ell\, 2^{-\beta J_\ell} \leq \ell (e + \ell)^{-\beta/2} \leq C_\beta$$

uniformly in d. This proves $\mu_{d,a}^{\mathrm{dy}} \leq C_\beta L \ell^{-\gamma}$. The Markov bound and the profile package are exactly Corollary 5.17.

If $\gamma = 1$ and $R > C_\beta L$, enlarge C_β in the preceding display only once and choose θ with $C_\beta L < \theta < R$. The exponential form of Corollary 5.17 then gives the displayed Bernstein bound. □

Corollary 5.28 (Empirical bad-normalizer criterion)**.** *Fix $A > 0$, $a > 0$, and $R \geq 1$. Let $B_i = T_d(\xi^{(i)})$, $1 \leq i \leq n$, be independent full Toeplitz matrices whose lag variables are independent, mean zero, unit variance, and have sub-Gaussian norm at most K. Let P_i be their symbols, put $N_d = \mathbb{E}\,\|T_d(\xi)\|_{\mathrm{op}}$, $\ell = \log(e+d)$, and*

$$M_a := \#\{1 \leq i \leq n : \|B_i\|_{\mathrm{op}} < aN_d\}.$$

There is a constant $C_{A,K,a,R}$ such that, whenever

$$n \leq d, \qquad n \geq C_{A,K,a,R}\ell,$$

one has

$$\mathbb{P}\left(\left\{\max_{\omega^d=1}\sum_{i=1}^{n}\frac{|P_i(\omega)|^2}{\|B_i\|_{\mathrm{op}}^2} > C_{A,K,a,R}\frac{n}{\ell}\right\}\cap\left\{M_a \leq R\frac{n}{\ell^2}\right\}\right) \leq C_{A,K,a,R}d^{-A}.$$

Whenever the displayed profile bound holds, independent signs have

$$\mathbb{P}_\varepsilon\left\{\max_{\omega^d=1}\left|\sum_{i=1}^{n}\varepsilon_i\frac{P_i(\omega)}{\|B_i\|_{\mathrm{op}}}\right| > C_{A,K,a,R}\sqrt{n}\right\} \leq d^{-A},$$

and

$$\mathbb{E}_\varepsilon\max_{\omega^d=1}\left|\sum_{i=1}^{n}\varepsilon_i\frac{P_i(\omega)}{\|B_i\|_{\mathrm{op}}}\right| \leq C_{A,K,a,R}\sqrt{n}.$$

Thus a deterministic sign vector with edge maximum $C_{A,K,a,R}\sqrt{n}$ exists. On the same profile event, simultaneously for every integer $k \geq 1$,

$$\max_{\omega^d=1}\mathbb{E}_\varepsilon\left|\sum_{i=1}^{n}\varepsilon_i\frac{P_i(\omega)}{\|B_i\|_{\mathrm{op}}}\right|^{2k} \leq \left(C_{A,K,a,R}k\,\frac{n}{\ell}\right)^k.$$

Proof. Let

$$G = \{i : \|B_i\|_{\mathrm{op}} \geq aN_d\}.$$

The expected-normalized edge-profile event of Proposition 5.13 gives

$$\max_{\omega^d=1}\sum_{i\in G}\frac{|P_i(\omega)|^2}{\|B_i\|_{\mathrm{op}}^2} \leq C_{A,K,a}\frac{n}{\ell}$$

outside an event of probability at most d^{-A}, after increasing constants. The same union bound and Paley–Zygmund/Chernoff Hilbert–Schmidt floor used in the proof of Theorem 5.14 give, outside another $C_{A,K}d^{-A}$ event,

$$\max_{i,\omega^d=1}|P_i(\omega)| \leq C_{A,K}\sqrt{d\ell}, \qquad \min_i\|B_i\|_{\mathrm{op}} \geq c_K\sqrt{d}.$$

Hence every index outside G contributes at most $C_{A,K}\ell$ to each edge node. On $\{M_a \leq Rn/\ell^2\}$, the full bad-normalizer contribution is therefore at most $C_{A,K,R}n/\ell$. Combining the good and bad contributions proves the displayed profile estimate. The random-sign tail, conditional expectation bound, deterministic certificate, and scalar moment estimates are Proposition 5.5, Corollary 5.9, and Proposition 5.7. □

Corollary 5.29 (Quantitative normalizer-tail bookkeeping)**.** *Fix $A > 0$ and $a > 0$. Let $B_i = T_d(\xi^{(i)})$, $1 \le i \le n$, be independent full Toeplitz matrices whose lag variables are independent, mean zero, unit variance, and have sub-Gaussian norm at most K. Let P_i be their symbols, put $N_d = \mathbb{E}\,\|T_d(\xi)\|_{\mathrm{op}}$, $\ell = \log(e+d)$, and*

$$p_d(a) = \mathbb{P}\{\|T_d(\xi)\|_{\mathrm{op}} < aN_d\}.$$

There is a constant $C_{A,K,a}$ such that, whenever

$$n \le d, \qquad n \ge C_{A,K,a}\ell,$$

one has

$$\mathbb{P}\left\{\max_{\omega^d=1}\sum_{i=1}^{n}\frac{|P_i(\omega)|^2}{\|B_i\|_{\mathrm{op}}^2} > C_{A,K,a}\frac{n}{\ell}\right\} \le C_{A,K,a}d^{-A} + p_d(a)\min\{n,\ell^2\}.$$

On the complement of this exceptional event, independent signs satisfy

$$\mathbb{P}_\varepsilon\left\{\max_{\omega^d=1}\left|\sum_{i=1}^{n}\varepsilon_i\frac{P_i(\omega)}{\|B_i\|_{\mathrm{op}}}\right| > C_{A,K,a}\sqrt{n}\right\} \le d^{-A},$$

and

$$\mathbb{E}_\varepsilon\max_{\omega^d=1}\left|\sum_{i=1}^{n}\varepsilon_i\frac{P_i(\omega)}{\|B_i\|_{\mathrm{op}}}\right| \le C_{A,K,a}\sqrt{n}.$$

Thus a deterministic sign vector with edge maximum $C_{A,K,a}\sqrt{n}$ exists. On the same event, simultaneously for every integer $k \ge 1$,

$$\max_{\omega^d=1}\mathbb{E}_\varepsilon\left|\sum_{i=1}^{n}\varepsilon_i\frac{P_i(\omega)}{\|B_i\|_{\mathrm{op}}}\right|^{2k} \le \left(C_{A,K,a}k\,\frac{n}{\ell}\right)^k.$$

Proof. Let

$$M_a := \#\{1 \le i \le n : \|B_i\|_{\mathrm{op}} < aN_d\}.$$

Since M_a is binomially dominated with mean $np_d(a)$, the bad-count event obeys

$$\mathbb{P}\{M_a > n/\ell^2\} \le p_d(a)\min\{n,\ell^2\}.$$

Indeed, if $n \ge \ell^2$ this is Markov's inequality, while if $n < \ell^2$ the event is $\{M_a \ge 1\}$, bounded by $np_d(a)$. Corollary 5.28, with $R = 1$, gives the profile, signing, expectation, and moment statements outside an additional $C_{A,K,a}d^{-A}$ event. Absorb constants to get the displayed bound. □

Corollary 5.30 (Quantitative random-search and stable-center package)**.** *Fix $A > 0$ and $a > 0$. In the setting of Corollary 5.29, write*

$$v_i(\omega) := \frac{P_i(\omega)}{\|B_i\|_{\mathrm{op}}}, \qquad \omega^d = 1, \qquad \ell = \log(e+d).$$

There is a constant $C_{A,K,a}$ such that, whenever

$$n \le d, \qquad n \ge C_{A,K,a}\ell,$$

outside a matrix event of probability at most

$$C_{A,K,a}d^{-A} + p_d(a)\min\{n, \ell^2\},$$

the following all hold conditionally on the matrices.

First, for every integer $M \geq 1$, if $\delta^{(1)}, \dots, \delta^{(M)}$ are independent uniform sign vectors, then for every $u \geq 0$,

$$\mathbb{P}_\delta\left\{\max_{1\leq s\leq M}\max_{\omega^d=1}\left|\sum_{i=1}^n \delta_i^{(s)} v_i(\omega)\right| > C_{A,K,a}\sqrt{n\left(1+\frac{\log(eM)+u}{\ell}\right)}\right\} \leq e^{-u}.$$

Second, for every integer $M \geq 1$,

$$\mathbb{P}_\delta\left\{\min_{1\leq s\leq M}\max_{\omega^d=1}\left|\sum_{i=1}^n \delta_i^{(s)} v_i(\omega)\right| > C_{A,K,a}\sqrt{n}\right\} \leq d^{-AM}.$$

More generally, if G_M denotes the number of sampled sign vectors whose edge maximum is at most $C_{A,K,a}\sqrt{n}$, then for every integer $1 \leq r \leq M$,

$$\mathbb{P}_\delta\{G_M \leq M - r\} \leq \binom{M}{r} d^{-Ar}.$$

Finally, with Rademacher probability at least $1 - d^{-A}$, a single random sign vector is the center of a Hamming ball of radius

$$r_0 = \lfloor c_{A,K,a}\ell \rfloor$$

on which every sign vector in the ball has edge maximum at most $C_{A,K,a}\sqrt{n}$. If H_M is the number of sampled signs among $\delta^{(1)}, \dots, \delta^{(M)}$ that are centers of such good Hamming balls, then for every integer $1 \leq r \leq M$,

$$\mathbb{P}_\delta\{H_M \leq M - r\} \leq \binom{M}{r} d^{-Ar}.$$

Proof. Corollary 5.29 gives, outside the displayed matrix exceptional event,

$$\max_{\omega^d=1}\sum_{i=1}^n |v_i(\omega)|^2 \leq C_{A,K,a}\frac{n}{\ell}.$$

Condition on any matrix realization satisfying this profile. The sampled-list bound and the binomial success-count estimate are exactly Corollary 5.6, after absorbing the profile constant into $C_{A,K,a}$. The Hamming-ball assertion and the sampled robust-center count are Corollary 5.10, again with the same absorption of constants. □

Corollary 5.31 (Polynomial-confidence normalized robust search)**.** *Fix $A > 0$, $B > 0$, and $a > 0$. In the setting of Corollary 5.29, let $\delta^{(1)}, \dots, \delta^{(M)}$ be independent uniform sign vectors, independent of the matrices, with $M \geq \lceil B \rceil$. There is a constant $C_{A,K,a}$ and a constant $c_{A,K,a} > 0$ such that, whenever*

$$n \leq d, \qquad n \geq C_{A,K,a}\ell, \qquad \ell = \log(e + d),$$

the joint matrix-and-sign probability that no sampled sign is a robust certificate is at most

$$C_{A,K,a}d^{-A} + p_d(a)\min\{n, \ell^2\} + d^{-B}.$$

Equivalently, outside that exceptional probability, some sampled $\delta^{(s)}$ is the center of a Hamming ball of radius

$$r_0 = \lfloor c_{A,K,a}\ell \rfloor$$

on which every sign vector ε' satisfies

$$\max_{\omega^d=1}\left|\sum_{i=1}^{n}\varepsilon'_i\frac{P_i(\omega)}{\|B_i\|_{\mathrm{op}}}\right| \le C_{A,K,a}\sqrt{n}.$$

Thus after the same normalizer-tail matrix event, a constant-size random list finds a logarithmically stable $O(\sqrt{n})$ certificate at any prescribed polynomial confidence level.

Proof. On the good matrix event of Corollary 5.29,

$$\max_{\omega^d=1}\sum_i\left|\frac{P_i(\omega)}{\|B_i\|_{\mathrm{op}}}\right|^2 \le C_{A,K,a}\frac{n}{\ell}.$$

Condition on such a realization and apply Corollary 5.11, absorbing the displayed profile constant into the constants. The conditional sign-search failure is at most d^{-B}. Adding the matrix exceptional probability from Corollary 5.29 proves the claim. □

Corollary 5.32 (Normalizer-tail sampled-list abundance)**.** *Fix $A > 0$, $B > 0$, $a > 0$, and a fraction $0 < \rho \le 1$. In the setting of Corollary 5.29, write*

$$\ell = \log(e + d), \qquad v_i(\omega) := \frac{P_i(\omega)}{\|B_i\|_{\mathrm{op}}}, \qquad \omega^d = 1.$$

Let $\delta^{(1)}, \dots, \delta^{(M)}$ be independent uniform sign vectors, independent of the matrices, where $M \ge 1$, and put

$$r_\rho = \lceil \rho M \rceil.$$

There are constants $C_{A,B,K,a}$ and $c_{A,B,K,a} > 0$ such that, whenever

$$n \le d, \qquad n \ge C_{A,B,K,a}\ell,$$

then, for every $u \ge 0$, with probability at least

$$1 - C_{A,B,K,a}\left(d^{-A} + p_d(a)\min\{n, \ell^2\}\right) - e^{-u} - 2\binom{M}{r_\rho}d^{-Br_\rho},$$

all three conclusions below hold simultaneously.

First, the whole sampled list obeys

$$\max_{1\le s\le M}\max_{\omega^d=1}\left|\sum_{i=1}^{n}\delta_i^{(s)}\frac{P_i(\omega)}{\|B_i\|_{\mathrm{op}}}\right| \le C_{A,B,K,a}\sqrt{n\left(1 + \frac{\log(eM) + u}{\ell}\right)}.$$

Second, at least $M - r_\rho + 1$ *sampled signs have edge maximum at most*

$$C_{A,B,K,a}\sqrt{n}.$$

Third, at least $M - r_\rho + 1$ *sampled signs are centers of Hamming balls of radius*

$$\lfloor c_{A,B,K,a}\ell \rfloor$$

on which every signing ε' *has edge maximum at most*

$$C_{A,B,K,a}\sqrt{n}.$$

In particular, at least a $1-\rho$ *fraction of the sampled signs are good, and at least a* $1-\rho$ *fraction are robust centers.*

Proof. By Corollary 5.29, outside a matrix event of probability at most

$$C_{A,K,a}d^{-A} + p_d(a)\min\{n, \ell^2\},$$

one has the edge profile

$$\max_{\omega^d=1}\sum_i |v_i(\omega)|^2 \le C_{A,K,a}\frac{n}{\ell}.$$

Condition on a matrix realization satisfying this profile. The sampled-list maximum follows from Corollary 5.6. Applying the same corollary with exponent B gives the good-sign count tail $\binom{M}{r_\rho}d^{-Br_\rho}$, after absorbing the profile constant into the displayed constants. Corollary 5.10, again with exponent B, gives the same tail for the robust-center count. A union bound over the matrix event and the three sign failures proves the displayed probability bound. Finally $M - r_\rho + 1 \ge M - \rho M$, which gives the fraction statements. □

Corollary 5.33 (Normalizer-tail polynomial-confidence robust fractions)**.** *Fix* $A > 0$, $S > 0$, $a > 0$, *and a fraction* $0 < \rho \le 1$. *Choose* $B > 0$ *and an integer* $M \ge 1$ *so that*

$$B \ge 2\log_2(e/\rho), \qquad M \ge \frac{2(S+1)}{B\rho}.$$

Write $\ell = \log(e+d)$. *In the setting of Corollary 5.29, suppose* $d \ge 2$, *and let* $\delta^{(1)}, \dots, \delta^{(M)}$ *be independent uniform sign vectors, independent of the matrices. There are constants* $C_{A,B,K,a}$ *and* $c_{A,B,K,a} > 0$ *such that, whenever*

$$n \le d, \qquad n \ge C_{A,B,K,a}\ell,$$

then, for every $u \ge 0$, *with probability at least*

$$1 - C_{A,B,K,a}\left(d^{-A} + p_d(a)\min\{n, \ell^2\}\right) - e^{-u} - d^{-S},$$

the whole sampled list obeys the entropy-sensitive maximum bound from Corollary 5.32, and at least a $1-\rho$ *fraction of the sampled signs are centers of radius*

$$\lfloor c_{A,B,K,a}\ell \rfloor$$

Hamming balls on which every signing ε' has edge maximum at most

$$C_{A,B,K,a}\sqrt{n}.$$

The same $1-\rho$ fraction also has edge maximum at most $C_{A,B,K,a}\sqrt{n}$.

In particular, taking $u = S\ell$, the same event has probability at least

$$1 - C_{A,B,K,a}\left(d^{-A} + p_d(a)\min\{n, \ell^2\}\right) - 2d^{-S},$$

and the whole sampled list has edge maximum at most

$$C_{A,B,K,a,S,\rho}\sqrt{n}.$$

Proof. Let $r_\rho = \lceil \rho M \rceil$. By Corollary 5.32, it remains only to simplify the binomial fraction tail. Since $r_\rho \geq \rho M$,

$$\binom{M}{r_\rho} \leq \left(\frac{eM}{r_\rho}\right)^{r_\rho} \leq (e/\rho)^{r_\rho}.$$

For $d \geq 2$ and $B \geq 2\log_2(e/\rho)$,

$$\binom{M}{r_\rho} d^{-Br_\rho} \leq d^{-Br_\rho/2}.$$

The lower bound on M gives $Br_\rho/2 \geq S+1$, so

$$2\binom{M}{r_\rho} d^{-Br_\rho} \leq 2d^{-(S+1)} \leq d^{-S}.$$

Substituting this into Corollary 5.32 proves the first assertion. If $u = S\ell$, then $e^{-u} \leq d^{-S}$, and

$$1 + \frac{\log(eM) + u}{\ell} \leq C_{S,\rho,B}$$

because M is fixed once S, ρ, B are fixed. This gives the constant-scale sampled-list maximum. □

Corollary 5.34 (Normalizer-tail polynomial-batch robust search)**.** *Fix $A > 0$, $a > 0$, $\beta \geq 0$, and $\sigma > 0$. Write $\ell = \log(e+d)$. In the setting of Corollary 5.29, let $\delta^{(1)}, \ldots, \delta^{(M)}$ be independent uniform sign vectors, independent of the matrices, with*

$$1 \leq M \leq d^\beta.$$

There are constants $C_{A,K,a,\beta,\sigma}$ and $c_{A,K,a,\beta,\sigma} > 0$ such that, whenever

$$n \leq d, \qquad n \geq C_{A,K,a,\beta,\sigma}\ell,$$

with probability at least

$$1 - C_{A,K,a,\beta,\sigma}\left(d^{-A} + p_d(a)\min\{n, \ell^2\}\right) - d^{-\sigma},$$

every sampled sign $\delta^{(s)}$, $1 \leq s \leq M$, is the center of a Hamming ball of radius

$$r_0 = \lfloor c_{A,K,a,\beta,\sigma}\ell \rfloor$$

on which every sign vector ε' satisfies

$$\max_{\omega^d=1}\left|\sum_{i=1}^{n}\varepsilon'_i\frac{P_i(\omega)}{\|B_i\|_{\mathrm{op}}}\right|\le C_{A,K,a,\beta,\sigma}\sqrt{n}.$$

In particular, the whole sampled batch itself has edge maximum at most

$$C_{A,K,a,\beta,\sigma}\sqrt{n}.$$

Proof. By Corollary 5.29, outside a matrix event of probability at most

$$C_{A,K,a}d^{-A}+p_d(a)\min\{n,\ell^2\},$$

one has the edge profile

$$\max_{\omega^d=1}\sum_i\left|\frac{P_i(\omega)}{\|B_i\|_{\mathrm{op}}}\right|^2\le C_{A,K,a}\frac{n}{\ell}.$$

Condition on a matrix realization satisfying this profile and apply Corollary 5.10 with sign exponent

$$B=\beta+\sigma.$$

The robust-center count fails with conditional sign probability at most

$$\binom{M}{1}d^{-(\beta+\sigma)}\le d^{-\sigma}.$$

On the complementary event $H_M\ge M$, every sampled sign is a robust center, and hence every sampled sign also has edge maximum at most the displayed constant times $\sqrt{n}$. Absorb the dependence on β,σ into the constants. □

Corollary 5.35 (Power-curve random-search and stable-center criterion)**.** *Fix $A>0$, $a>0$, $L>0$, $\gamma\ge 1$, $\beta>2$, and $R\ge 1$. In the setting of Corollary 5.27, assume*

$$F_{d,a}(t)\le L\,\ell^{-\gamma}t^{\beta}\qquad(0<t\le 1),\qquad \ell=\log(e+d).$$

Then there is a constant $C_{A,K,a,R}$ such that, whenever

$$n\le d,\qquad n\ge C_{A,K,a,R}\ell,$$

outside a matrix event of probability at most

$$C_{A,K,a,R}d^{-A}+\frac{C_\beta L}{R\,\ell^{\gamma-1}},$$

the following all hold conditionally on the matrices. With

$$v_i(\omega):=\frac{P_i(\omega)}{\|B_i\|_{\mathrm{op}}},\qquad \omega^d=1,$$

every M-sample of independent uniform sign vectors $\delta^{(1)},\dots,\delta^{(M)}$ satisfies, for every $u\ge 0$,

$$\mathbb{P}_\delta\left\{\max_{1\le s\le M}\max_{\omega^d=1}\left|\sum_{i=1}^{n}\delta_i^{(s)}v_i(\omega)\right|>C_{A,K,a,R}\sqrt{n\left(1+\frac{\log(eM)+u}{\ell}\right)}\right\}\le e^{-u}.$$

Moreover,

$$\mathbb{P}_\delta\left\{\min_{1\le s\le M}\max_{\omega^d=1}\left|\sum_{i=1}^n \delta_i^{(s)} v_i(\omega)\right| > C_{A,K,a,R}\sqrt{n}\right\} \le d^{-AM},$$

and if G_M counts sampled signs whose edge maximum is at most $C_{A,K,a,R}\sqrt{n}$, then

$$\mathbb{P}_\delta\{G_M \le M-r\} \le \binom{M}{r} d^{-Ar} \qquad (1\le r\le M).$$

Finally, with Rademacher probability at least $1-d^{-A}$, a random sign vector is the center of a Hamming ball of radius

$$r_0 = \lfloor c_{A,K,a,R}\ell \rfloor$$

on which every sign vector has edge maximum at most $C_{A,K,a,R}\sqrt{n}$; the sampled robust-center count obeys the same binomial tail

$$\mathbb{P}_\delta\{H_M \le M-r\} \le \binom{M}{r} d^{-Ar} \qquad (1\le r\le M).$$

If $\gamma = 1$ and $R > C_\beta L$, the matrix exceptional probability can instead be bounded by

$$C_{A,K,a,R,\beta,L} d^{-A} + \exp\left(-c_{R,\beta,L}\frac{n}{\ell^2}\right).$$

Proof. Corollary 5.27 gives the edge profile

$$\max_{\omega^d=1}\sum_{i=1}^n |v_i(\omega)|^2 \le C_{A,K,a,R}\frac{n}{\ell}$$

outside the displayed matrix event, with the exponential replacement in the critical-density case. Condition on any realization satisfying this profile. The sampled-list bound and success-count estimate follow from Corollary 5.6, while the Hamming-ball and sampled robust-center statements follow from Corollary 5.10, after absorbing the profile constant into $C_{A,K,a,R}$ and $c_{A,K,a,R}$. □

Corollary 5.36 (Power-curve sampled-list abundance)**.** *Fix $A>0$, $B>0$, $a>0$, $L>0$, $\gamma\ge 1$, $\beta>2$, $R\ge 1$, and a fraction $0<\rho\le 1$. In the setting of Corollary 5.27, assume*

$$F_{d,a}(t) \le L\,\ell^{-\gamma} t^\beta \qquad (0<t\le 1), \qquad \ell = \log(e+d).$$

Let $\delta^{(1)},\dots,\delta^{(M)}$ be independent uniform sign vectors, independent of the matrices, where $M\ge 1$, and put

$$r_\rho = \lceil \rho M \rceil.$$

There are constants $C_{A,B,K,a,R}$ and $c_{A,B,K,a,R}>0$ such that, whenever

$$n \le d, \qquad n \ge C_{A,B,K,a,R}\ell,$$

then, for every $u \geq 0$, with probability at least

$$1 - C_{A,B,K,a,R}\left(d^{-A} + \frac{C_\beta L}{R\ell^{\gamma-1}}\right) - e^{-u} - 2\binom{M}{r_\rho} d^{-Br_\rho},$$

all three conclusions below hold simultaneously.

First, the whole sampled list obeys

$$\max_{1\leq s\leq M}\max_{\omega^d=1}\left|\sum_{i=1}^{n}\delta_i^{(s)}\frac{P_i(\omega)}{\|B_i\|_{\mathrm{op}}}\right| \leq C_{A,B,K,a,R}\sqrt{n\left(1+\frac{\log(eM)+u}{\ell}\right)}.$$

Second, at least $M - r_\rho + 1$ sampled signs have edge maximum at most

$$C_{A,B,K,a,R}\sqrt{n}.$$

Third, at least $M - r_\rho + 1$ sampled signs are centers of Hamming balls of radius

$$\lfloor c_{A,B,K,a,R}\ell \rfloor$$

on which every signing ε' has edge maximum at most

$$C_{A,B,K,a,R}\sqrt{n}.$$

In particular, at least a $1-\rho$ fraction of the sampled signs are good, and at least a $1-\rho$ fraction are robust centers.

If $\gamma = 1$ and $R > C_\beta L$, then the same conclusion holds with the matrix part of the exceptional probability replaced by

$$C_{A,B,K,a,R,\beta,L}d^{-A} + \exp\left(-c_{R,\beta,L}\frac{n}{\ell^2}\right).$$

Proof. By Corollary 5.27, outside a matrix event of probability at most

$$C_{A,K,a,R}d^{-A} + \frac{C_\beta L}{R\ell^{\gamma-1}},$$

one has the edge profile

$$\max_{\omega^d=1}\sum_i\left|\frac{P_i(\omega)}{\|B_i\|_{\mathrm{op}}}\right|^2 \leq C_{A,K,a,R}\frac{n}{\ell}.$$

Condition on a matrix realization satisfying this profile. The sampled-list maximum follows from Corollary 5.6. Applying the same corollary with exponent B gives the good-sign count tail $\binom{M}{r_\rho}d^{-Br_\rho}$, after absorbing the profile constant into the displayed constants. Corollary 5.10, again with exponent B, gives the same tail for the robust-center count. A union bound over the matrix event and the three sign failures proves the displayed probability bound.

In the critical-density case, use the Bernstein replacement in Corollary 5.27 for the matrix event. Finally $M - r_\rho + 1 \geq M - \rho M$, which gives the fraction statements. □

Corollary 5.37 (Power-curve polynomial-confidence robust fractions)**.** *Fix $A > 0$, $S > 0$, $a > 0$, $L > 0$, $\gamma \geq 1$, $\beta > 2$, $R \geq 1$, and a fraction $0 < \rho \leq 1$. Choose $B > 0$ and an integer $M \geq 1$ so that*

$$B \geq 2\log_2(e/\rho), \qquad M \geq \frac{2(S+1)}{B\rho}.$$

In the setting of Corollary 5.27, suppose $d \geq 2$ and

$$F_{d,a}(t) \leq L\,\ell^{-\gamma}t^{\beta} \qquad (0 < t \leq 1), \qquad \ell = \log(e+d).$$

Let $\delta^{(1)}, \dots, \delta^{(M)}$ be independent uniform sign vectors, independent of the matrices. There are constants $C_{A,B,K,a,R}$ and $c_{A,B,K,a,R} > 0$ such that, whenever

$$n \leq d, \qquad n \geq C_{A,B,K,a,R}\ell,$$

then, for every $u \geq 0$, with probability at least

$$1 - C_{A,B,K,a,R}\left(d^{-A} + \frac{C_\beta L}{R\ell^{\gamma-1}}\right) - e^{-u} - d^{-S},$$

the whole sampled list obeys the entropy-sensitive maximum bound from Corollary 5.36, and at least a $1-\rho$ fraction of the sampled signs are centers of radius

$$\lfloor c_{A,B,K,a,R}\ell \rfloor$$

Hamming balls on which every signing ε' has edge maximum at most

$$C_{A,B,K,a,R}\sqrt{n}.$$

The same $1-\rho$ fraction also has edge maximum at most $C_{A,B,K,a,R}\sqrt{n}$.

In particular, taking $u = S\ell$, the same event has probability at least

$$1 - C_{A,B,K,a,R}\left(d^{-A} + \frac{C_\beta L}{R\ell^{\gamma-1}}\right) - 2d^{-S},$$

and the whole sampled list has edge maximum at most

$$C_{A,B,K,a,R,S,\rho}\sqrt{n}.$$

If $\gamma = 1$ and $R > C_\beta L$, then both conclusions remain valid with the matrix part of the exceptional probability replaced by

$$C_{A,B,K,a,R,\beta,L}d^{-A} + \exp\left(-c_{R,\beta,L}\frac{n}{\ell^2}\right).$$

Proof. Let $r_\rho = \lceil \rho M \rceil$. By Corollary 5.36, it remains only to simplify the binomial fraction tail. Since $r_\rho \geq \rho M$,

$$\binom{M}{r_\rho} \leq \left(\frac{eM}{r_\rho}\right)^{r_\rho} \leq (e/\rho)^{r_\rho}.$$

For $d \geq 2$ and $B \geq 2\log_2(e/\rho)$,

$$\binom{M}{r_\rho} d^{-Br_\rho} \leq d^{-Br_\rho/2}.$$

The lower bound on M gives $Br_\rho/2 \geq S+1$, hence

$$2\binom{M}{r_\rho} d^{-Br_\rho} \leq 2d^{-(S+1)} \leq d^{-S}.$$

Substituting this into Corollary 5.36 proves the first assertion, including the critical-density replacement. If $u = S\ell$, then $e^{-u} \leq d^{-S}$, and

$$1 + \frac{\log(eM) + u}{\ell} \leq C_{S,\rho,B}.$$

This gives the constant-scale sampled-list maximum. □

Corollary 5.38 (Power-curve polynomial-batch robust search)**.** *Fix $A > 0$, $a > 0$, $L > 0$, $\gamma \geq 1$, $\beta_0 > 2$, $R \geq 1$, $\beta_{\mathrm{b}} \geq 0$, and $\sigma > 0$. In the setting of Corollary 5.27, assume*

$$F_{d,a}(t) \leq L\,\ell^{-\gamma} t^{\beta_0} \qquad (0 < t \leq 1), \qquad \ell = \log(e+d).$$

Let $\delta^{(1)}, \dots, \delta^{(M)}$ be independent uniform sign vectors, independent of the matrices, with

$$1 \leq M \leq d^{\beta_{\mathrm{b}}}.$$

There are constants $C_{A,K,a,R,\beta_{\mathrm{b}},\sigma}$ and $c_{A,K,a,R,\beta_{\mathrm{b}},\sigma} > 0$ such that, whenever

$$n \leq d, \qquad n \geq C_{A,K,a,R,\beta_{\mathrm{b}},\sigma}\ell,$$

with probability at least

$$1 - C_{A,K,a,R,\beta_{\mathrm{b}},\sigma}\left(d^{-A} + \frac{C_{\beta_0} L}{R\ell^{\gamma-1}}\right) - d^{-\sigma},$$

every sampled sign $\delta^{(s)}$, $1 \leq s \leq M$, is the center of a Hamming ball of radius

$$r_0 = \lfloor c_{A,K,a,R,\beta_{\mathrm{b}},\sigma}\ell \rfloor$$

on which every sign vector ε' satisfies

$$\max_{\omega^d=1} \left| \sum_{i=1}^{n} \varepsilon'_i \frac{P_i(\omega)}{\|B_i\|_{\mathrm{op}}} \right| \leq C_{A,K,a,R,\beta_{\mathrm{b}},\sigma}\sqrt{n}.$$

In particular, the whole sampled batch itself has edge maximum at most

$$C_{A,K,a,R,\beta_{\mathrm{b}},\sigma}\sqrt{n}.$$

If $\gamma = 1$ and $R > C_{\beta_0} L$, then the same conclusion holds with the matrix part of the exceptional probability replaced by

$$C_{A,K,a,R,\beta_0,L,\beta_{\mathrm{b}},\sigma} d^{-A} + \exp\left(-c_{R,\beta_0,L}\frac{n}{\ell^2}\right).$$

Proof. By Corollary 5.27, outside the displayed power-curve matrix event one has the edge profile

$$\max_{\omega^d=1} \sum_i \left| \frac{P_i(\omega)}{\|B_i\|_{\mathrm{op}}} \right|^2 \leq C_{A,K,a,R}\frac{n}{\ell}.$$

Condition on a matrix realization satisfying this profile and apply Corollary 5.10 with sign exponent

$$B = \beta_{\mathrm{b}} + \sigma.$$

The robust-center count fails with conditional sign probability at most

$$\binom{M}{1} d^{-(\beta_{\mathrm{b}}+\sigma)} \le d^{-\sigma}.$$

On the complementary event every sampled sign is a robust center, and therefore also has edge maximum at most the displayed constant times $\sqrt{n}$. The critical-density assertion again uses the Bernstein matrix-event replacement in Corollary 5.27. □

Corollary 5.39 (Almost-sure power-curve search transfer)**.** *Fix $A > 0$, $a > 0$, $L > 0$, $\gamma \ge 1$, $\beta > 2$, and $R \ge 1$. There is a constant $C_{A,K,a,R}$ with the following property. Let $(d_m, n_m)_{m\ge1}$ be a triangular-array sequence with $n_m \le d_m$, put $\ell_m = \log(e + d_m)$, and assume*

$$n_m \ge C_{A,K,a,R}\ell_m \qquad (m \ge 1).$$

At each m, let $B_{m,i} = T_{d_m}(\xi^{(m,i)})$, $1 \le i \le n_m$, be independent full Toeplitz matrices with independent mean-zero, unit-variance lag variables of sub-Gaussian norm at most K, and let $P_{m,i}$ be their symbols. Suppose that, for every m,

$$F_{d_m,a}(t) := \mathbb{P}\left\{\|T_{d_m}(\xi)\|_{\mathrm{op}} < t\, a\, \mathbb{E}\,\|T_{d_m}(\xi)\|_{\mathrm{op}}\right\} \le L\,\ell_m^{-\gamma} t^{\beta} \qquad (0 < t \le 1).$$

If

$$\sum_{m=1}^{\infty} \left(d_m^{-A} + \ell_m^{1-\gamma}\right) < \infty,$$

then, with probability one over the matrix triangular array, all but finitely many m satisfy the full random-search and stable-center package of Corollary 5.35, with d, n, ℓ replaced by d_m, n_m, ℓ_m.

There is also a critical-density Bernstein form. If $\gamma = 1$ and $R > C_\beta L$, then the same almost-sure conclusion holds under the weaker summability condition

$$\sum_{m=1}^{\infty} \left(d_m^{-A} + \exp\left(-c_{R,\beta,L}\frac{n_m}{\ell_m^2}\right)\right) < \infty.$$

Consequently, along the full integer dimension sequence, if $A > 1$,

$$F_{d,a}(t) \le L\,\log^{-1}(e + d)\, t^{\beta} \qquad (0 < t \le 1,\ d \ge 1),$$

$\beta > 2$, $R > C_\beta L$, and if $n(d) \le d$ while, for all sufficiently large d,

$$n(d) \ge q\,\log^3(e + d) \qquad \text{for some } q > \frac{1}{c_{R,\beta,L}},$$

then almost surely all sufficiently large integer dimensions satisfy the same random-search and stable-center package.

Proof. Apply Corollary 5.35 at each m, enlarging $C_{A,K,a,R}$ if necessary. In the Markov form, the matrix exceptional probability is at most

$$C_{A,K,a,R} d_m^{-A} + C_\beta L R^{-1} \ell_m^{1-\gamma},$$

which is summable by hypothesis. The first Borel–Cantelli lemma gives an almost-sure event on which all but finitely many matrix arrays satisfy the finite power-curve search package.

In the critical-density case, the exponential replacement in Corollary 5.35 gives summable exceptional probabilities under the displayed Bernstein condition, so the same Borel–Cantelli argument applies. For the full-dimension consequence take $d_m = m$. Then $\sum_m m^{-A} < \infty$, and for all large m,

$$\exp\left(-c_{R,\beta,L} \frac{n(m)}{\log^2(e+m)}\right) \le (e+m)^{-c_{R,\beta,L} q},$$

which is summable because $c_{R,\beta,L} q > 1$. The finitely many initial dimensions do not affect the eventual almost-sure conclusion. □

Corollary 5.40 (Power-curve sampled robust certificates)**.** *Fix $A > 0$, $\alpha > 0$, $a > 0$, $L > 0$, $\gamma \ge 1$, $\beta > 2$, and $R \ge 1$. There is a constant $C_{A,K,a,R}$ with the following property. Let $(d_m, n_m)_{m \ge 1}$ be as in Corollary 5.39, with $\ell_m = \log(e + d_m)$, and suppose*

$$n_m \ge C_{A,K,a,R} \ell_m \qquad (m \ge 1).$$

Assume the same power-curve lower-tail bound

$$F_{d_m,a}(t) \le L\, \ell_m^{-\gamma} t^\beta \qquad (0 < t \le 1,\ m \ge 1).$$

For each m, sample M_m independent uniform sign vectors in $\{-1,1\}^{n_m}$, independently of the matrices, with

$$M_m \ge \lceil \alpha \rceil.$$

If

$$\sum_{m=1}^{\infty} \left(d_m^{-A} + \ell_m^{1-\gamma} + d_m^{-\alpha}\right) < \infty,$$

then, with probability one over the matrices and the sampled signs, all but finitely many m have a sampled sign which is the center of a Hamming ball of radius

$$r_m = \lfloor c_{A,K,a,R} \ell_m \rfloor$$

on which every signing ε' satisfies

$$\max_{\omega^{d_m}=1} \left| \sum_{i=1}^{n_m} \varepsilon_i' \frac{P_{m,i}(\omega)}{\|B_{m,i}\|_{\mathrm{op}}} \right| \le C_{A,K,a,R} \sqrt{n_m}.$$

In the critical-density case $\gamma = 1$, if $R > C_\beta L$, the same actual sampled-certificate conclusion holds under the weaker summability condition

$$\sum_{m=1}^{\infty} \left(d_m^{-A} + \exp\left(-c_{R,\beta,L} \frac{n_m}{\ell_m^2}\right) + d_m^{-\alpha}\right) < \infty.$$

Consequently, along the full integer dimension sequence, if $A > 1$, $\alpha > 1$,

$$F_{d,a}(t) \le L \log^{-1}(e+d)\, t^{\beta} \qquad (0 < t \le 1,\ d \ge 1),$$

$\beta > 2$, $R > C_\beta L$, *and*

$$n(d) \ge q \log^3(e+d) \qquad \text{for all large } d, \qquad q > \frac{1}{c_{R,\beta,L}},$$

then any fixed list size $M \ge \lceil \alpha \rceil$ *gives, almost surely, sampled logarithmically robust* $O(\sqrt{n(d)})$ *certificates for all sufficiently large integer dimensions.*

Proof. Apply Corollary 5.35 at each m. In the Markov form, outside a matrix event of probability at most

$$C_{A,K,a,R} d_m^{-A} + C_\beta L R^{-1} \ell_m^{1-\gamma},$$

the edge profile

$$\max_{\omega^{d_m}=1} \sum_{i=1}^{n_m} \left| \frac{P_{m,i}(\omega)}{\|B_{m,i}\|_{\mathrm{op}}} \right|^2 \le C_{A,K,a,R} \frac{n_m}{\ell_m}$$

holds. Condition on such a realization and apply Corollary 5.11, absorbing the profile constant. The sampled-sign failure is at most $d_m^{-\alpha}$. Therefore the total failure probability at index m is bounded by a constant times

$$d_m^{-A} + \ell_m^{1-\gamma} + d_m^{-\alpha},$$

which is summable by hypothesis. Borel–Cantelli gives the first assertion.

For $\gamma = 1$ and $R > C_\beta L$, use the Bernstein exceptional probability from Corollary 5.35 in place of the Markov term. This gives the second displayed summability condition. The full-dimension consequence is obtained by taking $d_m = m$, $n_m = n(m)$, and $M_m = M$. The series $\sum_m m^{-A}$ and $\sum_m m^{-\alpha}$ are finite, and the displayed lower bound on $n(m)$ makes

$$\exp\left(-c_{R,\beta,L} \frac{n(m)}{\log^2(e+m)} \right) \le (e+m)^{-c_{R,\beta,L} q}$$

summable. □

Corollary 5.41 (Almost-sure power-curve sampled-search counts)**.** *Fix* $A > 0$, $B > 0$, $a > 0$, $L > 0$, $\gamma \ge 1$, $\beta > 2$, *and* $R \ge 1$. *There are constants* $C_{A,B,K,a,R}$ *and* $c_{A,B,K,a,R} > 0$ *with the following property. Let* $(d_m, n_m)_{m \ge 1}$ *be a triangular-array sequence with* $n_m \le d_m$, *put* $\ell_m = \log(e + d_m)$, *and assume*

$$n_m \ge C_{A,B,K,a,R} \ell_m \qquad (m \ge 1).$$

At each m, *let* $B_{m,i} = T_{d_m}(\xi^{(m,i)})$, $1 \le i \le n_m$, *be independent full Toeplitz matrices with independent mean-zero, unit-variance lag variables of sub-Gaussian norm at most* K, *and let* $P_{m,i}$ *be their symbols. Suppose that, for every* m,

$$F_{d_m,a}(t) := \mathbb{P}\left\{ \|T_{d_m}(\xi)\|_{\mathrm{op}} < t\, a\, \mathbb{E}\, \|T_{d_m}(\xi)\|_{\mathrm{op}} \right\} \le L\, \ell_m^{-\gamma} t^{\beta} \qquad (0 < t \le 1).$$

Independently of the matrices, sample independent uniform sign vectors

$$\delta_m^{(1)},\dots,\delta_m^{(M_m)}\in\{-1,1\}^{n_m},\qquad M_m\geq 1.$$

Let $u_m\geq 0$ and $1\leq r_m\leq M_m$ be deterministic parameters. If

$$\sum_{m=1}^{\infty}\left(d_m^{-A}+\ell_m^{1-\gamma}+e^{-u_m}+\binom{M_m}{r_m}d_m^{-Br_m}\right)<\infty,$$

then, with probability one over matrices and sampled signs, all but finitely many m satisfy all of the following.

First, the whole sampled list obeys

$$\max_{1\leq s\leq M_m}\max_{\omega^{d_m}=1}\left|\sum_{i=1}^{n_m}\delta_{m,i}^{(s)}\frac{P_{m,i}(\omega)}{\|B_{m,i}\|_{\mathrm{op}}}\right|\leq C_{A,B,K,a,R}\sqrt{n_m\left(1+\frac{\log(eM_m)+u_m}{\ell_m}\right)}.$$

Second, at least M_m-r_m+1 sampled signs have edge maximum at most

$$C_{A,B,K,a,R}\sqrt{n_m}.$$

Third, at least M_m-r_m+1 sampled signs are centers of Hamming balls of radius

$$\lfloor c_{A,B,K,a,R}\ell_m\rfloor$$

on which every signing ε' has edge maximum at most

$$C_{A,B,K,a,R}\sqrt{n_m}.$$

In the critical-density case $\gamma=1$, if $R>C_\beta L$, the same conclusion holds under the weaker summability condition

$$\sum_{m=1}^{\infty}\left(d_m^{-A}+\exp\left(-c_{R,\beta,L}\frac{n_m}{\ell_m^2}\right)+e^{-u_m}+\binom{M_m}{r_m}d_m^{-Br_m}\right)<\infty.$$

Consequently, along the full integer dimension sequence, suppose $A>1$, $s>1$,

$$F_{d,a}(t)\leq L\,\log^{-1}(e+d)t^{\beta}\qquad(0<t\leq 1,\ d\geq 1),$$

$R>C_\beta L$, and

$$n(d)\leq d,\qquad n(d)\geq q\,\log^3(e+d)\qquad\text{for all large } d,\qquad q>\frac{1}{c_{R,\beta,L}}.$$

If $M_d\leq d^{\beta_{\mathrm{b}}}$ for some $\beta_{\mathrm{b}}\geq 0$ and the fixed sign-count exponent satisfies $B>\beta_{\mathrm{b}}+1$, then taking $r_d=1$ and

$$u_d=s\log(e+d)$$

gives an almost-sure event on which, for all sufficiently large integer dimensions, the whole sampled batch has edge maximum

$$O_{A,B,K,a,R,s,\beta_{\mathrm{b}}}(\sqrt{n(d)})$$

and every sampled sign in the batch is a robust center.

Proof. By Corollary 5.27, outside a matrix event of probability at most

$$C_{A,K,a,R} d_m^{-A} + C_\beta L R^{-1} \ell_m^{1-\gamma},$$

one has the edge profile

$$\max_{\omega^{d_m}=1} \sum_{i=1}^{n_m} \left| \frac{P_{m,i}(\omega)}{\|B_{m,i}\|_{\mathrm{op}}} \right|^2 \le C_{A,K,a,R} \frac{n_m}{\ell_m}.$$

Condition on a matrix realization satisfying this profile. The sampled-list maximum follows from Corollary 5.6. Applying that corollary with exponent B gives the good-sign count failure

$$\binom{M_m}{r_m} d_m^{-Br_m},$$

after absorbing the profile constant into $C_{A,B,K,a,R}$. The sampled robust-center count follows from Corollary 5.10, again with exponent B, with the same binomial tail. Hence the total failure probability at index m is bounded, up to a fixed constant, by the summand in the displayed Markov-form series. Borel–Cantelli gives the first assertion.

For $\gamma = 1$ and $R > C_\beta L$, use the critical-density replacement in Corollary 5.27 for the matrix event. The same Borel–Cantelli argument applies under the second displayed summability condition.

For the full integer-dimensional consequence take $d_m = m$, $n_m = n(m)$, and $u_m = s \log(e+m)$. The series $\sum_m m^{-A}$ and $\sum_m e^{-u_m}$ are finite because $A > 1$ and $s > 1$. The sample-size lower bound gives

$$\exp\left(-c_{R,\beta,L} \frac{n(m)}{\log^2(e+m)} \right) \le (e+m)^{-c_{R,\beta,L} q},$$

which is summable by the assumption on q. Finally, with $r_m = 1$,

$$\binom{M_m}{1} m^{-B} \le m^{\beta_{\mathrm{b}} - B},$$

and this is summable because $B > \beta_{\mathrm{b}} + 1$. The sampled-list maximum is $O(\sqrt{n(m)})$ since

$$\frac{\log(eM_m) + u_m}{\log(e+m)} \le C_{s,\beta_{\mathrm{b}}}$$

for all large m. With $r_m = 1$, the count conclusions say that every sampled sign is good and robust. □

Corollary 5.42 (Almost-sure power-curve robust fractions)**.** *Fix $A > 0$, $S > 0$, $a > 0$, $L > 0$, $\gamma \ge 1$, $\beta > 2$, $R \ge 1$, and a fraction $0 < \rho \le 1$. Choose $B > 0$ and an integer $M \ge 1$ so that*

$$B \ge 2 \log_2(e/\rho), \qquad M \ge \frac{2(S+1)}{B\rho}.$$

There are constants $C_{A,B,K,a,R}$ and $c_{A,B,K,a,R} > 0$ with the following property. Let $(d_m, n_m)_{m\ge 1}$ be a triangular-array sequence with

$$d_m \ge 2, \qquad n_m \le d_m, \qquad \ell_m = \log(e + d_m),$$

and assume

$$n_m \ge C_{A,B,K,a,R}\ell_m \qquad (m \ge 1).$$

At each m, let $B_{m,i} = T_{d_m}(\xi^{(m,i)})$, $1 \le i \le n_m$, be independent full Toeplitz matrices with independent mean-zero, unit-variance lag variables of sub-Gaussian norm at most K, and let $P_{m,i}$ be their symbols. Independently of the matrices, sample independent uniform sign vectors

$$\delta_m^{(1)}, \dots, \delta_m^{(M)} \in \{-1, 1\}^{n_m}.$$

Assume that, for every m,

$$F_{d_m,a}(t) := \mathbb{P}\left\{\|T_{d_m}(\xi)\|_{\mathrm{op}} < t\, a\, \mathbb{E}\,\|T_{d_m}(\xi)\|_{\mathrm{op}}\right\} \le L\,\ell_m^{-\gamma} t^{\beta} \qquad (0 < t \le 1),$$

and suppose

$$\sum_{m=1}^{\infty} \left(d_m^{-A} + \ell_m^{1-\gamma} + d_m^{-S}\right) < \infty.$$

Then, with probability one over the matrices and sampled signs, all but finitely many m satisfy both conclusions below.

First, the whole sampled list obeys

$$\max_{1\le s\le M} \max_{\omega^{d_m}=1} \left|\sum_{i=1}^{n_m} \delta_{m,i}^{(s)} \frac{P_{m,i}(\omega)}{\|B_{m,i}\|_{\mathrm{op}}}\right| \le C_{A,B,K,a,R,S,\rho}\sqrt{n_m}.$$

Second, at least a $1-\rho$ fraction of the M sampled signs are centers of Hamming balls of radius

$$\lfloor c_{A,B,K,a,R}\ell_m \rfloor$$

on which every signing ε' has edge maximum at most

$$C_{A,B,K,a,R}\sqrt{n_m}.$$

The same $1-\rho$ fraction also has edge maximum at most $C_{A,B,K,a,R}\sqrt{n_m}$.

In the critical-density case $\gamma = 1$, if $R > C_\beta L$, the same conclusion holds under the weaker summability condition

$$\sum_{m=1}^{\infty} \left(d_m^{-A} + \exp\left(-c_{R,\beta,L}\frac{n_m}{\ell_m^2}\right) + d_m^{-S}\right) < \infty.$$

Consequently, on any stretched-exponential dimension schedule

$$d_m \ge \exp(c_0 m^{\kappa}) \qquad (c_0, \kappa > 0),$$

the Markov-form conclusion holds for every $S > 0$ whenever $\gamma > 1 + 1/\kappa$. In the critical-density case $\gamma = 1$, the same conclusion holds on such schedules if either

$$n_m \ge q\,\ell_m^2 \log(e + m) \qquad \textit{for all large } m, \qquad q > \frac{1}{c_{R,\beta,L}},$$

or, more simply, if for some $\tau > 0$ and $Q > 0$,

$$n_m \geq Q\,\ell_m^{2+\tau} \qquad \text{for all large } m.$$

Proof. Put $u_m = S\ell_m$. Apply Corollary 5.37 at each dimension m. In the Markov form, the failure probability at m is bounded by a constant times

$$d_m^{-A} + \ell_m^{1-\gamma} + e^{-u_m} + d_m^{-S}.$$

Since $e^{-u_m} = (e + d_m)^{-S} \leq d_m^{-S}$, these probabilities are summable by hypothesis. Borel–Cantelli gives the eventual sampled-list and robust-fraction conclusions. The sampled-list bound is $O(\sqrt{n_m})$ because M is fixed and

$$1 + \frac{\log(eM) + u_m}{\ell_m} \leq C_{S,\rho,B}.$$

For $\gamma = 1$ and $R > C_\beta L$, use the critical-density replacement in Corollary 5.37; the same Borel–Cantelli argument applies with the displayed exponential summand.

On a stretched-exponential dimension schedule, $\sum_m d_m^{-A}$ and $\sum_m d_m^{-S}$ are finite for every $A, S > 0$, while $\ell_m \geq cm^\kappa$ for all large m. Thus

$$\sum_m \ell_m^{1-\gamma} \leq C \sum_m m^{-\kappa(\gamma-1)} < \infty$$

whenever $\gamma > 1 + 1/\kappa$. In the critical-density case, the lower bound $n_m \geq q\ell_m^2 \log(e + m)$ gives

$$\exp\left(-c_{R,\beta,L}\frac{n_m}{\ell_m^2}\right) \leq (e + m)^{-c_{R,\beta,L}q},$$

which is summable when $q > 1/c_{R,\beta,L}$. The alternative $n_m \geq Q\ell_m^{2+\tau}$ gives a stretched-exponential summand $\exp(-c'm^{\kappa\tau})$, also summable. □

Corollary 5.43 (Almost-sure power-curve polynomial batches)**.** *Fix $A > 0$, $a > 0$, $L > 0$, $\gamma \geq 1$, $\beta_0 > 2$, $R \geq 1$, $\beta_{\mathrm{b}} \geq 0$, and $\sigma > 0$. There are constants $C_{A,K,a,R,\beta_{\mathrm{b}},\sigma}$ and $c_{A,K,a,R,\beta_{\mathrm{b}},\sigma} > 0$ with the following property. Let $(d_m, n_m)_{m\geq 1}$ be a triangular-array sequence with*

$$d_m \geq 2, \qquad n_m \leq d_m, \qquad \ell_m = \log(e + d_m),$$

and assume

$$n_m \geq C_{A,K,a,R,\beta_{\mathrm{b}},\sigma}\ell_m \qquad (m \geq 1).$$

At each m, let $B_{m,i} = T_{d_m}(\xi^{(m,i)})$, $1 \leq i \leq n_m$, be independent full Toeplitz matrices with independent mean-zero, unit-variance lag variables of sub-Gaussian norm at most K, and let $P_{m,i}$ be their symbols. Independently of the matrices, sample M_m independent uniform sign vectors in $\{-1, 1\}^{n_m}$, where

$$1 \leq M_m \leq d_m^{\beta_{\mathrm{b}}}.$$

Assume that, for every m,

$$F_{d_m,a}(t) \leq L\,\ell_m^{-\gamma}t^{\beta_0} \qquad (0 < t \leq 1),$$

and suppose

$$\sum_{m=1}^{\infty}\left(d_m^{-A}+\ell_m^{1-\gamma}+d_m^{-\sigma}\right)<\infty.$$

Then, with probability one over the matrices and sampled signs, all but finitely many m have the following property: every sampled sign $\delta_m^{(s)}$, $1\le s\le M_m$, is the center of a Hamming ball of radius

$$\lfloor c_{A,K,a,R,\beta_{\mathrm{b}},\sigma}\ell_m\rfloor$$

on which every signing ε' satisfies

$$\max_{\omega^{d_m}=1}\left|\sum_{i=1}^{n_m}\varepsilon'_i\frac{P_{m,i}(\omega)}{\|B_{m,i}\|_{\mathrm{op}}}\right|\le C_{A,K,a,R,\beta_{\mathrm{b}},\sigma}\sqrt{n_m}.$$

In particular, the whole sampled batch itself has edge maximum

$$O_{A,K,a,R,\beta_{\mathrm{b}},\sigma}(\sqrt{n_m})$$

eventually almost surely.

In the critical-density case $\gamma=1$, if $R>C_{\beta_0}L$, the same conclusion holds under the weaker summability condition

$$\sum_{m=1}^{\infty}\left(d_m^{-A}+\exp\left(-c_{R,\beta_0,L}\frac{n_m}{\ell_m^2}\right)+d_m^{-\sigma}\right)<\infty.$$

Consequently, on any stretched-exponential dimension schedule

$$d_m\ge\exp(c_0m^{\kappa})\qquad(c_0,\kappa>0),$$

the Markov-form conclusion holds for every $\sigma>0$ whenever $\gamma>1+1/\kappa$. In the critical-density case $\gamma=1$, the same conclusion holds on such schedules under either of the sample-size alternatives

$$n_m\ge q\,\ell_m^2\log(e+m),\qquad q>\frac{1}{c_{R,\beta_0,L}},$$

for all large m, or

$$n_m\ge Q\,\ell_m^{2+\tau}$$

for all large m, with some $\tau>0$ and $Q>0$.

Proof. Apply Corollary 5.38 at each dimension. In the Markov form, the failure probability at m is bounded by a constant times

$$d_m^{-A}+\ell_m^{1-\gamma}+d_m^{-\sigma},$$

which is summable by hypothesis. Borel–Cantelli gives the eventual all-robust batch conclusion.

For $\gamma=1$ and $R>C_{\beta_0}L$, use the critical-density replacement in Corollary 5.38. The stated exponential summability then gives the same almost-sure conclusion. The stretched-exponential consequences are verified exactly as in Corollary 5.42: powers d_m^{-A} and $d_m^{-\sigma}$ are summable, $\ell_m^{1-\gamma}$ is summable when $\gamma>1+1/\kappa$, and the two critical sample-size lower bounds make $\exp(-cn_m/\ell_m^2)$ summable. □

Corollary 5.44 (Critical power-curve abundance on all integer dimensions)**.** *Fix $a > 0$, $L > 0$, $\beta > 2$, $R > C_\beta L$, and $A > 1$. Let $d = 1, 2, 3, \dots$, put $\ell_d = \log(e + d)$, and let $n(d) \le d$. At dimension d, let $B_{d,i} = T_d(\xi^{(d,i)})$, $1 \le i \le n(d)$, be independent full Toeplitz matrices with independent mean-zero, unit-variance lag variables of sub-Gaussian norm at most K, and let $P_{d,i}$ be their symbols. Assume the critical power-curve lower-tail bound*

$$F_{d,a}(t) := \mathbb{P}\left\{\|T_d(\xi)\|_{\mathrm{op}} < t\, a\, \mathbb{E}\,\|T_d(\xi)\|_{\mathrm{op}}\right\} \le L\,\ell_d^{-1} t^\beta \qquad (0 < t \le 1,\ d \ge 1),$$

and suppose that, for all sufficiently large d,

$$n(d) \ge q\,\ell_d^3 \qquad \text{with} \qquad q > \frac{1}{c_{R,\beta,L}}.$$

Then the following two all-integer conclusions hold.

First, fix $0 < \rho \le 1$, $S > 1$, choose $B > 0$ and an integer $M \ge 1$ such that

$$B \ge 2\log_2(e/\rho), \qquad M \ge \frac{2(S+1)}{B\rho},$$

and at every d sample M independent uniform sign vectors, independently of the matrices. With probability one over matrices and signs, all sufficiently large integer dimensions have sampled-list maximum

$$\max_{1\le s\le M}\max_{\omega^d=1}\left|\sum_{i=1}^{n(d)} \delta_{d,i}^{(s)}\frac{P_{d,i}(\omega)}{\|B_{d,i}\|_{\mathrm{op}}}\right| \le C_{A,B,K,a,R,S,\rho}\sqrt{n(d)},$$

and at least a $1-\rho$ fraction of the sampled signs are centers of Hamming balls of radius $\lfloor c_{A,B,K,a,R}\ell_d\rfloor$ on which every signing has edge maximum at most $C_{A,B,K,a,R}\sqrt{n(d)}$. The same fraction also has edge maximum at most $C_{A,B,K,a,R}\sqrt{n(d)}$.

Second, fix $\beta_{\mathrm{b}} \ge 0$ and $\sigma > 1$, and at dimension d sample any independent batch of signs with

$$1 \le M_d \le d^{\beta_{\mathrm{b}}}.$$

Then, with probability one over matrices and signs, all sufficiently large integer dimensions have every sampled sign as the center of a Hamming ball of radius $\lfloor c_{A,K,a,R,\beta_{\mathrm{b}},\sigma}\ell_d\rfloor$ on which every signing has edge maximum at most

$$C_{A,K,a,R,\beta_{\mathrm{b}},\sigma}\sqrt{n(d)}.$$

In particular, the whole sampled batch itself has edge maximum

$$O_{A,K,a,R,\beta_{\mathrm{b}},\sigma}(\sqrt{n(d)})$$

eventually almost surely.

Proof. The lower bound $n(d) \ge q\ell_d^3$ eventually dominates every fixed multiple of ℓ_d, so the admissibility hypotheses in the two source corollaries hold after discarding finitely many dimensions.

Apply Corollary 5.42 with $d_m = m$, $n_m = n(m)$, $\ell_m = \log(e+m)$, and $\gamma = 1$. The series $\sum_m m^{-A}$ and $\sum_m m^{-S}$ are finite because $A > 1$ and $S > 1$, while the lower bound on $n(m)$ gives

$$\exp\left(-c_{R,\beta,L}\frac{n(m)}{\ell_m^2}\right) \le (e+m)^{-c_{R,\beta,L}q}$$

for all large m, which is summable by the assumption on q. This proves the robust-fraction conclusion.

The all-robust polynomial-batch conclusion follows in the same way from Corollary 5.43: the only additional sampled-sign summand is $\sum_m m^{-\sigma}$, which is finite because $\sigma > 1$. □

Corollary 5.45 (Critical power-curve vanishing fractions on all integer dimensions)**.** *Fix $a > 0$, $L > 0$, $\beta > 2$, $R > C_\beta L$, $A > 1$, and $B > 1$. Fix $0 < \tau_0 < B - 1$. Let $d = 1, 2, 3, \ldots$, put $\ell_d = \log(e+d)$, and let $n(d) \le d$. At dimension d, let $B_{d,i} = T_d(\xi^{(d,i)})$, $1 \le i \le n(d)$, be independent full Toeplitz matrices with independent mean-zero, unit-variance lag variables of sub-Gaussian norm at most K, and let $P_{d,i}$ be their symbols. Assume the critical power-curve lower-tail bound*

$$F_{d,a}(t) := \mathbb{P}\left\{\|T_d(\xi)\|_{\mathrm{op}} < t\, a\, \mathbb{E}\,\|T_d(\xi)\|_{\mathrm{op}}\right\} \le L\,\ell_d^{-1}t^\beta \qquad (0 < t \le 1,\ d \ge 1),$$

and suppose that, for all sufficiently large d,

$$n(d) \ge q\,\ell_d^3 \qquad \text{with} \qquad q > \frac{1}{c_{R,\beta,L}}.$$

At each d, independently sample an arbitrary deterministic number $M_d \ge 1$ of uniform sign vectors in $\{-1,1\}^{n(d)}$. Let $u_d \ge 0$ satisfy

$$\sum_{d=1}^{\infty} e^{-u_d} < \infty,$$

and let $(\rho_d)_{d\ge1} \subset (0,1]$ satisfy

$$\rho_d \ge e\,d^{-\tau_0}$$

for all sufficiently large d. Then, with probability one over the matrices and sampled signs, all sufficiently large integer dimensions satisfy both conclusions below.

First, the whole sampled list obeys

$$\max_{1\le s\le M_d}\max_{\omega^d=1}\left|\sum_{i=1}^{n(d)}\delta_{d,i}^{(s)}\frac{P_{d,i}(\omega)}{\|B_{d,i}\|_{\mathrm{op}}}\right| \le C_{A,B,K,a,R}\sqrt{n(d)\left(1+\frac{\log(eM_d)+u_d}{\ell_d}\right)}.$$

Second, at least a $1-\rho_d$ fraction of the M_d sampled signs have edge maximum at most $C_{A,B,K,a,R}\sqrt{n(d)}$ and are centers of Hamming balls of radius

$$\lfloor c_{A,B,K,a,R}\ell_d \rfloor$$

on which every signing ε' has edge maximum at most

$$C_{A,B,K,a,R}\sqrt{n(d)}.$$

No upper bound on M_d is required. In particular, if $\rho_d \to 0$, then the robust-center fraction tends to one along the full integer dimension sequence.

Proof. Apply Corollary 5.41 with

$$d_m = m, \qquad n_m = n(m), \qquad r_m = \lceil \rho_m M_m \rceil,$$

and with the critical-density replacement. The eventual lower bound $n(m) \geq q\ell_m^3$ implies the admissibility condition $n_m \geq C_{A,B,K,a,R}\ell_m$ after discarding finitely many dimensions, and

$$\exp\left(-c_{R,\beta,L}\frac{n(m)}{\ell_m^2}\right) \leq (e+m)^{-c_{R,\beta,L}q}$$

is summable by the assumption on q. Also $\sum_m m^{-A} < \infty$ because $A > 1$, and the sampled-list tail is summable by the hypothesis on u_m.

It remains to check the binomial count tail. Since $r_m = \lceil \rho_m M_m \rceil$,

$$\binom{M_m}{r_m} \leq \left(\frac{e}{\rho_m}\right)^{r_m}.$$

The lower bound $\rho_m \geq e\, m^{-\tau_0}$ gives, for all large m,

$$\binom{M_m}{r_m} m^{-Br_m} \leq m^{-(B-\tau_0)r_m} \leq m^{-(B-\tau_0)}.$$

This is summable because $B - \tau_0 > 1$. The sampled-count theorem gives the displayed sampled-list maximum and at least $M_m - r_m + 1$ robust centers eventually. Finally,

$$M_m - r_m + 1 \geq M_m - \rho_m M_m,$$

so the lower bound is at least a $1 - \rho_m$ fraction. □

Corollary 5.46 (Sparse power-curve Borel windows)**.** *Fix $A > 0$, $a > 0$, $L > 0$, $\beta > 2$, and $R \geq 1$. There is a constant $C_{A,K,a,R}$ with the following property. Let $(d_m, n_m)_{m\geq 1}$ be a triangular-array sequence with $n_m \leq d_m$, $\ell_m = \log(e + d_m)$, and*

$$d_m \geq \exp(c_0 m^\kappa) \qquad (m \geq 1)$$

for some $c_0, \kappa > 0$. At each m, let the matrices be as in Corollary 5.39, and assume

$$n_m \geq C_{A,K,a,R}\ell_m \qquad (m \geq 1).$$

If, for some $\gamma > 1 + 1/\kappa$,

$$F_{d_m,a}(t) \leq L\,\ell_m^{-\gamma} t^\beta \qquad (0 < t \leq 1,\ m \geq 1),$$

then almost surely all but finitely many m satisfy the full random-search and stable-center package of Corollary 5.35.

In the critical-density case $\gamma = 1$, assume $R > C_\beta L$ and

$$F_{d_m,a}(t) \le L\,\ell_m^{-1} t^\beta \qquad (0 < t \le 1,\ m \ge 1).$$

Then the same eventual conclusion holds if either

$$n_m \ge q\,\ell_m^2 \log(e+m) \qquad \text{for all large } m, \qquad q > \frac{1}{c_{R,\beta,L}},$$

or, more simply, if for some $\rho > 0$ and $Q > 0$,

$$n_m \ge Q\,\ell_m^{2+\rho} \qquad \text{for all large } m.$$

Thus on stretched-exponential dimension schedules the critical power-curve density $F_{d,a}(t) \lesssim \ell^{-1} t^\beta$ needs only a superquadratic logarithmic sample window to give eventual randomized certificates.

Proof. Since $d_m \ge \exp(c_0 m^\kappa)$, the series $\sum_m d_m^{-A}$ is finite. Also $\ell_m \ge c m^\kappa$ for all large m. Hence

$$\sum_m \ell_m^{1-\gamma} \le C \sum_m m^{-\kappa(\gamma-1)} < \infty$$

when $\gamma > 1 + 1/\kappa$. The Markov form of Corollary 5.39 gives the first assertion.

For the critical-density case, the condition $n_m \ge q\ell_m^2 \log(e+m)$ gives

$$\exp\left(-c_{R,\beta,L}\frac{n_m}{\ell_m^2}\right) \le (e+m)^{-c_{R,\beta,L}q},$$

which is summable when $q > 1/c_{R,\beta,L}$. If instead $n_m \ge Q\ell_m^{2+\rho}$, then for all large m,

$$\exp\left(-c_{R,\beta,L}\frac{n_m}{\ell_m^2}\right) \le \exp(-c' m^{\kappa\rho}),$$

which is summable for every $\rho > 0$. The Bernstein form of Corollary 5.39 completes the proof. □

Corollary 5.47 (Sparse sampled power-curve Borel windows)**.** *Fix $A > 0$, $\alpha > 0$, $a > 0$, $L > 0$, $\beta > 2$, and $R \ge 1$. There is a constant $C_{A,K,a,R}$ with the following property. Let $(d_m, n_m)_{m\ge1}$ be a triangular-array sequence with $n_m \le d_m$, $\ell_m = \log(e + d_m)$, and*

$$d_m \ge \exp(c_0 m^\kappa) \qquad (m \ge 1)$$

for some $c_0, \kappa > 0$. At each m, let the matrices be as in Corollary 5.40, assume

$$n_m \ge C_{A,K,a,R}\ell_m \qquad (m \ge 1),$$

and sample M_m independent uniform sign vectors in $\{-1,1\}^{n_m}$, independently of the matrices, with

$$M_m \ge \lceil \alpha \rceil.$$

If, for some $\gamma > 1 + 1/\kappa$,

$$F_{d_m,a}(t) \le L\,\ell_m^{-\gamma} t^\beta \qquad (0 < t \le 1,\ m \ge 1),$$

then, with probability one over the matrices and sampled signs, all but finitely many m have a sampled sign which is the center of a Hamming ball of radius

$$r_m = \lfloor c_{A,K,a,R}\ell_m \rfloor$$

on which every signing ε' satisfies

$$\max_{\omega^{d_m}=1} \left| \sum_{i=1}^{n_m} \varepsilon'_i \frac{P_{m,i}(\omega)}{\|B_{m,i}\|_{\mathrm{op}}} \right| \le C_{A,K,a,R}\sqrt{n_m}.$$

In the critical-density case $\gamma = 1$, assume $R > C_\beta L$ and

$$F_{d_m,a}(t) \le L\,\ell_m^{-1} t^\beta \qquad (0 < t \le 1,\ m \ge 1).$$

Then the same actual sampled-certificate conclusion holds if either

$$n_m \ge q\,\ell_m^2 \log(e+m) \qquad \text{for all large } m, \qquad q > \frac{1}{c_{R,\beta,L}},$$

or, more simply, if for some $\rho > 0$ and $Q > 0$,

$$n_m \ge Q\,\ell_m^{2+\rho} \qquad \text{for all large } m.$$

Thus on stretched-exponential dimension schedules, every $\alpha > 0$ and every fixed list size $M \ge \lceil \alpha \rceil$ give eventual sampled robust certificates under the same sparse windows as Corollary 5.46.

Proof. Apply Corollary 5.40. The stretched-exponential lower bound on d_m makes both $\sum_m d_m^{-A}$ and $\sum_m d_m^{-\alpha}$ finite. Also $\ell_m \ge cm^\kappa$ for all large m, so

$$\sum_m \ell_m^{1-\gamma} \le C \sum_m m^{-\kappa(\gamma-1)} < \infty$$

when $\gamma > 1 + 1/\kappa$. This verifies the Markov-form summability condition of Corollary 5.40.

In the critical-density case, the lower bound $n_m \ge q\ell_m^2 \log(e+m)$ gives

$$\exp\left(-c_{R,\beta,L}\frac{n_m}{\ell_m^2}\right) \le (e+m)^{-c_{R,\beta,L}q},$$

which is summable when $q > 1/c_{R,\beta,L}$. The alternative $n_m \ge Q\ell_m^{2+\rho}$ gives

$$\exp\left(-c_{R,\beta,L}\frac{n_m}{\ell_m^2}\right) \le \exp(-c' m^{\kappa\rho})$$

for all large m, again summable. Together with the summability of $\sum_m d_m^{-A}$ and $\sum_m d_m^{-\alpha}$, this verifies the critical condition in Corollary 5.40. □

Corollary 5.48 (Sparse sampled-count power-curve Borel windows)**.** *Fix $A > 0$, $B > 0$, $a > 0$, $L > 0$, $\beta > 2$, $R \ge 1$, $c_0 > 0$, and $\kappa > 0$. There are constants $C_{A,B,K,a,R}$ and $c_{A,B,K,a,R} > 0$ with the following property. Let $(d_m, n_m)_{m\ge1}$ be a triangular-array sequence with $n_m \le d_m$, $\ell_m = \log(e+d_m)$, and*

$$d_m \ge \exp(c_0 m^\kappa), \qquad n_m \ge C_{A,B,K,a,R}\ell_m \qquad (m \ge 1).$$

At each m, take the independent full Toeplitz triangular array at dimension d_m, and independently sample M_m uniform sign vectors in $\{-1,1\}^{n_m}$, with

$M_m \ge 1$. *Let* $u_m \ge 0$ *and* $1 \le r_m \le M_m$ *be deterministic parameters. If, for some* $\gamma > 1 + 1/\kappa$,

$$F_{d_m,a}(t) \le L\,\ell_m^{-\gamma} t^{\beta} \qquad (0 < t \le 1,\ m \ge 1),$$

and

$$\sum_{m=1}^{\infty} \left(e^{-u_m} + \binom{M_m}{r_m} d_m^{-Br_m} \right) < \infty,$$

then, with probability one over the matrices and sampled signs, all but finitely many m *satisfy the sampled-list maximum bound, good-sign count, and robust-center count conclusions of Corollary 5.41.*

In the critical-density case $\gamma = 1$, *assume* $R > C_\beta L$ *and*

$$F_{d_m,a}(t) \le L\,\ell_m^{-1} t^{\beta} \qquad (0 < t \le 1,\ m \ge 1).$$

Then the same eventual sampled-count conclusion holds, under the same sign-summability condition, if either

$$n_m \ge q\,\ell_m^2 \log(e+m) \qquad \text{for all large } m, \qquad q > \frac{1}{c_{R,\beta,L}},$$

or, more simply, if for some $\rho > 0$ *and* $Q > 0$,

$$n_m \ge Q\,\ell_m^{2+\rho} \qquad \text{for all large } m.$$

In particular, let $s > 1$, *take* $u_m = s \log(e+m)$ *and* $r_m = 1$, *and suppose*

$$M_m \le d_m^{\beta_{\mathrm{b}}} \qquad (m \ge 1)$$

for some $\beta_{\mathrm{b}} \ge 0$. *Then every fixed sign-count exponent* $B > \beta_{\mathrm{b}}$ *gives an almost-sure event on which, for all sufficiently large* m, *every sampled sign in the batch is an* $O(\sqrt{n_m})$*-good robust center and the whole batch satisfies the displayed sampled-list maximum bound. More generally, if*

$$M_m \le \exp(b m^{\sigma})$$

for some $b > 0$, $0 < \sigma \le \kappa$, *then the same eventual all-sampled-sign conclusion holds whenever either* $\sigma < \kappa$, *or* $\sigma = \kappa$ *and* $b < Bc_0$.

Proof. Apply Corollary 5.41. The stretched-exponential lower bound on d_m makes $\sum_m d_m^{-A}$ finite. Also $\ell_m \ge c m^{\kappa}$ for all large m, so

$$\sum_m \ell_m^{1-\gamma} \le C \sum_m m^{-\kappa(\gamma-1)} < \infty$$

when $\gamma > 1 + 1/\kappa$. Together with the displayed sign summability, this verifies the Markov-form hypothesis of the sampled-count Borel theorem.

In the critical-density case, the lower bound $n_m \ge q\ell_m^2 \log(e+m)$ gives

$$\exp\left(-c_{R,\beta,L} \frac{n_m}{\ell_m^2} \right) \le (e+m)^{-c_{R,\beta,L} q},$$

which is summable when $q > 1/c_{R,\beta,L}$. The alternative $n_m \ge Q\ell_m^{2+\rho}$ gives

$$\exp\left(-c_{R,\beta,L} \frac{n_m}{\ell_m^2} \right) \le \exp(-c' m^{\kappa\rho})$$

for all large m, again summable. With the summability of $\sum_m d_m^{-A}$ and the displayed sign terms, this verifies the critical hypothesis in Corollary 5.41.

For the polynomial-in-d_m batch consequence,

$$\binom{M_m}{1} d_m^{-B} \le d_m^{\beta_{\mathrm{b}}-B},$$

which is summable on the stretched-exponential schedule when $B > \beta_{\mathrm{b}}$. For the $\exp(bm^\sigma)$ batch consequence,

$$\binom{M_m}{1} d_m^{-B} \le \exp\left(bm^\sigma - Bc_0 m^\kappa\right),$$

which is summable if $\sigma < \kappa$, or if $\sigma = \kappa$ and $b < Bc_0$. Finally, $\sum_m e^{-u_m} < \infty$ because $s > 1$. With $r_m = 1$, the count conclusion says all sampled signs are good robust centers eventually, and the comparison

$$\frac{\log(eM_m) + u_m}{\ell_m} = O(1)$$

in both displayed batch regimes gives the $O(\sqrt{n_m})$ sampled-list maximum. □

Corollary 5.49 (Sparse power-curve growing robust fractions)**.** *Fix $A > 0$, $a > 0$, $L > 0$, $\beta > 2$, $R \ge 1$, $c_0 > 0$, $\kappa > 0$, and $0 < \rho \le 1$. There are constants $C_{A,K,a,R}$ and $c_{A,K,a,R} > 0$ with the following property. Let $(d_m, n_m)_{m\ge1}$ be a triangular-array sequence with $n_m \le d_m$, $\ell_m = \log(e + d_m)$, and*

$$d_m \ge \exp(c_0 m^\kappa), \qquad n_m \ge C_{A,K,a,R}\ell_m \qquad (m \ge 1).$$

At each m, take the independent full Toeplitz triangular array at dimension d_m, writing $B_{m,i} = T_{d_m}(\xi^{(m,i)})$, $1 \le i \le n_m$, and $P_{m,i}$ for its symbol. Independently sample an arbitrary deterministic number $M_m \ge 1$ of uniform sign vectors

$$\delta_m^{(1)}, \dots, \delta_m^{(M_m)} \in \{-1, 1\}^{n_m}.$$

Let $u_m \ge 0$ be deterministic and assume

$$\sum_{m=1}^{\infty} e^{-u_m} < \infty.$$

If, for some $\gamma > 1 + 1/\kappa$,

$$F_{d_m,a}(t) \le L\,\ell_m^{-\gamma} t^\beta \qquad (0 < t \le 1,\ m \ge 1),$$

then, with probability one over the matrices and sampled signs, all but finitely many m satisfy both conclusions below.

First, the whole sampled list obeys the entropy-sensitive maximum bound

$$\max_{1\le s\le M_m} \max_{\omega^{d_m}=1} \left| \sum_{i=1}^{n_m} \delta_{m,i}^{(s)} \frac{P_{m,i}(\omega)}{\|B_{m,i}\|_{\mathrm{op}}} \right| \le C_{A,K,a,R} \sqrt{n_m \left(1 + \frac{\log(eM_m) + u_m}{\ell_m}\right)}.$$

Second, at least a $1-\rho$ fraction of the M_m sampled signs have edge maximum at most $C_{A,K,a,R}\sqrt{n_m}$ and are centers of Hamming balls of radius

$$\lfloor c_{A,K,a,R}\ell_m \rfloor$$

on which every signing ε' has edge maximum at most

$$C_{A,K,a,R}\sqrt{n_m}.$$

No upper bound on M_m is required for this robust-fraction conclusion.

In the critical-density case $\gamma = 1$, assume $R > C_\beta L$ and

$$F_{d_m,a}(t) \le L\,\ell_m^{-1}t^\beta \qquad (0 < t \le 1,\ m \ge 1).$$

Then the same eventual conclusions hold if either

$$n_m \ge q\,\ell_m^2\log(e+m) \qquad \text{for all large } m, \qquad q > \frac{1}{c_{R,\beta,L}},$$

or, more simply, if for some $\tau > 0$ and $Q > 0$,

$$n_m \ge Q\,\ell_m^{2+\tau} \qquad \text{for all large } m.$$

Proof. Apply Corollary 5.48 with sign-count exponent $B = 1$ and

$$r_m = \lceil \rho M_m \rceil.$$

The only additional summability condition to check is the binomial count tail. Since $r_m \ge \rho M_m$,

$$\binom{M_m}{r_m} d_m^{-r_m} \le \left(\frac{eM_m}{r_m}\right)^{r_m} d_m^{-r_m} \le \left(\frac{e}{\rho}d_m^{-1}\right)^{r_m}.$$

For all large m, the last display is at most $d_m^{-r_m/2}$, and hence at most $d_m^{-1/2}$, which is summable because $d_m \ge \exp(c_0 m^\kappa)$. The sampled-list tail is summable by the hypothesis on u_m. Corollary 5.48 therefore gives the entropy-sensitive list maximum and at least

$$M_m - r_m + 1 \ge (1-\rho)M_m$$

good robust centers eventually. The same argument applies in the critical density case, because the two displayed lower sample-window alternatives are exactly the critical alternatives in Corollary 5.48. □

Corollary 5.50 (Vanishing sparse power-curve discarded fractions)**.** *Fix $A > 0$, $B > 0$, $a > 0$, $L > 0$, $\beta > 2$, $R \ge 1$, $c_0 > 0$, $\kappa > 0$, and $0 < \tau_0 < B$. There are constants $C_{A,B,K,a,R}$ and $c_{A,B,K,a,R} > 0$ with the following property. Let $(d_m, n_m)_{m\ge1}$ be a triangular-array sequence with $n_m \le d_m$, $\ell_m = \log(e+d_m)$, and*

$$d_m \ge \exp(c_0 m^\kappa), \qquad n_m \ge C_{A,B,K,a,R}\ell_m \qquad (m \ge 1).$$

At each m, take the independent full Toeplitz triangular array at dimension d_m, writing $B_{m,i} = T_{d_m}(\xi^{(m,i)})$, $1 \le i \le n_m$, and $P_{m,i}$ for its symbol.

Independently sample an arbitrary deterministic number $M_m \geq 1$ of uniform sign vectors

$$\delta_m^{(1)}, \dots, \delta_m^{(M_m)} \in \{-1, 1\}^{n_m}.$$

Let $u_m \geq 0$ be deterministic and assume

$$\sum_{m=1}^{\infty} e^{-u_m} < \infty.$$

Let $(\rho_m)_{m \geq 1} \subset (0, 1]$ satisfy

$$\rho_m \geq e\, d_m^{-\tau_0}$$

for all sufficiently large m. If, for some $\gamma > 1 + 1/\kappa$,

$$F_{d_m,a}(t) \leq L\, \ell_m^{-\gamma} t^{\beta} \qquad (0 < t \leq 1,\ m \geq 1),$$

then, with probability one over the matrices and sampled signs, all but finitely many m satisfy both conclusions below.

First, the whole sampled list obeys

$$\max_{1 \leq s \leq M_m} \max_{\omega^{d_m}=1} \left| \sum_{i=1}^{n_m} \delta_{m,i}^{(s)} \frac{P_{m,i}(\omega)}{\|B_{m,i}\|_{\mathrm{op}}} \right| \leq C_{A,B,K,a,R} \sqrt{n_m \left(1 + \frac{\log(eM_m) + u_m}{\ell_m}\right)}.$$

Second, at least a $1 - \rho_m$ fraction of the M_m sampled signs have edge maximum at most $C_{A,B,K,a,R}\sqrt{n_m}$ and are centers of Hamming balls of radius

$$\lfloor c_{A,B,K,a,R} \ell_m \rfloor$$

on which every signing ε' has edge maximum at most

$$C_{A,B,K,a,R}\sqrt{n_m}.$$

No upper bound on M_m is required. In particular, if $\rho_m \to 0$, then the robust-center fraction tends to one along the sampled batches.

In the critical-density case $\gamma = 1$, assume $R > C_\beta L$ and

$$F_{d_m,a}(t) \leq L\, \ell_m^{-1} t^{\beta} \qquad (0 < t \leq 1,\ m \geq 1).$$

Then the same eventual conclusions hold if either

$$n_m \geq q\, \ell_m^2 \log(e + m) \qquad \text{for all large } m, \qquad q > \frac{1}{c_{R,\beta,L}},$$

or, more simply, if for some $\tau > 0$ and $Q > 0$,

$$n_m \geq Q\, \ell_m^{2+\tau} \qquad \text{for all large } m.$$

Proof. Apply Corollary 5.48 with sign-count exponent B and

$$r_m = \lceil \rho_m M_m \rceil.$$

The matrix summability hypotheses are exactly those verified in the proof of Corollary 5.49, and the sampled-list tail is summable by the hypothesis on u_m. It remains only to check the binomial count tail. Since $r_m = \lceil \rho_m M_m \rceil$,

$$\binom{M_m}{r_m} \leq \left(\frac{eM_m}{r_m}\right)^{r_m} \leq \left(\frac{e}{\rho_m}\right)^{r_m}.$$

The lower bound on ρ_m gives, for all large m,

$$\binom{M_m}{r_m} d_m^{-Br_m} \le d_m^{-(B-\tau_0)r_m} \le d_m^{-(B-\tau_0)}.$$

This is summable on the stretched-exponential schedule. Hence the sampled-count theorem applies. Finally,

$$M_m - r_m + 1 \ge M_m - \rho_m M_m,$$

so the good robust-center count is at least a $1-\rho_m$ fraction. The critical-density alternatives are exactly the alternatives already built into Corollary 5.48. □

Corollary 5.51 (Explicit lower-tail rates)**.** *Fix $A > 0$, $a > 0$, and write $\ell = \log(e+d)$. In the setting of Corollary 5.29, suppose*

$$p_d(a) := \mathbb{P}\{\|T_d(\xi)\|_{\mathrm{op}} < aN_d\} \le R_d.$$

Then, whenever $n \le d$ and $n \ge C_{A,K,a}\ell$, the realization-normalized edge profile fails with probability at most

$$C_{A,K,a} d^{-A} + R_d \min\{n, \ell^2\}.$$

In particular:

(i) *if $R_d \le C_0 \ell^{-B}$ with $B > 2$, then the exceptional probability is at most*

$$C_{A,K,a,C_0,B}(d^{-A} + \ell^{2-B});$$

(ii) *more generally, if $n \le C_1 \ell^{\gamma}$ and $R_d \le C_0 \ell^{-B}$, then the exceptional probability is at most*

$$C_{A,K,a,C_0,C_1,B,\gamma}(d^{-A} + \ell^{\min\{\gamma,2\}-B});$$

(iii) *if $R_d \le C_0 d^{-B}$ with $B > 0$, then for every $0 < B' < B$ the exceptional probability is at most*

$$C_{A,K,a,C_0,B,B'}(d^{-A} + d^{-B'}).$$

On the complement of the same exceptional event, independent signs have the $C_{A,K,a}\sqrt{n}$ edge bound with conditional probability at least $1 - d^{-A}$, sign-averaged edge maximum $C_{A,K,a}\sqrt{n}$, the same deterministic certificate, and scalar edge-grid moments bounded by $(C_{A,K,a} k\, n/\ell)^k$ for every $2k$-th moment.

Proof. The first display is exactly Corollary 5.29. The first polylogarithmic case follows by substituting $R_d \le C_0\ell^{-B}$ and using $\min\{n,\ell^2\} \le \ell^2$. The sample-window refinement uses $\min\{n,\ell^2\} \le C_1 \ell^{\min\{\gamma,2\}}$. For the polynomial case, enlarge the constant to cover bounded d; for large d, $\ell^2 \le C_{B,B'} d^{B-B'}$, so $R_d \min\{n,\ell^2\} \le C_{C_0,B,B'} d^{-B'}$. □

Corollary 5.52 (Explicit search-stability rates)**.** *Fix $A > 0$, $a > 0$, and write $\ell = \log(e+d)$. In the setting of Corollary 5.30, suppose*

$$p_d(a) := \mathbb{P}\{\|T_d(\xi)\|_{\mathrm{op}} < aN_d\} \le R_d.$$

Then, whenever $n \le d$ and $n \ge C_{A,K,a}\ell$, outside a matrix event of probability at most

$$C_{A,K,a}d^{-A} + R_d \min\{n, \ell^2\},$$

the full random-search and stable-center package of Corollary 5.30 holds. In particular:

(i) *if $R_d \le C_0\ell^{-B}$ with $B > 2$, this package fails with probability at most*

$$C_{A,K,a,C_0,B}(d^{-A} + \ell^{2-B});$$

(ii) *if $n \le C_1\ell^{\gamma}$ and $R_d \le C_0\ell^{-B}$, it fails with probability at most*

$$C_{A,K,a,C_0,C_1,B,\gamma}(d^{-A} + \ell^{\min\{\gamma,2\}-B});$$

(iii) *if $R_d \le C_0 d^{-B}$ with $B > 0$, then for every $0 < B' < B$ it fails with probability at most*

$$C_{A,K,a,C_0,B,B'}(d^{-A} + d^{-B'}).$$

Proof. Corollary 5.30 gives the first exceptional probability bound. The three displayed specializations are the same elementary substitutions as in Corollary 5.51. □

Corollary 5.53 (Almost-sure normalizer-tail transfer)**.** *Fix $A > 0$ and $a > 0$. Let $(d_m, n_m)_{m\ge1}$ be a sequence with $n_m \le d_m$, put $\ell_m = \log(e + d_m)$, and assume*

$$n_m \ge C_{A,K,a}\ell_m \qquad (m \ge 1).$$

For each m, let $B_{m,i} = T_{d_m}(\xi^{(m,i)})$, $1 \le i \le n_m$, be independent full Toeplitz matrices whose lag variables are independent, mean zero, unit variance, and have sub-Gaussian norm at most K, and let $P_{m,i}$ be their symbols. Let

$$p_{d_m}(a) = \mathbb{P}\{\|T_{d_m}(\xi)\|_{\mathrm{op}} < a\,\mathbb{E}\,\|T_{d_m}(\xi)\|_{\mathrm{op}}\}.$$

If

$$\sum_{m=1}^{\infty}\left(d_m^{-A} + p_{d_m}(a)\min\{n_m, \ell_m^2\}\right) < \infty,$$

then, with probability one, all but finitely many m satisfy

$$\max_{\omega^{d_m}=1}\sum_{i=1}^{n_m}\frac{|P_{m,i}(\omega)|^2}{\|B_{m,i}\|_{\mathrm{op}}^2} \le C_{A,K,a}\frac{n_m}{\ell_m}.$$

For those m, the signing, sign-averaged maximum, deterministic-certificate, and scalar moment conclusions of Corollary 5.29 hold with d, n, ℓ replaced by d_m, n_m, ℓ_m.

Proof. Apply Corollary 5.29 at each index m. The displayed summability assumption makes the profile exceptional probabilities summable. The first Borel–Cantelli lemma gives the almost-sure eventual profile bound, and the remaining conclusions are deterministic consequences of that same good profile event in Corollary 5.29. □

Corollary 5.54 (Almost-sure random-search and stable-center transfer)**.** *In the setting of Corollary 5.53, assume the same summability condition*

$$\sum_{m=1}^{\infty} \left(d_m^{-A} + p_{d_m}(a) \min\{n_m, \ell_m^2\} \right) < \infty.$$

For $\omega^{d_m} = 1$, put

$$v_{m,i}(\omega) := \frac{P_{m,i}(\omega)}{\|B_{m,i}\|_{\mathrm{op}}}.$$

Then, with probability one over the matrix triangular array, all but finitely many m have the following conditional sign-search properties.

For every integer $M \geq 1$, if $\delta^{(1)}, \dots, \delta^{(M)}$ are independent uniform signs in $\{-1, 1\}^{n_m}$, then for every $u \geq 0$,

$$\mathbb{P}_\delta \left\{ \max_{1 \leq s \leq M} \max_{\omega^{d_m}=1} \left| \sum_{i=1}^{n_m} \delta_i^{(s)} v_{m,i}(\omega) \right| > C_{A,K,a} \sqrt{n_m \left(1 + \frac{\log(eM) + u}{\ell_m} \right)} \right\} \leq e^{-u}.$$

Also,

$$\mathbb{P}_\delta \left\{ \min_{1 \leq s \leq M} \max_{\omega^{d_m}=1} \left| \sum_{i=1}^{n_m} \delta_i^{(s)} v_{m,i}(\omega) \right| > C_{A,K,a} \sqrt{n_m} \right\} \leq d_m^{-AM}.$$

If $G_{m,M}$ is the number of sampled sign vectors whose edge maximum is at most $C_{A,K,a}\sqrt{n_m}$, then for every integer $1 \leq r \leq M$,

$$\mathbb{P}_\delta\{G_{m,M} \leq M - r\} \leq \binom{M}{r} d_m^{-Ar}.$$

Finally, with Rademacher probability at least $1 - d_m^{-A}$, a random sign vector is the center of a Hamming ball of radius

$$r_m = \lfloor c_{A,K,a} \ell_m \rfloor$$

on which every signing has edge maximum at most $C_{A,K,a}\sqrt{n_m}$. If $H_{m,M}$ counts the sampled signs that are centers of such good balls, then for every integer $1 \leq r \leq M$,

$$\mathbb{P}_\delta\{H_{m,M} \leq M - r\} \leq \binom{M}{r} d_m^{-Ar}.$$

Proof. Apply Corollary 5.30 at each m. The same summability assumption makes its matrix exceptional probabilities summable. Borel–Cantelli gives an almost-sure event on which all but finitely many matrix realizations satisfy the finite-size search-stability package. Substituting d, n, ℓ by d_m, n_m, ℓ_m gives the displayed conditional statements. □

Corollary 5.55 (Almost-sure normalized sampled-search counts)**.** *Fix $A > 0$ and $a > 0$. Let $(d_m, n_m)_{m \geq 1}$ be a triangular-array sequence with*

$$d_m \geq 2, \qquad n_m \leq d_m, \qquad \ell_m = \log(e + d_m),$$

and assume

$$n_m \geq C_{A,K,a}\ell_m \qquad (m \geq 1).$$

At each m, let $B_{m,i} = T_{d_m}(\xi^{(m,i)})$, $1 \leq i \leq n_m$, be independent full Toeplitz matrices with independent mean-zero, unit-variance lag variables of sub-Gaussian norm at most K, and let $P_{m,i}$ be their symbols. Independently of the matrices, sample independent uniform sign vectors

$$\delta_m^{(1)}, \dots, \delta_m^{(M_m)} \in \{-1, 1\}^{n_m}, \qquad M_m \geq 1.$$

Let $u_m \geq 0$ and $1 \leq r_m \leq M_m$ be deterministic parameters, and put

$$p_{d_m}(a) = \mathbb{P}\{\|T_{d_m}(\xi)\|_{\mathrm{op}} < a\,\mathbb{E}\,\|T_{d_m}(\xi)\|_{\mathrm{op}}\}.$$

If

$$\sum_{m=1}^{\infty}\left[d_m^{-A} + p_{d_m}(a)\min\{n_m, \ell_m^2\} + e^{-u_m} + \binom{M_m}{r_m} d_m^{-Ar_m}\right] < \infty,$$

then, with probability one over the matrices and sampled signs, all but finitely many m satisfy all of the following.

First, the entire sampled list obeys

$$\max_{1\leq s\leq M_m}\max_{\omega^{d_m}=1}\left|\sum_{i=1}^{n_m}\delta_{m,i}^{(s)}\frac{P_{m,i}(\omega)}{\|B_{m,i}\|_{\mathrm{op}}}\right| \leq C_{A,K,a}\sqrt{n_m\left(1 + \frac{\log(eM_m) + u_m}{\ell_m}\right)}.$$

Second, at least $M_m - r_m + 1$ sampled signs have edge maximum at most $C_{A,K,a}\sqrt{n_m}$. Third, at least $M_m - r_m + 1$ sampled signs are centers of Hamming balls of radius

$$\lfloor c_{A,K,a}\ell_m \rfloor$$

on which every signing ε' has edge maximum at most $C_{A,K,a}\sqrt{n_m}$.

Consequently, along the full integer dimension sequence, suppose $s > 1$, $p_d(a) \leq C_0 d^{-B_0}$ with $B_0 > 1$, and $M_d \leq d^\beta$ for some $\beta \geq 0$. If $A > \beta + 1$, then taking $u_d = s\log(e + d)$ and $r_d = 1$ gives an almost-sure event on which, for all sufficiently large integer dimensions and every admissible window

$$C_{A,K,a}\log(e + d) \leq n(d) \leq d,$$

every sampled sign in the batch is a robust center and the whole batch satisfies the displayed sampled-list maximum bound.

More generally, on a polynomial dimension schedule $d_m \geq c_0 m^\kappa$, suppose $p_d(a) \leq C_0 d^{-B_0}$, $M_m \leq d_m^\beta$, and

$$B_0\kappa > 1, \qquad (A - \beta)\kappa > 1.$$

Then taking $u_m = s\log(e + m)$, $s > 1$, and $r_m = 1$ gives the same eventual all-good, all-robust sampled-batch conclusion for every admissible window

$$C_{A,K,a}\log(e + d_m) \leq n_m \leq d_m.$$

The sampled-list maximum is then $O_{A,K,a,s,\beta,\kappa,c_0}(\sqrt{n_m})$.

Proof. Apply Corollary 5.30 at each m, with

$$(d, n, \ell, M, u, r) = (d_m, n_m, \ell_m, M_m, u_m, r_m).$$

The matrix event contributes

$$C_{A,K,a} d_m^{-A} + p_{d_m}(a) \min\{n_m, \ell_m^2\}.$$

Conditioned on the complementary matrix event, the sampled-list maximum fails with probability at most e^{-u_m}, while the good-sign count and robust-center count each fail with probability at most

$$\binom{M_m}{r_m} d_m^{-Ar_m}.$$

Thus the joint matrix-and-sign failure probabilities are summable, up to a constant factor. The first Borel–Cantelli lemma gives an almost-sure event on which all three conclusions hold for all but finitely many m.

For the full-dimension consequence take $d_m = m$, $u_m = s \log(e + m)$, and $r_m = 1$. Then $\sum_m m^{-A} < \infty$, $\sum_m e^{-u_m} < \infty$, and

$$\binom{M_m}{1} m^{-A} \le m^{\beta - A},$$

which is summable because $A > \beta + 1$. The lower-normalizer contribution is bounded by a constant times $m^{-B_0} \log^2(e + m)$, hence is summable when $B_0 > 1$. With $r_m = 1$, the count lower bounds say every sampled sign is good and robust.

For the polynomial schedule, again set $u_m = s \log(e + m)$ and $r_m = 1$. The matrix tail d_m^{-A} is summable because $(A - \beta)\kappa > 1$ and $\beta \ge 0$. The lower-normalizer contribution is bounded by a constant times

$$m^{-B_0 \kappa} \log^2(e + m),$$

which is summable under $B_0 \kappa > 1$. The sampled-list tail is summable since $s > 1$, and the binomial count term satisfies

$$\binom{M_m}{1} d_m^{-A} \le d_m^{\beta - A} \le C_{\beta, A, c_0} \, m^{-(A-\beta)\kappa},$$

which is summable by hypothesis. Finally

$$\frac{\log(eM_m) + u_m}{\ell_m} \le C_{s, \beta, \kappa, c_0}$$

for all large m, so the entropy-sensitive sampled-list maximum reduces to the stated $O(\sqrt{n_m})$ bound. Finitely many initial indices are irrelevant. □

Corollary 5.56 (Summable logarithmic-tail sampled counts)**.** *Fix $A > 0$, $a > 0$, $C_0 < \infty$, $C_1 < \infty$, and $\gamma \ge 0$. Put*

$$\theta = \min\{\gamma, 2\}.$$

Suppose

$$p_d(a) := \mathbb{P}\{\|T_d(\xi)\|_{\mathrm{op}} < a \, \mathbb{E}\, \|T_d(\xi)\|_{\mathrm{op}}\} \le C_0 \big(\log(e + d)\big)^{-B} \qquad (d \ge 1).$$

Let $(d_m, n_m)_{m\ge 1}$ *satisfy, with* $\ell_m = \log(e + d_m)$,

$$d_m \ge 2, \qquad C_{A,K,a}\ell_m \le n_m \le \min\{d_m, C_1\ell_m^{\gamma}\}.$$

At each m, *take independent full Toeplitz samples at dimension* d_m, *and independently sample* M_m *uniform sign vectors in* $\{-1,1\}^{n_m}$, *with* $M_m \ge 1$. *Let* $u_m \ge 0$ *and* $1 \le r_m \le M_m$ *be deterministic parameters. If*

$$\sum_{m=1}^{\infty}\left(d_m^{-A} + \ell_m^{\theta-B} + e^{-u_m} + \binom{M_m}{r_m} d_m^{-Ar_m}\right) < \infty,$$

then, with probability one over the matrices and sampled signs, all but finitely many m *satisfy*

$$\max_{1\le s\le M_m}\max_{\omega^{d_m}=1}\left|\sum_{i=1}^{n_m}\delta_{m,i}^{(s)}\frac{P_{m,i}(\omega)}{\|B_{m,i}\|_{\mathrm{op}}}\right| \le C_{A,K,a}\sqrt{n_m\left(1+\frac{\log(eM_m)+u_m}{\ell_m}\right)}.$$

Moreover, at least $M_m - r_m + 1$ *sampled signs have edge maximum at most* $C_{A,K,a}\sqrt{n_m}$, *and at least* $M_m - r_m + 1$ *sampled signs are centers of Hamming balls of radius* $\lfloor c_{A,K,a}\ell_m\rfloor$ *on which every signing has edge maximum at most* $C_{A,K,a}\sqrt{n_m}$.

In particular, if $r_m = 1$, $M_m \le d_m^{\beta}$ *for some* $0 \le \beta < A$,

$$\sum_m\left(\ell_m^{\theta-B} + d_m^{\beta-A} + e^{-u_m}\right) < \infty, \qquad \sup_m \frac{u_m}{\ell_m} < \infty,$$

then eventually every sampled sign in the batch is an $O(\sqrt{n_m})$*-good robust center and the whole batch satisfies an* $O(\sqrt{n_m})$ *sampled-list maximum bound. If* $0 < \rho \le 1$, $0 < \tau_0 < A$, $r_m = \lceil \rho M_m\rceil$, *and*

$$\sum_m\left(\ell_m^{\theta-B} + d_m^{-(A-\tau_0)} + e^{-u_m}\right) < \infty,$$

then no upper bound on M_m *is needed to get, eventually almost surely, a* $1-\rho$ *fraction of sampled signs which are logarithmically robust centers; only the displayed sampled-list maximum keeps its entropy factor* $\log(eM_m)$.

Proof. The logarithmic sample window gives

$$\min\{n_m, \ell_m^2\} \le C_{C_1}\ell_m^{\theta}.$$

Hence

$$p_{d_m}(a)\min\{n_m, \ell_m^2\} \le C_{C_0,C_1}\ell_m^{\theta-B}.$$

The displayed summability hypothesis is therefore stronger than the hypothesis of Corollary 5.55, which gives the sampled-list maximum and count assertions.

For the polynomial-in-d_m batch consequence, $d_m^{-A} \le d_m^{\beta-A}$ and

$$\binom{M_m}{1} d_m^{-A} \le d_m^{\beta-A}.$$

The assumed boundedness of u_m/ℓ_m also gives $(\log(eM_m) + u_m)/\ell_m = O(1)$. Thus the entropy-sensitive sampled-list maximum has the $O(\sqrt{n_m})$ form,

and the count conclusion with $r_m = 1$ says every sampled sign is good and robust.

For $r_m = \lceil \rho M_m \rceil$,

$$\binom{M_m}{r_m} d_m^{-Ar_m} \le \left(\frac{e}{\rho} d_m^{-A}\right)^{r_m}.$$

For all large m this is at most

$$d_m^{-(A-\tau_0)r_m} \le d_m^{-(A-\tau_0)},$$

which is summable by hypothesis. Applying the first part gives the eventual $1-\rho$ robust-center fraction. □

Corollary 5.57 (Logarithmic-tail sampled-count schedule design)**.** *Fix $A > 0$, $a > 0$, $C_0 < \infty$, $C_1 < \infty$, and $\gamma \ge 0$. Put*

$$\theta = \min\{\gamma, 2\}.$$

Suppose

$$p_d(a) := \mathbb{P}\{\|T_d(\xi)\|_{\mathrm{op}} < a\,\mathbb{E}\,\|T_d(\xi)\|_{\mathrm{op}}\} \le C_0(\log(e+d))^{-B} \qquad (d \ge 1).$$

Let $h_m \ge 1$ be deterministic numbers such that

$$\sum_{m=1}^{\infty} h_m^{\theta - B} < \infty.$$

Let $(d_m, n_m)_{m \ge 1}$ satisfy, with $\ell_m = \log(e + d_m)$,

$$d_m \ge 2, \qquad \ell_m \ge h_m, \qquad C_{A,K,a}\ell_m \le n_m \le \min\{d_m, C_1 \ell_m^{\gamma}\}.$$

At each m, take independent full Toeplitz samples at dimension d_m, and independently sample M_m uniform sign vectors in $\{-1,1\}^{n_m}$, with $M_m \ge 1$. Let $u_m \ge 0$ and $1 \le r_m \le M_m$ be deterministic parameters. If

$$\sum_{m=1}^{\infty} \left(e^{-u_m} + \binom{M_m}{r_m} d_m^{-Ar_m} \right) < \infty,$$

then the sampled-list maximum, good-sign count, and robust-center count conclusions of Corollary 5.56 hold eventually almost surely.

In particular, if $r_m = 1$, $M_m \le d_m^{\beta}$ for some $0 \le \beta < A$, $\sum_m e^{-u_m} < \infty$, and

$$\sup_m \frac{u_m}{\ell_m} < \infty,$$

then eventually every sampled sign in the batch is an $O(\sqrt{n_m})$-good robust center and the whole batch satisfies an $O(\sqrt{n_m})$ sampled-list maximum bound. If $0 < \rho \le 1$, $r_m = \lceil \rho M_m \rceil$, and $\sum_m e^{-u_m} < \infty$, then no upper bound on M_m is needed to get, eventually almost surely, a $1-\rho$ fraction of sampled signs which are logarithmically robust centers; only the displayed sampled-list maximum keeps its entropy factor $\log(eM_m)$.

Proof. Since the displayed series has terms at least one unless $B > \theta$, the hypothesis forces $B > \theta$ and $h_m \to \infty$. Therefore, for every $c > 0$, there is $m_0(c)$ such that

$$e^{-ch_m} \le h_m^{\theta-B} \qquad (m \ge m_0(c)).$$

The lower bound $\ell_m \ge h_m$ gives

$$\ell_m^{\theta-B} \le h_m^{\theta-B}, \qquad d_m^{-c} \le C_c e^{-ch_m} \le C_c h_m^{\theta-B}$$

for all large m. Thus the matrix summands in Corollary 5.56 are summable, and the general sampled-count conclusion follows from the assumed sign-tail summability.

If $M_m \le d_m^{\beta}$ and $r_m = 1$, then

$$\binom{M_m}{1} d_m^{-A} \le d_m^{\beta-A} \le e^{-(A-\beta)h_m},$$

which is summable by the preceding comparison. Also $(\log(eM_m)+u_m)/\ell_m = O(1)$, so the sampled-list maximum is $O(\sqrt{n_m})$, and $r_m = 1$ makes every sampled sign good and robust.

For $r_m = \lceil \rho M_m \rceil$,

$$\binom{M_m}{r_m} d_m^{-Ar_m} \le \left(\frac{e}{\rho} d_m^{-A}\right)^{r_m}$$

for all large m. Hence, for any fixed $0 < \tau_0 < A$, the last display is at most

$$d_m^{-(A-\tau_0)} \le C_{A,\tau_0} e^{-(A-\tau_0)h_m}.$$

This is summable by the first paragraph, so the sampled-count theorem gives the robust-center fraction. □

Corollary 5.58 (Vanishing discarded fractions on logarithmic-tail schedules)**.** *In the setting of Corollary 5.57, fix $0 < \tau_0 < A$, and suppose $(\rho_m)_{m\ge1} \subset (0,1]$ satisfies*

$$\rho_m \ge e\, d_m^{-\tau_0} \qquad \textit{for all sufficiently large } m,$$

and set

$$r_m = \lceil \rho_m M_m \rceil.$$

If

$$\sum_m e^{-u_m} < \infty,$$

then, with probability one over the matrices and sampled signs, all but finitely many m satisfy the sampled-list maximum bound from Corollary 5.57, and at least a $1-\rho_m$ fraction of the M_m sampled signs are logarithmically robust centers. No upper bound on M_m is required. In particular, if $\rho_m \to 0$, then the robust-center fraction tends to one along the sampled batches, while the whole-list maximum retains only the entropy factor $\log(eM_m)$.

Proof. By Corollary 5.57, it is enough to check the binomial count-tail summability. Since $r_m = \lceil \rho_m M_m \rceil$,

$$\binom{M_m}{r_m} \le \left(\frac{eM_m}{r_m}\right)^{r_m} \le \left(\frac{e}{\rho_m}\right)^{r_m}.$$

The lower bound on ρ_m gives, for all large m,

$$\binom{M_m}{r_m} d_m^{-Ar_m} \le d_m^{-(A-\tau_0)r_m} \le d_m^{-(A-\tau_0)}.$$

The proof of Corollary 5.57 shows $\sum_m d_m^{-(A-\tau_0)} < \infty$ under the lower-profile hypothesis. The sampled-count theorem therefore applies. Finally,

$$M_m - r_m + 1 \ge M_m - \rho_m M_m,$$

so the count lower bound is at least a $1 - \rho_m$ fraction. □

Corollary 5.59 (Full-integer arbitrary-batch robust fractions)**.** *Fix $A > 1$, $a > 0$, $s > 1$, and $0 < \rho \le 1$. Suppose that*

$$p_d(a) := \mathbb{P}\{\|T_d(\xi)\|_{\mathrm{op}} < a\,\mathbb{E}\,\|T_d(\xi)\|_{\mathrm{op}}\} \le C_0 d^{-B_0} \qquad (d \ge 1)$$

for some $C_0 < \infty$ and $B_0 > 1$. Let $n(d)$ be any admissible full integer sample window with

$$C_{A,K,a} \log(e + d) \le n(d) \le d \qquad \text{for all large } d.$$

At each integer dimension d, let $B_{d,i} = T_d(\xi^{(d,i)})$, $1 \le i \le n(d)$, be independent full Toeplitz matrices with independent mean-zero, unit-variance lag variables of sub-Gaussian norm at most K, and let $P_{d,i}$ be their symbols. Independently sample an arbitrary deterministic number $M_d \ge 1$ of uniform sign vectors

$$\delta_d^{(1)}, \dots, \delta_d^{(M_d)} \in \{-1, 1\}^{n(d)}.$$

Then, with probability one over the matrices and sampled signs, all sufficiently large integer dimensions satisfy both conclusions below.

First, the whole sampled list obeys

$$\max_{1 \le t \le M_d} \max_{\omega^d = 1} \left| \sum_{i=1}^{n(d)} \delta_{d,i}^{(t)} \frac{P_{d,i}(\omega)}{\|B_{d,i}\|_{\mathrm{op}}} \right| \le C_{A,K,a} \sqrt{n(d) \left(1 + \frac{\log(eM_d) + s \log(e + d)}{\log(e + d)}\right)}.$$

Second, at least a $1 - \rho$ fraction of the M_d sampled signs have edge maximum at most $C_{A,K,a}\sqrt{n(d)}$ and are centers of Hamming balls of radius

$$\lfloor c_{A,K,a} \log(e + d) \rfloor$$

on which every signing ε' has edge maximum at most

$$C_{A,K,a}\sqrt{n(d)}.$$

No upper bound on M_d is required for this robust-fraction conclusion.

Proof. Apply Corollary 5.55 with $d_m = m$, $n_m = n(m)$,

$$u_m = s\log(e+m), \qquad r_m = \lceil \rho M_m \rceil,$$

where M_m denotes the sampled batch size at dimension m. The matrix summand $\sum_m m^{-A}$ is finite because $A > 1$, the sampled-list summand is finite because $s > 1$, and

$$p_m(a)\min\{n(m), \log^2(e+m)\} \le C_0 m^{-B_0}\log^2(e+m)$$

is summable because $B_0 > 1$. It remains to check the binomial count tail. Since $r_m \ge \rho M_m$,

$$\binom{M_m}{r_m} m^{-Ar_m} \le \left(\frac{eM_m}{r_m}\right)^{r_m} m^{-Ar_m} \le \left(\frac{e}{\rho}m^{-A}\right)^{r_m}.$$

Choose $0 < \tau_0 < A - 1$. For all large m, the last display is at most

$$m^{-(A-\tau_0)r_m} \le m^{-(A-\tau_0)},$$

which is summable because $A - \tau_0 > 1$. The sampled-count theorem gives the entropy-sensitive list maximum and at least

$$M_m - r_m + 1 \ge (1-\rho)M_m$$

good robust centers eventually. □

Corollary 5.60 (Full-integer power-vanishing robust fractions)**.** *Fix $A > 1$, $a > 0$, and $0 < \tau < A - 1$. Suppose that*

$$p_d(a) := \mathbb{P}\{\|T_d(\xi)\|_{\mathrm{op}} < a\,\mathbb{E}\,\|T_d(\xi)\|_{\mathrm{op}}\} \le C_0 d^{-B_0} \qquad (d \ge 1)$$

for some $C_0 < \infty$ and $B_0 > 1$. Let $n(d)$ be any admissible full integer sample window with

$$C_{A,K,a}\log(e+d) \le n(d) \le d \qquad \text{for all large } d.$$

At each integer dimension d, let $B_{d,i} = T_d(\xi^{(d,i)})$, $1 \le i \le n(d)$, be independent full Toeplitz matrices with independent mean-zero, unit-variance lag variables of sub-Gaussian norm at most K, and let $P_{d,i}$ be their symbols. Independently sample an arbitrary deterministic number $M_d \ge 1$ of uniform sign vectors

$$\delta_d^{(1)}, \dots, \delta_d^{(M_d)} \in \{-1, 1\}^{n(d)}.$$

Let $u_d \ge 0$ satisfy

$$\sum_{d=1}^{\infty} e^{-u_d} < \infty,$$

and let $(\rho_d)_{d\ge1} \subset (0, 1]$ satisfy

$$\rho_d \ge e\,d^{-\tau} \qquad \text{for all sufficiently large } d.$$

Then, with probability one over the matrices and sampled signs, all sufficiently large integer dimensions satisfy both conclusions below.

First, the whole sampled list obeys

$$\max_{1\le t\le M_d}\max_{\omega^d=1}\left|\sum_{i=1}^{n(d)}\delta_{d,i}^{(t)}\frac{P_{d,i}(\omega)}{\|B_{d,i}\|_{\mathrm{op}}}\right|\le C_{A,K,a}\sqrt{n(d)\left(1+\frac{\log(eM_d)+u_d}{\log(e+d)}\right)}.$$

Second, at least a $1-\rho_d$ *fraction of the* M_d *sampled signs have edge maximum at most* $C_{A,K,a}\sqrt{n(d)}$ *and are centers of Hamming balls of radius*

$$\lfloor c_{A,K,a}\log(e+d)\rfloor$$

on which every signing ε' *has edge maximum at most*

$$C_{A,K,a}\sqrt{n(d)}.$$

No upper bound on M_d *is required. In particular, if* $\rho_d\to 0$*, then the robust-center fraction tends to one along the full integer dimension sequence.*

Proof. Apply Corollary 5.55 with $d_m=m$, $n_m=n(m)$, and

$$r_m=\lceil\rho_m M_m\rceil,$$

where M_m, u_m, and ρ_m are the parameters at dimension m. The matrix summand $\sum_m m^{-A}$ is finite because $A>1$, the sampled-list summand is finite by hypothesis, and

$$p_m(a)\min\{n(m),\log^2(e+m)\}\le C_0 m^{-B_0}\log^2(e+m)$$

is summable because $B_0>1$. For the binomial count tail,

$$\binom{M_m}{r_m}\le\left(\frac{eM_m}{r_m}\right)^{r_m}\le\left(\frac{e}{\rho_m}\right)^{r_m}.$$

The lower bound $\rho_m\ge e\,m^{-\tau}$ gives, for all large m,

$$\binom{M_m}{r_m}m^{-Ar_m}\le m^{-(A-\tau)r_m}\le m^{-(A-\tau)},$$

which is summable because $A-\tau>1$. Hence the sampled-count theorem applies. Finally,

$$M_m-r_m+1\ge M_m-\rho_m M_m,$$

so the robust-center count is at least a $1-\rho_m$ fraction. □

Corollary 5.61 (Almost-sure normalized robust fractions)**.** *Fix* $A>0$*,* $S>0$*,* $a>0$*, and a fraction* $0<\rho\le 1$*. Suppose*

$$A\ge 2\log_2(e/\rho),$$

and choose an integer $M\ge 1$ *so that*

$$M\ge\frac{2(S+1)}{A\rho}.$$

Let $(d_m,n_m)_{m\ge1}$ *be a triangular-array sequence with*

$$d_m\ge 2,\qquad n_m\le d_m,\qquad \ell_m=\log(e+d_m),$$

and assume

$$n_m \geq C_{A,K,a}\ell_m \qquad (m \geq 1).$$

At each m, let $B_{m,i} = T_{d_m}(\xi^{(m,i)})$, $1 \leq i \leq n_m$, be independent full Toeplitz matrices with independent mean-zero, unit-variance lag variables of sub-Gaussian norm at most K, and let $P_{m,i}$ be their symbols. Independently of the matrices, sample independent uniform sign vectors

$$\delta_m^{(1)}, \dots, \delta_m^{(M)} \in \{-1, 1\}^{n_m}.$$

Put

$$p_{d_m}(a) = \mathbb{P}\{\|T_{d_m}(\xi)\|_{\mathrm{op}} < a\,\mathbb{E}\,\|T_{d_m}(\xi)\|_{\mathrm{op}}\}.$$

If

$$\sum_{m=1}^{\infty} \left(d_m^{-A} + p_{d_m}(a)\min\{n_m, \ell_m^2\} + d_m^{-S}\right) < \infty,$$

then, with probability one over the matrices and sampled signs, all but finitely many m satisfy both conclusions below.

First, the whole sampled list obeys

$$\max_{1\leq s\leq M}\max_{\omega^{d_m}=1}\left|\sum_{i=1}^{n_m}\delta_{m,i}^{(s)}\frac{P_{m,i}(\omega)}{\|B_{m,i}\|_{\mathrm{op}}}\right| \leq C_{A,K,a,S,\rho}\sqrt{n_m}.$$

Second, at least a $1-\rho$ fraction of the M sampled signs are centers of Hamming balls of radius

$$\lfloor c_{A,K,a}\ell_m \rfloor$$

on which every signing ε' has edge maximum at most

$$C_{A,K,a}\sqrt{n_m}.$$

The same $1-\rho$ fraction also has edge maximum at most $C_{A,K,a}\sqrt{n_m}$.

In particular, along the full integer dimension sequence, suppose $A > 1$, $S > 1$, and

$$p_d(a) \leq C_0 d^{-B_0} \qquad (d \geq 1)$$

for some $B_0 > 1$. Then any admissible window

$$C_{A,K,a}\log(e+d) \leq n(d) \leq d$$

has the same eventual robust-fraction conclusion almost surely. More generally, on polynomial dimension schedules $d_m \geq c_0 m^{\kappa}$, the same conclusion holds under the three summability inequalities

$$A\kappa > 1, \qquad B_0\kappa > 1, \qquad S\kappa > 1.$$

Proof. Let $r_\rho = \lceil \rho M \rceil$, and put $u_m = S\ell_m$. Apply the finite Corollary 5.30 at dimension m. The matrix exceptional probability contributes

$$C_{A,K,a}d_m^{-A} + p_{d_m}(a)\min\{n_m, \ell_m^2\}.$$

Outside that matrix event, the sampled-list maximum fails with conditional probability at most e^{-u_m}, while the good-sign count and robust-center count each fail with conditional probability at most

$$\binom{M}{r_\rho} d_m^{-Ar_\rho}.$$

It remains to bound these two sign terms by the displayed d_m^{-S} summand. Since $r_\rho \geq \rho M$,

$$\binom{M}{r_\rho} \leq \left(\frac{eM}{r_\rho}\right)^{r_\rho} \leq (e/\rho)^{r_\rho}.$$

For $d_m \geq 2$ and $A \geq 2\log_2(e/\rho)$,

$$\binom{M}{r_\rho} d_m^{-Ar_\rho} \leq d_m^{-Ar_\rho/2}.$$

The lower bound on M gives $Ar_\rho/2 \geq S+1$, hence

$$2\binom{M}{r_\rho} d_m^{-Ar_\rho} \leq 2d_m^{-(S+1)} \leq d_m^{-S}.$$

Also $e^{-u_m} = (e+d_m)^{-S} \leq d_m^{-S}$. Therefore the total matrix-and-sign failure probability is summable. Borel–Cantelli gives the eventual sampled-list maximum and the $1-\rho$ fraction of good and robust sampled signs. Since

$$1 + \frac{\log(eM) + u_m}{\ell_m} \leq C_{S,\rho,A},$$

the sampled-list maximum has the displayed $O(\sqrt{n_m})$ form.

The full-dimension polynomial-tail consequence follows by taking $d_m = m$: $\sum_m m^{-A}$, $\sum_m m^{-S}$, and $\sum_m m^{-B_0}\log^2(e+m)$ are finite under $A, S, B_0 > 1$. On polynomial dimension schedules the same estimates are bounded by constants times

$$m^{-A\kappa}, \qquad m^{-S\kappa}, \qquad m^{-B_0\kappa}\log^2(e+m),$$

which are summable under the three displayed inequalities. □

Corollary 5.62 (Almost-sure sampled robust certificates)**.** *Fix $A > 0$, $\alpha > 0$, and $a > 0$. Let $(d_m, n_m)_{m\geq 1}$ be a sequence with $n_m \leq d_m$, put $\ell_m = \log(e+d_m)$, and assume*

$$n_m \geq C_{A,K,a}\ell_m \qquad (m \geq 1).$$

At each m, let $B_{m,i} = T_{d_m}(\xi^{(m,i)})$, $1 \leq i \leq n_m$, be independent full Toeplitz matrices with independent mean-zero, unit-variance lag variables of sub-Gaussian norm at most K, and let $P_{m,i}$ be their symbols. Let

$$p_{d_m}(a) = \mathbb{P}\{\|T_{d_m}(\xi)\|_{\mathrm{op}} < a\,\mathbb{E}\,\|T_{d_m}(\xi)\|_{\mathrm{op}}\}.$$

For each m, sample M_m independent uniform sign vectors

$$\delta_m^{(1)}, \ldots, \delta_m^{(M_m)} \in \{-1,1\}^{n_m}, \qquad M_m \geq \lceil \alpha \rceil,$$

independently of the matrices. If

$$\sum_{m=1}^{\infty}\left(d_m^{-A}+p_{d_m}(a)\min\{n_m,\ell_m^2\}+d_m^{-\alpha}\right)<\infty,$$

then, with probability one over the matrices and the sampled signs, all but finitely many m have a sampled sign $\delta_m^{(s)}$ which is the center of a Hamming ball of radius

$$r_m=\lfloor c_{A,K,a}\ell_m\rfloor$$

on which every signing ε' satisfies

$$\max_{\omega^{d_m}=1}\left|\sum_{i=1}^{n_m}\varepsilon'_i\frac{P_{m,i}(\omega)}{\|B_{m,i}\|_{\mathrm{op}}}\right|\le C_{A,K,a}\sqrt{n_m}.$$

Thus the finite random search itself eventually returns logarithmically robust $O(\sqrt{n_m})$ certificates, rather than only a conditional certificate probability.

In particular, along the full integer dimension sequence, if $A>1$, $\alpha>1$, and

$$p_d(a)\le C_0d^{-B_0}\qquad(d\ge1)$$

for some $B_0>1$, then any admissible window

$$C_{A,K,a}\log(e+d)\le n(d)\le d$$

and any fixed list size $M\ge\lceil\alpha\rceil$ give, almost surely, robust sampled certificates for all sufficiently large integer dimensions.

Proof. Apply Corollary 5.31 at each m, with B there replaced by α. The probability that the m-th matrix array and its sampled sign list fail to contain the displayed robust certificate is at most

$$C_{A,K,a}d_m^{-A}+p_{d_m}(a)\min\{n_m,\ell_m^2\}+d_m^{-\alpha}.$$

The series of these probabilities is finite by hypothesis, so the first Borel–Cantelli lemma gives the eventual sampled-certificate conclusion.

For the full-dimension consequence take $d_m=m$, $n_m=n(m)$, and $M_m=M$. Then $\sum_m m^{-A}$ and $\sum_m m^{-\alpha}$ are finite, while

$$p_m(a)\min\{n(m),\log^2(e+m)\}\le C_0m^{-B_0}\log^2(e+m)$$

is summable because $B_0>1$. The general assertion applies. □

Corollary 5.63 (All-dimensions almost-sure transfer from polynomial lower tails)**.** *Fix $a>0$, $C_0<\infty$, and $A>1$. Suppose*

$$p_d(a):=\mathbb{P}\{\|T_d(\xi)\|_{\mathrm{op}}<a\,\mathbb{E}\,\|T_d(\xi)\|_{\mathrm{op}}\}\le C_0d^{-B}\qquad(d\ge1)$$

for some $B>1$. For each $d\ge1$, let $n(d)$ satisfy

$$C_{A,K,a}\log(e+d)\le n(d)\le d.$$

Then, for independent full Toeplitz samples $B_{d,i} = T_d(\xi^{(d,i)})$, $1 \le i \le n(d)$, at every integer dimension d, with symbols $P_{d,i}$, with probability one all but finitely many dimensions satisfy

$$\max_{\omega^d=1} \sum_{i=1}^{n(d)} \frac{|P_{d,i}(\omega)|^2}{\|B_{d,i}\|_{\mathrm{op}}^2} \le C_{A,K,a} \frac{n(d)}{\log(e+d)}.$$

For those d, the signing, sign-averaged maximum, deterministic-certificate, and scalar moment conclusions of Corollary 5.29 hold with n, ℓ replaced by $n(d), \log(e+d)$.

Proof. Apply Corollary 5.53 with $d_m = m$ and $n_m = n(m)$. The matrix-event part is summable because $A > 1$. Also

$$p_m(a) \min\{n_m, \log^2(e+m)\} \le C_0 m^{-B} \log^2(e+m),$$

which is summable because $B > 1$. Hence the general Borel–Cantelli transfer gives the eventual profile, and the deterministic consequences again come from Corollary 5.29. □

Corollary 5.64 (Sparse-scale almost-sure transfer)**.** *Fix $A > 0$, $a > 0$, $C_0 < \infty$, and $\lambda > 1$. Suppose either the polynomial tail*

$$p_d(a) := \mathbb{P}\{\|T_d(\xi)\|_{\mathrm{op}} < a\, \mathbb{E}\, \|T_d(\xi)\|_{\mathrm{op}}\} \le C_0 d^{-B} \qquad (d \ge 1).$$

holds for some $B > 0$, or the logarithmic tail

$$p_d(a) \le C_0 \big(\log(e+d)\big)^{-B} \qquad (d \ge 1)$$

holds for some $B > 3$. Let $(d_m, n_m)_{m \ge 1}$ satisfy

$$d_m \ge \lambda^m, \qquad C_{A,K,a} \log(e+d_m) \le n_m \le d_m.$$

Then, for independent full Toeplitz samples at the dimensions d_m, with probability one all but finitely many m satisfy

$$\max_{\omega^{d_m}=1} \sum_{i=1}^{n_m} \frac{|P_{m,i}(\omega)|^2}{\|B_{m,i}\|_{\mathrm{op}}^2} \le C_{A,K,a} \frac{n_m}{\log(e+d_m)}.$$

For those m, the signing, sign-averaged maximum, deterministic-certificate, and scalar moment conclusions of Corollary 5.29 hold with d, n, ℓ replaced by $d_m, n_m, \log(e+d_m)$.

Proof. By Corollary 5.53, it is enough to verify summability. Put $\ell_m = \log(e+d_m)$. We use $\min\{n_m, \ell_m^2\} \le \ell_m^2$. In the polynomial-tail case, for $x \ge 1$,

$$x^{-B} \big(\log(e+x)\big)^2 \le C_B x^{-B/2}.$$

Then

$$p_{d_m}(a) \min\{n_m, \ell_m^2\} \le C_{B,C_0} d_m^{-B/2} \le C_{B,C_0,\lambda} \lambda^{-Bm/2}.$$

In the logarithmic-tail case,

$$p_{d_m}(a) \min\{n_m, \ell_m^2\} \le C_0 \ell_m^{2-B} \le C_{B,C_0,\lambda} m^{2-B},$$

which is summable because $B > 3$. Also $d_m^{-A} \le \lambda^{-Am}$. The required series is therefore finite, and the Borel–Cantelli transfer applies. □

Corollary 5.65 (Sparse-scale sampled robust certificates)**.** *Fix $A > 0$, $\alpha > 0$, $a > 0$, $C_0 < \infty$, and $\lambda > 1$. Suppose either the polynomial tail*

$$p_d(a) := \mathbb{P}\{\|T_d(\xi)\|_{\mathrm{op}} < a\,\mathbb{E}\,\|T_d(\xi)\|_{\mathrm{op}}\} \le C_0 d^{-B} \qquad (d \ge 1)$$

holds for some $B > 0$, or the logarithmic tail

$$p_d(a) \le C_0\big(\log(e+d)\big)^{-B} \qquad (d \ge 1)$$

holds for some $B > 3$. Let $(d_m, n_m)_{m\ge1}$ satisfy

$$d_m \ge \lambda^m, \qquad C_{A,K,a}\log(e+d_m) \le n_m \le d_m.$$

For each m, sample M_m independent uniform sign vectors in $\{-1,1\}^{n_m}$, independently of the matrices, with

$$M_m \ge \lceil \alpha \rceil.$$

Then, with probability one over the matrices and sampled signs, all but finitely many m have a sampled sign which is the center of a Hamming ball of radius

$$r_m = \lfloor c_{A,K,a}\log(e+d_m)\rfloor$$

on which every signing ε' satisfies

$$\max_{\omega^{d_m}=1}\left|\sum_{i=1}^{n_m}\varepsilon'_i\frac{P_{m,i}(\omega)}{\|B_{m,i}\|_{\mathrm{op}}}\right| \le C_{A,K,a}\sqrt{n_m}.$$

Thus exponentially sparse dimensions need no lower bound on the sampled-list confidence exponent beyond $\alpha > 0$.

Proof. Apply Corollary 5.62. It remains to check

$$\sum_m \Big(d_m^{-A} + p_{d_m}(a)\min\{n_m, \ell_m^2\} + d_m^{-\alpha}\Big) < \infty, \qquad \ell_m = \log(e+d_m).$$

The exponential growth of d_m makes $\sum_m d_m^{-A}$ and $\sum_m d_m^{-\alpha}$ finite. The same estimates used in Corollary 5.64 show that the middle summand is summable: in the polynomial-tail case it is at most $C_{B,C_0,\lambda}\lambda^{-Bm/2}$, while in the logarithmic-tail case it is at most $C_{B,C_0,\lambda}m^{2-B}$, summable because $B > 3$. The sampled Borel–Cantelli transfer applies. □

Corollary 5.66 (Polynomial-scale almost-sure transfer from polynomial lower tails)**.** *Fix $a > 0$, $C_0 < \infty$, $c_0 > 0$, and $\kappa > 0$. Suppose*

$$p_d(a) := \mathbb{P}\{\|T_d(\xi)\|_{\mathrm{op}} < a\,\mathbb{E}\,\|T_d(\xi)\|_{\mathrm{op}}\} \le C_0 d^{-B} \qquad (d \ge 1)$$

for some $B > 0$. Let $A > 0$ satisfy

$$A\kappa > 1, \qquad B\kappa > 1.$$

Let $(d_m, n_m)_{m\ge1}$ satisfy

$$d_m \ge c_0 m^{\kappa}, \qquad C_{A,K,a}\log(e+d_m) \le n_m \le d_m.$$

Then, for independent full Toeplitz samples at the dimensions d_m, with probability one all but finitely many m satisfy

$$\max_{\omega^{d_m}=1} \sum_{i=1}^{n_m} \frac{|P_{m,i}(\omega)|^2}{\|B_{m,i}\|_{\mathrm{op}}^2} \le C_{A,K,a} \frac{n_m}{\log(e+d_m)}.$$

For those m, the signing, sign-averaged maximum, deterministic-certificate, and scalar moment conclusions of Corollary 5.29 hold with d, n, ℓ replaced by $d_m, n_m, \log(e+d_m)$.

Proof. By Corollary 5.53, it is enough to check

$$\sum_m \left(d_m^{-A} + p_{d_m}(a) \min\{n_m, \ell_m^2\} \right) < \infty, \qquad \ell_m := \log(e+d_m).$$

The first summand is at most $C_{A,c_0} m^{-A\kappa}$, which is summable by $A\kappa > 1$. For the second summand, $\min\{n_m, \ell_m^2\} \le \ell_m^2$, and

$$p_{d_m}(a)\ell_m^2 \le C_{B,C_0,c_0} m^{-B\kappa} (\log(e+m))^2.$$

This is summable by $B\kappa > 1$. The Borel–Cantelli transfer gives the eventual profile, and the remaining conclusions are again deterministic consequences of that profile in Corollary 5.29. □

Corollary 5.67 (Polynomial-scale sampled robust certificates)**.** *Fix $a > 0$, $C_0 < \infty$, $c_0 > 0$, $\kappa > 0$, and $A, \alpha > 0$. Suppose*

$$p_d(a) := \mathbb{P}\{\|T_d(\xi)\|_{\mathrm{op}} < a\, \mathbb{E}\, \|T_d(\xi)\|_{\mathrm{op}}\} \le C_0 d^{-B} \qquad (d \ge 1)$$

for some $B > 0$. Assume

$$A\kappa > 1, \qquad B\kappa > 1, \qquad \alpha\kappa > 1.$$

Let $(d_m, n_m)_{m\ge 1}$ satisfy

$$d_m \ge c_0 m^{\kappa}, \qquad C_{A,K,a} \log(e+d_m) \le n_m \le d_m.$$

At each m, sample M_m independent uniform sign vectors in $\{-1,1\}^{n_m}$, independently of the matrices, with

$$M_m \ge \lceil \alpha \rceil.$$

Then, with probability one over the matrices and sampled signs, all but finitely many m have a sampled sign which is the center of a Hamming ball of radius

$$r_m = \lfloor c_{A,K,a} \log(e+d_m) \rfloor$$

on which every signing ε' satisfies

$$\max_{\omega^{d_m}=1} \left| \sum_{i=1}^{n_m} \varepsilon_i' \frac{P_{m,i}(\omega)}{\|B_{m,i}\|_{\mathrm{op}}} \right| \le C_{A,K,a} \sqrt{n_m}.$$

Thus polynomially sparse dimension schedules need only $\alpha > 1/\kappa$ sampled-list confidence, rather than the all-integer condition $\alpha > 1$.

Proof. Apply Corollary 5.62. It is enough to verify

$$\sum_m \left(d_m^{-A} + p_{d_m}(a) \min\{n_m, \ell_m^2\} + d_m^{-\alpha} \right) < \infty, \qquad \ell_m = \log(e + d_m).$$

The first and third summands are bounded by constants times $m^{-A\kappa}$ and $m^{-\alpha\kappa}$, respectively, and are summable by hypothesis. For the middle term, $\min\{n_m, \ell_m^2\} \le \ell_m^2$, hence

$$p_{d_m}(a)\ell_m^2 \le C_{B,C_0,c_0} m^{-B\kappa} (\log(e+m))^2,$$

which is summable because $B\kappa > 1$. The general sampled-list Borel corollary gives the conclusion. □

Corollary 5.68 (Sparse almost-sure transfer in logarithmic sample windows)**.** *Fix $A > 0$, $a > 0$, $C_0 < \infty$, $C_1 < \infty$, $c_0 > 0$, $\kappa > 0$, and $\gamma \ge 0$. Suppose the logarithmic lower-normalizer tail satisfies*

$$p_d(a) := \mathbb{P}\{\|T_d(\xi)\|_{\mathrm{op}} < a\, \mathbb{E}\, \|T_d(\xi)\|_{\mathrm{op}}\} \le C_0 \big(\log(e+d)\big)^{-B} \qquad (d \ge 1)$$

for some

$$B > \min\{\gamma, 2\} + \frac{1}{\kappa}.$$

Let $(d_m, n_m)_{m \ge 1}$ satisfy, with $\ell_m = \log(e + d_m)$,

$$d_m \ge \exp(c_0 m^\kappa), \qquad C_{A,K,a}\ell_m \le n_m \le \min\{d_m, C_1 \ell_m^\gamma\}.$$

Then, for independent full Toeplitz samples at the dimensions d_m, with probability one all but finitely many m satisfy

$$\max_{\omega^{d_m}=1} \sum_{i=1}^{n_m} \frac{|P_{m,i}(\omega)|^2}{\|B_{m,i}\|_{\mathrm{op}}^2} \le C_{A,K,a} \frac{n_m}{\ell_m}.$$

For those m, the signing, sign-averaged maximum, deterministic-certificate, and scalar moment conclusions of Corollary 5.29 hold with d, n, ℓ replaced by d_m, n_m, ℓ_m.

Proof. By Corollary 5.53, it is enough to prove

$$\sum_m \left(d_m^{-A} + p_{d_m}(a) \min\{n_m, \ell_m^2\} \right) < \infty.$$

The first term is summable because $d_m \ge \exp(c_0 m^\kappa)$. For the second term,

$$\min\{n_m, \ell_m^2\} \le C_{C_1} \ell_m^{\min\{\gamma,2\}},$$

and hence

$$p_{d_m}(a) \min\{n_m, \ell_m^2\} \le C_{C_0,C_1} \ell_m^{\min\{\gamma,2\}-B} \le C_{C_0,C_1,c_0,\kappa} m^{\kappa(\min\{\gamma,2\}-B)}.$$

This is summable precisely under the displayed condition $B > \min\{\gamma, 2\} + 1/\kappa$. The Borel–Cantelli transfer then gives the eventual profile, and the remaining statements are the deterministic consequences of that profile in Corollary 5.29. □

Corollary 5.69 (Log-corrected sparse logarithmic-window transfer)**.** *Fix $A > 0$, $a > 0$, $C_0 < \infty$, $C_1 < \infty$, $c_0 > 0$, $\kappa > 0$, $\lambda \in \mathbb{R}$, and $\gamma \geq 0$. Put*

$$\theta = \min\{\gamma, 2\}.$$

Suppose the logarithmic lower-normalizer tail satisfies

$$p_d(a) := \mathbb{P}\{\|T_d(\xi)\|_{\mathrm{op}} < a\,\mathbb{E}\,\|T_d(\xi)\|_{\mathrm{op}}\} \leq C_0\big(\log(e+d)\big)^{-B} \qquad (d \geq 1),$$

where

$$\kappa(B-\theta) > 1 \qquad \text{or} \qquad \kappa(B-\theta) = 1 \ \text{ and } \ \lambda(B-\theta) > 1.$$

Let $(d_m, n_m)_{m\geq 1}$ satisfy, with

$$\ell_m = \log(e+d_m), \qquad L_m = \log(e+m),$$

the log-corrected sparse growth and logarithmic sample-window bounds

$$d_m \geq \exp(c_0 m^{\kappa} L_m^{\lambda}), \qquad C_{A,K,a}\ell_m \leq n_m \leq \min\{d_m, C_1 \ell_m^{\gamma}\}.$$

Then, for independent full Toeplitz samples at the dimensions d_m, with probability one all but finitely many m satisfy

$$\max_{\omega^{d_m}=1} \sum_{i=1}^{n_m} \frac{|P_{m,i}(\omega)|^2}{\|B_{m,i}\|_{\mathrm{op}}^2} \leq C_{A,K,a}\frac{n_m}{\ell_m}.$$

For those m, the signing, sign-averaged maximum, deterministic-certificate, and scalar moment conclusions of Corollary 5.29 hold with d, n, ℓ replaced by d_m, n_m, ℓ_m.

Moreover, for every $\alpha > 0$, if $M_m \geq \lceil \alpha \rceil$ independent uniform sign vectors are sampled at dimension m, then almost surely all but finitely many m contain a sampled sign which is the center of a Hamming ball of radius $\lfloor c_{A,K,a}\ell_m \rfloor$ on which every signing has edge maximum at most $C_{A,K,a}\sqrt{n_m}$. Finally, if $0 < \rho \leq 1$, $S > 0$, $A \geq 2\log_2(e/\rho)$, and

$$M \geq \frac{2(S+1)}{A\rho},$$

then a fixed list of M sampled signs has, eventually almost surely, a $1-\rho$ fraction of such logarithmically robust centers and the whole sampled list has $O_{A,K,a,S,\rho}(\sqrt{n_m})$ edge maximum.

Proof. By the logarithmic sample-window hypothesis,

$$\min\{n_m, \ell_m^2\} \leq C_{C_1}\ell_m^{\theta}.$$

The growth assumption gives $\ell_m \geq c_0 m^{\kappa} L_m^{\lambda}$ for all large m. Hence

$$\begin{aligned} p_{d_m}(a)\min\{n_m, \ell_m^2\} &\leq C_{C_0,C_1}\ell_m^{\theta-B} \\ &\leq C\, m^{-\kappa(B-\theta)} L_m^{-\lambda(B-\theta)}. \end{aligned}$$

The displayed condition on B is exactly the integral-test summability condition for this last series. Also, for every $r > 0$,

$$\sum_m d_m^{-r} \leq \sum_m \exp\{-r c_0 m^{\kappa} L_m^{\lambda}\} < \infty.$$

Therefore

$$\sum_m \left(d_m^{-A} + p_{d_m}(a)\min\{n_m, \ell_m^2\}\right) < \infty.$$

Corollary 5.53 gives the almost-sure eventual edge profile and its deterministic signing, average, certificate, and moment consequences.

Adding $\sum_m d_m^{-\alpha} < \infty$ and applying Corollary 5.62 gives the sampled robust-certificate conclusion. Adding $\sum_m d_m^{-S} < \infty$ and applying Corollary 5.61 gives the fixed-list robust-fraction conclusion under the displayed choices of A and M. □

Corollary 5.70 (Log-corrected sparse logarithmic-window sampled counts)**.** *Fix $A > 0$, $a > 0$, $C_0 < \infty$, $C_1 < \infty$, $c_0 > 0$, $\kappa > 0$, $\lambda \in \mathbb{R}$, and $\gamma \geq 0$. Put*

$$\theta = \min\{\gamma, 2\}.$$

Suppose

$$p_d(a) := \mathbb{P}\{\|T_d(\xi)\|_{\mathrm{op}} < a\,\mathbb{E}\,\|T_d(\xi)\|_{\mathrm{op}}\} \leq C_0\big(\log(e+d)\big)^{-B} \qquad (d \geq 1),$$

where

$$\kappa(B-\theta) > 1 \qquad \text{or} \qquad \kappa(B-\theta) = 1 \quad \text{and} \quad \lambda(B-\theta) > 1.$$

Let $(d_m, n_m)_{m\geq 1}$ satisfy, with

$$\ell_m = \log(e+d_m), \qquad L_m = \log(e+m),$$

$$d_m \geq \exp(c_0 m^\kappa L_m^\lambda), \qquad C_{A,K,a}\ell_m \leq n_m \leq \min\{d_m, C_1\ell_m^\gamma\}.$$

At each m, take independent full Toeplitz samples at dimension d_m, and independently sample M_m uniform sign vectors in $\{-1,1\}^{n_m}$, with $M_m \geq 1$. Let $u_m \geq 0$ and $1 \leq r_m \leq M_m$ be deterministic parameters. If

$$\sum_{m=1}^{\infty}\left(e^{-u_m} + \binom{M_m}{r_m} d_m^{-Ar_m}\right) < \infty,$$

then, with probability one over the matrices and sampled signs, all but finitely many m satisfy

$$\max_{1\leq s\leq M_m}\max_{\omega^{d_m}=1}\left|\sum_{i=1}^{n_m}\delta_{m,i}^{(s)}\frac{P_{m,i}(\omega)}{\|B_{m,i}\|_{\mathrm{op}}}\right| \leq C_{A,K,a}\sqrt{n_m\left(1+\frac{\log(eM_m)+u_m}{\ell_m}\right)}.$$

Moreover, at least $M_m - r_m + 1$ sampled signs have edge maximum at most $C_{A,K,a}\sqrt{n_m}$, and at least $M_m - r_m + 1$ sampled signs are centers of Hamming balls of radius $\lfloor c_{A,K,a}\ell_m\rfloor$ on which every signing has edge maximum at most $C_{A,K,a}\sqrt{n_m}$.

In particular, let $s > 1$, take $u_m = s\log(e+m)$, and take $r_m = 1$. If

$$M_m \leq d_m^\beta$$

for some $0 \leq \beta < A$, then eventually every sampled sign in the batch is an $O(\sqrt{n_m})$-good robust center and the whole batch satisfies an $O(\sqrt{n_m})$ sampled-list maximum bound. More generally, if

$$M_m \leq \exp(bm^\sigma L_m^\nu)$$

for some $b > 0$, $0 < \sigma \le \kappa$, and $\nu \in \mathbb{R}$, then the same eventual all-sampled-sign conclusion holds whenever

$$\sigma < \kappa, \qquad or \qquad \sigma = \kappa,\ \nu < \lambda, \qquad or \qquad \sigma = \kappa,\ \nu = \lambda,\ b < Ac_0.$$

Finally, for any $0 < \rho \le 1$, if $\sum_m e^{-u_m} < \infty$ and $r_m = \lceil \rho M_m \rceil$, then no upper bound on M_m is needed to get, eventually almost surely, a $1-\rho$ fraction of sampled signs which are logarithmically robust centers; only the displayed sampled-list maximum keeps its entropy factor $\log(eM_m)$.

Proof. As in the proof of Corollary 5.69, the hypotheses give

$$\sum_m \left(d_m^{-A} + p_{d_m}(a) \min\{n_m, \ell_m^2\} \right) < \infty.$$

Therefore Corollary 5.55 applies once the two displayed sign-tail summands are summable. This proves the general sampled-list maximum, good-sign count, and robust-center count assertions.

For the polynomial-in-d_m batch consequence,

$$\binom{M_m}{1} d_m^{-A} \le d_m^{\beta - A},$$

which is summable because $\beta < A$ and $d_m \ge \exp(c_0 m^\kappa L_m^\lambda)$. Also

$$\frac{\log(eM_m) + u_m}{\ell_m} \le C_{s,\beta,c_0,\kappa,\lambda}$$

for all large m, so the sampled-list maximum is $O(\sqrt{n_m})$.

For $M_m \le \exp(bm^\sigma L_m^\nu)$,

$$\binom{M_m}{1} d_m^{-A} \le \exp\{bm^\sigma L_m^\nu - Ac_0 m^\kappa L_m^\lambda\}.$$

The displayed alternatives make this last series summable. The same comparison shows $(\log(eM_m) + u_m)/\ell_m = O(1)$, so the sampled-list maximum again has the $O(\sqrt{n_m})$ form.

Finally, if $r_m = \lceil \rho M_m \rceil$, then

$$\binom{M_m}{r_m} d_m^{-Ar_m} \le \left(\frac{e}{\rho} d_m^{-A} \right)^{r_m}.$$

For every fixed $0 < \tau_0 < A$ and all large m, this is at most

$$d_m^{-(A-\tau_0)r_m} \le d_m^{-(A-\tau_0)},$$

which is summable on the log-corrected sparse schedule. The sampled-count theorem gives the stated $1-\rho$ robust-center fraction. □

Corollary 5.71 (Iterated-log sparse logarithmic-window sampled counts)**.** *Fix $A > 0$, $a > 0$, $C_0 < \infty$, $C_1 < \infty$, $c_0 > 0$, $\kappa > 0$, $\lambda, \mu \in \mathbb{R}$, and $\gamma \ge 0$. Put*

$$\theta = \min\{\gamma, 2\}.$$

Suppose

$$p_d(a) := \mathbb{P}\{\|T_d(\xi)\|_{\mathrm{op}} < a\,\mathbb{E}\,\|T_d(\xi)\|_{\mathrm{op}}\} \le C_0\big(\log(e+d)\big)^{-B} \qquad (d \ge 1),$$

and write $q = B - \theta$. *Assume*

$$\kappa q > 1,$$
$$\text{or} \quad \kappa q = 1 \quad \text{and} \quad \lambda q > 1,$$
$$\text{or} \quad \kappa q = 1,\ \lambda q = 1, \quad \text{and} \quad \mu q > 1.$$

Let $(d_m, n_m)_{m\ge1}$ *satisfy, with*

$$\ell_m = \log(e+d_m), \qquad L_m = \log(e+m), \qquad H_m = \log(e+L_m),$$
$$d_m \ge \exp(c_0 m^\kappa L_m^\lambda H_m^\mu), \qquad C_{A,K,a}\ell_m \le n_m \le \min\{d_m, C_1\ell_m^\gamma\}.$$

At each m, *take independent full Toeplitz samples at dimension* d_m, *and independently sample* M_m *uniform sign vectors in* $\{-1,1\}^{n_m}$, *with* $M_m \ge 1$. *Let* $u_m \ge 0$ *and* $1 \le r_m \le M_m$ *be deterministic parameters. If*

$$\sum_{m=1}^{\infty}\left(e^{-u_m} + \binom{M_m}{r_m} d_m^{-Ar_m}\right) < \infty,$$

then, with probability one over the matrices and sampled signs, all but finitely many m *satisfy*

$$\max_{1\le s\le M_m}\max_{\omega^{d_m}=1}\left|\sum_{i=1}^{n_m}\delta_{m,i}^{(s)}\frac{P_{m,i}(\omega)}{\|B_{m,i}\|_{\mathrm{op}}}\right| \le C_{A,K,a}\sqrt{n_m\left(1+\frac{\log(eM_m)+u_m}{\ell_m}\right)}.$$

Moreover, at least $M_m - r_m + 1$ *sampled signs have edge maximum at most* $C_{A,K,a}\sqrt{n_m}$, *and at least* $M_m - r_m + 1$ *sampled signs are centers of Hamming balls of radius* $\lfloor c_{A,K,a}\ell_m\rfloor$ *on which every signing has edge maximum at most* $C_{A,K,a}\sqrt{n_m}$.

In particular, let $s > 1$, *take* $u_m = s\log(e+m)$, *and take* $r_m = 1$. *If*

$$M_m \le d_m^\beta$$

for some $0 \le \beta < A$, *then eventually every sampled sign in the batch is an* $O(\sqrt{n_m})$*-good robust center and the whole batch satisfies an* $O(\sqrt{n_m})$ *sampled-list maximum bound. More generally, if*

$$M_m \le \exp(bm^\sigma L_m^\nu H_m^\eta)$$

for some $b > 0$, $0 < \sigma \le \kappa$, *and* $\nu, \eta \in \mathbb{R}$, *then the same eventual all-sampled-sign conclusion holds whenever*

$$\sigma < \kappa, \qquad \text{or} \qquad \sigma = \kappa,\ \nu < \lambda, \qquad \text{or} \qquad \sigma = \kappa,\ \nu = \lambda,\ \eta < \mu,$$

or

$$\sigma = \kappa, \qquad \nu = \lambda, \qquad \eta = \mu, \qquad b < Ac_0.$$

Finally, for any $0 < \rho \le 1$, *if* $\sum_m e^{-u_m} < \infty$ *and* $r_m = \lceil \rho M_m\rceil$, *then no upper bound on* M_m *is needed to get, eventually almost surely, a* $1-\rho$ *fraction of sampled signs which are logarithmically robust centers; only the displayed sampled-list maximum keeps its entropy factor* $\log(eM_m)$.

Proof. The matrix summability is the only new point. Since $\min\{n_m, \ell_m^2\} \le C_1' \ell_m^\theta$, the lower bound on d_m gives

$$p_{d_m}(a) \min\{n_m, \ell_m^2\} \le C m^{-\kappa q} L_m^{-\lambda q} H_m^{-\mu q}.$$

The displayed alternatives are exactly the integral-test conditions making this series summable. Also $d_m^{-A} \le \exp(-c m^\kappa L_m^\lambda H_m^\mu)$ is summable. Hence Corollary 5.55 applies once the two sign-tail summands are summable, proving the general sampled-list maximum and count assertions.

If $M_m \le d_m^\beta$, then

$$\binom{M_m}{1} d_m^{-A} \le d_m^{\beta - A},$$

which is summable for $\beta < A$, and $(\log(eM_m) + u_m)/\ell_m = O(1)$. If $M_m \le \exp(b m^\sigma L_m^\nu H_m^\eta)$, then

$$\binom{M_m}{1} d_m^{-A} \le \exp\{b m^\sigma L_m^\nu H_m^\eta - A c_0 m^\kappa L_m^\lambda H_m^\mu\}.$$

The lexicographic alternatives make this series summable, and give the same boundedness of $(\log(eM_m) + u_m)/\ell_m$. Finally, with $r_m = \lceil \rho M_m \rceil$,

$$\binom{M_m}{r_m} d_m^{-A r_m} \le \left(\frac{e}{\rho} d_m^{-A}\right)^{r_m}$$

for all large m. For every fixed $0 < \tau_0 < A$, this is eventually at most

$$d_m^{-(A - \tau_0)},$$

which is summable. The sampled-count theorem gives the claimed robust-center fraction. □

Corollary 5.72 (Sampled logarithmic-window sparse transfer)**.** *Fix $A > 0$, $\alpha > 0$, $a > 0$, $C_0 < \infty$, $C_1 < \infty$, $c_0 > 0$, $\kappa > 0$, and $\gamma \ge 0$. Suppose*

$$p_d(a) := \mathbb{P}\{\|T_d(\xi)\|_{\mathrm{op}} < a\, \mathbb{E}\, \|T_d(\xi)\|_{\mathrm{op}}\} \le C_0 \big(\log(e + d)\big)^{-B} \qquad (d \ge 1)$$

for some

$$B > \min\{\gamma, 2\} + \frac{1}{\kappa}.$$

Let $(d_m, n_m)_{m \ge 1}$ satisfy, with $\ell_m = \log(e + d_m)$,

$$d_m \ge \exp(c_0 m^\kappa), \qquad C_{A,K,a} \ell_m \le n_m \le \min\{d_m, C_1 \ell_m^\gamma\}.$$

For each m, sample M_m independent uniform sign vectors in $\{-1, 1\}^{n_m}$, independently of the matrices, with

$$M_m \ge \lceil \alpha \rceil.$$

Then, with probability one over the matrices and sampled signs, all but finitely many m have a sampled sign which is the center of a Hamming ball of radius

$$r_m = \lfloor c_{A,K,a} \ell_m \rfloor$$

on which every signing ε' satisfies

$$\max_{\omega^{d_m}=1}\left|\sum_{i=1}^{n_m}\varepsilon'_i\frac{P_{m,i}(\omega)}{\|B_{m,i}\|_{\mathrm{op}}}\right|\le C_{A,K,a}\sqrt{n_m}.$$

Thus the logarithmic-window threshold of Corollary 5.68 also gives actual sampled robust certificates; on stretched-exponential schedules no restriction beyond $\alpha>0$ is needed for the sampled-list confidence exponent.

Proof. Apply Corollary 5.62. It is enough to prove

$$\sum_m\left(d_m^{-A}+p_{d_m}(a)\min\{n_m,\ell_m^2\}+d_m^{-\alpha}\right)<\infty.$$

The first and third terms are summable because $d_m\ge\exp(c_0m^\kappa)$. The middle term is summable by the same estimate used in Corollary 5.68:

$$p_{d_m}(a)\min\{n_m,\ell_m^2\}\le C\,m^{\kappa(\min\{\gamma,2\}-B)},$$

and the exponent is smaller than -1 by the displayed hypothesis on B. The sampled Borel–Cantelli transfer gives the conclusion. □

Corollary 5.73 (Sparse logarithmic-window robust fractions)**.** *Fix $A>0$, $S>0$, $a>0$, $C_0<\infty$, $C_1<\infty$, $c_0>0$, $\kappa>0$, $\gamma\ge0$, and a fraction $0<\rho\le1$. Assume*

$$A\ge2\log_2(e/\rho),$$

and choose an integer $M\ge1$ so that

$$M\ge\frac{2(S+1)}{A\rho}.$$

Suppose

$$p_d(a):=\mathbb{P}\{\|T_d(\xi)\|_{\mathrm{op}}<a\,\mathbb{E}\,\|T_d(\xi)\|_{\mathrm{op}}\}\le C_0\big(\log(e+d)\big)^{-B}\qquad(d\ge1)$$

for some

$$B>\min\{\gamma,2\}+\frac{1}{\kappa}.$$

Let $(d_m,n_m)_{m\ge1}$ satisfy, with $\ell_m=\log(e+d_m)$,

$$d_m\ge\exp(c_0m^\kappa),\qquad C_{A,K,a}\ell_m\le n_m\le\min\{d_m,C_1\ell_m^\gamma\}.$$

At each m, take independent full Toeplitz samples at dimension d_m, and independently sample M uniform sign vectors in $\{-1,1\}^{n_m}$. Then, with probability one over the matrices and sampled signs, all but finitely many m satisfy both conclusions below.

First, the whole sampled list obeys

$$\max_{1\le s\le M}\max_{\omega^{d_m}=1}\left|\sum_{i=1}^{n_m}\delta_{m,i}^{(s)}\frac{P_{m,i}(\omega)}{\|B_{m,i}\|_{\mathrm{op}}}\right|\le C_{A,K,a,S,\rho}\sqrt{n_m}.$$

Second, at least a $1-\rho$ fraction of the M sampled signs are centers of Hamming balls of radius

$$\lfloor c_{A,K,a}\ell_m\rfloor$$

on which every signing ε' has edge maximum at most

$$C_{A,K,a}\sqrt{n_m}.$$

The same $1-\rho$ fraction also has edge maximum at most $C_{A,K,a}\sqrt{n_m}$. Thus, on stretched-exponential logarithmic-window schedules, the robust-fraction confidence exponent has no lower summability restriction beyond $S>0$.

Proof. Apply Corollary 5.61. It is enough to check

$$\sum_m \left(d_m^{-A} + p_{d_m}(a)\min\{n_m,\ell_m^2\} + d_m^{-S}\right) < \infty.$$

The stretched-exponential lower bound on d_m makes both $\sum_m d_m^{-A}$ and $\sum_m d_m^{-S}$ finite, for every positive A and S. The middle term is controlled by the same estimate used in Corollary 5.68:

$$p_{d_m}(a)\min\{n_m,\ell_m^2\} \le C\, m^{\kappa(\min\{\gamma,2\}-B)}.$$

The displayed hypothesis on B makes the exponent smaller than -1, so this series is finite. The general robust-fraction Borel theorem gives the eventual sampled-list maximum and the $1-\rho$ robust-center fraction. □

Corollary 5.74 (Sparse logarithmic-window sampled counts)**.** *Fix $A>0$, $a>0$, $C_0<\infty$, $C_1<\infty$, $c_0>0$, $\kappa>0$, and $\gamma\ge 0$. Suppose*

$$p_d(a) := \mathbb{P}\{\|T_d(\xi)\|_{\mathrm{op}} < a\,\mathbb{E}\,\|T_d(\xi)\|_{\mathrm{op}}\} \le C_0\big(\log(e+d)\big)^{-B} \qquad (d\ge 1)$$

for some

$$B > \min\{\gamma,2\} + \frac{1}{\kappa}.$$

Let $(d_m,n_m)_{m\ge1}$ satisfy, with $\ell_m=\log(e+d_m)$,

$$d_m \ge \exp(c_0 m^\kappa), \qquad C_{A,K,a}\ell_m \le n_m \le \min\{d_m, C_1\ell_m^\gamma\}.$$

At each m, take independent full Toeplitz samples at dimension d_m, and independently sample M_m uniform sign vectors in $\{-1,1\}^{n_m}$, with $M_m\ge 1$. Let $u_m\ge 0$ and $1\le r_m\le M_m$ be deterministic parameters. If

$$\sum_{m=1}^{\infty}\left(e^{-u_m} + \binom{M_m}{r_m} d_m^{-Ar_m}\right) < \infty,$$

then, with probability one over the matrices and sampled signs, all but finitely many m satisfy the sampled-list maximum bound, good-sign count, and robust-center count conclusions of Corollary 5.55.

In particular, let $s>1$, take $u_m = s\log(e+m)$ and $r_m=1$, and suppose

$$M_m \le d_m^{\beta} \qquad (m\ge 1)$$

for some $0\le\beta<A$. Then eventually every sampled sign in the batch is an $O(\sqrt{n_m})$-good robust center and the whole batch satisfies the displayed sampled-list maximum bound. More generally, if

$$M_m \le \exp(bm^\sigma)$$

for some $b > 0$, $0 < \sigma \le \kappa$, then the same eventual all-sampled-sign conclusion holds whenever either $\sigma < \kappa$, or $\sigma = \kappa$ and $b < Ac_0$.

Proof. Apply Corollary 5.55. The matrix summand $\sum_m d_m^{-A}$ is finite because $d_m \ge \exp(c_0 m^\kappa)$. The normalizer-tail summand is finite by the estimate from Corollary 5.68:

$$p_{d_m}(a)\min\{n_m, \ell_m^2\} \le C\, m^{\kappa(\min\{\gamma,2\}-B)},$$

and the exponent is less than -1. The two sign-tail summands are finite by the displayed hypothesis, so the sampled-count Borel theorem applies.

For the polynomial-in-d_m batch consequence,

$$\binom{M_m}{1} d_m^{-A} \le d_m^{\beta-A},$$

which is summable on the stretched-exponential schedule when $\beta < A$. Also

$$\frac{\log(eM_m) + u_m}{\ell_m} \le C_{s,\beta,c_0,\kappa}$$

for all large m, so the sampled-list maximum is $O(\sqrt{n_m})$. For the $\exp(bm^\sigma)$ batch consequence,

$$\binom{M_m}{1} d_m^{-A} \le \exp\left(bm^\sigma - Ac_0 m^\kappa\right),$$

which is summable if $\sigma < \kappa$, or if $\sigma = \kappa$ and $b < Ac_0$. The same comparison with ℓ_m gives the $O(\sqrt{n_m})$ sampled-list maximum. □

Corollary 5.75 (Sparse logarithmic-window growing robust fractions)**.** *Fix $A > 0$, $a > 0$, $C_0 < \infty$, $C_1 < \infty$, $c_0 > 0$, $\kappa > 0$, $\gamma \ge 0$, and $0 < \rho \le 1$. Suppose*

$$p_d(a) := \mathbb{P}\{\|T_d(\xi)\|_{\mathrm{op}} < a\,\mathbb{E}\,\|T_d(\xi)\|_{\mathrm{op}}\} \le C_0\big(\log(e+d)\big)^{-B} \qquad (d \ge 1)$$

for some

$$B > \min\{\gamma, 2\} + \frac{1}{\kappa}.$$

Let $(d_m, n_m)_{m\ge1}$ satisfy, with $\ell_m = \log(e + d_m)$,

$$d_m \ge \exp(c_0 m^\kappa), \qquad C_{A,K,a}\ell_m \le n_m \le \min\{d_m, C_1 \ell_m^\gamma\}.$$

At each m, let $B_{m,i} = T_{d_m}(\xi^{(m,i)})$, $1 \le i \le n_m$, be independent full Toeplitz matrices with independent mean-zero, unit-variance lag variables of sub-Gaussian norm at most K, and let $P_{m,i}$ be their symbols. Independently sample an arbitrary deterministic number $M_m \ge 1$ of uniform sign vectors

$$\delta_m^{(1)}, \dots, \delta_m^{(M_m)} \in \{-1, 1\}^{n_m}.$$

Let $u_m \ge 0$ be deterministic and assume

$$\sum_{m=1}^{\infty} e^{-u_m} < \infty.$$

Then, with probability one over the matrices and sampled signs, all but finitely many m satisfy both conclusions below.

First, the whole sampled list obeys

$$\max_{1\le s\le M_m}\max_{\omega^{d_m}=1}\left|\sum_{i=1}^{n_m}\delta_{m,i}^{(s)}\frac{P_{m,i}(\omega)}{\|B_{m,i}\|_{\mathrm{op}}}\right|\le C_{A,K,a}\sqrt{n_m\left(1+\frac{\log(eM_m)+u_m}{\ell_m}\right)}.$$

Second, at least a $1-\rho$ fraction of the M_m sampled signs have edge maximum at most $C_{A,K,a}\sqrt{n_m}$ and are centers of Hamming balls of radius

$$\lfloor c_{A,K,a}\ell_m\rfloor$$

on which every signing ε' has edge maximum at most

$$C_{A,K,a}\sqrt{n_m}.$$

No upper bound on M_m is required for this robust-fraction conclusion.

Proof. Apply Corollary 5.74 with

$$r_m=\lceil\rho M_m\rceil.$$

The only additional summability condition to check is the binomial count tail. Since $r_m\ge\rho M_m$,

$$\binom{M_m}{r_m}d_m^{-Ar_m}\le\left(\frac{eM_m}{r_m}\right)^{r_m}d_m^{-Ar_m}\le\left(\frac{e}{\rho}d_m^{-A}\right)^{r_m}.$$

For every fixed $0<\tau_0<A$ and all large m, the last display is at most

$$d_m^{-(A-\tau_0)r_m}\le d_m^{-(A-\tau_0)},$$

which is summable on the stretched-exponential schedule. The sampled-list tail is summable by the hypothesis on u_m. The sampled count theorem therefore gives the entropy-sensitive sampled-list maximum and at least

$$M_m-r_m+1\ge(1-\rho)M_m$$

robust centers eventually. □

Corollary 5.76 (Tunable bounded-density Chernoff transfer)**.** *Fix $A>0$, $a>0$, and constants $R\ge1$ and $0<\theta<R$. Write $\ell=\log(e+d)$. In the setting of Corollary 5.29, suppose*

$$p_d(a):=\mathbb{P}\{\|T_d(\xi)\|_{\mathrm{op}}<aN_d\}\le\frac{\theta}{\ell^2}.$$

Set

$$I(R,\theta):=R\log\frac{R}{\theta}-R+\theta>0.$$

Then there is a constant $C_{A,K,a,R}$ such that, whenever

$$n\le d,\qquad n\ge C_{A,K,a,R}\ell,$$

one has

$$\mathbb{P}\left\{\max_{\omega^d=1}\sum_{i=1}^{n}\frac{|P_i(\omega)|^2}{\|B_i\|_{\mathrm{op}}^2}>C_{A,K,a,R}\frac{n}{\ell}\right\}\le C_{A,K,a,R}d^{-A}+\exp\left(-I(R,\theta)\frac{n}{\ell^2}\right).$$

On the complement of this exceptional event, independent signs have the $C_{A,K,a,R}\sqrt{n}$ edge bound with conditional probability at least $1-d^{-A}$, sign-averaged edge maximum $C_{A,K,a,R}\sqrt{n}$, a deterministic signing certificate, and scalar edge-grid moments bounded by $(C_{A,K,a,R}k\,n/\ell)^k$ for every $2k$-th moment.

Proof. Let

$$M_a := \#\{1 \le i \le n : \|B_i\|_{\mathrm{op}} < aN_d\}.$$

Corollary 5.28 gives the full profile and signing package on $\{M_a \le Rn/\ell^2\}$, outside a $C_{A,K,a,R}d^{-A}$ event. It remains only to bound the bad-count excess. The random variable M_a is binomially dominated by a binomial variable with success probability $p_d(a)$. If $R/\ell^2 \ge 1$, then $\{M_a > Rn/\ell^2\}$ is empty. Otherwise $\theta/\ell^2 < R/\ell^2 < 1$, and monotonicity in the success probability plus Chernoff's bound for $X \sim \mathrm{Bin}(n, \theta/\ell^2)$ gives

$$\mathbb{P}\{M_a > Rn/\ell^2\} \le \mathbb{P}\{X > Rn/\ell^2\} \le \exp\left(-\frac{n}{\ell^2}\left(R\log\frac{R}{\theta} - R + \theta\right)\right).$$

Combining this with the empirical criterion proves the claim. □

Corollary 5.77 (Almost-sure Chernoff profile transfer)**.** *Fix $A > 0$, $a > 0$, and constants $R \ge 1$ and $0 < \theta < R$. There is a constant $C_{A,K,a,R}$ with the following property. Let $(d_m, n_m)_{m\ge1}$ be a triangular-array sequence with $n_m \le d_m$, put $\ell_m = \log(e + d_m)$, and assume*

$$n_m \ge C_{A,K,a,R}\ell_m \qquad (m \ge 1).$$

At each m, let $B_{m,i} = T_{d_m}(\xi^{(m,i)})$, $1 \le i \le n_m$, be independent full Toeplitz matrices whose lag variables are independent, mean zero, unit variance, and have sub-Gaussian norm at most K, and let $P_{m,i}$ be their symbols. Suppose

$$p_{d_m}(a) := \mathbb{P}\{\|T_{d_m}(\xi)\|_{\mathrm{op}} < a\,\mathbb{E}\,\|T_{d_m}(\xi)\|_{\mathrm{op}}\} \le \frac{\theta}{\ell_m^2} \qquad (m \ge 1),$$

and set $I(R,\theta) = R\log(R/\theta) - R + \theta$. If

$$\sum_{m=1}^{\infty}\left(d_m^{-A} + \exp\left(-I(R,\theta)\frac{n_m}{\ell_m^2}\right)\right) < \infty,$$

then, with probability one over the matrix triangular array, all but finitely many m satisfy

$$\max_{\omega^{d_m}=1}\sum_{i=1}^{n_m}\frac{|P_{m,i}(\omega)|^2}{\|B_{m,i}\|_{\mathrm{op}}^2} \le C_{A,K,a,R}\frac{n_m}{\ell_m}.$$

For those m, the independent-random-sign tail, sign-averaged edge maximum, deterministic signing certificate, and scalar edge-grid moment conclusions of Corollary 5.76 hold with d, n, ℓ replaced by d_m, n_m, ℓ_m.

In particular, if $L \geq 1$, $\eta > 0$, *and* $p_{d_m}(a) \leq L/\ell_m^2$, *then the same conclusion holds under*

$$\sum_{m=1}^{\infty} \left(d_m^{-A} + \exp\left(-Lh(\eta) \frac{n_m}{\ell_m^2} \right) \right) < \infty, \qquad h(\eta) = (1+\eta)\log(1+\eta) - \eta.$$

Consequently, along the full integer dimension sequence, if $A > 1$,

$$p_d(a) \leq \frac{L}{\log^2(e+d)} \qquad (d \geq 1),$$

and if $n(d) \leq d$ *while, for all sufficiently large* d,

$$n(d) \geq q \, \log^3(e+d) \qquad \text{for some } q > \frac{1}{Lh(\eta)},$$

then almost surely all sufficiently large integer dimensions satisfy the same edge-profile, signing, and scalar-moment package. For $\eta = 1$, *the threshold is* $q > 1/(L(2\log 2 - 1))$.

Proof. Apply Corollary 5.76 at each m, with the constant $C_{A,K,a,R}$ enlarged if necessary. The displayed summability condition makes the matrix exceptional probabilities summable. The first Borel–Cantelli lemma therefore gives an almost-sure event on which all but finitely many m satisfy the finite Chernoff profile and signing package.

The relative-excess version is obtained by taking $R = (1+\eta)L$ and $\theta = L$, so $I((1+\eta)L, L) = Lh(\eta)$. For the all-dimensions consequence take $d_m = m$. Then $\sum_m m^{-A} < \infty$, and for all large m,

$$\exp\left(-Lh(\eta) \frac{n(m)}{\log^2(e+m)} \right) \leq (e+m)^{-Lh(\eta)q},$$

which is summable because $Lh(\eta)q > 1$. The finitely many initial dimensions are irrelevant to the eventual almost-sure conclusion. □

Corollary 5.78 (Tunable Chernoff search-stability transfer)**.** *Fix* $A > 0$, $a > 0$, *and constants* $R \geq 1$ *and* $0 < \theta < R$. *Write* $\ell = \log(e+d)$. *In the setting of Corollary 5.76, suppose*

$$p_d(a) := \mathbb{P}\{\|T_d(\xi)\|_{\mathrm{op}} < aN_d\} \leq \frac{\theta}{\ell^2}.$$

Set $I(R,\theta) = R\log(R/\theta) - R + \theta$. *Then there is a constant* $C_{A,K,a,R}$ *such that, whenever*

$$n \leq d, \qquad n \geq C_{A,K,a,R}\ell,$$

outside a matrix event of probability at most

$$C_{A,K,a,R} d^{-A} + \exp\left(-I(R,\theta) \frac{n}{\ell^2} \right),$$

the full random-search and stable-center package of Corollary 5.30 holds, with $C_{A,K,a}$ *and* $c_{A,K,a}$ *replaced by constants depending on* (A, K, a, R).

In particular, if $p_d(a) \le L/\ell^2$ *with* $L \ge 1$*, then for every* $\eta > 0$ *the same package holds outside probability*

$$C_{A,K,a,L,\eta} d^{-A} + \exp\left(-Lh(\eta)\frac{n}{\ell^2}\right), \qquad h(\eta) = (1+\eta)\log(1+\eta) - \eta.$$

Taking $\eta = 1$*, there are constants* $C_{A,K,a,L}, c_L > 0$ *such that the package holds outside probability*

$$C_{A,K,a,L} d^{-A} + \exp\left(-c_L \frac{n}{\ell^2}\right).$$

Proof. Let $M_a = \#\{i : \|B_i\|_{\mathrm{op}} < aN_d\}$. On the event $\{M_a \le Rn/\ell^2\}$, Corollary 5.28 gives the edge profile

$$\max_{\omega^d=1} \sum_{i=1}^{n} \frac{|P_i(\omega)|^2}{\|B_i\|_{\mathrm{op}}^2} \le C_{A,K,a,R} \frac{n}{\ell}$$

outside a $C_{A,K,a,R} d^{-A}$ matrix event. Conditioning on any realization with this profile, Corollaries 5.6 and 5.10 give the sampled-list, success-count, Hamming-ball, and sampled robust-center statements exactly as in Corollary 5.30, with constants depending on R.

The same Chernoff calculation as in the proof of Corollary 5.76 gives

$$\mathbb{P}\{M_a > Rn/\ell^2\} \le \exp\left(-I(R,\theta)\frac{n}{\ell^2}\right).$$

This proves the first assertion. The relative-excess specialization is obtained by taking $R = (1+\eta)L$ and $\theta = L$, so $I((1+\eta)L, L) = Lh(\eta)$. The final display is the case $\eta = 1$, with $c_L = L(2\log 2 - 1)$, after absorbing constants. □

Corollary 5.79 (Chernoff joint sampled-search package)**.** *Fix* $A > 0$*,* $B > 0$*,* $a > 0$*, and constants* $R \ge 1$ *and* $0 < \theta < R$*. Write* $\ell = \log(e+d)$*. In the setting of Corollary 5.78, suppose*

$$p_d(a) := \mathbb{P}\{\|T_d(\xi)\|_{\mathrm{op}} < aN_d\} \le \frac{\theta}{\ell^2}.$$

Let $\delta^{(1)}, \dots, \delta^{(M)}$ *be independent uniform sign vectors, independent of the matrices, where* $M \ge 1$*. Put*

$$v_i(\omega) := \frac{P_i(\omega)}{\|B_i\|_{\mathrm{op}}}, \qquad \omega^d = 1,$$

and set $I(R,\theta) = R\log(R/\theta) - R + \theta$*. There are constants* $C_{A,B,K,a,R}$ *and* $c_{A,B,K,a,R} > 0$ *such that, whenever*

$$n \le d, \qquad n \ge C_{A,B,K,a,R}\ell,$$

the following joint matrix-and-sign estimates hold.

First, for every $u \geq 0$,

$$\mathbb{P}\Bigg\{ \max_{1\leq s\leq M} \max_{\omega^d=1} \left|\sum_{i=1}^n \delta_i^{(s)} v_i(\omega)\right| > C_{A,B,K,a,R}\sqrt{n\left(1+\frac{\log(eM)+u}{\ell}\right)}\Bigg\} \leq C_{A,B,K,a,R} d^{-A} + \exp\left(-I(R,\theta)\frac{n}{\ell^2}\right) + e^{-u}.$$

Second, let

$$G_M := \#\left\{1 \leq s \leq M : \max_{\omega^d=1} \left|\sum_{i=1}^n \delta_i^{(s)} v_i(\omega)\right| \leq C_{A,B,K,a,R}\sqrt{n}\right\}.$$

Then, for every integer $1 \leq r \leq M$,

$$\mathbb{P}\{G_M \leq M - r\} \leq C_{A,B,K,a,R} d^{-A} + \exp\left(-I(R,\theta)\frac{n}{\ell^2}\right) + \binom{M}{r} d^{-Br}.$$

Finally, call a sampled sign a robust center if it is the center of a Hamming ball of radius

$$r_0 = \lfloor c_{A,B,K,a,R}\ell \rfloor$$

on which every sign vector ε' satisfies

$$\max_{\omega^d=1} \left|\sum_{i=1}^n \varepsilon_i' v_i(\omega)\right| \leq C_{A,B,K,a,R}\sqrt{n}.$$

If H_M is the number of robust centers among $\delta^{(1)}, \dots, \delta^{(M)}$, then for every integer $1 \leq r \leq M$,

$$\mathbb{P}\{H_M \leq M - r\} \leq C_{A,B,K,a,R} d^{-A} + \exp\left(-I(R,\theta)\frac{n}{\ell^2}\right) + \binom{M}{r} d^{-Br}.$$

In particular, if $p_d(a) \leq L/\ell^2$ with $L \geq 1$, then for every $\eta > 0$ all three estimates remain true with the Chernoff term replaced by

$$\exp\left(-Lh(\eta)\frac{n}{\ell^2}\right), \qquad h(\eta) = (1+\eta)\log(1+\eta) - \eta,$$

after allowing the constants to depend on L and η.

Proof. On the good matrix event in the proof of Corollary 5.78,

$$\max_{\omega^d=1} \sum_{i=1}^n |v_i(\omega)|^2 \leq C_{A,K,a,R}\frac{n}{\ell}.$$

That good event fails with probability at most

$$C_{A,K,a,R} d^{-A} + \exp\left(-I(R,\theta)\frac{n}{\ell^2}\right).$$

Condition on any matrix realization satisfying the profile. The sampled-list maximum follows from Corollary 5.6, and the good-sign count follows from

the same corollary with exponent B. The robust-center count follows from Corollary 5.10, again with exponent B. Absorbing the profile constant into the displayed constants gives the three conditional sign estimates; adding the matrix exceptional probability gives the joint estimates. The relative-excess specialization is $R = (1+\eta)L$ and $\theta = L$. □

Corollary 5.80 (Chernoff sampled-list abundance)**.** *Fix $A > 0$, $B > 0$, $a > 0$, constants $R \geq 1$ and $0 < \theta < R$, and a fraction $0 < \rho \leq 1$. Write $\ell = \log(e+d)$. In the setting of Corollary 5.78, suppose*

$$p_d(a) := \mathbb{P}\{\|T_d(\xi)\|_{\mathrm{op}} < aN_d\} \leq \frac{\theta}{\ell^2}.$$

Let $\delta^{(1)}, \dots, \delta^{(M)}$ be independent uniform sign vectors, independent of the matrices, where $M \geq 1$, and put

$$r_\rho = \lceil \rho M \rceil, \qquad I(R,\theta) = R\log(R/\theta) - R + \theta.$$

There are constants $C_{A,B,K,a,R}$ and $c_{A,B,K,a,R} > 0$ such that, whenever

$$n \leq d, \qquad n \geq C_{A,B,K,a,R}\ell,$$

then, for every $u \geq 0$, with probability at least

$$1 - C_{A,B,K,a,R}\left[d^{-A} + \exp\left(-I(R,\theta)\frac{n}{\ell^2}\right)\right] - e^{-u} - 2\binom{M}{r_\rho} d^{-Br_\rho},$$

all three conclusions below hold simultaneously.

First, the whole sampled list obeys

$$\max_{1\leq s\leq M}\max_{\omega^d=1}\left|\sum_{i=1}^{n}\delta_i^{(s)}\frac{P_i(\omega)}{\|B_i\|_{\mathrm{op}}}\right| \leq C_{A,B,K,a,R}\sqrt{n\left(1+\frac{\log(eM)+u}{\ell}\right)}.$$

Second, at least $M - r_\rho + 1$ sampled signs have edge maximum at most $C_{A,B,K,a,R}\sqrt{n}$. Third, at least $M - r_\rho + 1$ sampled signs are centers of Hamming balls of radius

$$\lfloor c_{A,B,K,a,R}\ell \rfloor$$

on which every signing ε' has edge maximum at most $C_{A,B,K,a,R}\sqrt{n}$. In particular, at least a $1-\rho$ fraction of the sampled signs are good, and at least a $1-\rho$ fraction are robust centers.

If $p_d(a) \leq L/\ell^2$ with $L \geq 1$, then for every $\eta > 0$ the same conclusion holds with the Chernoff term replaced by

$$\exp\left(-Lh(\eta)\frac{n}{\ell^2}\right), \qquad h(\eta) = (1+\eta)\log(1+\eta) - \eta,$$

after allowing the constants to depend on L and η.

Proof. Apply Corollary 5.79 with

$$(M, r) = (M, r_\rho).$$

The sampled-list maximum fails with probability at most the matrix exception plus e^{-u}. The good-sign and robust-center count conclusions each fail with

probability at most the same matrix exception plus $\binom{M}{r_\rho} d^{-Br_\rho}$. A union bound gives the displayed simultaneous probability after absorbing the three copies of the matrix exception into $C_{A,B,K,a,R}$. Since

$$M - r_\rho + 1 \geq M - \rho M,$$

the count lower bounds imply the stated $1-\rho$ fraction conclusions. The relative-excess form again takes $R = (1+\eta)L$ and $\theta = L$. □

Corollary 5.81 (Chernoff polynomial-confidence robust fractions)**.** *Fix $A > 0$, $S > 0$, $a > 0$, constants $R \geq 1$ and $0 < \theta < R$, and a fraction $0 < \rho \leq 1$. Choose $B > 0$ and an integer $M \geq 1$ so that*

$$B \geq 2\log_2(e/\rho), \qquad M \geq \frac{2(S+1)}{B\rho}.$$

Write $\ell = \log(e+d)$. In the setting of Corollary 5.78, suppose $d \geq 2$ and

$$p_d(a) := \mathbb{P}\{\|T_d(\xi)\|_{\mathrm{op}} < aN_d\} \leq \frac{\theta}{\ell^2}.$$

Let $\delta^{(1)}, \dots, \delta^{(M)}$ be independent uniform sign vectors, independent of the matrices. Set $I(R,\theta) = R\log(R/\theta) - R + \theta$. There are constants $C_{A,B,K,a,R}$ and $c_{A,B,K,a,R} > 0$ such that, whenever

$$n \leq d, \qquad n \geq C_{A,B,K,a,R}\ell,$$

then, for every $u \geq 0$, with probability at least

$$1 - C_{A,B,K,a,R}\left[d^{-A} + \exp\left(-I(R,\theta)\frac{n}{\ell^2}\right)\right] - e^{-u} - d^{-S},$$

the whole sampled list obeys the entropy-sensitive maximum bound from Corollary 5.80, and at least a $1-\rho$ fraction of the sampled signs are centers of radius

$$\lfloor c_{A,B,K,a,R}\ell \rfloor$$

Hamming balls on which every signing ε' has edge maximum at most

$$C_{A,B,K,a,R}\sqrt{n}.$$

In particular, taking $u = S\ell$, the same event has probability at least

$$1 - C_{A,B,K,a,R}\left[d^{-A} + \exp\left(-I(R,\theta)\frac{n}{\ell^2}\right)\right] - 2d^{-S},$$

and the whole sampled list has edge maximum at most $C_{A,B,K,a,R,S,\rho}\sqrt{n}$.

If $p_d(a) \leq L/\ell^2$ with $L \geq 1$, then for every $\eta > 0$ the same conclusions hold with the Chernoff term replaced by

$$\exp\left(-Lh(\eta)\frac{n}{\ell^2}\right), \qquad h(\eta) = (1+\eta)\log(1+\eta) - \eta,$$

after allowing the constants to depend on L and η.

Proof. Let $r_\rho = \lceil \rho M \rceil$. The only point not already contained in Corollary 5.80 is the simplification of the binomial fraction tail. Since $r_\rho \geq \rho M$,

$$\binom{M}{r_\rho} \leq \left(\frac{eM}{r_\rho}\right)^{r_\rho} \leq (e/\rho)^{r_\rho}.$$

For $d \geq 2$ and $B \geq 2\log_2(e/\rho)$, this gives

$$\binom{M}{r_\rho} d^{-Br_\rho} \leq d^{-Br_\rho/2}.$$

The lower bound on M implies $Br_\rho/2 \geq S+1$, hence

$$2\binom{M}{r_\rho} d^{-Br_\rho} \leq 2d^{-(S+1)} \leq d^{-S}.$$

Substituting this into Corollary 5.80 proves the first assertion. If $u = S\ell$, then $e^{-u} = (e+d)^{-S} \leq d^{-S}$, while

$$1 + \frac{\log(eM) + u}{\ell} \leq C_{S,\rho,B}$$

because M is fixed once (S, ρ, B) are fixed. This gives the constant-scale sampled-list bound. The relative-excess statement is the same specialization $R = (1+\eta)L$, $\theta = L$. □

Corollary 5.82 (Chernoff polynomial-batch robust search)**.** *Fix $A > 0$, $a > 0$, constants $R \geq 1$ and $0 < \theta < R$, and parameters $\beta \geq 0$ and $\sigma > 0$. Write $\ell = \log(e+d)$. In the setting of Corollary 5.78, suppose*

$$p_d(a) := \mathbb{P}\{\|T_d(\xi)\|_{\mathrm{op}} < aN_d\} \leq \frac{\theta}{\ell^2}.$$

Let $\delta^{(1)}, \ldots, \delta^{(M)}$ be independent uniform sign vectors, independent of the matrices, with

$$1 \leq M \leq d^\beta.$$

Set $I(R,\theta) = R\log(R/\theta) - R + \theta$. There are constants $C_{A,K,a,R,\beta,\sigma}$ and $c_{A,K,a,R,\beta,\sigma} > 0$ such that, whenever

$$n \leq d, \qquad n \geq C_{A,K,a,R,\beta,\sigma}\ell,$$

with probability at least

$$1 - C_{A,K,a,R,\beta,\sigma}d^{-A} - \exp\left(-I(R,\theta)\frac{n}{\ell^2}\right) - d^{-\sigma},$$

every sampled sign $\delta^{(s)}$, $1 \leq s \leq M$, is the center of a Hamming ball of radius

$$r_0 = \lfloor c_{A,K,a,R,\beta,\sigma}\ell \rfloor$$

on which every sign vector ε' satisfies

$$\max_{\omega^d=1} \left| \sum_{i=1}^n \varepsilon'_i \frac{P_i(\omega)}{\|B_i\|_{\mathrm{op}}} \right| \leq C_{A,K,a,R,\beta,\sigma}\sqrt{n}.$$

In particular, the whole sampled batch itself has edge maximum at most $C_{A,K,a,R,\beta,\sigma}\sqrt{n}$.

If $p_d(a) \le L/\ell^2$ *with* $L \ge 1$*, then for every* $\eta > 0$ *the same conclusion holds with the displayed exceptional probability replaced by*

$$C_{A,K,a,L,\eta,\beta,\sigma} d^{-A} + \exp\left(-Lh(\eta)\frac{n}{\ell^2}\right) + d^{-\sigma}, \qquad h(\eta) = (1+\eta)\log(1+\eta) - \eta.$$

Proof. Apply Corollary 5.79 with sign-count exponent

$$B = \beta + \sigma$$

and with $r = 1$ in the robust-center count. The probability that at least one of the M sampled signs is not a robust center is at most

$$C_{A,B,K,a,R} d^{-A} + \exp\left(-I(R,\theta)\frac{n}{\ell^2}\right) + Md^{-B}.$$

Since $M \le d^{\beta}$ and $B = \beta + \sigma$, the last term is at most $d^{-\sigma}$. Relabeling the constants gives the first assertion. Since the center itself is contained in its Hamming ball, the robust-center conclusion also gives the displayed bound for every sampled sign. The relative-excess version is again the specialization $R = (1+\eta)L$ and $\theta = L$. □

Corollary 5.83 (Chernoff polynomial-confidence robust search)**.** *Fix* $A > 0$, $B > 0$, $a > 0$*, and constants* $R \ge 1$ *and* $0 < \theta < R$*. Write* $\ell = \log(e + d)$*. In the setting of Corollary 5.78, suppose*

$$p_d(a) := \mathbb{P}\{\|T_d(\xi)\|_{\mathrm{op}} < aN_d\} \le \frac{\theta}{\ell^2}.$$

Let $\delta^{(1)}, \dots, \delta^{(M)}$ *be independent uniform sign vectors, independent of the matrices, with* $M \ge \lceil B \rceil$*. Set*

$$I(R,\theta) = R\log(R/\theta) - R + \theta.$$

There are constants $C_{A,K,a,R}$ *and* $c_{A,K,a,R} > 0$ *such that, whenever*

$$n \le d, \qquad n \ge C_{A,K,a,R}\ell,$$

the joint matrix-and-sign probability that no sampled sign is a robust certificate is at most

$$C_{A,K,a,R} d^{-A} + \exp\left(-I(R,\theta)\frac{n}{\ell^2}\right) + d^{-B}.$$

Equivalently, outside that exceptional probability, some sampled $\delta^{(s)}$ *is the center of a Hamming ball of radius*

$$r_0 = \lfloor c_{A,K,a,R}\ell \rfloor$$

on which every sign vector ε' *satisfies*

$$\max_{\omega^d=1} \left| \sum_{i=1}^{n} \varepsilon'_i \frac{P_i(\omega)}{\|B_i\|_{\mathrm{op}}} \right| \le C_{A,K,a,R}\sqrt{n}.$$

In particular, if $p_d(a) \le L/\ell^2$ with $L \ge 1$, then for every $\eta > 0$ the same conclusion holds with the displayed exceptional probability replaced by

$$C_{A,K,a,L,\eta} d^{-A} + \exp\left(-Lh(\eta)\frac{n}{\ell^2}\right) + d^{-B}, \qquad h(\eta) = (1+\eta)\log(1+\eta) - \eta.$$

Proof. On the good matrix event in the proof of Corollary 5.78,

$$\max_{\omega^d=1} \sum_{i=1}^{n} \frac{|P_i(\omega)|^2}{\|B_i\|_{\mathrm{op}}^2} \le C_{A,K,a,R}\frac{n}{\ell}.$$

That good event fails with probability at most

$$C_{A,K,a,R} d^{-A} + \exp\left(-I(R,\theta)\frac{n}{\ell^2}\right).$$

Condition on any matrix realization satisfying the profile. Applying Corollary 5.11, and absorbing the profile constant into $C_{A,K,a,R}$ and $c_{A,K,a,R}$, shows that the sampled signs fail to contain a robust Hamming-ball center with conditional probability at most d^{-B}. Adding the matrix exceptional probability proves the first claim. The relative-excess specialization is $R = (1+\eta)L$ and $\theta = L$, so $I((1+\eta)L, L) = Lh(\eta)$. □

Corollary 5.84 (Almost-sure Chernoff search-stability transfer)**.** *Fix $A > 0$, $a > 0$, and constants $R \ge 1$ and $0 < \theta < R$. There is a constant $C_{A,K,a,R}$ with the following property. Let $(d_m, n_m)_{m\ge1}$ be a triangular-array sequence with $n_m \le d_m$, put $\ell_m = \log(e + d_m)$, and assume*

$$n_m \ge C_{A,K,a,R}\ell_m \qquad (m \ge 1).$$

At each m, let $B_{m,i} = T_{d_m}(\xi^{(m,i)})$, $1 \le i \le n_m$, be independent full Toeplitz matrices with independent mean-zero, unit-variance lag variables of sub-Gaussian norm at most K, and let $P_{m,i}$ be their symbols. Suppose

$$p_{d_m}(a) := \mathbb{P}\{\|T_{d_m}(\xi)\|_{\mathrm{op}} < a\,\mathbb{E}\,\|T_{d_m}(\xi)\|_{\mathrm{op}}\} \le \frac{\theta}{\ell_m^2} \qquad (m \ge 1),$$

and set $I(R,\theta) = R\log(R/\theta) - R + \theta$. If

$$\sum_{m=1}^{\infty}\left(d_m^{-A} + \exp\left(-I(R,\theta)\frac{n_m}{\ell_m^2}\right)\right) < \infty,$$

then, with probability one over the matrix triangular array, all but finitely many m satisfy the full random-search and stable-center package of Corollary 5.30, with d, n, ℓ replaced by d_m, n_m, ℓ_m and constants depending on (A, K, a, R).

In particular, if $L \ge 1$, $\eta > 0$, and $p_{d_m}(a) \le L/\ell_m^2$, then the same conclusion holds under

$$\sum_{m=1}^{\infty}\left(d_m^{-A} + \exp\left(-Lh(\eta)\frac{n_m}{\ell_m^2}\right)\right) < \infty, \qquad h(\eta) = (1+\eta)\log(1+\eta) - \eta.$$

Consequently, along the full integer dimension sequence, if $A > 1$,

$$p_d(a) \le \frac{L}{\log^2(e+d)} \qquad (d \ge 1),$$

and if $n(d) \le d$ *while, for all sufficiently large* d,

$$n(d) \ge q \, \log^3(e+d) \qquad \text{for some } q > \frac{1}{Lh(\eta)},$$

then almost surely all sufficiently large integer dimensions satisfy the same random-search and stable-center package. For $\eta = 1$, *the threshold is* $q > 1/(L(2\log 2 - 1))$.

Proof. Apply Corollary 5.78 at each m, with the constant $C_{A,K,a,R}$ enlarged if necessary. The displayed summability condition makes the matrix exceptional probabilities summable, so the first Borel–Cantelli lemma gives an almost-sure event on which all but finitely many matrix realizations satisfy the finite Chernoff search package. Substituting d, n, ℓ by d_m, n_m, ℓ_m gives the first assertion.

The relative-excess version is the specialization $R = (1+\eta)L$ and $\theta = L$, exactly as in Corollary 5.78. For the full-dimension consequence take $d_m = m$. Then $\sum_m m^{-A} < \infty$ because $A > 1$, and the sample-size assumption gives

$$\exp\left(-Lh(\eta)\frac{n(m)}{\log^2(e+m)}\right) \le (e+m)^{-Lh(\eta)q}$$

for all large m, which is summable because $Lh(\eta)q > 1$. Finite initial dimensions do not affect the almost-sure eventual conclusion. □

Corollary 5.85 (Almost-sure sampled Chernoff robust certificates)**.** *Fix* $A > 0$, $\alpha > 0$, $a > 0$, *and constants* $R \ge 1$ *and* $0 < \theta < R$. *There are constants* $C_{A,K,a,R}$ *and* $c_{A,K,a,R} > 0$ *with the following property. Let* $(d_m, n_m)_{m\ge1}$ *be a triangular-array sequence with* $n_m \le d_m$, *put* $\ell_m = \log(e+d_m)$, *and assume*

$$n_m \ge C_{A,K,a,R}\ell_m \qquad (m \ge 1).$$

At each m, *let* $B_{m,i} = T_{d_m}(\xi^{(m,i)})$, $1 \le i \le n_m$, *be independent full Toeplitz matrices with independent mean-zero, unit-variance lag variables of sub-Gaussian norm at most* K, *and let* $P_{m,i}$ *be their symbols. Independently of the matrices, sample independent uniform sign vectors*

$$\delta_m^{(1)}, \dots, \delta_m^{(M_m)} \in \{-1,1\}^{n_m}, \qquad M_m \ge \lceil \alpha \rceil.$$

Suppose

$$p_{d_m}(a) := \mathbb{P}\{\|T_{d_m}(\xi)\|_{\mathrm{op}} < a\,\mathbb{E}\,\|T_{d_m}(\xi)\|_{\mathrm{op}}\} \le \frac{\theta}{\ell_m^2} \qquad (m \ge 1),$$

and set $I(R,\theta) = R\log(R/\theta) - R + \theta$. *If*

$$\sum_{m=1}^{\infty}\left(d_m^{-A} + \exp\left(-I(R,\theta)\frac{n_m}{\ell_m^2}\right) + d_m^{-\alpha}\right) < \infty,$$

then, with probability one over the matrices and sampled signs, all but finitely many m have some sampled sign $\delta_m^{(s_m)}$ which is the center of a Hamming ball of radius

$$r_m = \lfloor c_{A,K,a,R}\ell_m \rfloor$$

on which every signing ε' satisfies

$$\max_{\omega^{d_m}=1} \left| \sum_{i=1}^{n_m} \varepsilon'_i \frac{P_{m,i}(\omega)}{\|B_{m,i}\|_{\mathrm{op}}} \right| \le C_{A,K,a,R}\sqrt{n_m}.$$

In particular, if $L \ge 1$, $\eta > 0$, and $p_{d_m}(a) \le L/\ell_m^2$, then the same conclusion holds under

$$\sum_{m=1}^{\infty} \left(d_m^{-A} + \exp\left(-Lh(\eta)\frac{n_m}{\ell_m^2} \right) + d_m^{-\alpha} \right) < \infty, \qquad h(\eta) = (1+\eta)\log(1+\eta) - \eta.$$

Consequently, along the full integer dimension sequence, if $A > 1$, $\alpha > 1$,

$$p_d(a) \le \frac{L}{\log^2(e+d)} \qquad (d \ge 1),$$

and if $n(d) \le d$ while, for all sufficiently large d,

$$n(d) \ge q\,\log^3(e+d) \qquad \text{for some } q > \frac{1}{Lh(\eta)},$$

then $M_d \ge \lceil \alpha \rceil$ sampled signs per dimension eventually contain such a robust certificate almost surely. For $\eta = 1$, the threshold is $q > 1/(L(2\log 2 - 1))$.

Proof. For each m, apply Corollary 5.78. Outside a matrix event of probability at most

$$C_{A,K,a,R} d_m^{-A} + \exp\left(-I(R,\theta)\frac{n_m}{\ell_m^2} \right),$$

its proof gives the edge profile

$$\max_{\omega^{d_m}=1} \sum_{i=1}^{n_m} \frac{|P_{m,i}(\omega)|^2}{\|B_{m,i}\|_{\mathrm{op}}^2} \le C_{A,K,a,R}\frac{n_m}{\ell_m}.$$

Conditioned on any matrix realization with this profile, Corollary 5.11, with $B = \alpha$, says that the sampled signs fail to contain a robust Hamming-ball center with probability at most $d_m^{-\alpha}$, after absorbing the profile constant into $C_{A,K,a,R}$ and $c_{A,K,a,R}$. Hence the joint matrix-and-sign failure probability at dimension m is bounded by the m-th summand in the first displayed series, up to constants. Borel–Cantelli gives the eventual sampled-certificate conclusion.

The relative-excess specialization is $R = (1+\eta)L$ and $\theta = L$, so $I((1+\eta)L, L) = Lh(\eta)$. For the full-dimension consequence take $d_m = m$. Then $\sum_m m^{-A} < \infty$ and $\sum_m m^{-\alpha} < \infty$, while the sample-size lower bound gives

$$\exp\left(-Lh(\eta)\frac{n(m)}{\log^2(e+m)} \right) \le (e+m)^{-Lh(\eta)q}$$

for all large m, which is summable because $Lh(\eta)q > 1$. The finitely many initial dimensions are irrelevant. □

Corollary 5.86 (Almost-sure Chernoff sampled-search counts)**.** *Fix $A > 0$, $B > 0$, $a > 0$, and constants $R \geq 1$ and $0 < \theta < R$. There are constants $C_{A,B,K,a,R}$ and $c_{A,B,K,a,R} > 0$ with the following property. Let $(d_m, n_m)_{m\geq 1}$ be a triangular-array sequence with $n_m \leq d_m$, put $\ell_m = \log(e + d_m)$, and assume*

$$n_m \geq C_{A,B,K,a,R}\ell_m \qquad (m \geq 1).$$

At each m, let $B_{m,i} = T_{d_m}(\xi^{(m,i)})$, $1 \leq i \leq n_m$, be independent full Toeplitz matrices with independent mean-zero, unit-variance lag variables of sub-Gaussian norm at most K, and let $P_{m,i}$ be their symbols. Independently of the matrices, sample independent uniform sign vectors

$$\delta_m^{(1)}, \ldots, \delta_m^{(M_m)} \in \{-1, 1\}^{n_m}, \qquad M_m \geq 1.$$

Let $u_m \geq 0$ and $1 \leq r_m \leq M_m$ be deterministic parameters. Suppose

$$p_{d_m}(a) := \mathbb{P}\{\|T_{d_m}(\xi)\|_{\mathrm{op}} < a\,\mathbb{E}\,\|T_{d_m}(\xi)\|_{\mathrm{op}}\} \leq \frac{\theta}{\ell_m^2} \qquad (m \geq 1),$$

set $I(R, \theta) = R\log(R/\theta) - R + \theta$, and assume

$$\sum_{m=1}^{\infty}\left[d_m^{-A} + \exp\left(-I(R,\theta)\frac{n_m}{\ell_m^2}\right) + e^{-u_m} + \binom{M_m}{r_m} d_m^{-Br_m}\right] < \infty.$$

Then, with probability one over the matrices and sampled signs, all but finitely many m satisfy all of the following.

First, the entire sampled list obeys

$$\max_{1\leq s\leq M_m}\max_{\omega^{d_m}=1}\left|\sum_{i=1}^{n_m}\delta_{m,i}^{(s)}\frac{P_{m,i}(\omega)}{\|B_{m,i}\|_{\mathrm{op}}}\right| \leq C_{A,B,K,a,R}\sqrt{n_m\left(1 + \frac{\log(eM_m) + u_m}{\ell_m}\right)}.$$

Second, at least $M_m - r_m + 1$ of the sampled signs have edge maximum at most $C_{A,B,K,a,R}\sqrt{n_m}$. Third, at least $M_m - r_m + 1$ sampled signs are centers of Hamming balls of radius

$$\lfloor c_{A,B,K,a,R}\ell_m \rfloor$$

on which every signing ε' has edge maximum at most $C_{A,B,K,a,R}\sqrt{n_m}$.

In particular, if $L \geq 1$, $\eta > 0$, and $p_{d_m}(a) \leq L/\ell_m^2$, then the same conclusion holds with the Chernoff summand replaced by

$$\exp\left(-Lh(\eta)\frac{n_m}{\ell_m^2}\right), \qquad h(\eta) = (1+\eta)\log(1+\eta) - \eta.$$

Consequently, along the full integer dimension sequence, suppose $A > 1$, $s > 1$, $L \geq 1$, $\eta > 0$, and $p_d(a) \leq L/\log^2(e + d)$. If

$$n(d) \geq q\,\log^3(e + d) \quad \text{for all large } d, \qquad q > \frac{1}{Lh(\eta)},$$

and if a batch size satisfies $M_d \leq d^\beta$ for some $\beta \geq 0$, then for every $B > \beta+1$, taking $r_d = 1$ and $u_d = s\log(e + d)$ gives an almost-sure event on which,

for all sufficiently large integer dimensions, every sampled sign in the batch is a robust center and the whole batch satisfies the displayed sampled-list maximum bound.

Proof. Apply Corollary 5.79 at each m. In that application use

$$(d, n, \ell, M, u, r) = (d_m, n_m, \ell_m, M_m, u_m, r_m).$$

The failure probabilities for the sampled-list maximum, the good-sign count, and the robust-center count are bounded by a constant times the summand in the displayed series, after absorbing the two identical binomial count terms into the constant. The first Borel–Cantelli lemma therefore gives an almost-sure event on which all three conclusions hold for all but finitely many m.

The relative-excess form is obtained from the same specialization $R = (1+\eta)L$, $\theta = L$, so $I((1+\eta)L, L) = Lh(\eta)$. For the full-dimension consequence take $d_m = m$, $u_m = s\log(e+m)$, and $r_m = 1$. Then $\sum_m m^{-A} < \infty$, $\sum_m e^{-u_m} < \infty$, and

$$\binom{M_m}{1} m^{-B} \le m^{\beta - B},$$

which is summable when $B > \beta + 1$. The lower bound on $n(m)$ gives

$$\exp\left(-Lh(\eta)\frac{n(m)}{\log^2(e+m)}\right) \le (e+m)^{-Lh(\eta)q}$$

for all large m, and this is summable by the assumption on q. Thus the general Borel statement applies. With $r_m = 1$, the bounds $G_{M_m} \ge M_m$ and $H_{M_m} \ge M_m$ say that every sampled sign in the batch is good and robust. □

Corollary 5.87 (All-integer Chernoff vanishing robust fractions)**.** *Fix $A > 1$, $B > 1$, $a > 0$, $L \ge 1$, $\eta > 0$, and $0 < \tau_0 < B - 1$, and put*

$$h(\eta) = (1+\eta)\log(1+\eta) - \eta.$$

There are constants $C_{A,B,K,a,L,\eta}$ and $c_{A,B,K,a,L,\eta} > 0$ with the following property. Suppose

$$p_d(a) := \mathbb{P}\{\|T_d(\xi)\|_{\mathrm{op}} < a\,\mathbb{E}\,\|T_d(\xi)\|_{\mathrm{op}}\} \le \frac{L}{\log^2(e+d)} \qquad (d \ge 1).$$

Let $n(d) \le d$ and assume that, for all sufficiently large d,

$$n(d) \ge q\,\log^3(e+d) \qquad \text{for some } q > \frac{1}{Lh(\eta)}.$$

At dimension d, let $B_{d,i} = T_d(\xi^{(d,i)})$, $1 \le i \le n(d)$, be independent full Toeplitz matrices with independent mean-zero, unit-variance lag variables of sub-Gaussian norm at most K, and let $P_{d,i}$ be their symbols. Independently

of the matrices, sample an arbitrary deterministic number $M_d \geq 1$ of uniform sign vectors in $\{-1,1\}^{n(d)}$. Let $u_d \geq 0$ satisfy

$$\sum_{d=1}^{\infty} e^{-u_d} < \infty,$$

and let $(\rho_d)_{d\geq 1} \subset (0,1]$ satisfy

$$\rho_d \geq e\, d^{-\tau_0}$$

for all sufficiently large d. Then, with probability one over the matrices and sampled signs, all sufficiently large integer dimensions satisfy both conclusions below.

First, the whole sampled list obeys

$$\max_{1\leq s\leq M_d} \max_{\omega^d=1} \left| \sum_{i=1}^{n(d)} \delta_{d,i}^{(s)} \frac{P_{d,i}(\omega)}{\|B_{d,i}\|_{\mathrm{op}}} \right| \leq C_{A,B,K,a,L,\eta} \sqrt{n(d)\left(1+\frac{\log(eM_d)+u_d}{\log(e+d)}\right)}.$$

Second, at least a $1-\rho_d$ fraction of the M_d sampled signs have edge maximum at most $C_{A,B,K,a,L,\eta}\sqrt{n(d)}$ and are centers of Hamming balls of radius

$$\lfloor c_{A,B,K,a,L,\eta} \log(e+d) \rfloor$$

on which every signing ε' has edge maximum at most

$$C_{A,B,K,a,L,\eta}\sqrt{n(d)}.$$

No upper bound on M_d is required. In particular, if $\rho_d \to 0$, then the robust-center fraction tends to one along all integer dimensions.

Proof. Apply Corollary 5.86 with

$$d_m = m, \qquad n_m = n(m), \qquad R = (1+\eta)L, \qquad \theta = L, \qquad r_m = \lceil \rho_m M_m \rceil.$$

The eventual lower bound $n(m) \geq q\log^3(e+m)$ implies $n_m \geq C_{A,B,K,a,L,\eta}\ell_m$ after discarding finitely many dimensions. The matrix summands are summable because $A > 1$ and

$$\exp\left(-Lh(\eta)\frac{n(m)}{\log^2(e+m)}\right) \leq (e+m)^{-Lh(\eta)q}$$

for all large m, with $Lh(\eta)q > 1$. The sampled-list tail is summable by the hypothesis on u_m.

For the binomial count tail,

$$\binom{M_m}{r_m} \leq \left(\frac{e}{\rho_m}\right)^{r_m}.$$

The lower bound $\rho_m \geq e\, m^{-\tau_0}$ gives, for all large m,

$$\binom{M_m}{r_m} m^{-Br_m} \leq m^{-(B-\tau_0)r_m} \leq m^{-(B-\tau_0)},$$

which is summable because $B-\tau_0>1$. The sampled-count theorem therefore gives the sampled-list maximum and at least M_m-r_m+1 good robust centers eventually. Since

$$M_m-r_m+1\ge M_m-\rho_m M_m,$$

this is at least a $1-\rho_m$ fraction. □

Corollary 5.88 (Almost-sure Chernoff robust fractions)**.** *Fix $A>0$, $S>0$, $a>0$, constants $R\ge 1$ and $0<\theta<R$, and a fraction $0<\rho\le 1$. Choose $B>0$ and an integer $M\ge 1$ so that*

$$B\ge 2\log_2(e/\rho),\qquad M\ge\frac{2(S+1)}{B\rho}.$$

There are constants $C_{A,B,K,a,R}$ and $c_{A,B,K,a,R}>0$ with the following property. Let $(d_m,n_m)_{m\ge1}$ be a triangular-array sequence with

$$d_m\ge 2,\qquad n_m\le d_m,\qquad \ell_m=\log(e+d_m),$$

and assume

$$n_m\ge C_{A,B,K,a,R}\ell_m\qquad(m\ge1).$$

At each m, let $B_{m,i}=T_{d_m}(\xi^{(m,i)})$, $1\le i\le n_m$, be independent full Toeplitz matrices with independent mean-zero, unit-variance lag variables of sub-Gaussian norm at most K, and let $P_{m,i}$ be their symbols. Independently of the matrices, sample independent uniform sign vectors

$$\delta_m^{(1)},\dots,\delta_m^{(M)}\in\{-1,1\}^{n_m}.$$

Suppose

$$p_{d_m}(a):=\mathbb{P}\{\|T_{d_m}(\xi)\|_{\mathrm{op}}<a\,\mathbb{E}\,\|T_{d_m}(\xi)\|_{\mathrm{op}}\}\le\frac{\theta}{\ell_m^2}\qquad(m\ge1),$$

set $I(R,\theta)=R\log(R/\theta)-R+\theta$, and assume

$$\sum_{m=1}^{\infty}\left[d_m^{-A}+\exp\left(-I(R,\theta)\frac{n_m}{\ell_m^2}\right)+d_m^{-S}\right]<\infty.$$

Then, with probability one over the matrices and sampled signs, all but finitely many m satisfy both conclusions below.

First, the whole sampled list obeys

$$\max_{1\le s\le M}\max_{\omega^{d_m}=1}\left|\sum_{i=1}^{n_m}\delta_{m,i}^{(s)}\frac{P_{m,i}(\omega)}{\|B_{m,i}\|_{\mathrm{op}}}\right|\le C_{A,B,K,a,R,S,\rho}\sqrt{n_m}.$$

Second, at least a $1-\rho$ fraction of the M sampled signs are centers of Hamming balls of radius

$$\lfloor c_{A,B,K,a,R}\ell_m\rfloor$$

on which every signing ε' has edge maximum at most

$$C_{A,B,K,a,R}\sqrt{n_m}.$$

The same $1-\rho$ fraction also has edge maximum at most $C_{A,B,K,a,R}\sqrt{n_m}$.

In particular, if $L \geq 1$, $\eta > 0$, and $p_{d_m}(a) \leq L/\ell_m^2$, then the same conclusion holds with the Chernoff summand replaced by

$$\exp\left(-Lh(\eta)\frac{n_m}{\ell_m^2}\right), \qquad h(\eta) = (1+\eta)\log(1+\eta) - \eta.$$

Consequently, along the full integer dimension sequence, suppose $A > 1$, $S > 1$, $L \geq 1$, $\eta > 0$, and $p_d(a) \leq L/\log^2(e+d)$. If

$$n(d) \geq q\,\log^3(e+d) \quad \textit{for all large } d, \qquad q > \frac{1}{Lh(\eta)},$$

then these $M = O_{\rho,S,B}(1)$ sampled signs per dimension eventually have a $1-\rho$ fraction of robust centers almost surely.

Proof. Let $r_\rho = \lceil \rho M \rceil$ and $u_m = S\ell_m$. We apply Corollary 5.86 with the constant batch size $M_m = M$, count parameter $r_m = r_\rho$, and sampled-list parameter u_m. It remains only to check the summability of the two sign terms.

As in Corollary 5.81,

$$\binom{M}{r_\rho} \leq (e/\rho)^{r_\rho}.$$

Since $d_m \geq 2$ and $B \geq 2\log_2(e/\rho)$,

$$\binom{M}{r_\rho} d_m^{-Br_\rho} \leq d_m^{-Br_\rho/2}.$$

The lower bound on M gives $Br_\rho/2 \geq S+1$, hence

$$\binom{M}{r_\rho} d_m^{-Br_\rho} \leq d_m^{-(S+1)} \leq d_m^{-S}.$$

Also $e^{-u_m} = (e+d_m)^{-S} \leq d_m^{-S}$. Therefore the summability hypothesis of Corollary 5.86 is satisfied.

That corollary gives the sampled-list maximum with

$$1 + \frac{\log(eM) + u_m}{\ell_m} = 1 + S + \frac{\log(eM)}{\ell_m} \leq C_{S,\rho,B},$$

because M is fixed once the displayed parameters are fixed. It also gives at least $M - r_\rho + 1$ good sampled signs and at least $M - r_\rho + 1$ robust centers. Since $M - r_\rho + 1 \geq M - \rho M$, both lower bounds are at least a $1-\rho$ fraction of the sampled list.

The relative-excess assertion follows from the same specialization $R = (1+\eta)L$, $\theta = L$. For the full-dimension consequence take $d_m = m$. Then $\sum_m m^{-A} < \infty$ and $\sum_m m^{-S} < \infty$, while the sample-size lower bound implies

$$\exp\left(-Lh(\eta)\frac{n(m)}{\log^2(e+m)}\right) \leq (e+m)^{-Lh(\eta)q}$$

for all large m, which is summable because $Lh(\eta)q > 1$. The finitely many initial dimensions do not affect the almost-sure eventual conclusion. □

Corollary 5.89 (Polynomial-scale Chernoff robust fractions)**.** *Fix $A > 0$, $S > 0$, $a > 0$, constants $R \geq 1$ and $0 < \theta < R$, a polynomial dimension exponent $\kappa > 0$, a constant $c_0 > 0$, and a fraction $0 < \rho \leq 1$. Choose $B > 0$ and an integer $M \geq 1$ so that*

$$B \geq 2\log_2(e/\rho), \qquad M \geq \frac{2(S+1)}{B\rho}.$$

Assume

$$A\kappa > 1, \qquad S\kappa > 1.$$

There are constants $C_{A,B,K,a,R}$ and $c_{A,B,K,a,R} > 0$ with the following property. Let $(d_m, n_m)_{m\geq 1}$ be a triangular-array sequence with

$$d_m \geq 2, \qquad d_m \geq c_0 m^{\kappa}, \qquad n_m \leq d_m, \qquad \ell_m = \log(e + d_m),$$

and assume

$$n_m \geq C_{A,B,K,a,R}\ell_m \qquad (m \geq 1).$$

At each m, let $B_{m,i} = T_{d_m}(\xi^{(m,i)})$, $1 \leq i \leq n_m$, be independent full Toeplitz matrices with independent mean-zero, unit-variance lag variables of sub-Gaussian norm at most K, and let $P_{m,i}$ be their symbols. Independently of the matrices, sample independent uniform sign vectors

$$\delta_m^{(1)}, \dots, \delta_m^{(M)} \in \{-1, 1\}^{n_m}.$$

Suppose

$$p_{d_m}(a) := \mathbb{P}\{\|T_{d_m}(\xi)\|_{\mathrm{op}} < a\,\mathbb{E}\,\|T_{d_m}(\xi)\|_{\mathrm{op}}\} \leq \frac{\theta}{\ell_m^2} \qquad (m \geq 1),$$

put $I(R, \theta) = R\log(R/\theta) - R + \theta$, and assume that, for all sufficiently large m,

$$n_m \geq q\,\ell_m^2 \log(e + m) \qquad \text{for some } q > \frac{1}{I(R,\theta)}.$$

Then, with probability one over the matrices and sampled signs, all but finitely many m satisfy both conclusions below.

First, the whole sampled list obeys

$$\max_{1\leq s\leq M} \max_{\omega^{d_m}=1} \left| \sum_{i=1}^{n_m} \delta_{m,i}^{(s)} \frac{P_{m,i}(\omega)}{\|B_{m,i}\|_{\mathrm{op}}} \right| \leq C_{A,B,K,a,R,S,\rho}\sqrt{n_m}.$$

Second, at least a $1 - \rho$ fraction of the sampled signs are centers of Hamming balls of radius

$$\lfloor c_{A,B,K,a,R}\ell_m \rfloor$$

on which every signing ε' has edge maximum at most

$$C_{A,B,K,a,R}\sqrt{n_m}.$$

The same $1 - \rho$ fraction also has edge maximum at most $C_{A,B,K,a,R}\sqrt{n_m}$.

In particular, if $L \geq 1$, $\eta > 0$, and $p_{d_m}(a) \leq L/\ell_m^2$, then the same conclusion holds under

$$n_m \geq q\,\ell_m^2 \log(e + m) \qquad \text{for all large } m, \qquad q > \frac{1}{Lh(\eta)},$$

where $h(\eta) = (1+\eta)\log(1+\eta) - \eta$, after allowing the constants to depend on L and η. Thus polynomial dimension schedules $d_m \gtrsim m^{\kappa}$ need only $S > 1/\kappa$ robust fraction confidence, rather than the all-integer $S > 1$ condition.

Proof. Apply Corollary 5.88. It is enough to verify

$$\sum_m \left[d_m^{-A} + \exp\left(-I(R,\theta)\frac{n_m}{\ell_m^2} \right) + d_m^{-S} \right] < \infty.$$

The polynomial lower bound on d_m gives

$$d_m^{-A} \le C_{A,c_0} m^{-A\kappa}, \qquad d_m^{-S} \le C_{S,c_0} m^{-S\kappa},$$

and these two series are finite by $A\kappa > 1$ and $S\kappa > 1$. For all large m, the sample-window assumption gives

$$\exp\left(-I(R,\theta)\frac{n_m}{\ell_m^2} \right) \le (e+m)^{-I(R,\theta)q},$$

which is summable because $I(R,\theta)q > 1$. The general Borel-fraction corollary gives the eventual sampled-list maximum and robust-center fraction. The relative-excess assertion is the specialization $R = (1+\eta)L$, $\theta = L$, for which $I(R,\theta) = Lh(\eta)$. □

Corollary 5.90 (Critical-log Chernoff Borel criterion)**.** *Fix $A > 0$, $a > 0$, $L \ge 1$, and $\eta > 0$, and put*

$$h(\eta) = (1+\eta)\log(1+\eta) - \eta.$$

There is a constant $C_{A,K,a,L,\eta}$ with the following property. Let $(d_m, n_m)_{m\ge 1}$ be any triangular-array sequence with $n_m \le d_m$, $\ell_m = \log(e + d_m)$, and

$$n_m \ge C_{A,K,a,L,\eta}\ell_m \qquad (m \ge 1).$$

Suppose

$$p_{d_m}(a) := \mathbb{P}\{\|T_{d_m}(\xi)\|_{\mathrm{op}} < a\,\mathbb{E}\,\|T_{d_m}(\xi)\|_{\mathrm{op}}\} \le \frac{L}{\ell_m^2} \qquad (m \ge 1),$$

and

$$\sum_{m=1}^{\infty} d_m^{-A} < \infty.$$

If, for all sufficiently large m,

$$n_m \ge q\,\ell_m^2 \log(e+m) \qquad \text{for some } q > \frac{1}{Lh(\eta)},$$

then, with probability one over the matrix triangular array, all but finitely many m satisfy both the edge-profile/signing/scalar-moment package of Corollary 5.77 and the random-search/stable-center package of Corollary 5.84.

Proof. The relative-excess summability condition in Corollaries 5.77 and 5.84 is

$$\sum_m \left(d_m^{-A} + \exp\left(-Lh(\eta)\frac{n_m}{\ell_m^2} \right) \right) < \infty.$$

The first summand is summable by hypothesis. The lower bound on n_m gives, for all sufficiently large m,

$$\exp\left(-Lh(\eta)\frac{n_m}{\ell_m^2}\right) \le (e+m)^{-Lh(\eta)q},$$

and the exponent is strictly larger than one. Hence the displayed series is finite, and the two Chernoff Borel transfers apply. □

Corollary 5.91 (Stretched-scale Chernoff Borel window)**.** *Fix $A > 0$, $a > 0$, $L \ge 1$, $\eta > 0$, $c_0 > 0$, and $\kappa > 0$. Put $h(\eta) = (1+\eta)\log(1+\eta) - \eta$. There is a constant $C_{A,K,a,L,\eta}$ with the following property. Suppose*

$$p_d(a) := \mathbb{P}\{\|T_d(\xi)\|_{\mathrm{op}} < a\,\mathbb{E}\,\|T_d(\xi)\|_{\mathrm{op}}\} \le \frac{L}{\log^2(e+d)} \qquad (d \ge 1).$$

Let $(d_m, n_m)_{m\ge1}$ satisfy, with $\ell_m = \log(e+d_m)$,

$$d_m \ge \exp(c_0 m^\kappa), \qquad C_{A,K,a,L,\eta}\ell_m \le n_m \le d_m.$$

Assume in addition that at least one of the following two lower sample-window conditions holds for all sufficiently large m:

$$n_m \ge C_1 \ell_m^{2+\rho} \qquad \text{for some } C_1 > 0 \text{ and } \rho > 0,$$

or

$$n_m \ge q\,\ell_m^2 \log(e+m) \qquad \text{for some } q > \frac{1}{Lh(\eta)}.$$

Then, for the independent full Toeplitz triangular array at dimensions d_m, with probability one all but finitely many m satisfy the realization-normalized edge-profile, random-sign, sign-averaged, deterministic-certificate, and scalar edge-grid moment package of Corollary 5.77. On the same eventual matrix array, the full random-search and stable-center package of Corollary 5.84 also holds.

Proof. By Corollaries 5.77 and 5.84, it is enough to check

$$\sum_m \left(d_m^{-A} + \exp\left(-Lh(\eta)\frac{n_m}{\ell_m^2}\right)\right) < \infty.$$

The first summand is summable because $d_m \ge \exp(c_0 m^\kappa)$. If $n_m \ge C_1\ell_m^{2+\rho}$, then for all large m,

$$\exp\left(-Lh(\eta)\frac{n_m}{\ell_m^2}\right) \le \exp(-Lh(\eta)C_1\ell_m^\rho) \le \exp(-c\,m^{\kappa\rho}),$$

which is summable. If instead $n_m \ge q\ell_m^2\log(e+m)$, then

$$\exp\left(-Lh(\eta)\frac{n_m}{\ell_m^2}\right) \le (e+m)^{-Lh(\eta)q},$$

which is summable by the displayed lower bound on q. The two Borel–Cantelli transfers give the asserted eventual profile and search packages. □

Corollary 5.92 (Sparse sampled Chernoff Borel window)**.** *Fix $A > 0$, $\alpha > 0$, $a > 0$, $L \geq 1$, $\eta > 0$, $c_0 > 0$, and $\kappa > 0$. Put $h(\eta) = (1+\eta)\log(1+\eta) - \eta$. There are constants $C_{A,K,a,L,\eta}$ and $c_{A,K,a,L,\eta} > 0$ with the following property. Suppose*

$$p_d(a) := \mathbb{P}\{\|T_d(\xi)\|_{\mathrm{op}} < a\,\mathbb{E}\,\|T_d(\xi)\|_{\mathrm{op}}\} \leq \frac{L}{\log^2(e+d)} \qquad (d \geq 1).$$

Let $(d_m, n_m)_{m\geq 1}$ satisfy, with $\ell_m = \log(e+d_m)$,

$$d_m \geq \exp(c_0 m^{\kappa}), \qquad C_{A,K,a,L,\eta}\ell_m \leq n_m \leq d_m.$$

At each m, take the independent full Toeplitz triangular array at dimension d_m, and independently sample M_m uniform sign vectors in $\{-1,1\}^{n_m}$, with

$$M_m \geq \lceil \alpha \rceil.$$

Assume in addition that at least one of the following two lower sample-window conditions holds for all sufficiently large m:

$$n_m \geq C_1 \ell_m^{2+\rho} \qquad \text{for some } C_1 > 0 \text{ and } \rho > 0,$$

or

$$n_m \geq q\,\ell_m^2 \log(e+m) \qquad \text{for some } q > \frac{1}{Lh(\eta)}.$$

Then, with probability one over the matrices and sampled signs, all but finitely many m have some sampled sign which is the center of a Hamming ball of radius

$$r_m = \lfloor c_{A,K,a,L,\eta}\ell_m \rfloor$$

on which every signing ε' satisfies

$$\max_{\omega^{d_m}=1} \left| \sum_{i=1}^{n_m} \varepsilon_i' \frac{P_{m,i}(\omega)}{\|B_{m,i}\|_{\mathrm{op}}} \right| \leq C_{A,K,a,L,\eta}\sqrt{n_m}.$$

Thus on stretched-exponential dimension schedules, the sparse Chernoff Borel window gives actual sampled robust certificates from $O_\alpha(1)$ sign trials for every $\alpha > 0$.

Proof. Apply Corollary 5.85 with $R = (1+\eta)L$ and $\theta = L$. Then

$$I((1+\eta)L, L) = Lh(\eta).$$

It is enough to check

$$\sum_m \left(d_m^{-A} + \exp\left(-Lh(\eta)\frac{n_m}{\ell_m^2} \right) + d_m^{-\alpha} \right) < \infty.$$

The stretched-exponential lower bound on d_m makes both $\sum_m d_m^{-A}$ and $\sum_m d_m^{-\alpha}$ finite. If $n_m \geq C_1\ell_m^{2+\rho}$, then the Chernoff summand is bounded for all large m by

$$\exp(-c\,m^{\kappa\rho}),$$

which is summable. If instead $n_m \geq q\ell_m^2\log(e+m)$, then

$$\exp\left(-Lh(\eta)\frac{n_m}{\ell_m^2} \right) \leq (e+m)^{-Lh(\eta)q},$$

which is summable by the lower bound on q. The sampled Chernoff Borel transfer gives the asserted eventual sampled robust certificates. □

Corollary 5.93 (Sparse Chernoff robust-fraction windows)**.** *Fix $A > 0$, $S > 0$, $a > 0$, $L \geq 1$, $\eta > 0$, $c_0 > 0$, $\kappa > 0$, and a fraction $0 < \rho \leq 1$. Choose $B > 0$ and an integer $M \geq 1$ so that*

$$B \geq 2\log_2(e/\rho), \qquad M \geq \frac{2(S+1)}{B\rho}.$$

Put $h(\eta) = (1+\eta)\log(1+\eta) - \eta$. There are constants $C_{A,B,K,a,L,\eta}$ and $c_{A,B,K,a,L,\eta} > 0$ with the following property. Suppose

$$p_d(a) := \mathbb{P}\{\|T_d(\xi)\|_{\mathrm{op}} < a\,\mathbb{E}\,\|T_d(\xi)\|_{\mathrm{op}}\} \leq \frac{L}{\log^2(e+d)} \qquad (d \geq 1).$$

Let $(d_m, n_m)_{m\geq 1}$ satisfy, with $\ell_m = \log(e + d_m)$,

$$d_m \geq 2, \qquad d_m \geq \exp(c_0 m^{\kappa}), \qquad C_{A,B,K,a,L,\eta}\ell_m \leq n_m \leq d_m.$$

At each m, take the independent full Toeplitz triangular array at dimension d_m, and independently sample M uniform sign vectors in $\{-1,1\}^{n_m}$. Assume in addition that at least one of the following two lower sample-window conditions holds for all sufficiently large m:

$$n_m \geq C_1\ell_m^{2+\zeta} \qquad \text{for some } C_1 > 0 \text{ and } \zeta > 0,$$

or

$$n_m \geq q\,\ell_m^2\log(e+m) \qquad \text{for some } q > \frac{1}{Lh(\eta)}.$$

Then, with probability one over the matrices and sampled signs, all but finitely many m satisfy both conclusions below.

First, the whole sampled list obeys

$$\max_{1\leq s\leq M}\max_{\omega^{d_m}=1}\left|\sum_{i=1}^{n_m}\delta_{m,i}^{(s)}\frac{P_{m,i}(\omega)}{\|B_{m,i}\|_{\mathrm{op}}}\right| \leq C_{A,B,K,a,L,\eta,S,\rho}\sqrt{n_m}.$$

Second, at least a $1-\rho$ fraction of the sampled signs are centers of Hamming balls of radius

$$\lfloor c_{A,B,K,a,L,\eta}\ell_m\rfloor$$

on which every signing ε' has edge maximum at most

$$C_{A,B,K,a,L,\eta}\sqrt{n_m}.$$

The same $1-\rho$ fraction also has edge maximum at most $C_{A,B,K,a,L,\eta}\sqrt{n_m}$. Thus, on stretched-exponential dimension schedules, the constant-size robust-fraction test has no lower summability restriction on S beyond $S > 0$.

Proof. Apply Corollary 5.88 with

$$R = (1+\eta)L, \qquad \theta = L.$$

Then $I(R,\theta) = Lh(\eta)$. It is enough to check

$$\sum_m \left[d_m^{-A} + \exp\left(-Lh(\eta)\frac{n_m}{\ell_m^2}\right) + d_m^{-S} \right] < \infty.$$

The stretched-exponential lower bound on d_m makes both $\sum_m d_m^{-A}$ and $\sum_m d_m^{-S}$ finite, for every positive A and S. If $n_m \geq C_1 \ell_m^{2+\zeta}$, then the Chernoff summand is bounded for all large m by

$$\exp(-c\, m^{\kappa\zeta}),$$

which is summable. If instead $n_m \geq q\ell_m^2 \log(e+m)$, then

$$\exp\left(-Lh(\eta)\frac{n_m}{\ell_m^2}\right) \leq (e+m)^{-Lh(\eta)q},$$

which is summable by the lower bound on q. The Borel-fraction transfer gives the asserted eventual sampled-list maximum and robust-center fraction. □

Corollary 5.94 (Sparse Chernoff sampled-count windows)**.** *Fix $A > 0$, $B > 0$, $a > 0$, $L \geq 1$, $\eta > 0$, $c_0 > 0$, and $\kappa > 0$. Put $h(\eta) = (1+\eta)\log(1+\eta) - \eta$. There are constants $C_{A,B,K,a,L,\eta}$ and $c_{A,B,K,a,L,\eta} > 0$ with the following property. Suppose*

$$p_d(a) := \mathbb{P}\{\|T_d(\xi)\|_{\mathrm{op}} < a\, \mathbb{E}\, \|T_d(\xi)\|_{\mathrm{op}}\} \leq \frac{L}{\log^2(e+d)} \qquad (d \geq 1).$$

Let $(d_m, n_m)_{m\geq 1}$ satisfy, with $\ell_m = \log(e+d_m)$,

$$d_m \geq \exp(c_0 m^\kappa), \qquad C_{A,B,K,a,L,\eta}\ell_m \leq n_m \leq d_m.$$

At each m, take the independent full Toeplitz triangular array at dimension d_m, and independently sample M_m uniform sign vectors in $\{-1,1\}^{n_m}$, with $M_m \geq 1$. Let $u_m \geq 0$ and $1 \leq r_m \leq M_m$ be deterministic parameters. Assume in addition that at least one of the following two lower sample-window conditions holds for all sufficiently large m:

$$n_m \geq C_1 \ell_m^{2+\rho} \qquad \text{for some } C_1 > 0 \text{ and } \rho > 0,$$

or

$$n_m \geq q\, \ell_m^2 \log(e+m) \qquad \text{for some } q > \frac{1}{Lh(\eta)}.$$

If

$$\sum_{m=1}^{\infty} \left(e^{-u_m} + \binom{M_m}{r_m} d_m^{-Br_m} \right) < \infty,$$

then, with probability one over the matrices and sampled signs, all but finitely many m satisfy the sampled-list maximum bound, good-sign count, and robust-center count conclusions of Corollary 5.86, with constants depending on (A,B,K,a,L,η).

In particular, let $s > 1$ and take $u_m = s\log(e+m)$ and $r_m = 1$. If

$$M_m \leq d_m^\beta \qquad (m \geq 1)$$

for some $\beta \geq 0$, then every $B > \beta$ gives an almost-sure event on which, for all sufficiently large m, every sampled sign in the batch is an $O(\sqrt{n_m})$-good robust center and the whole batch satisfies the displayed sampled-list maximum bound. More generally, if

$$M_m \leq \exp(bm^\sigma)$$

for some $b > 0$, $0 < \sigma \leq \kappa$, then the same eventual all-sampled-sign conclusion holds whenever either $\sigma < \kappa$, or $\sigma = \kappa$ and $b < Bc_0$.

Proof. Apply Corollary 5.86 with $R = (1+\eta)L$ and $\theta = L$. The stretched-exponential lower bound on d_m makes $\sum_m d_m^{-A}$ finite. The same computation used in Corollary 5.91 shows that either sample-window hypothesis makes

$$\sum_m \exp\left(-Lh(\eta)\frac{n_m}{\ell_m^2}\right)$$

finite. The two extra sign summands are summable by hypothesis, so the sampled-count Borel theorem applies.

For the polynomial-in-d_m batch consequence,

$$\binom{M_m}{1} d_m^{-B} \leq d_m^{\beta-B},$$

which is summable on the stretched-exponential schedule when $B > \beta$. For the $\exp(bm^\sigma)$ batch consequence,

$$\binom{M_m}{1} d_m^{-B} \leq \exp\left(bm^\sigma - Bc_0 m^\kappa\right),$$

which is summable if $\sigma < \kappa$, or if $\sigma = \kappa$ and $b < Bc_0$. Finally, $\sum_m e^{-u_m} < \infty$ because $s > 1$. With $r_m = 1$, the count conclusion says all M_m sampled signs are good and robust centers eventually. □

Corollary 5.95 (Sparse Chernoff growing robust fractions)**.** *Fix $A > 0$, $a > 0$, $L \geq 1$, $\eta > 0$, $c_0 > 0$, $\kappa > 0$, and $0 < \rho \leq 1$. Put $h(\eta) = (1+\eta)\log(1+\eta) - \eta$. There are constants $C_{A,K,a,L,\eta}$ and $c_{A,K,a,L,\eta} > 0$ with the following property. Suppose*

$$p_d(a) := \mathbb{P}\{\|T_d(\xi)\|_{\mathrm{op}} < a\,\mathbb{E}\,\|T_d(\xi)\|_{\mathrm{op}}\} \leq \frac{L}{\log^2(e+d)} \qquad (d \geq 1).$$

Let $(d_m, n_m)_{m\geq 1}$ satisfy, with $\ell_m = \log(e + d_m)$,

$$d_m \geq \exp(c_0 m^\kappa), \qquad C_{A,K,a,L,\eta}\ell_m \leq n_m \leq d_m.$$

At each m, take the independent full Toeplitz triangular array at dimension d_m, writing $B_{m,i} = T_{d_m}(\xi^{(m,i)})$, $1 \leq i \leq n_m$, and $P_{m,i}$ for its symbol. Independently sample an arbitrary deterministic number $M_m \geq 1$ of uniform sign vectors

$$\delta_m^{(1)}, \ldots, \delta_m^{(M_m)} \in \{-1, 1\}^{n_m}.$$

Let $u_m \geq 0$ be deterministic and assume

$$\sum_{m=1}^{\infty} e^{-u_m} < \infty.$$

Assume in addition that at least one of the following two lower sample-window conditions holds for all sufficiently large m:

$$n_m \geq C_1 \ell_m^{2+\zeta} \qquad \text{for some } C_1 > 0 \text{ and } \zeta > 0,$$

or

$$n_m \geq q\, \ell_m^2 \log(e+m) \qquad \text{for some } q > \frac{1}{Lh(\eta)}.$$

Then, with probability one over the matrices and sampled signs, all but finitely many m satisfy both conclusions below.

First, the whole sampled list obeys

$$\max_{1\leq s\leq M_m} \max_{\omega^{d_m}=1} \left| \sum_{i=1}^{n_m} \delta_{m,i}^{(s)} \frac{P_{m,i}(\omega)}{\|B_{m,i}\|_{\mathrm{op}}} \right| \leq C_{A,K,a,L,\eta} \sqrt{n_m \left(1 + \frac{\log(eM_m) + u_m}{\ell_m}\right)}.$$

Second, at least a $1-\rho$ fraction of the M_m sampled signs have edge maximum at most $C_{A,K,a,L,\eta}\sqrt{n_m}$ and are centers of Hamming balls of radius

$$\lfloor c_{A,K,a,L,\eta} \ell_m \rfloor$$

on which every signing ε' has edge maximum at most

$$C_{A,K,a,L,\eta}\sqrt{n_m}.$$

No upper bound on M_m is required for this robust-fraction conclusion.

Proof. Apply Corollary 5.94 with sign-count exponent 1 and

$$r_m = \lceil \rho M_m \rceil.$$

The matrix and Chernoff summability hypotheses are exactly those verified in the proof of Corollary 5.94. The sampled-list tail is summable by the hypothesis on u_m. For the binomial count tail, $r_m \geq \rho M_m$ gives

$$\binom{M_m}{r_m} d_m^{-r_m} \leq \left(\frac{eM_m}{r_m}\right)^{r_m} d_m^{-r_m} \leq \left(\frac{e}{\rho} d_m^{-1}\right)^{r_m}.$$

For all large m, this is at most $d_m^{-r_m/2} \leq d_m^{-1/2}$, which is summable on the stretched-exponential schedule. The sampled-count theorem therefore gives the entropy-sensitive sampled-list maximum and at least

$$M_m - r_m + 1 \geq (1-\rho) M_m$$

robust centers eventually. □

Corollary 5.96 (Vanishing sparse Chernoff discarded fractions)**.** *Fix $A > 0$, $B > 0$, $a > 0$, $L \geq 1$, $\eta > 0$, $c_0 > 0$, and $\kappa > 0$. Fix $0 < \tau_0 < B$.*

Put $h(\eta) = (1+\eta)\log(1+\eta) - \eta$. *There are constants* $C_{A,B,K,a,L,\eta}$ *and* $c_{A,B,K,a,L,\eta} > 0$ *with the following property. Suppose*

$$p_d(a) := \mathbb{P}\{\|T_d(\xi)\|_{\mathrm{op}} < a\,\mathbb{E}\,\|T_d(\xi)\|_{\mathrm{op}}\} \le \frac{L}{\log^2(e+d)} \qquad (d \ge 1).$$

Let $(d_m, n_m)_{m\ge 1}$ *satisfy, with* $\ell_m = \log(e+d_m)$,

$$d_m \ge \exp(c_0 m^{\kappa}), \qquad C_{A,B,K,a,L,\eta}\ell_m \le n_m \le d_m.$$

At each m, *take the independent full Toeplitz triangular array at dimension* d_m, *writing* $B_{m,i} = T_{d_m}(\xi^{(m,i)})$, $1 \le i \le n_m$, *and* $P_{m,i}$ *for its symbol. Independently sample an arbitrary deterministic number* $M_m \ge 1$ *of uniform sign vectors*

$$\delta_m^{(1)}, \dots, \delta_m^{(M_m)} \in \{-1,1\}^{n_m}.$$

Let $u_m \ge 0$ *be deterministic and assume*

$$\sum_{m=1}^{\infty} e^{-u_m} < \infty.$$

Let $(\rho_m)_{m\ge 1} \subset (0,1]$ *satisfy*

$$\rho_m \ge e\, d_m^{-\tau_0}$$

for all sufficiently large m. *Assume in addition that at least one of the following two lower sample-window conditions holds for all sufficiently large* m*:*

$$n_m \ge C_1 \ell_m^{2+\zeta} \qquad \text{for some } C_1 > 0 \text{ and } \zeta > 0,$$

or

$$n_m \ge q\,\ell_m^2 \log(e+m) \qquad \text{for some } q > \frac{1}{Lh(\eta)}.$$

Then, with probability one over the matrices and sampled signs, all but finitely many m *satisfy both conclusions below.*

First, the whole sampled list obeys

$$\max_{1\le s\le M_m} \max_{\omega^{d_m}=1} \left| \sum_{i=1}^{n_m} \delta_{m,i}^{(s)} \frac{P_{m,i}(\omega)}{\|B_{m,i}\|_{\mathrm{op}}} \right| \le C_{A,B,K,a,L,\eta} \sqrt{n_m \left(1 + \frac{\log(eM_m) + u_m}{\ell_m}\right)}.$$

Second, at least a $1-\rho_m$ *fraction of the* M_m *sampled signs have edge maximum at most* $C_{A,B,K,a,L,\eta}\sqrt{n_m}$ *and are centers of Hamming balls of radius*

$$\lfloor c_{A,B,K,a,L,\eta}\ell_m \rfloor$$

on which every signing ε' *has edge maximum at most*

$$C_{A,B,K,a,L,\eta}\sqrt{n_m}.$$

No upper bound on M_m *is required. In particular, if* $\rho_m \to 0$, *then the robust-center fraction tends to one along the sampled batches.*

Proof. Apply Corollary 5.94 with sign-count exponent B and

$$r_m = \lceil \rho_m M_m \rceil.$$

The matrix and Chernoff summability hypotheses are exactly those verified in the proof of Corollary 5.95, and the sampled-list tail is summable by the hypothesis on u_m. For the binomial count tail,

$$\binom{M_m}{r_m} \le \left(\frac{eM_m}{r_m}\right)^{r_m} \le \left(\frac{e}{\rho_m}\right)^{r_m}.$$

The lower bound on ρ_m gives, for all large m,

$$\binom{M_m}{r_m} d_m^{-Br_m} \le d_m^{-(B-\tau_0)r_m} \le d_m^{-(B-\tau_0)},$$

which is summable on the stretched-exponential schedule. The sampled-count theorem therefore applies. Since

$$M_m - r_m + 1 \ge M_m - \rho_m M_m,$$

the robust-center count is at least a $1-\rho_m$ fraction. □

Corollary 5.97 (Relative-excess bounded lower-tail density)**.** *Fix $A > 0$, $a > 0$, $L \ge 1$, and $\eta > 0$, and write $\ell = \log(e+d)$. In the setting of Corollary 5.29, suppose*

$$p_d(a) := \mathbb{P}\{\|T_d(\xi)\|_{\mathrm{op}} < aN_d\} \le \frac{L}{\ell^2}.$$

Set

$$h(\eta) := (1+\eta)\log(1+\eta) - \eta > 0.$$

Then there is a constant $C_{A,K,a,L,\eta}$ such that, whenever

$$n \le d, \qquad n \ge C_{A,K,a,L,\eta}\ell,$$

one has

$$\mathbb{P}\left\{\max_{\omega^d=1}\sum_{i=1}^{n}\frac{|P_i(\omega)|^2}{\|B_i\|_{\mathrm{op}}^2} > C_{A,K,a,L,\eta}\frac{n}{\ell}\right\} \le C_{A,K,a,L,\eta}d^{-A} + \exp\left(-Lh(\eta)\frac{n}{\ell^2}\right).$$

On the complement of this exceptional event, independent signs have the $C_{A,K,a,L,\eta}\sqrt{n}$ edge bound with conditional probability at least $1-d^{-A}$, sign-averaged edge maximum $C_{A,K,a,L,\eta}\sqrt{n}$, a deterministic signing certificate, and scalar edge-grid moments bounded by $(C_{A,K,a,L,\eta}k\,n/\ell)^k$ for every $2k$-th moment.

Proof. Apply Corollary 5.76 with $\theta = L$ and $R = (1+\eta)L$. Since $L \ge 1$, the empirical budget parameter satisfies $R \ge 1$, and

$$I((1+\eta)L, L) = (1+\eta)L\log(1+\eta) - \eta L = Lh(\eta).$$

□

Corollary 5.98 (Large-sample bounded lower-tail density)**.** *Fix $A > 0$, $a > 0$, and $L \geq 1$, and write $\ell = \log(e + d)$. In the setting of Corollary 5.29, suppose*

$$p_d(a) := \mathbb{P}\{\|T_d(\xi)\|_{\mathrm{op}} < aN_d\} \leq \frac{L}{\ell^2}.$$

Then there are constants $C_{A,K,a,L}, c_L > 0$ such that, whenever

$$n \leq d, \qquad n \geq C_{A,K,a,L}\ell,$$

one has

$$\mathbb{P}\left\{\max_{\omega^d=1} \sum_{i=1}^{n} \frac{|P_i(\omega)|^2}{\|B_i\|_{\mathrm{op}}^2} > C_{A,K,a,L}\frac{n}{\ell}\right\} \leq C_{A,K,a,L}d^{-A} + \exp\left(-c_L\frac{n}{\ell^2}\right).$$

On the complement of this exceptional event, independent signs have

$$\mathbb{P}_\varepsilon\left\{\max_{\omega^d=1}\left|\sum_{i=1}^{n}\varepsilon_i\frac{P_i(\omega)}{\|B_i\|_{\mathrm{op}}}\right| > C_{A,K,a,L}\sqrt{n}\right\} \leq d^{-A},$$

and

$$\mathbb{E}_\varepsilon \max_{\omega^d=1}\left|\sum_{i=1}^{n}\varepsilon_i\frac{P_i(\omega)}{\|B_i\|_{\mathrm{op}}}\right| \leq C_{A,K,a,L}\sqrt{n}.$$

Thus a deterministic sign vector with edge maximum $C_{A,K,a,L}\sqrt{n}$ exists. On the same event, simultaneously for every integer $k \geq 1$,

$$\max_{\omega^d=1}\mathbb{E}_\varepsilon\left|\sum_{i=1}^{n}\varepsilon_i\frac{P_i(\omega)}{\|B_i\|_{\mathrm{op}}}\right|^{2k} \leq \left(C_{A,K,a,L}k\,\frac{n}{\ell}\right)^k.$$

Consequently, if $n/\ell^2 \to \infty$, the realization-normalized edge profile holds with probability tending to one. If in addition $d/n \to \infty$, the corresponding full Toeplitz/Hankel contractions have $O_{K,a,L}(\sqrt{n})$ discrepancy with probability tending to one.

Proof. Apply Corollary 5.97 with $\eta = 1$. The displayed profile estimate and the signing, expectation, and moment statements follow with $c_L = L(2\log 2 - 1)$, after absorbing constants. The final Toeplitz/Hankel discrepancy conclusion follows from Proposition 5.1 and the Toeplitz–Hankel reversal. □

Theorem 5.99 (Convex-concentrated normalizers)**.** *Assume in addition to the hypotheses of Theorem 5.14 that the coefficient law is either real or complex. Write $q = 1$ in the real case and $q = 2$ in the complex case, after identifying $\mathbb{C}$ with $\mathbb{R}^2$. Assume the product law satisfies a dimension-free convex concentration inequality: there is C_{cc} such that for every m, every convex 1-Lipschitz $F : (\mathbb{R}^q)^m \to \mathbb{R}$, and independent copies $X_1, \ldots, X_m$ of ξ,*

$$\mathbb{P}\{F(X) - \mathbb{E}F(X) \leq -t\} \leq \exp(-t^2/C_{\mathrm{cc}}) \qquad (t \geq 0).$$

Then the lower-tail hypothesis of Theorem 5.14 holds for some $a = a(K, C_{\mathrm{cc}}) > 0$. Consequently, if $n \le d \le e^{o(n)}$ and $d/n \to \infty$, the realization-normalized full Toeplitz contractions $B_i/\|B_i\|_{\mathrm{op}}$ satisfy

$$\mathrm{disc}_{\mathrm{op}}(B_1/\|B_1\|_{\mathrm{op}}, \dots, B_n/\|B_n\|_{\mathrm{op}}) = O_{K,C_{\mathrm{cc}}}(\sqrt{n})$$

with probability tending to one. The same conclusion holds for full Hankel contractions. In particular the conclusion applies to standard real or complex Gaussian coefficients, Rademacher coefficients, and more generally bounded real or complex coefficient laws with mean zero and variance one.

Proof. Let $X = (\xi_{-(d-1)}, \dots, \xi_{d-1}) \in (\mathbb{R}^q)^{2d-1}$ and

$$F(X) = \|T_d(\xi)\|_{\mathrm{op}}.$$

The function F is convex, and for two lag vectors x, y,

$$\begin{aligned} |F(x) - F(y)| &\le \|T_d(x-y)\|_{\mathrm{op}} \\ &\le \sup_\theta \left| \sum_{|k|<d} (x_k - y_k) e^{ik\theta} \right| \\ &\le \sqrt{2d-1}\, \|x-y\|_2 . \end{aligned}$$

Here $\|x-y\|_2$ is the Euclidean norm on $(\mathbb{R}^q)^{2d-1}$. Thus $F/\sqrt{2d-1}$ is convex 1-Lipschitz. By the convex concentration hypothesis and Meckes' lower bound $N_d = \mathbb{E}F \ge c_K \sqrt{d \log(e+d)}$,

$$\mathbb{P}\{F < N_d/2\} \le \exp\left(-c_{K,C_{\mathrm{cc}}} \frac{N_d^2}{d}\right) \le (e+d)^{-\beta}$$

for some $\beta = \beta(K, C_{\mathrm{cc}}) > 0$. This is $o(\log^{-2}(e+d))$, so Theorem 5.14 applies with $a = 1/2$.

Real and complex Gaussian laws satisfy the displayed convex concentration by Gaussian isoperimetry on the corresponding Euclidean space. Rademacher and bounded real or complex product laws satisfy it by Talagrand's convex concentration theorem for product spaces [5]. The Hankel case again follows by Toeplitz–Hankel reversal. □

Corollary 5.100 (Strongly log-concave coefficient laws)**.** *Let the real coefficient law have mean zero, variance one, and density $\exp(-V(x))\,dx/Z$ on $\mathbb{R}$ with $V'' \ge \rho > 0$. More generally, in the complex case identify $\mathbb{C}$ with $\mathbb{R}^2$, assume the law has mean zero, $\mathbb{E}|\xi|^2 = 1$, and density $\exp(-V(z))\,dz/Z$ with $\nabla^2 V \ge \rho I_2$. Then the conclusion of Theorem 5.99 holds, with constants depending only on ρ and the variance normalization. Thus, whenever*

$$n \le d \le e^{o(n)}, \qquad d/n \to \infty,$$

independent realization-normalized full Toeplitz contractions drawn from this law satisfy

$$\mathrm{disc}_{\mathrm{op}}(B_1/\|B_1\|_{\mathrm{op}}, \dots, B_n/\|B_n\|_{\mathrm{op}}) = O_\rho(\sqrt{n})$$

with probability tending to one, and the same holds for full Hankel contractions.

In the triangular-array range

$$n_m \le d_m, \qquad d_m/n_m \to \infty, \qquad \log d_m = o(n_m),$$

the sampled operator-certificate, robust-fraction, polynomial-batch, and almost-sure sampled-batch conclusions of Corollaries 5.138, 5.139, and 5.140 also apply to these strongly log-concave laws.

Proof. By the Bakry–Emery/log-Sobolev concentration theorem, product measures with $\nabla^2 V \ge \rho I$ satisfy dimension-free Gaussian concentration for every Lipschitz function, and hence the one-sided convex concentration hypothesis in Theorem 5.99; see, for example, [4]. The same concentration applied to linear coordinate functionals gives a sub-Gaussian coefficient norm depending only on ρ and the variance normalization. Theorem 5.99 therefore gives the displayed realization-normalized Toeplitz and Hankel discrepancy conclusion. The final sampled-certificate assertions are exactly the three cited corollaries applied with the convex-concentration constant supplied by the same log-Sobolev estimate. □

Corollary 5.101 (Finite convolutions of convex-concentrated laws)**.** *Fix $s \ge 1$. Let $\eta^{(1)}, \dots, \eta^{(s)}$ be independent real coefficient variables, or independent complex coefficient variables identified with $\mathbb{R}^2$. Assume each law is centered, sub-Gaussian, and satisfies the dimension-free convex concentration hypothesis of Theorem 5.99, with constants independent of the product dimension. For fixed scalars $a_1, \dots, a_s$, set*

$$\zeta := \sum_{r=1}^{s} a_r \eta^{(r)}, \qquad \xi := \frac{\zeta}{(\mathbb{E}|\zeta|^2)^{1/2}},$$

assuming $\mathbb{E}|\zeta|^2 > 0$. Then the coefficient law ξ satisfies the one-matrix lower-normalizer tail required in Theorem 5.14: for some $a > 0$ and $\beta > 0$,

$$\mathbb{P}\{\|T_d(\xi)\|_{\mathrm{op}} < a\,\mathbb{E}\,\|T_d(\xi)\|_{\mathrm{op}}\} \le C(1+d)^{-\beta}.$$

Consequently, whenever

$$n \le d \le e^{o(n)}, \qquad d/n \to \infty,$$

the realization-normalized full Toeplitz and full Hankel Matrix Spencer bounds hold for independent matrices with coefficient law ξ:

$$\mathrm{disc}_{\mathrm{op}}(B_1/\|B_1\|_{\mathrm{op}}, \dots, B_n/\|B_n\|_{\mathrm{op}}) = O(\sqrt{n})$$

with probability tending to one. The constants depend only on the number of summands, the coefficients a_r, the variance normalization, and the sub-Gaussian and convex-concentration constants of the input laws.

The same finite sampled operator-certificate, robust-fraction, polynomial-batch, and almost-sure sampled-batch conclusions as in Corollaries 5.138,

5.139, and 5.140 hold for this finite convolution class. In particular, Gaussian plus Rademacher noise, Gaussian plus bounded noise, and strongly-log-concave plus bounded noise are covered after centering and variance normalization.

Proof. Write $\sigma^2 = \mathbb{E}|\zeta|^2$. Let

$$X^{(r)} = (\eta^{(r)}_{-(d-1)}, \dots, \eta^{(r)}_{d-1})$$

be the independent lag vector for the r-th input law, and define

$$F(X^{(1)}, \dots, X^{(s)}) := \left\| T_d\left(\sigma^{-1} \sum_{r=1}^{s} a_r X^{(r)}\right)\right\|_{\mathrm{op}}.$$

The map F is convex jointly in all blocks. If all blocks except the r-th are fixed, then

$$|F(\dots, x, \dots) - F(\dots, y, \dots)| \le \frac{|a_r|}{\sigma}\sqrt{2d-1}\,\|x-y\|_2\,.$$

Let $N_d = \mathbb{E}F$, and for $0 \le r \le s$ put

$$H_r(X^{(1)}, \dots, X^{(r)}) := \mathbb{E}\Big[F(X^{(1)}, \dots, X^{(s)}) \mid X^{(1)}, \dots, X^{(r)}\Big].$$

Thus $H_0 = N_d$ and $H_s = F$. For each fixed $(X^{(1)}, \dots, X^{(r-1)})$, the function

$$x \mapsto H_r(X^{(1)}, \dots, X^{(r-1)}, x)$$

is convex and $(|a_r|/\sigma)\sqrt{2d-1}$-Lipschitz, because it is an average of functions with that property. Moreover

$$H_{r-1} = \mathbb{E}_{X^{(r)}} H_r(X^{(1)}, \dots, X^{(r)})$$

with the previous blocks fixed. Hence the convex concentration hypothesis for the r-th input law gives, uniformly in the previous blocks,

$$\mathbb{P}\{H_r - H_{r-1} \le -u \mid X^{(1)}, \dots, X^{(r-1)}\} \le \exp\left(-\frac{c_r \sigma^2 u^2}{|a_r|^2 d}\right),$$

with the evident omission of terms for $a_r = 0$. If $F - N_d \le -t$, then for some r, $H_r - H_{r-1} \le -t/s$. A union bound therefore gives

$$\mathbb{P}\{F < N_d - t\} \le \sum_{r:a_r\neq 0} \exp\left(-\frac{c_r \sigma^2 t^2}{s^2 |a_r|^2 d}\right).$$

The coefficient law ξ is centered, variance one, and sub-Gaussian, with norm depending only on the displayed data. Meckes' lower bound therefore gives

$$N_d = \mathbb{E}\,\|T_d(\xi)\|_{\mathrm{op}} \ge c\sqrt{d\log(e+d)}.$$

Taking $t = N_d/2$ in the preceding tail estimate yields

$$\mathbb{P}\{F < N_d/2\} \le C(1+d)^{-\beta}$$

for some $\beta > 0$. This is the lower-normalizer input in Theorem 5.14, with $a = 1/2$, and since $(1+d)^{-\beta}\log^2(e+d) \to 0$, all conclusions of that theorem apply in the stated full-bandwidth range.

The sampled and robust-center assertions follow by the same proofs as Corollaries 5.138, 5.139, and 5.140: those arguments use only the polynomial lower-normalizer tail just proved, the normalizer-tail search corollaries, and the deterministic edge-to-operator transfer. □

Corollary 5.102 (Bounded Gaussian scale-mixture coefficient laws)**.** *Let g be a standard real Gaussian variable, or a standard complex Gaussian variable normalized by $\mathbb{E}|g|^2 = 1$. Let $S \geq 0$ be independent of g and suppose*

$$0 < s_- \leq S \leq s_+ < \infty \qquad \text{almost surely.}$$

After the variance normalization

$$\xi := \frac{Sg}{(\mathbb{E}S^2)^{1/2}},$$

the coefficient law ξ satisfies the one-matrix lower-normalizer tail required in Theorem 5.14: for some $a > 0$, $\beta > 0$, and $C < \infty$, depending only on s_+/s_- and on the real or complex field,

$$\mathbb{P}\{\|T_d(\xi)\|_{\mathrm{op}} < a\,\mathbb{E}\,\|T_d(\xi)\|_{\mathrm{op}}\} \leq C(1+d)^{-\beta}.$$

Consequently, whenever

$$n \leq d \leq e^{o(n)}, \qquad d/n \to \infty,$$

the realization-normalized full Toeplitz and full Hankel Matrix Spencer bounds hold for independent matrices with coefficient law ξ:

$$\mathrm{disc}_{\mathrm{op}}(B_1/\|B_1\|_{\mathrm{op}}, \ldots, B_n/\|B_n\|_{\mathrm{op}}) = O_{s_+/s_-}(\sqrt{n})$$

with probability tending to one. The sampled operator-certificate, robust-fraction, polynomial-batch, and almost-sure sampled-batch conclusions of Corollaries 5.138, 5.139, and 5.140 hold for this class as well.

Proof. The normalization changes s_- and s_+ by a common factor, so we assume $\mathbb{E}S^2 = 1$. Let S_k, $|k| < d$, be the independent volatility variables and condition on their realized values $\sigma = (\sigma_k)$. For the Gaussian lag vector $G = (g_k)$, set

$$F_\sigma(G) := \left\| T_d((\sigma_k g_k)_{|k|<d}) \right\|_{\mathrm{op}}.$$

For two Gaussian-coordinate vectors x, y,

$$|F_\sigma(x) - F_\sigma(y)| \leq \sup_\theta \left| \sum_{|k|<d} \sigma_k (x_k - y_k) e^{ik\theta} \right| \leq s_+\sqrt{2d-1}\,\|x-y\|_2.$$

Thus Gaussian concentration gives

$$\mathbb{P}_G\{F_\sigma \leq \mathbb{E}_G F_\sigma - t\} \leq \exp\left(-c\frac{t^2}{s_+^2 d}\right).$$

It remains to compare the conditional mean to the unconditional normalizer. Since $\sigma_k^2 = s_-^2 + \tau_k^2$ with $\tau_k \geq 0$, the conditional Gaussian vector $(\sigma_k g_k)$ has the same law as

$$(s_- g'_k + \tau_k h_k)_{|k|<d},$$

where g' and h are independent standard Gaussian vectors. By Jensen's inequality, conditioning on g',

$$\mathbb{E}_h \left\| T_d(s_- g' + \tau h) \right\|_{\mathrm{op}} \geq \left\| T_d(s_- g') \right\|_{\mathrm{op}}.$$

Therefore

$$\mathbb{E}_G F_\sigma \geq s_- \, \mathbb{E} \left\| T_d(g') \right\|_{\mathrm{op}} \geq c_0 s_- \sqrt{d \log(e+d)}$$

by the standard Gaussian Toeplitz lower bound. The matching sub-Gaussian upper bound gives

$$N_d := \mathbb{E} \left\| T_d(\xi) \right\|_{\mathrm{op}} \leq C_0 s_+ \sqrt{d \log(e+d)}.$$

Choose $a = c_0 s_- / (4 C_0 s_+)$. Then, for every σ,

$$a N_d \leq \frac{1}{2} \mathbb{E}_G F_\sigma.$$

The conditional concentration estimate with $t = \frac{1}{2}\mathbb{E}_G F_\sigma$ yields

$$\mathbb{P}_G\{F_\sigma < a N_d\} \leq \exp\left(-c \frac{s_-^2}{s_+^2} \log(e+d) \right) \leq C(1+d)^{-\beta}.$$

Integrating over σ proves the displayed lower-normalizer tail.

The law ξ is centered, variance one, and sub-Gaussian with constants depending only on s_+/s_-. Theorem 5.14 gives the realization-normalized Toeplitz conclusion, and the Hankel conclusion follows by Toeplitz–Hankel reversal. The sampled certificate assertions use the same proofs as the cited convex-normalizer sampled corollaries, replacing their convex-concentration lower-tail input by the polynomial lower tail just proved. □

Theorem 5.103 (Heterogeneous lower-normalizer libraries)**.** *Fix $K, a, \beta, L > 0$. For each pair (d, n), let $B_i = T_d(\xi^{(i)})$, $1 \leq i \leq n$, be independent full Toeplitz matrices. The lag variables inside the i-th matrix are independent copies of a real or complex law $\mu_i = \mu_{d,n,i}$, but the laws may vary with i, d, n. Assume every μ_i is centered, has variance one, has sub-Gaussian norm at most K, and satisfies the uniform one-matrix lower-tail bound*

$$\mathbb{P}_{\mu_i}\left\{ \left\| T_d(\mu_i) \right\|_{\mathrm{op}} < a\, N_{d,i} \right\} \leq L(1+d)^{-\beta}, \qquad N_{d,i} := \mathbb{E}_{\mu_i} \left\| T_d(\mu_i) \right\|_{\mathrm{op}}.$$

Let P_i be the symbol of B_i, and put $\ell = \log(e+d)$. If $n \leq d$ and $\ell = o(n)$, then

$$\max_{\omega^d = 1} \sum_{i=1}^{n} \frac{|P_i(\omega)|^2}{\|B_i\|_{\mathrm{op}}^2} \leq C_{K,a,\beta,L} \frac{n}{\ell}$$

with probability tending to one. More quantitatively, for every $A > 0$, once $n \geq C_{A,K,a,\beta,L}\ell$, the exceptional probability is at most

$$C_{A,K,a,\beta,L} d^{-A} + L(1+d)^{-\beta} \min\{n, \ell^2\}.$$

Consequently independent signs have edge maximum $O_{K,a,\beta,L}(\sqrt{n})$ with high conditional probability, the sign-averaged edge maximum is $O_{K,a,\beta,L}(\sqrt{n})$, and a deterministic signing with that edge bound exists. If in addition $d/n \to \infty$, then

$$\operatorname{disc}_{\mathrm{op}}(B_1/\|B_1\|_{\mathrm{op}}, \dots, B_n/\|B_n\|_{\mathrm{op}}) = O_{K,a,\beta,L}(\sqrt{n})$$

with probability tending to one, and the same conclusion holds for the corresponding full Hankel contractions.

In particular, the realization-normalized theorem is stable under arbitrary heterogeneous selection from any fixed, or dimension-uniform, library of coefficient laws covered by Theorem 5.99, Corollary 5.100, or Corollaries 5.101 and 5.102, as long as the displayed parameters are uniform. The matrices need not be identically distributed. Matrix-level mixtures over a finite such library are included after conditioning on the selected laws.

Proof. The proof of Theorem 5.14 is unchanged except for the bookkeeping of the deterministic normalizing scales. We give the details because they are the point of the heterogeneous extension.

First we replace Proposition 5.13 by its non-identical version. For a fixed edge node $\omega^d = 1$,

$$P_i(\omega) = \sum_{|k|<d} \xi_k^{(i)} \omega^k$$

is centered sub-Gaussian with norm at most $C_K\sqrt{d}$, uniformly in i. Meckes' lower bound, applied to each law μ_i, gives

$$N_{d,i} \geq c_K\sqrt{d\ell}.$$

Thus

$$Y_i(\omega) := \frac{|P_i(\omega)|^2}{N_{d,i}^2}$$

are independent nonnegative sub-exponential variables with $\mathbb{E}Y_i(\omega) \leq C_K/\ell$ and $\|Y_i(\omega)\|_{\psi_1} \leq C_K/\ell$, uniformly in i. Bernstein's inequality for independent, not necessarily identically distributed, sub-exponential variables gives, after increasing $C_{A,K}$,

$$\mathbb{P}\left\{\sum_{i=1}^n Y_i(\omega) > C_{A,K}\frac{n}{\ell}\right\} \leq \exp(-c_{A,K}n).$$

Since $n \geq C_{A,K}\ell$, a union bound over the d edge nodes gives

$$\max_{\omega^d=1} \sum_{i=1}^n \frac{|P_i(\omega)|^2}{N_{d,i}^2} \leq C_{A,K}\frac{n}{\ell}$$

outside a d^{-A} event.

Let

$$G = \{i : \|B_i\|_{\mathrm{op}} \geq aN_{d,i}\}, \qquad M = |G^c|.$$

The uniform lower-tail hypothesis gives

$$\mathbb{E}M \leq nL(1+d)^{-\beta}.$$

If $n \geq \ell^2$, Markov's inequality gives

$$\mathbb{P}\{M > n/\ell^2\} \leq L(1+d)^{-\beta}\ell^2,$$

while if $n < \ell^2$, the event $M > n/\ell^2$ forces $M \geq 1$, and is bounded by $L(1+d)^{-\beta}n$. Hence the bad-normalizer count contributes the displayed second exceptional term.

On the expected-profile event,

$$\max_{\omega^d=1} \sum_{i\in G} \frac{|P_i(\omega)|^2}{\|B_i\|_{\mathrm{op}}^2} \leq a^{-2} \max_{\omega^d=1} \sum_{i=1}^{n} \frac{|P_i(\omega)|^2}{N_{d,i}^2} \leq C_{A,K,a}\frac{n}{\ell}.$$

The samples in G^c are controlled by the same crude root-value and Hilbert–Schmidt floor used in Theorem 5.14. Uniform sub-Gaussianity gives, outside another d^{-A} event,

$$\max_{i\leq n,\ \omega^d=1} |P_i(\omega)| \leq C_{A,K}\sqrt{d\ell}.$$

Also, since each law has variance one and sub-Gaussian norm at most K, a uniform Paley–Zygmund and Chernoff argument gives

$$\|B_i\|_{\mathrm{op}} \geq c_K\sqrt{d} \qquad (1 \leq i \leq n)$$

outside another d^{-A} event. Therefore every bad sample contributes at most $C_{A,K}\ell$ to each edge node, and the event $M \leq n/\ell^2$ gives the total bad contribution $C_{A,K}n/\ell$.

Combining the good and bad contributions proves the heterogeneous edge profile. The conditional random-sign bound, sign average, and deterministic certificate then follow from Proposition 5.5 and Corollary 5.9. If $d/n \to \infty$, Proposition 5.1 transfers the edge bound to full Toeplitz operator norm, and the Hankel statement follows by Toeplitz–Hankel reversal.

The final library statement follows because Theorem 5.99 and Corollaries 5.100 and 5.101 and 5.102 give the required polynomial one-matrix lower tail with constants depending only on their displayed parameters. If the law of each matrix is first selected from a finite library, conditioning on the selected laws reduces the model to the deterministic heterogeneous array just proved. □

Corollary 5.104 (Matrix-level radial scalings)**.** *Let $A_1, \dots, A_n$ be any full Toeplitz or full Hankel array for which one of the realization-normalized conclusions above applies, and assume $\|A_i\|_{\mathrm{op}} > 0$ almost surely. Let $R_1, \dots, R_n$ be arbitrary positive scalar random variables on the same probability space; no independence, identical-distribution, boundedness, or moment assumption is imposed on the R_i. Put*

$$\widetilde{A}_i := R_i A_i.$$

Then every realization-normalized conclusion for the array A_i holds unchanged for the radially scaled array $\widetilde{A}_i$. In particular, all full Toeplitz/Hankel $O(\sqrt{n})$ discrepancy, sampled operator-certificate, robust-fraction, polynomial-batch, and almost-sure sampled-batch conclusions proved above remain valid after

multiplying the covered matrices by arbitrary positive, possibly heavy-tailed, matrix-level amplitudes.

Proof. The assertion is pathwise. Since $R_i > 0$,

$$\frac{\widetilde{A}_i}{\left\|\widetilde{A}_i\right\|_{\mathrm{op}}} = \frac{R_i A_i}{R_i \left\|A_i\right\|_{\mathrm{op}}} = \frac{A_i}{\left\|A_i\right\|_{\mathrm{op}}} \qquad (1 \le i \le n).$$

Thus the normalized matrix array, and therefore every discrepancy value, edge-profile event, sampled-sign certificate, Hamming-stability statement, and Borel–Cantelli conclusion that depends only on that normalized array, is identical for A_i and $\widetilde{A}_i$. □

Theorem 5.105 (Heterogeneous average lower-tail budget)**.** *Fix $K, a > 0$. For each pair (d, n), let $B_i = T_d(\xi^{(i)})$, $1 \le i \le n$, be independent full Toeplitz matrices. The lag variables inside the i-th matrix are independent copies of a real or complex law $\mu_i = \mu_{d,n,i}$, but the laws may vary with i, d, n. Assume every μ_i is centered, has variance one, and has sub-Gaussian norm at most K. Put*

$$N_{d,i} := \mathbb{E}_{\mu_i} \left\|T_d(\mu_i)\right\|_{\mathrm{op}}, \qquad p_{d,i}(a) := \mathbb{P}_{\mu_i}\{\left\|T_d(\mu_i)\right\|_{\mathrm{op}} < a N_{d,i}\},$$

and define the aggregate lower-tail budget

$$\Pi_{d,n}(a) := \sum_{i=1}^{n} p_{d,i}(a).$$

Let P_i be the symbol of B_i, and put $\ell = \log(e + d)$. Assume $n \le d$, $\ell = o(n)$, and

$$\Pi_{d,n}(a) \min\left\{1, \frac{\ell^2}{n}\right\} = o(1).$$

Then

$$\max_{\omega^d = 1} \sum_{i=1}^{n} \frac{|P_i(\omega)|^2}{\|B_i\|_{\mathrm{op}}^2} \le C_{K,a} \frac{n}{\ell}$$

with probability tending to one. More quantitatively, for every $A > 0$, once $n \ge C_{A,K,a}\ell$, the exceptional probability is at most

$$C_{A,K,a} d^{-A} + \Pi_{d,n}(a) \min\left\{1, \frac{\ell^2}{n}\right\}.$$

Consequently independent signs have edge maximum $O_{K,a}(\sqrt{n})$ with high conditional probability, the sign-averaged edge maximum is $O_{K,a}(\sqrt{n})$, and a deterministic signing with that edge bound exists. If also $d/n \to \infty$, then

$$\mathrm{disc}_{\mathrm{op}}(B_1/\left\|B_1\right\|_{\mathrm{op}}, \dots, B_n/\left\|B_n\right\|_{\mathrm{op}}) = O_{K,a}(\sqrt{n})$$

with probability tending to one, and the same conclusion holds for full Hankel contractions.

The sampled-certificate conclusions of Corollary 5.108 remain valid under this average-budget hypothesis after replacing the matrix term $L\ell_m^2(1+d_m)^{-\beta}$ *by*

$$\Pi_m(a)\min\left\{1,\frac{\ell_m^2}{n_m}\right\},\qquad \Pi_m(a):=\sum_{i=1}^{n_m}p_{d_m,i}(a).$$

In particular, the almost-sure sampled-batch conclusion holds whenever the corresponding series with this replacement is summable.

Proof. The proof is the heterogeneous proof with the bad-normalizer count kept in its aggregate form. First, the non-identical expected-normalized profile still holds. For each fixed $\omega^d=1$,

$$Y_i(\omega):=\frac{|P_i(\omega)|^2}{N_{d,i}^2}$$

are independent nonnegative sub-exponential variables with

$$\mathbb{E}Y_i(\omega)\le C_K/\ell,\qquad \|Y_i(\omega)\|_{\psi_1}\le C_K/\ell,$$

uniformly in i, because $P_i(\omega)$ has sub-Gaussian norm $C_K\sqrt{d}$ and Meckes' lower bound gives $N_{d,i}\ge c_K\sqrt{d\ell}$. Bernstein's inequality and a union bound over the d edge nodes give

$$\max_{\omega^d=1}\sum_{i=1}^{n}\frac{|P_i(\omega)|^2}{N_{d,i}^2}\le C_{A,K}\frac{n}{\ell}$$

outside a d^{-A} event.

Let

$$G=\{i:\|B_i\|_{\mathrm{op}}\ge aN_{d,i}\},\qquad M=|G^c|\,.$$

Then $\mathbb{E}M=\Pi_{d,n}(a)$. If $n\ge\ell^2$, Markov's inequality gives

$$\mathbb{P}\{M>n/\ell^2\}\le\Pi_{d,n}(a)\frac{\ell^2}{n};$$

if $n<\ell^2$, the same event forces $M\ge1$, so it has probability at most $\Pi_{d,n}(a)$. This is the displayed budget term.

On the expected-profile event, the good-normalizer contribution is bounded by

$$\max_{\omega^d=1}\sum_{i\in G}\frac{|P_i(\omega)|^2}{\|B_i\|_{\mathrm{op}}^2}\le a^{-2}\max_{\omega^d=1}\sum_{i=1}^{n}\frac{|P_i(\omega)|^2}{N_{d,i}^2}\le C_{A,K,a}\frac{n}{\ell}.$$

For the bad indices, the same uniform root-value and Hilbert–Schmidt floor argument used in Theorem 5.103 gives, outside another d^{-A} event,

$$\max_{i\le n,\ \omega^d=1}|P_i(\omega)|\le C_{A,K}\sqrt{d\ell},\qquad \|B_i\|_{\mathrm{op}}\ge c_K\sqrt{d}\quad(1\le i\le n).$$

Thus each bad sample contributes at most $C_{A,K}\ell$ to each edge node, and on $M\le n/\ell^2$ the total bad contribution is at most $C_{A,K}n/\ell$. Combining the two parts proves the profile.

The random-sign, average, deterministic-signing, Toeplitz, and Hankel conclusions then follow from Proposition 5.5, Corollary 5.9, Proposition 5.1, and Toeplitz–Hankel reversal, exactly as before. The sampled-certificate statements use the same proof as Corollary 5.108, with the matrix exceptional term replaced by the aggregate budget just proved, followed by the same Borel–Cantelli argument in the almost-sure case. □

Theorem 5.106 (Heterogeneous dyadic lower-tail budget)**.** *Fix $K, a > 0$. In the setting of Theorem 5.105, put $\ell = \log(e+d)$ and*

$$J_\ell := \left\lceil \frac{1}{2}\log_2(e+\ell) \right\rceil.$$

For each i, define the one-matrix dyadic depth variable

$$Y^{\mathrm{dy}}_{d,i,a} := \sum_{0\le r<J_\ell} 4^{r+1}\mathbf{1}\Big\{2^{-(r+1)}aN_{d,i} \le \|T_d(\mu_i)\|_{\mathrm{op}} < 2^{-r}aN_{d,i}\Big\} + \ell\,\mathbf{1}\Big\{\|T_d(\mu_i)\|_{\mathrm{op}} < 2^{-J_\ell}aN_{d,i}\Big\},$$

where the probability and expectation are with respect to an independent draw from the law μ_i. Put

$$\Delta^{\mathrm{dy}}_{d,n}(a) := \sum_{i=1}^{n} \mathbb{E}_{\mu_i} Y^{\mathrm{dy}}_{d,i,a}.$$

Assume

$$n \le d, \qquad \ell = o(n), \qquad \frac{\ell}{n}\Delta^{\mathrm{dy}}_{d,n}(a) = o(1).$$

Then

$$\max_{\omega^d=1} \sum_{i=1}^{n} \frac{|P_i(\omega)|^2}{\|B_i\|^2_{\mathrm{op}}} \le C_{K,a}\frac{n}{\ell}$$

with probability tending to one. More quantitatively, for every $A > 0$ and $R \ge 1$, once $n \ge C_{A,K,a}\ell$,

$$\mathbb{P}\left\{\max_{\omega^d=1} \sum_{i=1}^{n} \frac{|P_i(\omega)|^2}{\|B_i\|^2_{\mathrm{op}}} > C_{A,K,a,R}\frac{n}{\ell}\right\} \le C_{A,K,a,R}d^{-A} + \frac{\ell\,\Delta^{\mathrm{dy}}_{d,n}(a)}{Rn}.$$

On the complementary matrix event, independent signs have edge maximum $O_{K,a,R}(\sqrt{n})$ with high conditional probability, the sign-averaged edge maximum is $O_{K,a,R}(\sqrt{n})$, and a deterministic signing with that edge bound exists. If $d/n \to \infty$, then the same $O_{K,a,R}(\sqrt{n})$ conclusion holds for the corresponding realization-normalized full Toeplitz and full Hankel contractions.

There is also a large-sample exponential form. If $0 < \theta < R$ and

$$\Delta^{\mathrm{dy}}_{d,n}(a) \le \theta\frac{n}{\ell},$$

then the preceding exceptional probability may be replaced by

$$C_{A,K,a,R,\theta}d^{-A} + \exp\left(-c_{K,a,R,\theta}\frac{n}{\ell^2}\right).$$

The sampled-certificate conclusions of Corollary 5.108 remain valid under this dyadic-budget hypothesis. In the Markov form, replace the matrix term $L\ell_m^2(1+d_m)^{-\beta}$ *by*

$$\frac{\ell_m\,\Delta_m^{\mathrm{dy}}(a)}{Rn_m}, \qquad \Delta_m^{\mathrm{dy}}(a) := \sum_{i=1}^{n_m} \mathbb{E}Y_{d_m,i,a}^{\mathrm{dy}}.$$

In the Bernstein form, under $\Delta_m^{\mathrm{dy}}(a) \le \theta n_m/\ell_m$ *with* $0<\theta<R$, *replace that matrix term by* $\exp(-cn_m/\ell_m^2)$. *The corresponding almost-sure sampled-batch conclusions hold under the same replacements in the summability hypotheses.*

Proof. The proof is the heterogeneous expected-profile argument with the dyadic bad normalizer budget kept instead of a single cutoff count.

For a fixed edge node, the variables

$$\frac{|P_i(\omega)|^2}{N_{d,i}^2}$$

are independent sub-exponential variables with mean and ψ_1-norm $O_K(1/\ell)$, uniformly in i. Hence, as in the proof of Theorem 5.105,

$$\max_{\omega^d=1} \sum_{i=1}^{n} \frac{|P_i(\omega)|^2}{N_{d,i}^2} \le C_{A,K}\frac{n}{\ell}$$

outside an event of probability at most $C_{A,K}d^{-A}$.

For the realized matrices, define

$$W_a^{\mathrm{dy}} := \sum_{i=1}^{n} \sum_{0\le r<J_\ell} 4^{r+1}\mathbf{1}\Big\{2^{-(r+1)}aN_{d,i} \le \|B_i\|_{\mathrm{op}} < 2^{-r}aN_{d,i}\Big\} + \ell\sum_{i=1}^{n}\mathbf{1}\Big\{\|B_i\|_{\mathrm{op}} < 2^{-J_\ell}aN_{d,i}\Big\}.$$

Let $G=\{i : \|B_i\|_{\mathrm{op}} \ge aN_{d,i}\}$. On the expected-profile event, the contribution of G is at most $C_{A,K,a}n/\ell$. Intersect this with the uniform root-value and Hilbert–Schmidt floor event

$$\max_{i\le n,\ \omega^d=1} |P_i(\omega)| \le C_{A,K}\sqrt{d\ell}, \qquad \|B_i\|_{\mathrm{op}} \ge c_K\sqrt{d} \quad (1\le i\le n),$$

which also fails with probability at most $C_{A,K}d^{-A}$. Since $N_{d,i}\ge c_K\sqrt{d\ell}$, this event gives

$$\max_{i,\omega^d=1}\frac{|P_i(\omega)|^2}{N_{d,i}^2} \le C_{A,K}.$$

Thus an index in the r-th dyadic bin contributes at most $C_{A,K,a}4^{r+1}$, while an index in the final bottom bin contributes at most $C_{A,K}\ell$ by the Hilbert–Schmidt floor. Therefore

$$\max_{\omega^d=1}\sum_{i=1}^{n}\frac{|P_i(\omega)|^2}{\|B_i\|_{\mathrm{op}}^2} \le C_{A,K,a}\left(\frac{n}{\ell}+W_a^{\mathrm{dy}}\right)$$

outside the same d^{-A}-probability matrix event.

The variables defining W_a^{dy} are independent and have expectations $\mathbb{E}Y_{d,i,a}^{\mathrm{dy}}$, so

$$\mathbb{E}W_a^{\mathrm{dy}} = \Delta_{d,n}^{\mathrm{dy}}(a).$$

Markov's inequality gives

$$\mathbb{P}\left\{W_a^{\mathrm{dy}} > R\frac{n}{\ell}\right\} \le \frac{\ell\,\Delta_{d,n}^{\mathrm{dy}}(a)}{Rn}.$$

On the complement, the displayed profile holds with constant depending on R, which proves the quantitative Markov form and the high-probability statement under $\ell\Delta_{d,n}^{\mathrm{dy}}(a)/n \to 0$.

For the exponential form, note that $0 \le Y_{d,i,a}^{\mathrm{dy}} \le C\ell$, since $4^{J_\ell} \le C(e+\ell)$. If $\Delta_{d,n}^{\mathrm{dy}}(a) \le \theta n/\ell$, then $\mathbb{E}W_a^{\mathrm{dy}} \le \theta n/\ell$. Bernstein's inequality for independent bounded, not necessarily identically distributed, variables gives

$$\mathbb{P}\left\{W_a^{\mathrm{dy}} > R\frac{n}{\ell}\right\} \le \exp\left(-c_{R,\theta}\frac{n}{\ell^2}\right),$$

after changing constants to absorb the bound $C\ell$.

Conditioned on the resulting edge-profile event, the random-sign tail, sign average, deterministic signing, scalar moment package, and robust Hamming-ball search statements are exactly those of Proposition 5.5, Corollary 5.9, and the sampled-search corollaries. Proposition 5.1 and Toeplitz–Hankel reversal transfer the edge certificates to full Toeplitz and full Hankel operator norm when $d/n \to \infty$. The sampled and almost-sure statements follow by replacing the matrix exceptional term in the proof of Corollary 5.108 by the Markov or Bernstein bound just proved and then applying the same Borel–Cantelli argument. □

Corollary 5.107 (Heterogeneous inverse-moment lower-normalizer budget)**.** *Fix $K, a > 0$. In the setting of Theorem 5.106, define*

$$X_{d,i} := \|T_d(\mu_i)\|_{\mathrm{op}}$$

for an independent draw from μ_i, and put

$$\nu_{d,i,a}^{(2)} := \mathbb{E}_{\mu_i}\left[\mathbf{1}_{\{X_{d,i}<aN_{d,i}\}}\min\left\{\left(\frac{aN_{d,i}}{X_{d,i}}\right)^2, \ell\right\}\right], \qquad \ell = \log(e+d),$$

where the minimum is interpreted as ℓ on the event $X_{d,i} = 0$. Set

$$\mathcal{V}_{d,n}^{(2)}(a) := \sum_{i=1}^{n} \nu_{d,i,a}^{(2)}.$$

If

$$n \le d, \qquad \ell = o(n), \qquad \frac{\ell}{n}\mathcal{V}_{d,n}^{(2)}(a) = o(1),$$

then the realization-normalized edge profile of Theorem 5.106 holds with probability tending to one, and hence the $O_{K,a}(\sqrt{n})$ Toeplitz/Hankel Matrix Spencer conclusion holds whenever $d/n \to \infty$.

More quantitatively, for every $A > 0$ and $R \geq 1$, once $n \geq C_{A,K,a}\ell$, the profile exceptional probability is at most

$$C_{A,K,a,R} d^{-A} + \frac{C\ell\, \mathcal{V}_{d,n}^{(2)}(a)}{Rn}.$$

On the complementary matrix event, the random-sign tail, sign-averaged maximum, deterministic certificate, scalar moment package, and sampled operator certificates inherited in Theorem 5.106 hold with the same replacement. If $0 < \theta < R$ and

$$C\, \mathcal{V}_{d,n}^{(2)}(a) \leq \theta \frac{n}{\ell},$$

then the matrix exceptional term may instead be replaced by

$$C_{A,K,a,R,\theta} d^{-A} + \exp\left(-c_{K,a,R,\theta} \frac{n}{\ell^2}\right).$$

More generally, for $p > 0$ define

$$I_{d,i,a}^{(p)} := \mathbb{E}_{\mu_i}\left[\mathbf{1}_{\{X_{d,i} < aN_{d,i}\}} \left(\frac{aN_{d,i}}{X_{d,i}}\right)^p\right], \qquad \mathcal{I}_{d,n}^{(p)}(a) := \sum_{i=1}^{n} I_{d,i,a}^{(p)}.$$

Then

$$\mathcal{V}_{d,n}^{(2)}(a) \leq \ell^{(2-p)_+/2} \mathcal{I}_{d,n}^{(p)}(a).$$

Thus the same conclusion follows from

$$\frac{\ell^{1+(2-p)_+/2}}{n} \mathcal{I}_{d,n}^{(p)}(a) = o(1),$$

with the corresponding quantitative replacements.

Proof. For each i, let $Z_i = aN_{d,i}/X_{d,i}$ on $\{X_{d,i} < aN_{d,i}\}$. The same pointwise bin estimate as in the proof of Corollary 5.18 gives

$$Y_{d,i,a}^{\mathrm{dy}} \leq C\, \mathbf{1}_{\{X_{d,i} < aN_{d,i}\}} \min\{Z_i^2, \ell\}.$$

Indeed, on the r-th finite dyadic bin one has $Z_i^2 > 4^r$, so the bin weight 4^{r+1} is at most $4Z_i^2$, while $r < J_\ell$ also gives $4^{r+1} \leq C\ell$. On the final bottom bin, the bottom weight ℓ is bounded by $\min\{Z_i^2, \ell\}$. Taking expectations and summing over i therefore yields

$$\Delta_{d,n}^{\mathrm{dy}}(a) \leq C\, \mathcal{V}_{d,n}^{(2)}(a).$$

The Markov and Bernstein assertions now follow directly from Theorem 5.106.

For the inverse-p comparison, on $\{X_{d,i} < aN_{d,i}\}$ we have $Z_i > 1$. If $p \geq 2$, then $\min\{Z_i^2, \ell\} \leq Z_i^p$. If $0 < p < 2$, then $\min\{Z_i^2, \ell\} \leq \ell^{1-p/2} Z_i^p$, separately on the ranges $Z_i \leq \sqrt{\ell}$ and $Z_i > \sqrt{\ell}$. Taking expectations and summing gives the displayed aggregate inverse-p bound. □

Corollary 5.108 (Heterogeneous sampled operator certificates)**.** *In the setting of Theorem 5.103, let (d_m, n_m) be any sequence satisfying*

$$n_m \le d_m, \qquad d_m/n_m \to \infty, \qquad \log d_m = o(n_m),$$

and put $\ell_m = \log(e + d_m)$. For each m, let $B_{m,i} = T_{d_m}(\xi^{(m,i)})$, $1 \le i \le n_m$, be independent full Toeplitz matrices whose coefficient laws $\mu_{m,i}$ may vary with i and m, but satisfy the uniform hypotheses of Theorem 5.103 with the same (K, a, β, L). Set

$$A_{m,i} := \frac{B_{m,i}}{\|B_{m,i}\|_{\mathrm{op}}}, \qquad \widetilde{A}_{m,i} := J_{d_m} A_{m,i}.$$

Independently of the matrices, sample independent uniform sign vectors in $\{-1, 1\}^{n_m}$.

First, for every fixed $B > 0$, if $M_m \ge \lceil B \rceil$, then with probability tending to one some sampled sign $\delta_m^{(s)}$ is the center of a Hamming ball of radius

$$\lfloor c_B \ell_m \rfloor$$

on which every sign vector ε' satisfies

$$\left\| \sum_{i=1}^{n_m} \varepsilon'_i A_{m,i} \right\|_{\mathrm{op}} \le C_B \sqrt{n_m}, \qquad \left\| \sum_{i=1}^{n_m} \varepsilon'_i \widetilde{A}_{m,i} \right\|_{\mathrm{op}} \le C_B \sqrt{n_m}.$$

More quantitatively, for every $A_0 > 0$, the edge-certificate failure before the deterministic edge-to-operator transfer is at most

$$C_{A_0,B,K,a,\beta,L} d_m^{-A_0} + L \ell_m^2 (1 + d_m)^{-\beta} + d_m^{-B}$$

for all sufficiently large m.

Second, fix $0 < \rho \le 1$, $S > 0$, and a deterministic integer $M \ge 1$. If M signs are sampled at level m, then with probability tending to one at least $M - \lceil \rho M \rceil + 1$ of them are centers of Hamming balls of radius $\lfloor c_{\rho,S,M} \ell_m \rfloor$ on which the two operator-norm bounds above hold with $C_{\rho,S,M}$ in place of C_B. The edge failure is at most

$$C_{\rho,S,M,A_0,K,a,\beta,L} d_m^{-A_0} + L \ell_m^2 (1 + d_m)^{-\beta} + C_{\rho,S,M} d_m^{-S}.$$

Third, for every polynomial batch exponent $\theta \ge 0$ and confidence exponent $\sigma > 0$, if $1 \le M_m \le d_m^\theta$, then with probability tending to one every sampled sign is the center of such an operator-norm robust Hamming ball, with constants depending only on $\theta, \sigma, K, a, \beta, L$. The corresponding edge failure is at most

$$C_{\theta,\sigma,A_0,K,a,\beta,L} d_m^{-A_0} + L \ell_m^2 (1 + d_m)^{-\beta} + d_m^{-\sigma}.$$

Finally, let $M_m \ge 1$, $1 \le r_m \le M_m$, and $S > 0$ be deterministic. If for some $A_0 > 0$

$$\sum_{m=1}^{\infty} \left[d_m^{-A_0} + \ell_m^2 (1 + d_m)^{-\beta} + \binom{M_m}{r_m} d_m^{-S r_m} \right] < \infty,$$

then, almost surely over the matrices and sampled signs, all but finitely many m have at least $M_m - r_m + 1$ sampled signs which are centers of

Toeplitz/Hankel operator-norm robust Hamming balls of radius $\lfloor c\ell_m \rfloor$. In particular, on every stretched-exponential dimension schedule $d_m \geq \exp(c_0 m^\kappa)$, every sampled sign in every polynomial batch $M_m \leq d_m^\theta$ is eventually such a robust center almost surely, after choosing any $S > \theta$ and taking $r_m = 1$.

Proof. Theorem 5.103, applied with exponent A_0, gives a matrix event of probability at least

$$1 - C_{A_0,K,a,\beta,L} d_m^{-A_0} - L\ell_m^2 (1 + d_m)^{-\beta}$$

on which

$$V_m^2 := \max_{\omega^{d_m}=1} \sum_{i=1}^{n_m} \frac{|P_{m,i}(\omega)|^2}{\|B_{m,i}\|_{\mathrm{op}}^2} \leq C_{A_0,K,a,\beta,L} \frac{n_m}{\ell_m}.$$

On this matrix event, Corollary 5.11 gives the finite-search Hamming ball, with the additional sign failure d_m^{-B}. Corollary 5.2 transfers that whole Hamming ball from edge maximum to realization-normalized full Toeplitz and full Hankel operator norm.

For the robust-fraction statement, apply Corollary 5.10 on the same matrix event with sign exponent $2S$ and $r = \lceil \rho M \rceil$. The sign failure is bounded by $\binom{M}{r} d_m^{-2Sr} \leq C_{\rho,S,M} d_m^{-S}$ for all large m, and the deterministic transfer is again Corollary 5.2.

For the polynomial batch statement, use Corollary 5.10 with sign exponent $\theta + \sigma + 1$ and union bound over the $M_m \leq d_m^\theta$ sampled signs; the sign failure is at most $d_m^{-\sigma}$ after increasing constants. Transfer the resulting family of Hamming balls by Corollary 5.2.

The almost-sure statement is the same argument with Borel–Cantelli. At level m, Corollary 5.10, with sign exponent S, gives edge robust centers except on an event of probability at most the displayed summand up to a fixed multiplicative constant. Summability gives the eventual edge statement almost surely, and Corollary 5.2 transfers the eventual Hamming balls to Toeplitz and Hankel operator norm. For the stretched-exponential consequence with $r_m = 1$,

$$\binom{M_m}{1} d_m^{-S} \leq d_m^{\theta - S},$$

which is summable when $S > \theta$; the matrix terms are also summable on a stretched-exponential schedule for every $A_0, \beta > 0$. □

Corollary 5.109 (Heterogeneous inverse-moment sampled operator schedules)**.** *Let (d_m, n_m) be any sequence satisfying*

$$n_m \leq d_m, \qquad d_m/n_m \to \infty, \qquad \log d_m = o(n_m),$$

and put $\ell_m = \log(e + d_m)$. For each m, let $B_{m,i} = T_{d_m}(\xi^{(m,i)})$, $1 \leq i \leq n_m$, be independent full Toeplitz matrices. The coefficient law $\mu_{m,i}$ may vary with m, i, but all laws are centered, variance one, and have sub-Gaussian norm at

most K. Define $N_{m,i} = \mathbb{E}_{\mu_{m,i}} \|T_{d_m}(\mu_{m,i})\|_{\mathrm{op}}$, $X_{m,i} = \|T_{d_m}(\mu_{m,i})\|_{\mathrm{op}}$ for an independent draw, and

$$\mathcal{V}_m^{(2)}(a) := \sum_{i=1}^{n_m} \mathbb{E}_{\mu_{m,i}} \left[\mathbf{1}_{\{X_{m,i} < aN_{m,i}\}} \min\left\{ \left(\frac{aN_{m,i}}{X_{m,i}} \right)^2, \ell_m \right\} \right],$$

with the minimum interpreted as ℓ_m on $X_{m,i} = 0$. Set

$$A_{m,i} := \frac{B_{m,i}}{\|B_{m,i}\|_{\mathrm{op}}}, \qquad \widetilde{A}_{m,i} := J_{d_m} A_{m,i}.$$

Independently of the matrices, sample $M_m \geq 1$ independent uniform sign vectors. Fix $R \geq 1$, $S > 0$, and deterministic integers $1 \leq r_m \leq M_m$. If, for some $A_0 > 0$,

$$\sum_{m=1}^{\infty} \left[d_m^{-A_0} + \frac{\ell_m \, \mathcal{V}_m^{(2)}(a)}{R n_m} + \binom{M_m}{r_m} d_m^{-S r_m} \right] < \infty,$$

then, almost surely over the matrices and sampled signs, all but finitely many m have at least $M_m - r_m + 1$ sampled signs which are centers of Hamming balls of radius $\lfloor c\ell_m \rfloor$ on which every sign vector ε' satisfies

$$\left\| \sum_{i=1}^{n_m} \varepsilon_i' A_{m,i} \right\|_{\mathrm{op}} \leq C\sqrt{n_m}, \qquad \left\| \sum_{i=1}^{n_m} \varepsilon_i' \widetilde{A}_{m,i} \right\|_{\mathrm{op}} \leq C\sqrt{n_m}.$$

The constants depend only on K, a, R, S, A_0.

The Markov matrix term may be replaced by a Bernstein term. If $0 < \theta < R$,

$$C_* \mathcal{V}_m^{(2)}(a) \leq \theta \frac{n_m}{\ell_m} \qquad \text{eventually,}$$

where C_* is the absolute constant from Corollary 5.107, and

$$\sum_{m=1}^{\infty} \left[d_m^{-A_0} + \exp\left(-c_{K,a,R,\theta} \frac{n_m}{\ell_m^2} \right) + \binom{M_m}{r_m} d_m^{-S r_m} \right] < \infty,$$

then the same almost-sure sampled operator-certificate conclusion holds.

More generally, for $p > 0$ one may replace $\mathcal{V}_m^{(2)}(a)$ in the Markov summability condition by

$$\ell_m^{(2-p)_+/2} \mathcal{I}_m^{(p)}(a), \qquad \mathcal{I}_m^{(p)}(a) := \sum_{i=1}^{n_m} \mathbb{E}_{\mu_{m,i}} \left[\mathbf{1}_{\{X_{m,i} < aN_{m,i}\}} \left(\frac{aN_{m,i}}{X_{m,i}} \right)^p \right].$$

Thus a sufficient Markov summability condition is

$$\sum_{m} \left[d_m^{-A_0} + \frac{\ell_m^{1+(2-p)_+/2} \mathcal{I}_m^{(p)}(a)}{R n_m} + \binom{M_m}{r_m} d_m^{-S r_m} \right] < \infty.$$

Proof. Corollary 5.107, applied at level m, gives an edge-profile matrix event whose failure probability is bounded by the displayed Markov matrix terms, and on which

$$\max_{\omega^{d_m}=1} \sum_{i=1}^{n_m} \frac{|P_{m,i}(\omega)|^2}{\|B_{m,i}\|_{\mathrm{op}}^2} \le C \frac{n_m}{\ell_m}.$$

On this event, Corollary 5.10, with sign exponent S, says that at least $M_m - r_m + 1$ sampled signs are logarithmic-radius edge-robust centers except with additional probability $\binom{M_m}{r_m} d_m^{-Sr_m}$. Since $d_m/n_m \to \infty$ and $\log d_m = o(n_m)$, Corollary 5.2 transfers these edge Hamming balls to realization-normalized full Toeplitz and full Hankel operator norm. The displayed summability and Borel–Cantelli give the Markov branch.

The Bernstein branch is the same proof using the Bernstein matrix failure term from Corollary 5.107. The final inverse-p assertion is the aggregate inverse-p comparison in that same corollary. □

Corollary 5.110 (Heterogeneous inverse-moment operator robust fractions)**.** *In the setting and notation of Corollary 5.109, suppose that for some $A_0 > 0$, $R \ge 1$, and $a > 0$,*

$$\sum_{m=1}^{\infty} \left[d_m^{-A_0} + \frac{\ell_m \, \mathcal{V}_m^{(2)}(a)}{R n_m} \right] < \infty.$$

For each m, let $\delta_m^{(1)}, \dots, \delta_m^{(M_m)}$ be any deterministic number $M_m \ge 1$ of independent uniform sampled sign vectors, independent of the matrices.

First, let $0 < \rho \le 1$ and $q > 0$ satisfy

$$\sum_{m=1}^{\infty} d_m^{-q} < \infty.$$

Then, almost surely, all but finitely many m have at least a $1 - \rho$ fraction of the sampled signs as centers of Hamming balls of radius $\lfloor c_{\rho,q} \ell_m \rfloor$ on which every sign vector ε' satisfies

$$\left\| \sum_{i=1}^{n_m} \varepsilon_i' A_{m,i} \right\|_{\mathrm{op}} \le C_{\rho,q} \sqrt{n_m}, \qquad \left\| \sum_{i=1}^{n_m} \varepsilon_i' \widetilde{A}_{m,i} \right\|_{\mathrm{op}} \le C_{\rho,q} \sqrt{n_m}.$$

Second, let $q > 0$, $\tau > 0$, and let $0 < \rho_m \le 1$ be deterministic with

$$\rho_m \ge e \, d_m^{-\tau} \qquad \text{eventually,} \qquad \sum_{m=1}^{\infty} d_m^{-q} < \infty.$$

Then the preceding conclusion holds with ρ replaced by ρ_m: almost surely all but finitely many sampled batches have at least a $1 - \rho_m$ fraction of their signs as Toeplitz/Hankel operator-norm robust centers, with constants depending only on K, a, R, A_0, q, τ.

Third, let $\theta \ge 0$ and $q > 0$ satisfy $\sum_m d_m^{-q} < \infty$. If $M_m \le d_m^{\theta}$ eventually, then almost surely every sampled sign in every sufficiently large batch is such a robust center, with constants depending only on K, a, R, A_0, θ, q.

The same three conclusions hold if the displayed Markov matrix-budget summability is replaced by the Bernstein matrix budget in Corollary 5.109. They also hold under the inverse-p sufficient budget obtained by replacing $\mathcal{V}_m^{(2)}(a)$ *with* $\ell_m^{(2-p)_+/2}\mathcal{I}_m^{(p)}(a)$.

Proof. For the fixed-fraction assertion set

$$r_m = \lceil \rho M_m \rceil$$

in Corollary 5.109 and use sign exponent $2q$. Since

$$\binom{M_m}{r_m} \le \left(\frac{eM_m}{r_m}\right)^{r_m} \le \left(\frac{e}{\rho}\right)^{r_m},$$

we have, for all sufficiently large m,

$$\binom{M_m}{r_m} d_m^{-2qr_m} \le d_m^{-qr_m} \le d_m^{-q}.$$

The assumed matrix summability and $\sum_m d_m^{-q} < \infty$ therefore give the summability hypothesis of Corollary 5.109. Its conclusion gives at least $M_m - r_m + 1 \ge (1-\rho)M_m$ robust sampled signs.

For the vanishing-fraction assertion set

$$r_m = \lceil \rho_m M_m \rceil$$

and use sign exponent $q+\tau$. The eventual lower bound $\rho_m \ge e d_m^{-\tau}$ gives

$$\binom{M_m}{r_m} \le \left(\frac{e}{\rho_m}\right)^{r_m} \le d_m^{\tau r_m},$$

and hence

$$\binom{M_m}{r_m} d_m^{-(q+\tau)r_m} \le d_m^{-qr_m} \le d_m^{-q}.$$

Again the schedule corollary applies, and $M_m - r_m + 1 \ge (1-\rho_m)M_m$.

For the polynomial-batch assertion take $r_m = 1$ and sign exponent $\theta + q$. Then

$$\binom{M_m}{1} d_m^{-(\theta+q)} \le d_m^{-q}$$

eventually, so the schedule corollary says that $M_m - r_m + 1 = M_m$ sampled signs are robust centers.

The Bernstein and inverse-p variants are identical, using the corresponding matrix-budget replacements already built into Corollary 5.109. □

Corollary 5.111 (Heterogeneous inverse-moment sampled-list operator maxima)**.** *In the setting and notation of Corollary 5.109, write*

$$v_{m,i}(\omega) := \frac{P_{m,i}(\omega)}{\|B_{m,i}\|_{\mathrm{op}}}, \qquad \omega^{d_m} = 1.$$

Denote the sampled sign vectors by

$$\delta_m^{(1)}, \dots, \delta_m^{(M_m)} \in \{-1, 1\}^{n_m}.$$

Let $u_m \geq 0$, $S > 0$, and $1 \leq r_m \leq M_m$ be deterministic. If, for some $A_0 > 0$ and $R \geq 1$,

$$\sum_{m=1}^{\infty} \left[d_m^{-A_0} + \frac{\ell_m \, \mathcal{V}_m^{(2)}(a)}{R n_m} + e^{-u_m} + \binom{M_m}{r_m} d_m^{-S r_m} \right] < \infty,$$

then, almost surely over the matrices and sampled signs, all but finitely many m satisfy the following three conclusions.

First, the whole sampled list has entropy-sensitive edge maximum

$$\max_{1 \leq s \leq M_m} \max_{\omega^{d_m}=1} \left| \sum_{i=1}^{n_m} \delta_{m,i}^{(s)} v_{m,i}(\omega) \right| \leq C \sqrt{n_m \left(1 + \frac{\log(eM_m) + u_m}{\ell_m} \right)}.$$

Second, the same sampled signs have the full Toeplitz/Hankel operator-norm maximum

$$\max_{1 \leq s \leq M_m} \left\| \sum_{i=1}^{n_m} \delta_{m,i}^{(s)} A_{m,i} \right\|_{\mathrm{op}} \leq C \sqrt{n_m \left(1 + \frac{\log(eM_m) + u_m}{\ell_m} \right)},$$

$$\max_{1 \leq s \leq M_m} \left\| \sum_{i=1}^{n_m} \delta_{m,i}^{(s)} \widetilde{A}_{m,i} \right\|_{\mathrm{op}} \leq C \sqrt{n_m \left(1 + \frac{\log(eM_m) + u_m}{\ell_m} \right)}.$$

Third, at least $M_m - r_m + 1$ sampled signs are centers of Hamming balls of radius $\lfloor c\ell_m \rfloor$ on which every sign vector ε' satisfies the $O(\sqrt{n_m})$ Toeplitz and Hankel operator-norm bounds from Corollary 5.109. The constants depend only on K, a, R, S, A_0.

In particular, if the matrix budget

$$\sum_{m} \left[d_m^{-A_0} + \frac{\ell_m \, \mathcal{V}_m^{(2)}(a)}{R n_m} \right]$$

is finite and $\sum_m d_m^{-q} < \infty$, then taking $u_m = q\ell_m$ gives the same whole-list operator maximum with

$$1 + \frac{\log(eM_m) + u_m}{\ell_m} \leq C_q \left(1 + \frac{\log(eM_m)}{\ell_m} \right).$$

Under this same summability, every polynomial batch $M_m \leq d_m^{\theta}$ is eventually entirely robust and has whole-list Toeplitz/Hankel operator maximum $O_{\theta,q}(\sqrt{n_m})$.

The Markov matrix-budget term may be replaced by the Bernstein matrix budget in Corollary 5.109, and the inverse-p sufficient budget is obtained by replacing $\mathcal{V}_m^{(2)}(a)$ with $\ell_m^{(2-p)_+/2} \mathcal{I}_m^{(p)}(a)$.

Proof. Corollary 5.107, applied at level m, gives an edge-profile matrix event whose failure probability is bounded by the displayed Markov matrix terms, and on which

$$\max_{\omega^{d_m}=1} \sum_{i=1}^{n_m} |v_{m,i}(\omega)|^2 \leq C_0 \frac{n_m}{\ell_m}.$$

Condition on such a matrix realization. Corollary 5.6 with parameter u_m gives the displayed whole-list edge maximum except with conditional probability at most e^{-u_m}. On the same matrix realization, Corollary 5.10, with exponent S, gives at least $M_m - r_m + 1$ logarithmic-radius robust centers except with conditional probability at most $\binom{M_m}{r_m} d_m^{-Sr_m}$. The robust Hamming balls transfer to Toeplitz and Hankel operator norm by Corollary 5.2.

It remains only to transfer the whole-list maximum for the sampled signs themselves. The deterministic proof of Proposition 5.1 gives, uniformly in each sampled sign $\delta_m^{(s)}$,

$$\left\| \sum_i \delta_{m,i}^{(s)} A_{m,i} \right\|_{\mathrm{op}} \le C_{\mathrm{red}} \max_{\omega^{d_m}=1} \left| \sum_i \delta_{m,i}^{(s)} v_{m,i}(\omega) \right| + E_m,$$

where $E_m = o(\sqrt{n_m})$. The entropy-sensitive right side is always at least a constant multiple of $\sqrt{n_m}$, so E_m is absorbed after increasing C and discarding finitely many m. The Hankel estimate is the same because multiplication by J_{d_m} preserves operator norm and converts Toeplitz matrices to Hankel matrices.

The displayed summability and Borel–Cantelli give the Markov branch. For the polynomial-batch consequence, take $r_m = 1$, $u_m = q\ell_m$, and $S = \theta + q$; then

$$e^{-u_m} \le d_m^{-q}, \qquad \binom{M_m}{1} d_m^{-S} \le d_m^{-q}$$

eventually, while the entropy factor is bounded by a constant depending on θ, q. The Bernstein and inverse-p variants are the corresponding matrix-budget replacements from Corollary 5.109. □

Corollary 5.112 (Concrete heterogeneous batch-maximal operator schedules)**.** *In the setting and notation of Corollary 5.111, put*

$$\mathfrak{b}_m(a, R) := \frac{\ell_m \, \mathcal{V}_m^{(2)}(a)}{R n_m}.$$

Assume one of the following Markov-form matrix schedules.

(i) $d_m = m$, $A_0 > 1$, *and*

$$\mathfrak{b}_m(a, R) \le C_0 m^{-B} \qquad \text{eventually for some } B > 1.$$

(ii) $d_m \ge c_0 m^{\kappa}$ *for some* $c_0, \kappa > 0$,

$A_0\kappa > 1$, $\qquad \mathfrak{b}_m(a, R) \le C_0 d_m^{-B} \qquad$ *eventually for some* B *with* $B\kappa > 1$.

(iii) $d_m \ge \exp(c_0 m^{\kappa})$ *for some* $c_0, \kappa > 0$, $A_0 > 0$, *and*

$$\mathfrak{b}_m(a, R) \le C_0 \ell_m^{-\eta} \qquad \text{eventually for some } \eta > 1/\kappa.$$

Equivalently, the last display follows from $\mathcal{V}_m^{(2)}(a)/n_m \le C_1 \ell_m^{-B}$ *with* $B > 1 + 1/\kappa$.

Let $q > 0$ be chosen so that

$$\sum_m d_m^{-q} < \infty,$$

that is, $q > 1$ in (i)*, $q\kappa > 1$ in* (ii)*, and any $q > 0$ in* (iii)*. Set $u_m = q\ell_m$.*

Then all the following conclusions hold almost surely over the matrices and sampled signs. First, for any deterministic sampled batch sizes M_m, any fixed $0 < \rho \le 1$, and $r_m = \lceil \rho M_m \rceil$, all but finitely many m have at least a $1 - \rho$ fraction of sampled signs as Toeplitz/Hankel operator-norm robust centers, and the whole sampled list has the entropy-sensitive edge and operator-norm maximum from Corollary 5.111.

Second, for arbitrary M_m, any $\tau > 0$, and deterministic $\rho_m \in (0, 1]$ satisfying

$$\rho_m \ge e\, d_m^{-\tau} \qquad \text{eventually,}$$

the same whole-list maximum holds and at least a $1 - \rho_m$ fraction of sampled signs are robust centers eventually.

Third, if $M_m \le d_m^{\theta}$ eventually for some $\theta \ge 0$, then every sampled sign in every sufficiently large batch is a Toeplitz/Hankel operator-norm robust center and

$$\max_{1 \le s \le M_m} \left\| \sum_{i=1}^{n_m} \delta_{m,i}^{(s)} A_{m,i} \right\|_{\mathrm{op}} + \max_{1 \le s \le M_m} \left\| \sum_{i=1}^{n_m} \delta_{m,i}^{(s)} \widetilde{A}_{m,i} \right\|_{\mathrm{op}} \le C_{\theta,q} \sqrt{n_m}$$

for all sufficiently large m.

The same conclusions hold in the Bernstein branch. Namely, suppose that for some $0 < \theta_0 < R$,

$$C_* \mathcal{V}_m^{(2)}(a) \le \theta_0 \frac{n_m}{\ell_m} \qquad \text{eventually,}$$

where C_* is the absolute constant from Corollary 5.107, and that

$$\sum_m \left[d_m^{-A_0} + \exp\left(-c_{K,a,R,\theta_0} \frac{n_m}{\ell_m^2} \right) \right] < \infty.$$

Then the same three sampled-batch conclusions hold after adding the same summable sign-tail power $\sum_m d_m^{-q}$. In particular this Bernstein matrix summability is guaranteed along all integer dimensions if $A_0 > 1$ and

$$n(d) \ge Q \log^3(e + d) \quad \text{eventually with} \quad Q > 1/c_{K,a,R,\theta_0},$$

along polynomial schedules $d_m \ge c_0 m^{\kappa}$ if $A_0 \kappa > 1$ and

$$n_m \ge Q \ell_m^2 \log(e + m) \quad \text{eventually with} \quad Q > 1/c_{K,a,R,\theta_0},$$

and along stretched-exponential schedules $d_m \ge \exp(c_0 m^{\kappa})$ under the same $\ell_m^2 \log(e + m)$ lower bound, or under the simpler condition

$$n_m \ge Q \ell_m^{2+\tau} \qquad \text{eventually}$$

for any $Q, \tau > 0$.

Proof. In each of the three Markov schedules,

$$\sum_m d_m^{-A_0} < \infty, \qquad \sum_m \mathfrak{b}_m(a,R) < \infty.$$

For (i) this is the pair of p-series assumptions $A_0 > 1$ and $B > 1$. For (ii), $d_m^{-A_0} \le Cm^{-A_0\kappa}$ and $\mathfrak{b}_m(a,R) \le Cm^{-B\kappa}$. For (iii), $\sum_m d_m^{-A_0} < \infty$, while

$$\mathfrak{b}_m(a,R) \le C\ell_m^{-\eta} \le C'm^{-\kappa\eta},$$

which is summable because $\kappa\eta > 1$. The displayed average inverse-moment condition implies $\mathfrak{b}_m(a,R) \le C\ell_m^{1-B}$, so it is the special case $\eta = B-1$.

Now $u_m = q\ell_m$ gives $e^{-u_m} \le d_m^{-q}$, up to changing finitely many terms. For fixed ρ, take $S = 2q$ and $r_m = \lceil \rho M_m \rceil$. The standard binomial estimate gives

$$\binom{M_m}{r_m} d_m^{-Sr_m} \le d_m^{-q}$$

for all large m. For the vanishing-fraction assertion, take $r_m = \lceil \rho_m M_m \rceil$ and $S = q + \tau$. Since $\rho_m \ge ed_m^{-\tau}$,

$$\binom{M_m}{r_m} d_m^{-Sr_m} \le d_m^{-q}.$$

For polynomial batches, take $r_m = 1$ and $S = \theta + q$, so

$$\binom{M_m}{1} d_m^{-S} \le d_m^{-q}.$$

Thus in all three cases the summability hypothesis of Corollary 5.111 holds. In the polynomial-batch case,

$$\frac{\log(eM_m) + u_m}{\ell_m} = O_{\theta,q}(1),$$

which gives the displayed $O(\sqrt{n_m})$ operator maximum.

The Bernstein statement is the same invocation of Corollary 5.111 with its Bernstein matrix-budget replacement. The all-integer lower bound gives

$$\exp\left(-c\frac{n(d)}{\log^2(e+d)}\right) \le (e+d)^{-cQ},$$

which is summable when $Q > 1/c$. The polynomial-schedule lower bound gives

$$\exp\left(-c\frac{n_m}{\ell_m^2}\right) \le (e+m)^{-cQ},$$

and $Q > 1/c$ makes this summable; $A_0\kappa > 1$ handles the $d_m^{-A_0}$ term. On stretched-exponential schedules the same $\ell_m^2 \log(e+m)$ lower bound gives the same summable power of m, while $n_m \ge Q\ell_m^{2+\tau}$ gives

$$\exp\left(-cQ\ell_m^{\tau}\right) \le \exp(-c'm^{\kappa\tau}),$$

which is summable. □

Corollary 5.113 (Concrete heterogeneous inverse-p batch-maximal schedules)**.** *In the setting and notation of Corollary 5.111, fix $p > 0$ and set*

$$\sigma_p := 1 + \frac{(2-p)_+}{2}, \qquad \mathfrak{i}_m^{(p)}(a,R) := \frac{\ell_m^{\sigma_p}\mathcal{I}_m^{(p)}(a)}{Rn_m},$$

where $\mathcal{I}_m^{(p)}(a)$ is the aggregate inverse-p lower-normalizer mass from Corollary 5.109. Assume one of the following inverse-p Markov schedules.

(i) *$d_m = m$, $A_0 > 1$, and*

$$\mathfrak{i}_m^{(p)}(a,R) \le C_0 m^{-B} \qquad \text{eventually for some } B > 1.$$

It is enough, more concretely, that

$$\frac{\mathcal{I}_m^{(p)}(a)}{n_m} \le C_1 m^{-B} \qquad \text{eventually for some } B > 1.$$

(ii) *$d_m \ge c_0 m^{\kappa}$ for some $c_0, \kappa > 0$,*

$A_0\kappa > 1$, $\mathfrak{i}_m^{(p)}(a,R) \le C_0 d_m^{-B}$ *eventually for some B with $B\kappa > 1$.*

It is enough that

$$\frac{\mathcal{I}_m^{(p)}(a)}{n_m} \le C_1 d_m^{-B} \qquad \text{eventually for some } B \text{ with } B\kappa > 1.$$

(iii) *$d_m \ge \exp(c_0 m^{\kappa})$ for some $c_0, \kappa > 0$, $A_0 > 0$, and*

$$\mathfrak{i}_m^{(p)}(a,R) \le C_0 \ell_m^{-\eta} \qquad \text{eventually for some } \eta > 1/\kappa.$$

It is enough that

$$\frac{\mathcal{I}_m^{(p)}(a)}{n_m} \le C_1 \ell_m^{-B} \qquad \text{eventually for some } B > \sigma_p + 1/\kappa.$$

Let $q > 0$ satisfy $\sum_m d_m^{-q} < \infty$, equivalently $q > 1$ in (i)*, $q\kappa > 1$ in* (ii)*, and any $q > 0$ in* (iii)*. Set $u_m = q\ell_m$. Then the three sampled-batch conclusions of Corollary 5.112 hold: arbitrary sampled batches have eventual fixed robust fractions, arbitrary sampled batches have eventual vanishing discarded fractions whenever $\rho_m \ge e d_m^{-\tau}$, and polynomial sampled batches $M_m \le d_m^{\theta}$ are eventually entirely Toeplitz/Hankel operator-norm robust with whole-list operator maximum $O_{\theta,q}(\sqrt{n_m})$.*

Proof. The inverse-p replacement in Corollary 5.111 makes $\mathfrak{i}_m^{(p)}(a,R)$ the matrix term in the Markov summability condition. Hence it is enough to verify

$$\sum_m d_m^{-A_0} < \infty, \qquad \sum_m \mathfrak{i}_m^{(p)}(a,R) < \infty.$$

The displayed $\mathfrak{i}_m^{(p)}$-assumptions give this exactly as in Corollary 5.112. The concrete average inverse-p assumptions also give it: on all integer dimensions, $\mathfrak{i}_m^{(p)}(a,R) \le C\ell_m^{\sigma_p} m^{-B}$, which is summable for $B > 1$; on polynomial

schedules, $\mathfrak{i}_m^{(p)}(a,R) \le C\ell_m^{\sigma_p} m^{-B\kappa}$, which is summable for $B\kappa > 1$; and on stretched-exponential schedules,

$$\mathfrak{i}_m^{(p)}(a,R) \le C\ell_m^{\sigma_p - B} \le C' m^{-\kappa(B-\sigma_p)},$$

which is summable when $B > \sigma_p + 1/\kappa$.

The sign-tail choices are identical to the proof of Corollary 5.112: with $u_m = q\ell_m$, $e^{-u_m} \le d_m^{-q}$ up to finitely many terms; use $S = 2q$ and $r_m = \lceil \rho M_m \rceil$ for fixed fractions, $S = q + \tau$ and $r_m = \lceil \rho_m M_m \rceil$ for vanishing fractions, and $S = \theta + q$, $r_m = 1$, for polynomial batches. The summability hypothesis of Corollary 5.111 follows in each case, and the polynomial-batch entropy factor is $O_{\theta,q}(1)$. □

Corollary 5.114 (Random-environment inverse-p batch-maximal schedules)**.** *Let (d_m, n_m) be deterministic and satisfy*

$$n_m \le d_m, \qquad d_m/n_m \to \infty, \qquad \log d_m = o(n_m),$$

and put $\ell_m = \log(e + d_m)$. Let $\mathcal{G}$ be an environment σ-field. For each m, i, let the coefficient law $\mu_{m,i}$ be $\mathcal{G}$-measurable; assume almost surely that all selected laws are centered, variance one, and have sub-Gaussian norm at most K. Conditional on $\mathcal{G}$, draw the Toeplitz matrices

$$B_{m,i} = T_{d_m}(\xi^{(m,i)}), \qquad 1 \le i \le n_m,$$

independently from the selected laws, and, for deterministic $M_m \ge 1$, draw the sampled sign lists independently of the matrices. Define $N_{m,i}$, $X_{m,i}$, and $\mathcal{I}_m^{(p)}(a)$ conditionally from the selected laws as in Corollary 5.109.

Fix $a > 0$, $R \ge 1$, and $p > 0$, and set

$$\sigma_p := 1 + \frac{(2-p)_+}{2}.$$

Suppose there are deterministic $\alpha_m \ge 0$ such that

$$\sum_m \mathbb{P}\left\{ \frac{\mathcal{I}_m^{(p)}(a)}{n_m} > \alpha_m \right\} < \infty.$$

Assume, in addition, one of the following deterministic schedules:

(i) *$d_m = m$, $A_0 > 1$, and*

$$\alpha_m \le C_0 m^{-B} \qquad \text{eventually for some } B > 1.$$

(ii) *$d_m \ge c_0 m^\kappa$ for some $c_0, \kappa > 0$,*

$$A_0\kappa > 1, \qquad \alpha_m \le C_0 d_m^{-B} \qquad \text{eventually for some } B \text{ with } B\kappa > 1.$$

(iii) *$d_m \ge \exp(c_0 m^\kappa)$ for some $c_0, \kappa > 0$, $A_0 > 0$, and*

$$\alpha_m \le C_0 \ell_m^{-B} \qquad \text{eventually for some } B > \sigma_p + 1/\kappa.$$

Then, for every $q > 0$ with $\sum_m d_m^{-q} < \infty$ and with $u_m = q\ell_m$, the three sampled-batch conclusions of Corollary 5.113 hold almost surely over the joint environment, matrix, and sampled-sign randomness. Thus random law

selection and random sparse contamination are allowed whenever their empirical aggregate inverse-p lower-normalizer budget misses the corresponding deterministic schedule only on a summable set of dimensions.

Proof. By Borel–Cantelli,

$$\frac{\mathcal{I}_m^{(p)}(a)}{n_m} \le \alpha_m$$

for all but finitely many m, on an environment event of probability one. Fix an environment in this full-measure event. The selected coefficient laws are now deterministic, uniformly centered, variance one, and K-sub-Gaussian, and the displayed deterministic bounds on α_m put the environment in one of the three average inverse-p schedules of Corollary 5.113. That corollary therefore gives the stated fixed-fraction, vanishing-fraction, and polynomial-batch conclusions with conditional probability one over the matrices and sampled signs. Integrating these conditional probability-one statements over the full-measure environment event gives the joint almost-sure conclusion. □

Corollary 5.115 (Expectation criteria for random-environment inverse-p schedules)**.** *In the setting and notation of Corollary 5.114, put*

$$\beta_m := \mathbb{E}\left[\frac{\mathcal{I}_m^{(p)}(a)}{n_m}\right], \qquad \sigma_p := 1 + \frac{(2-p)_+}{2}.$$

Assume one of the following expectation schedules.

(i) $d_m = m$, $A_0 > 1$, *and*

$$\sum_m m^B \beta_m < \infty \qquad \text{for some } B > 1.$$

In particular, $\beta_m = O(m^{-B_*})$ *with* $B_* > 2$ *suffices.*

(ii) $d_m \ge c_0 m^\kappa$ *for some* $c_0, \kappa > 0$, $A_0\kappa > 1$, *and*

$$\sum_m d_m^B \beta_m < \infty \qquad \text{for some } B \text{ with } B\kappa > 1.$$

In particular, $\beta_m = O(d_m^{-B_*})$ *with* $B_*\kappa > 2$ *suffices.*

(iii) $d_m \ge \exp(c_0 m^\kappa)$ *for some* $c_0, \kappa > 0$, $A_0 > 0$, *and*

$$\sum_m \ell_m^B \beta_m < \infty \qquad \text{for some } B > \sigma_p + 1/\kappa.$$

In particular, $\beta_m = O(\ell_m^{-B_*})$ *with*

$$B_* > \sigma_p + 2/\kappa$$

suffices.

Then, for every $q > 0$ *with* $\sum_m d_m^{-q} < \infty$ *and with* $u_m = q\ell_m$, *the fixed-fraction, vanishing-fraction, and all-robust polynomial-batch conclusions of Corollary 5.114 hold almost surely.*

More concretely, suppose the environment selects, for each m, i, *a random catalog index* $\Lambda_{m,i}$ *in a countable set* $\mathcal{J}_{m,i}$, *possibly with arbitrary dependence*

among all indices, and then sets $\mu_{m,i} = \mu_{m,i,\Lambda_{m,i}}$. *Assume every catalog law is centered, variance one, and* K*-sub-Gaussian, and write*

$$\pi_{m,i,j} := \mathbb{P}\{\Lambda_{m,i} = j\}.$$

For

$$\iota^{(p)}_{m,i,j}(a) := \mathbb{E}_{\mu_{m,i,j}}\left[\mathbf{1}_{\{X_{m,i,j}<aN_{m,i,j}\}}\left(\frac{aN_{m,i,j}}{X_{m,i,j}}\right)^p\right],$$

where $N_{m,i,j} = \mathbb{E}_{\mu_{m,i,j}} X_{m,i,j}$ *and* $X_{m,i,j} = \|T_{d_m}(\mu_{m,i,j})\|_{\mathrm{op}}$ *for an independent draw from* $\mu_{m,i,j}$, *one has*

$$\beta_m = \frac{1}{n_m}\sum_{i=1}^{n_m}\sum_{j\in\mathcal{J}_{m,i}} \pi_{m,i,j}\,\iota^{(p)}_{m,i,j}(a).$$

Thus the preceding tests use only marginal catalog weights; no independence among the environment selections is required.

Proof. For the first assertion set $\alpha_m = m^{-B}$, d_m^{-B}, or ℓ_m^{-B} in the three cases. Markov's inequality gives

$$\mathbb{P}\left\{\frac{\mathcal{I}^{(p)}_m(a)}{n_m} > \alpha_m\right\} \le \frac{\beta_m}{\alpha_m},$$

so the displayed expectation summability hypotheses imply the summable environment-tail hypothesis of Corollary 5.114. The deterministic bounds on α_m are exactly the three schedules in that corollary, and the stated sampled-batch conclusions follow.

The power-decay sufficient conditions are obtained by leaving one summable power for Markov. On all integer dimensions choose $1 < B < B_* - 1$. On polynomial schedules choose

$$1/\kappa < B < B_* - 1/\kappa,$$

which is possible when $B_*\kappa > 2$. On stretched-exponential schedules choose

$$\sigma_p + 1/\kappa < B < B_* - 1/\kappa,$$

which is possible when $B_* > \sigma_p + 2/\kappa$. Since $\ell_m \ge cm^\kappa$ in the last case, each displayed expectation series is finite.

For the catalog formula, conditional on a realization of the selected indices, $\mathcal{I}^{(p)}_m(a) = \sum_i \iota^{(p)}_{m,i,\Lambda_{m,i}}(a)$. Taking expectations and using nonnegativity gives the displayed identity by linearity and monotone convergence. This uses only the one-dimensional marginal probabilities $\pi_{m,i,j}$. □

Corollary 5.116 (Bernstein criteria for independent random-catalog schedules)**.** *Work in the countable catalog setting of Corollary 5.115. For each* m, *assume that the catalog indices* $\Lambda_{m,1},\dots,\Lambda_{m,n_m}$ *are independent, and put*

$$Z_{m,i} := \iota^{(p)}_{m,i,\Lambda_{m,i}}(a), \qquad \beta_m := \frac{1}{n_m}\sum_{i=1}^{n_m}\mathbb{E}Z_{m,i}.$$

Suppose that for deterministic $L_m > 0$ and $\alpha_m > 0$,

$$0 \le Z_{m,i} \le L_m \quad \text{almost surely for all } i, \qquad \beta_m \le \frac{\alpha_m}{2} \qquad \text{eventually,}$$

and

$$\sum_m \exp\left(-c_{\mathrm{B}} \frac{n_m \alpha_m}{L_m}\right) < \infty$$

for a sufficiently small absolute constant $c_{\mathrm{B}} > 0$. If, in addition, α_m satisfies one of the three deterministic inverse-p schedules in Corollary 5.114, then, for every $q > 0$ with $\sum_m d_m^{-q} < \infty$ and $u_m = q\ell_m$, the fixed-fraction, vanishing-fraction, and all-robust polynomial-batch conclusions of that corollary hold almost surely.

In particular, for polynomial schedules $d_m \ge c_0 m^{\kappa}$, it is enough to have, for some B with $B\kappa > 1$,

$$\beta_m \le c_1 d_m^{-B}, \qquad n_m \ge Q L_m d_m^B \log(e+m)$$

eventually, with Q sufficiently large. For stretched-exponential schedules $d_m \ge \exp(c_0 m^{\kappa})$, it is enough to have, for some $B > 1 + (2-p)_+/2 + 1/\kappa$,

$$\beta_m \le c_1 \ell_m^{-B}, \qquad n_m \ge Q L_m \ell_m^B \log(e+m)$$

eventually, again with Q sufficiently large. The same conclusion also holds if the logarithmic lower bound is replaced by any stronger condition making $\sum_m \exp(-c_{\mathrm{B}} n_m \alpha_m / L_m)$ finite.

Proof. For each m, the selected environment budget is

$$\frac{\mathcal{I}_m^{(p)}(a)}{n_m} = \frac{1}{n_m} \sum_{i=1}^{n_m} Z_{m,i}.$$

The variables $Z_{m,i}$ are independent, nonnegative, and bounded by L_m. Their total mean is $n_m \beta_m$, and

$$\sum_i \operatorname{Var} Z_{m,i} \le \sum_i \mathbb{E} Z_{m,i}^2 \le L_m \sum_i \mathbb{E} Z_{m,i} = L_m n_m \beta_m.$$

If $\beta_m \le \alpha_m/2$, then Bernstein's inequality gives

$$\begin{aligned} \mathbb{P}\left\{ \frac{1}{n_m} \sum_i Z_{m,i} > \alpha_m \right\} &\le \exp\left(-\frac{n_m^2 (\alpha_m - \beta_m)^2}{2 L_m n_m \beta_m + \frac{2}{3} L_m n_m (\alpha_m - \beta_m)} \right) \\ &\le \exp\left(-c_{\mathrm{B}} \frac{n_m \alpha_m}{L_m} \right) \end{aligned}$$

after reducing c_{B}, if necessary. The assumed exponential series is therefore exactly the summable environment-tail hypothesis of Corollary 5.114. Applying that corollary with the given deterministic schedule for α_m proves the first assertion.

For the polynomial and stretched-exponential displays, take $\alpha_m = 2c_1 d_m^{-B}$ or $\alpha_m = 2c_1 \ell_m^{-B}$, respectively, enlarging constants harmlessly. The lower bounds on n_m make the Bernstein tail at most a summable power of $e+m$,

while the displayed exponents are exactly the deterministic inverse-p schedule thresholds. □

Corollary 5.117 (Freedman criteria for predictable random-catalog schedules)**.** *Work in the countable catalog setting of Corollary 5.115. For each m, let*

$$\mathcal{F}_{m,0} \subset \mathcal{F}_{m,1} \subset \cdots \subset \mathcal{F}_{m,n_m}$$

be a filtration such that $\Lambda_{m,i}$ is $\mathcal{F}_{m,i}$-measurable, and put

$$Z_{m,i} := \iota^{(p)}_{m,i,\Lambda_{m,i}}(a).$$

Assume that for deterministic $L_m > 0$, $\alpha_m > 0$, and deterministic envelopes $b_{m,i} \geq 0$,

$$0 \leq Z_{m,i} \leq L_m \quad \text{almost surely,} \qquad \mathbb{E}[Z_{m,i} \mid \mathcal{F}_{m,i-1}] \leq b_{m,i}$$

for all i, and

$$\frac{1}{n_m}\sum_{i=1}^{n_m} b_{m,i} \leq \frac{\alpha_m}{2} \qquad \text{eventually.}$$

If

$$\sum_m \exp\left(-c_{\mathrm{F}}\frac{n_m\alpha_m}{L_m}\right) < \infty$$

for a sufficiently small absolute constant $c_{\mathrm{F}} > 0$, and if α_m satisfies one of the three deterministic inverse-p schedules in Corollary 5.114, then, for every $q > 0$ with $\sum_m d_m^{-q} < \infty$ and $u_m = q\ell_m$, the fixed-fraction, vanishing-fraction, and all-robust polynomial-batch conclusions of that corollary hold almost surely.

In particular, the concrete polynomial and stretched-exponential schedule displays of Corollary 5.116 remain valid with β_m replaced by

$$\bar{\beta}_m := \frac{1}{n_m}\sum_{i=1}^{n_m} b_{m,i}.$$

Thus adaptive, sequential, or without-replacement law-catalog exposure is covered whenever its one-step predictable inverse-p budget obeys the same bounded-catalog schedule.

Proof. Let

$$\mu_{m,i} := \mathbb{E}[Z_{m,i} \mid \mathcal{F}_{m,i-1}], \qquad D_{m,i} := Z_{m,i} - \mu_{m,i}.$$

Then $D_{m,i}$ is a martingale difference and $D_{m,i} \leq L_m$. Its predictable quadratic variation satisfies

$$\sum_i \mathbb{E}[D_{m,i}^2 \mid \mathcal{F}_{m,i-1}] \leq \sum_i \mathbb{E}[Z_{m,i}^2 \mid \mathcal{F}_{m,i-1}] \leq L_m \sum_i \mu_{m,i} \leq \frac{1}{2}L_m n_m \alpha_m$$

eventually. On the same event, if

$$\frac{1}{n_m}\sum_i Z_{m,i} > \alpha_m,$$

then

$$\sum_i D_{m,i} > n_m\alpha_m - \sum_i \mu_{m,i} \geq \frac{1}{2} n_m \alpha_m.$$

Freedman's inequality therefore gives

$$\mathbb{P}\left\{ \frac{1}{n_m} \sum_i Z_{m,i} > \alpha_m \right\} \leq \exp\left(-c_{\mathrm{F}} \frac{n_m \alpha_m}{L_m} \right)$$

after reducing c_{F}, if necessary. The assumed exponential series gives the summable environment-tail hypothesis of Corollary 5.114. The deterministic inverse-p schedule for α_m then gives the sampled-batch conclusions. The concrete schedules are the same substitutions as in Corollary 5.116. □

Corollary 5.118 (Truncated Freedman criteria for predictable random-catalog schedules)**.** *Work in the countable catalog setting of Corollary 5.115, with the filtration convention of Corollary 5.117. Put*

$$Z_{m,i} := \iota^{(p)}_{m,i,\Lambda_{m,i}}(a).$$

Assume $Z_{m,i} < \infty$ almost surely. Let $L_m > 0$, $\alpha_m > 0$, and deterministic envelopes $b_{m,i}, r_{m,i} \geq 0$ satisfy

$$\mathbb{E}[Z_{m,i} \wedge L_m \mid \mathcal{F}_{m,i-1}] \leq b_{m,i}, \qquad \mathbb{E}[(Z_{m,i} - L_m)_+ \mid \mathcal{F}_{m,i-1}] \leq r_{m,i}$$

for all i. Write

$$\bar{b}_m := \frac{1}{n_m} \sum_{i=1}^{n_m} b_{m,i}, \qquad \bar{r}_m := \frac{1}{n_m} \sum_{i=1}^{n_m} r_{m,i}.$$

Suppose that $\bar{b}_m \leq \alpha_m/4$ eventually and that

$$\sum_m \left[\exp\left(-c_{\mathrm{TF}} \frac{n_m \alpha_m}{L_m} \right) + \frac{\bar{r}_m}{\alpha_m} \right] < \infty$$

for a sufficiently small absolute constant $c_{\mathrm{TF}} > 0$. If α_m satisfies one of the three deterministic inverse-p schedules in Corollary 5.114, then, for every $q > 0$ with $\sum_m d_m^{-q} < \infty$ and $u_m = q\ell_m$, the fixed-fraction, vanishing-fraction, and all-robust polynomial-batch conclusions of that corollary hold almost surely.

In particular, the concrete polynomial and stretched-exponential displays of Corollary 5.116 remain valid for unbounded predictable catalogs with β_m replaced by the truncated predictable budget $\bar{b}_m$, the half-budget condition replaced by $\bar{b}_m \leq \alpha_m/4$, and the added excess summability $\sum_m \bar{r}_m/\alpha_m < \infty$. Thus, for example, on polynomial schedules $d_m \geq c_0 m^\kappa$, it is enough to have

$$\bar{b}_m \leq c_1 d_m^{-B}, \qquad n_m \geq Q L_m d_m^B \log(e+m), \qquad \sum_m d_m^B \bar{r}_m < \infty$$

for some B with $B\kappa > 1$, and on stretched-exponential schedules $d_m \geq \exp(c_0 m^\kappa)$, it is enough to have

$$\bar{b}_m \leq c_1 \ell_m^{-B}, \qquad n_m \geq Q L_m \ell_m^B \log(e+m), \qquad \sum_m \ell_m^B \bar{r}_m < \infty$$

for some $B > 1 + (2-p)_+/2 + 1/\kappa$, with Q sufficiently large.

Proof. Let

$$W_{m,i} := Z_{m,i} \wedge L_m, \qquad R_{m,i} := (Z_{m,i} - L_m)_+,$$

and put $\mu_{m,i} := \mathbb{E}[W_{m,i} \mid \mathcal{F}_{m,i-1}]$. The martingale differences $D_{m,i} := W_{m,i} - \mu_{m,i}$ satisfy $D_{m,i} \le L_m$, and, eventually,

$$\sum_i \mathbb{E}[D_{m,i}^2 \mid \mathcal{F}_{m,i-1}] \le \sum_i \mathbb{E}[W_{m,i}^2 \mid \mathcal{F}_{m,i-1}] \le L_m \sum_i \mu_{m,i} \le \frac{1}{4} L_m n_m \alpha_m.$$

If $n_m^{-1} \sum_i W_{m,i} > \alpha_m/2$, then

$$\sum_i D_{m,i} > \frac{1}{2} n_m \alpha_m - \sum_i \mu_{m,i} \ge \frac{1}{4} n_m \alpha_m.$$

Freedman's inequality therefore gives

$$\mathbb{P}\left\{ \frac{1}{n_m} \sum_i W_{m,i} > \frac{\alpha_m}{2} \right\} \le \exp\left(-c_{\mathrm{TF}} \frac{n_m \alpha_m}{L_m} \right)$$

after reducing c_{TF}, if necessary. Since $Z_{m,i} = W_{m,i} + R_{m,i}$, Markov's inequality gives

$$\begin{aligned} \mathbb{P}\left\{ \frac{1}{n_m} \sum_i Z_{m,i} > \alpha_m \right\} &\le \mathbb{P}\left\{ \frac{1}{n_m} \sum_i W_{m,i} > \frac{\alpha_m}{2} \right\} \\ &\quad + \mathbb{P}\left\{ \frac{1}{n_m} \sum_i R_{m,i} > \frac{\alpha_m}{2} \right\} \\ &\le \exp\left(-c_{\mathrm{TF}} \frac{n_m \alpha_m}{L_m} \right) + \frac{2\bar{r}_m}{\alpha_m}. \end{aligned}$$

The displayed summability is therefore the summable environment-tail hypothesis of Corollary 5.114. The deterministic inverse-p schedule for α_m gives the sampled batch conclusions. The polynomial and stretched-exponential criteria are the same substitutions as in the bounded Freedman corollary, with the displayed excess series added. □

Corollary 5.119 (Conditional-moment criteria for unbounded predictable catalogs)**.** *Work in the setting of Corollary 5.118. Fix $s > 1$. Assume that, for deterministic $b_{m,i}, h_{m,i} \ge 0$,*

$$\mathbb{E}[Z_{m,i} \mid \mathcal{F}_{m,i-1}] \le b_{m,i}, \qquad \mathbb{E}[Z_{m,i}^s \mid \mathcal{F}_{m,i-1}] \le h_{m,i}$$

for all i, and put

$$\bar{b}_m := \frac{1}{n_m} \sum_{i=1}^{n_m} b_{m,i}, \qquad \bar{h}_m := \frac{1}{n_m} \sum_{i=1}^{n_m} h_{m,i}.$$

Let $L_m > 0$ and $\alpha_m > 0$ be deterministic. Suppose that

$$\bar{b}_m \le \frac{\alpha_m}{4} \qquad \textit{eventually},$$

and that

$$\sum_m \left[\exp\left(-c_{\mathrm{M}}\frac{n_m\alpha_m}{L_m}\right)+\frac{\bar h_m}{L_m^{s-1}\alpha_m}\right]<\infty$$

for a sufficiently small absolute constant $c_{\mathrm{M}}>0$. If α_m satisfies one of the three deterministic inverse-p schedules in Corollary 5.114, then, for every $q>0$ with $\sum_m d_m^{-q}<\infty$ and $u_m=q\ell_m$, the fixed-fraction, vanishing-fraction, and all-robust polynomial-batch conclusions of that corollary hold almost surely.

In concrete schedules, this gives a direct tail-moment substitute for the bounded-catalog hypotheses. Along all integer dimensions $d_m=m$, with $A_0>1$, it is enough to have, for some $B>1$,

$$\bar b_m\le c_1 m^{-B},\qquad n_m\ge QL_m m^B\log(e+m),\qquad \sum_m\frac{m^B\bar h_m}{L_m^{s-1}}<\infty.$$

Along polynomial schedules $d_m\ge c_0 m^\kappa$, with $A_0\kappa>1$, it is enough to have, for some B with $B\kappa>1$,

$$\bar b_m\le c_1 d_m^{-B},\qquad n_m\ge QL_m d_m^B\log(e+m),\qquad \sum_m\frac{d_m^B\bar h_m}{L_m^{s-1}}<\infty.$$

Along stretched-exponential schedules $d_m\ge\exp(c_0m^\kappa)$, it is enough to have, for some $B>1+(2-p)_+/2+1/\kappa$,

$$\bar b_m\le c_1\ell_m^{-B},\qquad n_m\ge QL_m\ell_m^B\log(e+m),\qquad \sum_m\frac{\ell_m^B\bar h_m}{L_m^{s-1}}<\infty,$$

with Q sufficiently large in each display.

Proof. For $W_{m,i}=Z_{m,i}\wedge L_m$, the first moment assumption gives

$$\mathbb{E}[W_{m,i}\mid\mathcal{F}_{m,i-1}]\le\mathbb{E}[Z_{m,i}\mid\mathcal{F}_{m,i-1}]\le b_{m,i}.$$

Also, for $z\ge0$,

$$(z-L_m)_+\le L_m^{1-s}z^s.$$

Hence

$$\mathbb{E}[(Z_{m,i}-L_m)_+\mid\mathcal{F}_{m,i-1}]\le L_m^{1-s}h_{m,i}.$$

Corollary 5.118 therefore applies with excess envelope $r_{m,i}=L_m^{1-s}h_{m,i}$. Its summability condition is precisely the displayed exponential series plus $\sum_m\bar h_m/(L_m^{s-1}\alpha_m)<\infty$, and it gives the sampled-batch conclusions.

The concrete schedule displays follow by taking $\alpha_m=4c_1m^{-B}$, $4c_1d_m^{-B}$, or $4c_1\ell_m^{-B}$, respectively, and increasing Q, if necessary, so that the exponential Freedman series is summable. The remaining displayed series is exactly the excess-tail series in the three cases. □

Corollary 5.120 (Automatic truncation for moment-predictable catalogs)**.** *Work in the setting of Corollary 5.119. Let $\lambda_m>0$ be deterministic and suppose that*

$$\sum_m\exp(-c_{\mathrm{AT}}\lambda_m)<\infty$$

for a sufficiently small absolute constant $c_{\mathrm{AT}} > 0$. Put

$$L_m := \frac{n_m \alpha_m}{\lambda_m}.$$

If

$$\bar{b}_m \le \frac{\alpha_m}{4} \qquad \text{eventually,} \qquad \sum_m \frac{\lambda_m^{s-1} \bar{h}_m}{n_m^{s-1} \alpha_m^s} < \infty,$$

and if α_m satisfies one of the three deterministic inverse-p schedules in Corollary 5.114, then, for every $q > 0$ with $\sum_m d_m^{-q} < \infty$ and $u_m = q\ell_m$, the fixed-fraction, vanishing-fraction, and all-robust polynomial-batch conclusions of that corollary hold almost surely.

In particular, choose $\lambda_m = A \log(e+m)$ with A sufficiently large. Then the same conclusions hold in each of the following concrete regimes.

(i) *$d_m = m$, $A_0 > 1$, $\bar{b}_m \le c_1 m^{-B}$ for some $B > 1$, and*

$$\sum_m \frac{m^{Bs} (\log(e+m))^{s-1} \bar{h}_m}{n_m^{s-1}} < \infty.$$

(ii) *$d_m \ge c_0 m^{\kappa}$, $A_0 \kappa > 1$, $\bar{b}_m \le c_1 d_m^{-B}$ for some B with $B\kappa > 1$, and*

$$\sum_m \frac{d_m^{Bs} (\log(e+m))^{s-1} \bar{h}_m}{n_m^{s-1}} < \infty.$$

(iii) *$d_m \ge \exp(c_0 m^{\kappa})$, $A_0 > 0$, $\bar{b}_m \le c_1 \ell_m^{-B}$ for some $B > 1 + (2-p)_+/2 + 1/\kappa$, and*

$$\sum_m \frac{\ell_m^{Bs} (\log(e+m))^{s-1} \bar{h}_m}{n_m^{s-1}} < \infty.$$

Proof. Apply Corollary 5.119 with the displayed L_m. The exponential term is exactly $\exp(-c_{\mathrm{AT}} \lambda_m)$, after reducing c_{AT} if necessary, and

$$\frac{\bar{h}_m}{L_m^{s-1} \alpha_m} = \frac{\lambda_m^{s-1} \bar{h}_m}{n_m^{s-1} \alpha_m^s}.$$

Thus the stated hypotheses are precisely the moment-Freedman summability hypotheses. The sampled-batch conclusions follow from that corollary.

For the concrete regimes, take $\alpha_m = 4c_1 m^{-B}$, $4c_1 d_m^{-B}$, or $4c_1 \ell_m^{-B}$, respectively. A large enough fixed A makes $\sum_m (e+m)^{-c_{\mathrm{AT}} A}$ summable, and the remaining three series are the displayed moment series, up to harmless constants. □

Corollary 5.121 (Active-menu automatic truncation for predictable catalogs)**.** *Fix finite active menu sizes $J_m \ge 1$. For each $1 \le r \le J_m$, work in the predictable countable-catalog setting of Corollary 5.117, with possibly r-dependent catalogs and filtrations, and set*

$$Z_{m,i}^{(r)} := \iota^{(p)}_{m,i,\Lambda_{m,i}^{(r)}}(a).$$

Assume every catalog law in the active menu is centered, variance one, and K-sub-Gaussian. Fix $s > 1$. Suppose that deterministic envelopes $b_{m,i}^{(r)}, h_{m,i}^{(r)} \geq 0$ satisfy

$$\mathbb{E}[Z_{m,i}^{(r)} \mid \mathcal{F}_{m,i-1}^{(r)}] \leq b_{m,i}^{(r)}, \qquad \mathbb{E}[(Z_{m,i}^{(r)})^s \mid \mathcal{F}_{m,i-1}^{(r)}] \leq h_{m,i}^{(r)}$$

for all active m, i, r. Put

$$\bar{b}_m^* := \max_{1 \leq r \leq J_m} \frac{1}{n_m} \sum_{i=1}^{n_m} b_{m,i}^{(r)}, \qquad H_m := \sum_{r=1}^{J_m} \frac{1}{n_m} \sum_{i=1}^{n_m} h_{m,i}^{(r)}.$$

Let $\alpha_m > 0$ satisfy one of the three deterministic inverse-p schedules in Corollary 5.114. If deterministic $\lambda_m > 0$ obey

$$\bar{b}_m^* \leq \frac{\alpha_m}{4} \qquad \textit{eventually}$$

and

$$\sum_m \left[J_m \exp(-c_{\mathrm{AM}} \lambda_m) + \frac{\lambda_m^{s-1} H_m}{n_m^{s-1} \alpha_m^s} \right] < \infty$$

for a sufficiently small absolute constant $c_{\mathrm{AM}} > 0$, then, almost surely over the environment,

$$\max_{1 \leq r \leq J_m} \frac{1}{n_m} \sum_{i=1}^{n_m} Z_{m,i}^{(r)} \leq \alpha_m$$

for all sufficiently large m.

Consequently, if the active menu also satisfies the summability needed to run the matrix and sampled-sign Borel–Cantelli steps uniformly,

$$\sum_m J_m d_m^{-A_0} < \infty, \qquad \sum_m J_m d_m^{-q} < \infty,$$

and $u_m = q\ell_m$, then the fixed-fraction, vanishing-fraction, and all-robust polynomial-batch Toeplitz/Hankel conclusions of Corollary 5.114 hold simultaneously for every active $r \leq J_m$, for all sufficiently large m, over the joint environment, matrix, and sampled-sign randomness.

In particular, one may take

$$\lambda_m = A \log(e + m + J_m)$$

with A large enough, in which case the exponential menu term is summable for every finite J_m. The remaining automatic-truncation tests are then the single moment-series conditions

$$\sum_m \frac{m^{Bs} (\log(e + m + J_m))^{s-1} H_m}{n_m^{s-1}} < \infty$$

in the all-integer schedule $d_m = m$, $\alpha_m \asymp m^{-B}$, $B > 1$;

$$\sum_m \frac{d_m^{Bs} (\log(e + m + J_m))^{s-1} H_m}{n_m^{s-1}} < \infty$$

in the polynomial schedule $d_m \geq c_0 m^\kappa$, $\alpha_m \asymp d_m^{-B}$, $B\kappa > 1$*; and*

$$\sum_m \frac{\ell_m^{Bs}(\log(e+m+J_m))^{s-1}H_m}{n_m^{s-1}} < \infty$$

in the stretched-exponential schedule $d_m \geq \exp(c_0 m^\kappa)$, $\alpha_m \asymp \ell_m^{-B}$, $B > 1 + (2-p)_+/2 + 1/\kappa$*, together with the displayed active-menu matrix/sign summability for simultaneous sampled certificates.*

Proof. For a fixed active pair (m,r), apply Corollary 5.120 with the truncation level $L_m = n_m\alpha_m/\lambda_m$. Its proof gives the finite-dimensional tail estimate

$$\mathbb{P}\left\{\frac{1}{n_m}\sum_{i=1}^{n_m} Z_{m,i}^{(r)} > \alpha_m\right\} \leq \exp(-c_{\mathrm{AM}}\lambda_m) + \frac{C_s\lambda_m^{s-1}\bar{h}_{m,r}}{n_m^{s-1}\alpha_m^s},$$

where

$$\bar{h}_{m,r} := \frac{1}{n_m}\sum_{i=1}^{n_m} h_{m,i}^{(r)}.$$

After decreasing c_{AM} and absorbing C_s, summing over $r \leq J_m$ gives

$$\mathbb{P}\left\{\max_{1\leq r\leq J_m}\frac{1}{n_m}\sum_i Z_{m,i}^{(r)} > \alpha_m\right\} \leq J_m e^{-c_{\mathrm{AM}}\lambda_m} + \frac{\lambda_m^{s-1}H_m}{n_m^{s-1}\alpha_m^s}.$$

The assumed summability and Borel–Cantelli prove the simultaneous environment budget bound.

On that full-measure environment event, each active catalog row satisfies the deterministic inverse-p hypothesis. Repeating the Borel–Cantelli proof of Corollary 5.113 over the finite active menu only multiplies the matrix and sampled-sign tail terms by J_m; the two displayed summability assumptions make those enlarged series finite. The same fixed-fraction, vanishing-fraction, and polynomial-batch conclusions therefore hold for all active rows eventually.

Finally, since

$$J_m(e+m+J_m)^{-c_{\mathrm{AM}}A} \leq (e+m+J_m)^{1-c_{\mathrm{AM}}A} \leq (e+m)^{1-c_{\mathrm{AM}}A},$$

choosing A with $c_{\mathrm{AM}}A > 2$ makes the exponential menu series summable. Substituting the three displayed choices of α_m gives the three concrete moment series, up to harmless constants. □

Corollary 5.122 (Concrete active-prefix automatic-truncation schedules)**.** *Work in the setting and notation of Corollary 5.121, and let*

$$\sigma_p := 1 + \frac{(2-p)_+}{2}.$$

Assume $\bar{b}_m^* \leq \alpha_m/4$ *eventually. Then the simultaneous active-menu sampled Toeplitz/Hankel conclusions of that corollary hold in each of the following concrete regimes, with* $u_m = q\ell_m$*.*

(i) *All integer dimensions. Suppose* $d_m = m$, $J_m \le C_J m^{\Gamma}$, *and choose exponents*

$$A_0 > \Gamma + 1, \qquad q > \Gamma + 1, \qquad B > 1.$$

Let $\alpha_m = c_\alpha m^{-B}$. *It is enough that*

$$\sum_m \frac{m^{Bs}(\log(e+m))^{s-1} H_m}{n_m^{s-1}} < \infty.$$

(ii) *Polynomial dimension schedules. Suppose*

$$d_m \ge c_0 m^{\kappa}, \qquad J_m \le C_J d_m^{\Gamma},$$

and choose

$$A_0 > \Gamma + \frac{1}{\kappa}, \qquad q > \Gamma + \frac{1}{\kappa}, \qquad B\kappa > 1.$$

Let $\alpha_m = c_\alpha d_m^{-B}$. *It is enough that*

$$\sum_m \frac{d_m^{Bs} \ell_m^{s-1} H_m}{n_m^{s-1}} < \infty.$$

(iii) *Stretched-exponential dimensions with polynomial-in-d active prefixes. Suppose*

$$d_m \ge \exp(c_0 m^{\kappa}), \qquad J_m \le C_J d_m^{\Gamma},$$

and choose

$$A_0 > \Gamma, \qquad q > \Gamma, \qquad B > \sigma_p + \frac{1}{\kappa}.$$

Let $\alpha_m = c_\alpha \ell_m^{-B}$. *It is enough that*

$$\sum_m \frac{\ell_m^{Bs+s-1} H_m}{n_m^{s-1}} < \infty.$$

(iv) *Stretched-exponential dimensions with logarithmic active prefixes. Suppose*

$$d_m \ge \exp(c_0 m^{\kappa}), \qquad J_m \le C_J \ell_m^{\Gamma},$$

and choose

$$A_0 > 0, \qquad q > 0, \qquad B > \sigma_p + \frac{1}{\kappa}.$$

Let $\alpha_m = c_\alpha \ell_m^{-B}$. *It is enough that*

$$\sum_m \frac{\ell_m^{Bs}(\log(e+m))^{s-1} H_m}{n_m^{s-1}} < \infty.$$

Here $C_J, c_\alpha, c_0, \kappa$ *are fixed positive constants and* $\Gamma \ge 0$ *is fixed in the regimes where they appear. In each case the conclusion is simultaneous for all active* $r \le J_m$, *and hence applies to a countable predictable catalog library whenever level* m *only exposes its first* J_m *active streams.*

Proof. In every case take

$$\lambda_m = A \log(e + m + J_m)$$

with A large enough. The exponential menu term in Corollary 5.121 is then summable. It remains to check the active matrix/sign tails and the displayed moment series.

In (i), $J_m d_m^{-A_0} \le C m^{\Gamma - A_0}$ and $J_m d_m^{-q} \le C m^{\Gamma - q}$, so the two active-menu tail series are finite under the stated inequalities. Also $\log(e + m + J_m) \le C \log(e + m)$, and the moment series of Corollary 5.121 is dominated by the displayed series after substituting $\alpha_m = c_\alpha m^{-B}$. The condition $B > 1$ is the all-integer deterministic inverse-p schedule.

In (ii),

$$J_m d_m^{-A_0} \le C d_m^{\Gamma - A_0} \le C' m^{-\kappa(A_0 - \Gamma)}$$

and similarly for q. The displayed bounds on A_0 and q make both series summable. Since $d_m \ge c_0 m^\kappa$ and $J_m \le C_J d_m^\Gamma$, $\log(e + m + J_m) \le C \ell_m$. Substituting $\alpha_m = c_\alpha d_m^{-B}$ gives the displayed moment series, and $B\kappa > 1$ is the deterministic inverse-p threshold.

In (iii), the two active-menu tail terms are bounded by $C d_m^{\Gamma - A_0}$ and $C d_m^{\Gamma - q}$, which are summable because d_m grows stretched-exponentially and $A_0, q > \Gamma$. Again $\log(e + m + J_m) \le C \ell_m$. With $\alpha_m = c_\alpha \ell_m^{-B}$, this gives the series $\sum_m \ell_m^{Bs+s-1} H_m / n_m^{s-1}$, and $B > \sigma_p + 1/\kappa$ is the stretched-exponential inverse-p threshold.

In (iv), $J_m d_m^{-A_0}$ and $J_m d_m^{-q}$ are summable for every positive A_0, q, since $J_m \le C \ell_m^\Gamma$ is only polynomial in ℓ_m while d_m grows stretched-exponentially. Also $\log(e + m + J_m) \le C \log(e + m)$. The active-menu moment summability is therefore dominated by the stated moment series after substituting $\alpha_m = c_\alpha \ell_m^{-B}$. Applying Corollary 5.121 proves all four cases. □

Corollary 5.123 (Random active-prefix counts)**.** *Work with a countable predictable catalog library indexed by $r \ge 1$, in the setting of Corollary 5.121. Let N_m be a nonnegative integer-valued random active count on the same probability space. Let $J_m \ge 1$ be deterministic. Suppose that*

$$\sum_m \mathbb{P}\{N_m > J_m\} < \infty.$$

If the deterministic prefix $1 \le r \le J_m$ satisfies the hypotheses of Corollary 5.121, then the three sampled Toeplitz/Hankel conclusions of that corollary hold simultaneously for every randomly active stream $1 \le r \le N_m$, for all sufficiently large m, almost surely. The same statement holds if the deterministic prefix is checked through any of the concrete schedules in Corollary 5.122.

The overflow condition has the following direct moment tests. For any $t > 0$, it is enough that

$$\sum_m \frac{\mathbb{E}[(1 + N_m)^t]}{(1 + J_m)^t} < \infty.$$

Consequently, for deterministic constants $C, \eta \geq 0$, the overflow series is summable in the following typical cases.

(i) *If $d_m = m$, $J_m = \lceil m^\Gamma \rceil$, and $\mathbb{E}[(1+N_m)^t] \leq Cm^\eta$, then it is enough that*
$$t\Gamma > \eta + 1.$$

(ii) *If $d_m \geq c_0 m^\kappa$, $J_m = \lceil d_m^\Gamma \rceil$, and $\mathbb{E}[(1+N_m)^t] \leq C d_m^\eta$, then it is enough that*
$$\kappa(t\Gamma - \eta) > 1.$$

(iii) *If $d_m \geq \exp(c_0 m^\kappa)$, $J_m = \lceil d_m^\Gamma \rceil$, and $\mathbb{E}[(1+N_m)^t] \leq C d_m^\eta$, then it is enough that*
$$t\Gamma > \eta.$$

(iv) *If $d_m \geq \exp(c_0 m^\kappa)$, $J_m = \lceil \ell_m^\Gamma \rceil$, and $\mathbb{E}[(1+N_m)^t] \leq C \ell_m^\eta$, then it is enough that*
$$\kappa(t\Gamma - \eta) > 1.$$

Thus random proposal counts, adaptive search frontiers, and random catalog occupancy prefixes may be used in place of deterministic active-prefix sizes whenever their overflow tails are summable.

Proof. By Borel–Cantelli, $N_m \leq J_m$ for all sufficiently large m, almost surely. On the same probability-one event on which Corollary 5.121 gives the simultaneous conclusions for all deterministic prefix streams $1 \leq r \leq J_m$, those conclusions automatically hold for the smaller random set $1 \leq r \leq N_m$. The concrete schedule version is the same argument after using Corollary 5.122 to verify the deterministic prefix hypotheses.

For the moment tests, Markov's inequality gives
$$\mathbb{P}\{N_m > J_m\} \leq \frac{\mathbb{E}[(1+N_m)^t]}{(1+J_m)^t}.$$
The first displayed summability therefore implies the overflow condition. The four rate tests are the corresponding p-series or stretched-exponential substitutions: in (i) the summand is $O(m^{\eta - t\Gamma})$; in (ii) it is $O(d_m^{\eta - t\Gamma}) \leq O(m^{-\kappa(t\Gamma-\eta)})$; in (iii) it is $O(d_m^{-(t\Gamma-\eta)})$, hence summable for $t\Gamma > \eta$; and in (iv) it is $O(\ell_m^{\eta - t\Gamma}) \leq O(m^{-\kappa(t\Gamma-\eta)})$. □

Corollary 5.124 (Sub-Poisson active-count overflow tests)**.** *Work in the setting of Corollary 5.123. Let $\mathcal{H}_m$ be a σ-field and let $\nu_m \geq 0$ be deterministic. Assume that, for every $\theta \geq 0$,*
$$\mathbb{E}[\exp(\theta N_m) \mid \mathcal{H}_m] \leq \exp\Big(\nu_m(e^\theta - 1)\Big) \qquad \text{almost surely.}$$
If
$$J_m \geq 2\nu_m \qquad \text{eventually,} \qquad \sum_m \exp(-c_{\mathrm{SP}} J_m) < \infty$$
for an absolute constant $c_{\mathrm{SP}} > 0$, then
$$\sum_m \mathbb{P}\{N_m > J_m\} < \infty.$$

Thus the overflow hypothesis of Corollary 5.123 holds, so any deterministic prefix certified by Corollary 5.121 or Corollary 5.122 remains valid with the random count N_m.

The hypothesis includes the standard proposal-count models. It holds if, conditional on $\mathcal{H}_m$, N_m is Poisson with mean at most ν_m. It also holds if

$$N_m = \sum_{a \in \mathcal{A}_m} \eta_{m,a},$$

where the $\eta_{m,a}$ are conditionally independent Bernoulli variables and

$$\sum_{a \in \mathcal{A}_m} \mathbb{P}\{\eta_{m,a} = 1 \mid \mathcal{H}_m\} \le \nu_m \qquad \text{almost surely.}$$

Thus the random active-prefix theorem covers Poisson proposal counts, independent occupancy catalogs, and conditionally independent adaptive frontier exposures whenever the deterministic prefix J_m is at least twice the conditional mean envelope and the displayed exponential series is summable.

In particular, the overflow condition is automatic whenever

$$J_m \ge Q \log(e + m)$$

with $Q > 1/c_{\mathrm{SP}}$ and $J_m \ge 2\nu_m$ eventually. The deterministic prefix choices $J_m = \lceil m^\Gamma \rceil$, $J_m = \lceil d_m^\Gamma \rceil$, or $J_m = \lceil \ell_m^\Gamma \rceil$ with $\Gamma > 0$ therefore have summable sub-Poisson overflow tails in the corresponding all-integer, polynomial, and stretched-exponential schedules.

Proof. Taking $\theta = \log 2$ in the conditional moment generating function and using Markov's inequality gives

$$\begin{aligned} \mathbb{P}\{N_m > J_m \mid \mathcal{H}_m\} &\le \exp(\nu_m - J_m \log 2) \\ &\le \exp\Big(-(\log 2 - \tfrac{1}{2}) J_m\Big) \end{aligned}$$

whenever $J_m \ge 2\nu_m$. After integrating over $\mathcal{H}_m$, this gives the claimed overflow bound with $c_{\mathrm{SP}} = \log 2 - \frac{1}{2}$, or any smaller absolute constant. Corollary 5.123 then gives the random-count conclusions.

For a conditionally Poisson count with mean $\mu_m \le \nu_m$, the displayed moment generating function is the Poisson identity with μ_m bounded by ν_m. For the Bernoulli occupancy model,

$$\mathbb{E}[\exp(\theta N_m) \mid \mathcal{H}_m] = \prod_{a \in \mathcal{A}_m} \Big(1 + p_{m,a}(e^\theta - 1)\Big),$$

where $p_{m,a} = \mathbb{P}\{\eta_{m,a} = 1 \mid \mathcal{H}_m\}$. Since $1 + x \le e^x$, this product is at most

$$\exp\left((e^\theta - 1) \sum_a p_{m,a}\right) \le \exp\Big(\nu_m (e^\theta - 1)\Big).$$

Finally, $J_m \ge Q \log(e + m)$ gives $\exp(-c_{\mathrm{SP}} J_m) \le (e + m)^{-c_{\mathrm{SP}} Q}$, which is summable when $Q > 1/c_{\mathrm{SP}}$; the stated polynomial and logarithmic prefix examples eventually dominate such a logarithm whenever $\Gamma > 0$ in the relevant schedule. □

Corollary 5.125 (Predictable active-intensity overflow tests)**.** *Work in the setting of Corollary 5.123. Let $\mathcal{H}_m \subseteq \mathcal{G}_{m,0} \subseteq \mathcal{G}_{m,1} \subseteq \cdots \subseteq \mathcal{G}_{m,K_m}$ be a filtration and let $\eta_{m,a} \in \{0,1\}$, $1 \le a \le K_m$, be $\mathcal{G}_{m,a}$-measurable activation indicators. Put*

$$N_m = \sum_{a=1}^{K_m} \eta_{m,a}, \qquad p_{m,a} := \mathbb{E}[\eta_{m,a} \mid \mathcal{G}_{m,a-1}].$$

Assume that, for a deterministic $\nu_m \ge 0$,

$$\sum_{a=1}^{K_m} p_{m,a} \le \nu_m \qquad \text{almost surely.}$$

Then the hypothesis of Corollary 5.124 holds for N_m. Hence

$$J_m \ge 2\nu_m \text{ eventually}, \qquad \sum_m \exp(-c_{\mathrm{SP}} J_m) < \infty$$

imply

$$\sum_m \mathbb{P}\{N_m > J_m\} < \infty.$$

Consequently all deterministic active-prefix Toeplitz/Hankel sampled certificates certified by Corollary 5.121 or Corollary 5.122 remain valid for every adaptively active stream $1 \le r \le N_m$, eventually almost surely.

The same conclusion holds for countably many proposals $a \ge 1$ whenever the partial predictable intensities are almost surely bounded by ν_m. Thus independence is not needed for the overflow step: adaptive sequential frontiers, online candidate filters, without-replacement exposure schemes, and stopping-rule proposal lists are covered once their predictable total activation intensity is dominated by ν_m.

Proof. Fix m and $\theta \ge 0$. Since $\eta_{m,a} \in \{0,1\}$,

$$\mathbb{E}[\exp(\theta \eta_{m,a}) \mid \mathcal{G}_{m,a-1}] = 1 + p_{m,a}(e^\theta - 1) \le \exp\Big(p_{m,a}(e^\theta - 1)\Big).$$

Therefore

$$M_j := \exp\left(\theta \sum_{a \le j} \eta_{m,a} - (e^\theta - 1) \sum_{a \le j} p_{m,a}\right), \qquad 0 \le j \le K_m,$$

is a nonnegative supermartingale. Since $\mathcal{H}_m \subseteq \mathcal{G}_{m,0}$,

$$\mathbb{E}[M_{K_m} \mid \mathcal{H}_m] \le 1.$$

Using the predictable-intensity bound,

$$\begin{aligned}\mathbb{E}[\exp(\theta N_m) \mid \mathcal{H}_m] &\le \exp\Big(\nu_m(e^\theta - 1)\Big)\, \mathbb{E}[M_{K_m} \mid \mathcal{H}_m] \\ &\le \exp\Big(\nu_m(e^\theta - 1)\Big).\end{aligned}$$

This is exactly the sub-Poisson hypothesis of Corollary 5.124, which gives the summable overflow and random active-prefix conclusions.

For countably many proposals, apply the finite argument to the first K indicators and let $K \to \infty$ by conditional monotone convergence. □

Corollary 5.126 (Random predictable active-intensity envelopes)**.** *Use the notation of Corollary 5.125, but replace the deterministic almost-sure intensity bound by the predictable total intensity*

$$V_m := \sum_{a=1}^{K_m} p_{m,a}.$$

Let $\nu_m \geq 0$ and $J_m \geq 1$ be deterministic. If

$$\sum_m \mathbb{P}\{V_m > \nu_m\} < \infty, \qquad J_m \geq 2\nu_m \text{ eventually},$$

and

$$\sum_m \exp(-c_{\mathrm{SP}} J_m) < \infty,$$

then

$$\sum_m \mathbb{P}\{N_m > J_m\} < \infty.$$

Consequently the random active-prefix conclusions of Corollary 5.123 hold: every deterministic active-prefix Toeplitz/Hankel sampled certificate verified through Corollary 5.121 or Corollary 5.122 remains valid for all randomly active streams $1 \leq r \leq N_m$, eventually almost surely.

The countable-proposal version is the same with

$$V_m := \sum_{a \geq 1} p_{m,a},$$

provided this sum is finite outside the displayed summable intensity-overflow event. Thus the active-count step may be certified by a separate random environment or degree event: adaptive proposal graphs, random candidate pools, online thinning rules, and stopping-rule frontiers only pay the probability that their predictable total intensity exceeds the deterministic envelope ν_m.

Proof. First suppose $K_m < \infty$, and let $E_m = \{V_m \leq \nu_m\}$. With

$$M_j = \exp\left(\theta \sum_{a \leq j} \eta_{m,a} - (e^\theta - 1) \sum_{a \leq j} p_{m,a}\right)$$

as in the proof of Corollary 5.125, $(M_j)_{j \leq K_m}$ is a nonnegative supermartingale and $\mathbb{E}M_{K_m} \leq 1$. Hence, for $\theta = \log 2$,

$$\begin{aligned}
\mathbb{P}\{N_m > J_m,\ E_m\} &\leq e^{-\theta J_m} \mathbb{E}\Big[1_{E_m} e^{\theta N_m}\Big] \\
&= e^{-\theta J_m} \mathbb{E}\Big[1_{E_m} \exp\Big((e^\theta - 1)V_m\Big)\, M_{K_m}\Big] \\
&\leq \exp(\nu_m - J_m \log 2).
\end{aligned}$$

When $J_m \geq 2\nu_m$, this is at most $\exp(-c_{\mathrm{SP}} J_m)$, with the same absolute constant as in Corollary 5.124. Therefore

$$\mathbb{P}\{N_m > J_m\} \leq \mathbb{P}\{V_m > \nu_m\} + \exp(-c_{\mathrm{SP}} J_m),$$

and the displayed summability assumptions give the overflow condition. Corollary 5.123 then gives the stated active-prefix conclusions.

For countably many proposals, apply the finite bound to the first K indicators, with $V_{m,K} = \sum_{a \leq K} p_{m,a} \leq V_m$, and let $K \to \infty$ by monotone convergence. □

Corollary 5.127 (Moment tests for random predictable active intensities)**.** *In the setting of Corollary 5.126, let $t > 0$. If*

$$\sum_m \frac{\mathbb{E}[(1+V_m)^t]}{(1+\nu_m)^t} < \infty, \qquad J_m \geq 2\nu_m \text{ eventually},$$

and

$$\sum_m \exp(-c_{\mathrm{SP}} J_m) < \infty,$$

then

$$\sum_m \mathbb{P}\{N_m > J_m\} < \infty.$$

Hence every deterministic active-prefix Toeplitz/Hankel sampled certificate verified through Corollary 5.121 or Corollary 5.122 remains valid for all randomly active streams $1 \leq r \leq N_m$, eventually almost surely.

The moment condition is explicit in the standard growth regimes. For deterministic $C, \eta \geq 0$ and $c_\nu > 0$, the intensity-envelope overflow series is summable in the following cases.

(i) *If $d_m = m$, $\nu_m = c_\nu m^\Gamma$, and $\mathbb{E}[(1+V_m)^t] \leq C m^\eta$, it is enough that*

$$t\Gamma > \eta + 1.$$

(ii) *If $d_m \geq c_0 m^\kappa$, $\nu_m = c_\nu d_m^\Gamma$, and $\mathbb{E}[(1+V_m)^t] \leq C d_m^\eta$, it is enough that*

$$\kappa(t\Gamma - \eta) > 1.$$

(iii) *If $d_m \geq \exp(c_0 m^\kappa)$, $\nu_m = c_\nu d_m^\Gamma$, and $\mathbb{E}[(1+V_m)^t] \leq C d_m^\eta$, it is enough that*

$$t\Gamma > \eta.$$

(iv) *If $d_m \geq \exp(c_0 m^\kappa)$, $\nu_m = c_\nu \ell_m^\Gamma$, and $\mathbb{E}[(1+V_m)^t] \leq C \ell_m^\eta$, it is enough that*

$$\kappa(t\Gamma - \eta) > 1.$$

For any deterministic prefix $J_m \geq 2\nu_m$ with $\sum_m e^{-c_{\mathrm{SP}} J_m} < \infty$, these tests give the corresponding random active-count certificates.

Proof. Markov's inequality gives

$$\mathbb{P}\{V_m > \nu_m\} \leq \frac{\mathbb{E}[(1+V_m)^t]}{(1+\nu_m)^t}.$$

Thus the displayed moment summability verifies the intensity-envelope exception series in Corollary 5.126. That corollary then gives the active-count overflow and the active-prefix certificate conclusions.

The four rate checks are the same substitutions as in the moment tests for random active-prefix counts. In (i) the summand is $O(m^{\eta-t\Gamma})$; in (ii) it is $O(d_m^{\eta-t\Gamma}) \le O(m^{-\kappa(t\Gamma-\eta)})$; in (iii) it is stretched-exponentially summable when $t\Gamma > \eta$; and in (iv) it is $O(\ell_m^{\eta-t\Gamma}) \le O(m^{-\kappa(t\Gamma-\eta)})$. □

Corollary 5.128 (Freedman tests for random predictable active intensities)**.** *In the setting of Corollary 5.126, suppose that the predictable total intensity admits a random-environment majorant*

$$V_m \le \sum_{b=1}^{B_m} X_{m,b}.$$

Let $\mathcal{R}_{m,0} \subseteq \mathcal{R}_{m,1} \subseteq \cdots \subseteq \mathcal{R}_{m,B_m}$ be a filtration and suppose that $X_{m,b}$ is $\mathcal{R}_{m,b}$-measurable, $0 \le X_{m,b} \le L_m$, and

$$\sum_{b=1}^{B_m} \mathbb{E}[X_{m,b} \mid \mathcal{R}_{m,b-1}] \le \mu_m, \qquad \sum_{b=1}^{B_m} \mathbb{E}[X_{m,b}^2 \mid \mathcal{R}_{m,b-1}] \le \sigma_m^2$$

almost surely, for deterministic $\mu_m, \sigma_m, L_m \ge 0$. Let $\nu_m > \mu_m$, put $u_m = \nu_m - \mu_m$, and assume that

$$\sum_m \exp\left(-c_{\mathrm{FI}} \min\left\{\frac{u_m^2}{\sigma_m^2}, \frac{u_m}{L_m}\right\}\right) < \infty,$$

with the usual convention that a ratio with zero denominator is $+\infty$ when the numerator is positive. If also

$$J_m \ge 2\nu_m \text{ eventually}, \qquad \sum_m \exp(-c_{\mathrm{SP}} J_m) < \infty,$$

then

$$\sum_m \mathbb{P}\{N_m > J_m\} < \infty.$$

Therefore the deterministic active-prefix Toeplitz/Hankel sampled certificates verified through Corollary 5.121 or Corollary 5.122 hold for all randomly active streams $1 \le r \le N_m$, eventually almost surely.

In particular, if

$$\mu_m \le \nu_m/2, \qquad \sigma_m^2 \le L_m \nu_m,$$

then it is enough that

$$\sum_m \exp(-c'_{\mathrm{FI}} \nu_m / L_m) < \infty$$

and $J_m \ge 2\nu_m$ eventually. For unit-bounded intensity contributions $L_m = 1$, any sufficiently large multiple of $\log(e+m)$ for ν_m and J_m gives a summable envelope failure. Thus random candidate-pool degrees, adaptive proposal loads, and online thinning intensities may be certified by predictable mean, square-function, and jump envelopes rather than by moment growth assumptions.

Proof. Let

$$m_{m,b} := \mathbb{E}[X_{m,b} \mid \mathcal{R}_{m,b-1}], \qquad Y_{m,b} := X_{m,b} - m_{m,b}.$$

Then $(Y_{m,b})_b$ is a martingale-difference sequence, $|Y_{m,b}| \le L_m$, and its conditional variance process is bounded by

$$\sum_b \mathbb{E}[Y_{m,b}^2 \mid \mathcal{R}_{m,b-1}] \le \sum_b \mathbb{E}[X_{m,b}^2 \mid \mathcal{R}_{m,b-1}] \le \sigma_m^2.$$

If $V_m > \nu_m = \mu_m + u_m$, then

$$\sum_b Y_{m,b} \ge \sum_b X_{m,b} - \sum_b m_{m,b} > u_m.$$

Scalar Freedman's inequality therefore gives

$$\mathbb{P}\{V_m > \nu_m\} \le \exp\left(-c_{\mathrm{FI}} \min\left\{\frac{u_m^2}{\sigma_m^2}, \frac{u_m}{L_m}\right\}\right)$$

for an absolute constant $c_{\mathrm{FI}} > 0$. The displayed summability verifies the intensity-envelope exception series in Corollary 5.126, which gives the active-count overflow and the active-prefix conclusions.

If $\mu_m \le \nu_m/2$, then $u_m \ge \nu_m/2$. Under $\sigma_m^2 \le L_m \nu_m$, the Freedman exponent is bounded below by a constant multiple of ν_m/L_m, giving the simplified criterion. □

Corollary 5.129 (Truncated Freedman tests for unbounded predictable active intensities)**.** *In the setting of Corollary 5.126, suppose that the predictable total intensity admits an adapted nonnegative majorant*

$$V_m \le \sum_{b=1}^{B_m} X_{m,b}.$$

Let $\mathcal{R}_{m,0} \subseteq \mathcal{R}_{m,1} \subseteq \cdots \subseteq \mathcal{R}_{m,B_m}$ be a filtration and suppose that $X_{m,b}$ is $\mathcal{R}_{m,b}$-measurable and finite almost surely. Fix a deterministic truncation level $L_m > 0$, and write

$$W_{m,b} := X_{m,b} \wedge L_m, \qquad R_{m,b} := (X_{m,b} - L_m)_+.$$

Assume that deterministic numbers $\mu_m, \sigma_m, \rho_m \ge 0$ satisfy

$$\sum_{b=1}^{B_m} \mathbb{E}[W_{m,b} \mid \mathcal{R}_{m,b-1}] \le \mu_m, \qquad \sum_{b=1}^{B_m} \mathbb{E}[W_{m,b}^2 \mid \mathcal{R}_{m,b-1}] \le \sigma_m^2,$$

and

$$\sum_{b=1}^{B_m} \mathbb{E}[R_{m,b} \mid \mathcal{R}_{m,b-1}] \le \rho_m$$

almost surely. Let $\nu_m > \mu_m$, put $u_m = \nu_m - \mu_m$, and assume that

$$\sum_m \left[\exp\left(-c_{\mathrm{TFI}} \min\left\{\frac{u_m^2}{\sigma_m^2}, \frac{u_m}{L_m}\right\}\right) + \frac{\rho_m}{u_m}\right] < \infty,$$

with the usual convention that a ratio with zero denominator is $+\infty$ when the numerator is positive. If also

$$J_m \geq 2\nu_m \text{ eventually}, \qquad \sum_m \exp(-c_{\mathrm{SP}} J_m) < \infty,$$

then

$$\sum_m \mathbb{P}\{N_m > J_m\} < \infty.$$

Consequently every deterministic active-prefix Toeplitz/Hankel sampled certificate verified through Corollary 5.121 or Corollary 5.122 holds for all randomly active streams $1 \leq r \leq N_m$, eventually almost surely.

In particular, if

$$\mu_m \leq \nu_m/2, \qquad \sigma_m^2 \leq L_m \nu_m,$$

then it is enough that

$$\sum_m \left[\exp(-c'_{\mathrm{TFI}} \nu_m / L_m) + \frac{\rho_m}{\nu_m}\right] < \infty$$

and $J_m \geq 2\nu_m$ eventually. A local conditional-moment check is obtained as follows: if $s > 1$ and

$$\sum_{b=1}^{B_m} \mathbb{E}[X_{m,b}^s \mid \mathcal{R}_{m,b-1}] \leq H_m \qquad \text{almost surely},$$

then one may take $\rho_m = H_m / L_m^{s-1}$. Hence unbounded random candidate-pool degrees, heavy-tailed proposal loads, and adaptive thinning intensities are covered once the truncated predictable mean, truncated square-function, and excess series are summable.

Proof. Let

$$w_{m,b} := \mathbb{E}[W_{m,b} \mid \mathcal{R}_{m,b-1}], \qquad D_{m,b} := W_{m,b} - w_{m,b}.$$

Then $(D_{m,b})_b$ is a martingale-difference sequence, $|D_{m,b}| \leq L_m$, and

$$\sum_b \mathbb{E}[D_{m,b}^2 \mid \mathcal{R}_{m,b-1}] \leq \sum_b \mathbb{E}[W_{m,b}^2 \mid \mathcal{R}_{m,b-1}] \leq \sigma_m^2.$$

If $\sum_b X_{m,b} > \nu_m = \mu_m + u_m$, then either $\sum_b R_{m,b} > u_m/2$ or

$$\sum_b W_{m,b} > \mu_m + u_m/2.$$

On the latter event,

$$\sum_b D_{m,b} = \sum_b W_{m,b} - \sum_b w_{m,b} > u_m/2.$$

Scalar Freedman's inequality therefore gives

$$\mathbb{P}\left\{\sum_b W_{m,b} > \mu_m + u_m/2\right\} \leq \exp\left(-c_{\mathrm{TFI}} \min\left\{\frac{u_m^2}{\sigma_m^2}, \frac{u_m}{L_m}\right\}\right)$$

after reducing the absolute constant c_{TFI}. Markov's inequality and the predictable excess bound give

$$\mathbb{P}\left\{\sum_b R_{m,b} > u_m/2\right\} \le \frac{2}{u_m}\sum_b \mathbb{E}R_{m,b} \le \frac{2\rho_m}{u_m}.$$

Absorbing the harmless factor 2 into the summability assumption proves

$$\sum_m \mathbb{P}\{V_m > \nu_m\} < \infty.$$

Corollary 5.126 then gives the active-count overflow and the active-prefix certificate conclusions.

If $\mu_m \le \nu_m/2$, then $u_m \ge \nu_m/2$, and the hypothesis $\sigma_m^2 \le L_m\nu_m$ lowers the Freedman exponent by a constant multiple of ν_m/L_m. Finally, for $x \ge 0$, $(x - L_m)_+ \le x^s/L_m^{s-1}$; applying this conditionally and summing gives the stated choice $\rho_m = H_m/L_m^{s-1}$. □

Corollary 5.130 (Automatic moment truncation for predictable active intensities)**.** *In the setting of Corollary 5.129, fix $s > 1$. Suppose that deterministic $\nu_m > 0$, $H_m \ge 0$, and $\lambda_m > 0$ satisfy*

$$\sum_{b=1}^{B_m} \mathbb{E}[X_{m,b} \mid \mathcal{R}_{m,b-1}] \le \frac{\nu_m}{2}, \qquad \sum_{b=1}^{B_m} \mathbb{E}[X_{m,b}^s \mid \mathcal{R}_{m,b-1}] \le H_m$$

almost surely. If

$$\sum_m \left[\exp(-c_{\mathrm{ATI}}\lambda_m) + \frac{H_m\lambda_m^{s-1}}{\nu_m^s}\right] < \infty,$$

and

$$J_m \ge 2\nu_m \text{ eventually}, \qquad \sum_m \exp(-c_{\mathrm{SP}}J_m) < \infty,$$

then

$$\sum_m \mathbb{P}\{N_m > J_m\} < \infty.$$

Consequently all deterministic active-prefix Toeplitz/Hankel sampled certificates verified through Corollary 5.121 or Corollary 5.122 hold for all randomly active streams $1 \le r \le N_m$, eventually almost surely.

With $\lambda_m = A\log(e+m)$ and A sufficiently large, the criterion is explicit in the standard growth regimes. For deterministic $C, \eta \ge 0$ and $c_\nu > 0$, the active-intensity overflow series is summable in the following cases.

(i) *If $d_m = m$, $\nu_m = c_\nu m^\Gamma$, and $H_m \le Cm^\eta$, it is enough that*

$$s\Gamma > \eta + 1.$$

(ii) *If $d_m \ge c_0 m^\kappa$, $\nu_m = c_\nu d_m^\Gamma$, and $H_m \le Cd_m^\eta$, it is enough that*

$$\kappa(s\Gamma - \eta) > 1.$$

(iii) *If $d_m \ge \exp(c_0 m^\kappa)$, $\nu_m = c_\nu d_m^\Gamma$, and $H_m \le Cd_m^\eta$, it is enough that*

$$s\Gamma > \eta.$$

(iv) *If* $d_m \ge \exp(c_0 m^\kappa)$, $\nu_m = c_\nu \ell_m^\Gamma$, *and* $H_m \le C\ell_m^\eta$, *it is enough that*

$$\kappa(s\Gamma - \eta) > 1.$$

Thus a single conditional s*-moment envelope and a logarithmic truncation parameter certify heavy-tailed adaptive proposal-load intensities in the same polynomial, stretched-exponential, and logarithmic-prefix schedules as the bounded-load Freedman tests.*

Proof. Put

$$L_m := \frac{\nu_m}{\lambda_m}, \qquad W_{m,b} := X_{m,b} \wedge L_m, \qquad R_{m,b} := (X_{m,b} - L_m)_+.$$

Then

$$\sum_b \mathbb{E}[W_{m,b} \mid \mathcal{R}_{m,b-1}] \le \sum_b \mathbb{E}[X_{m,b} \mid \mathcal{R}_{m,b-1}] \le \frac{\nu_m}{2}.$$

Since $W_{m,b}^2 \le L_m W_{m,b}$,

$$\sum_b \mathbb{E}[W_{m,b}^2 \mid \mathcal{R}_{m,b-1}] \le L_m \sum_b \mathbb{E}[W_{m,b} \mid \mathcal{R}_{m,b-1}] \le L_m \nu_m.$$

Also $(x - L_m)_+ \le x^s / L_m^{s-1}$, so

$$\sum_b \mathbb{E}[R_{m,b} \mid \mathcal{R}_{m,b-1}] \le \frac{H_m}{L_m^{s-1}}.$$

The half-budget truncated-Freedman corollary therefore applies with

$$\rho_m = \frac{H_m}{L_m^{s-1}}.$$

Its two summability terms become

$$\exp(-c'_{\mathrm{TFI}} \nu_m / L_m) = \exp(-c'_{\mathrm{TFI}} \lambda_m), \qquad \frac{\rho_m}{\nu_m} = \frac{H_m \lambda_m^{s-1}}{\nu_m^s}.$$

After renaming the absolute constant, the displayed summability assumption gives the random active-count overflow and the active-prefix certificate conclusions.

For the concrete regimes, $\lambda_m = A \log(e+m)$ makes the exponential series summable when A is large enough. In (i) the excess term is

$$O\Big(m^{\eta - s\Gamma} (\log(e+m))^{s-1}\Big),$$

which is summable if $s\Gamma > \eta + 1$. In (ii) it is

$$O\Big(d_m^{\eta - s\Gamma} (\log(e+m))^{s-1}\Big) \le O\Big(m^{-\kappa(s\Gamma - \eta)} (\log(e+m))^{s-1}\Big),$$

which is summable under the displayed condition. In (iii) it is stretched-exponentially summable when $s\Gamma > \eta$. In (iv) it is

$$O\Big(\ell_m^{\eta - s\Gamma} (\log(e+m))^{s-1}\Big) \le O\Big(m^{-\kappa(s\Gamma - \eta)} (\log(e+m))^{s-1}\Big),$$

which is summable when $\kappa(s\Gamma - \eta) > 1$. □

Corollary 5.131 (Automatic prefixes from predictable active-intensity moments)**.** *Work in the setting of Corollary 5.130. Fix $s > 1$, and suppose that deterministic $\nu_m > 0$ and $H_m \geq 0$ satisfy*

$$\sum_{b=1}^{B_m} \mathbb{E}[X_{m,b} \mid \mathcal{R}_{m,b-1}] \leq \frac{\nu_m}{2}, \qquad \sum_{b=1}^{B_m} \mathbb{E}[X_{m,b}^s \mid \mathcal{R}_{m,b-1}] \leq H_m$$

almost surely. Choose constants $A, Q > 0$ large enough that

$$c_{\mathrm{ATI}} A > 1, \qquad c_{\mathrm{SP}} Q > 1,$$

and put

$$\lambda_m := A \log(e+m), \qquad J_m := \lceil \max\{2\nu_m,\ Q \log(e+m)\} \rceil .$$

If

$$\sum_m \frac{H_m (\log(e+m))^{s-1}}{\nu_m^s} < \infty,$$

then

$$\sum_m \mathbb{P}\{N_m > J_m\} < \infty.$$

Consequently every deterministic active-prefix Toeplitz/Hankel sampled certificate verified through Corollary 5.121 or Corollary 5.122 for the prefix $1 \leq r \leq J_m$ holds for all randomly active streams $1 \leq r \leq N_m$, eventually almost surely.

The prefix growth is automatic in the usual schedules. If $\nu_m \leq C_\nu m^\Gamma$ with $\Gamma > 0$, then $J_m \leq C m^\Gamma$, so the all-integer active-prefix schedule applies with the same menu-growth exponent. If $d_m \geq c_0 m^\kappa$ and $\nu_m \leq C_\nu d_m^\Gamma$ with $\Gamma > 0$, then $J_m \leq C d_m^\Gamma$. If $d_m \geq \exp(c_0 m^\kappa)$ and $\nu_m \leq C_\nu d_m^\Gamma$ with $\Gamma > 0$, the same polynomial-in-d active-prefix growth holds, while $\nu_m \leq C_\nu \ell_m^\Gamma$ with $\Gamma > 0$ gives $J_m \leq C \ell_m^\Gamma$. In those four regimes, the corresponding active-prefix schedule of Corollary 5.122 can be used with the same A_0, q, and catalog-moment thresholds, while the overflow part is reduced to the one displayed moment series above.

Proof. Apply Corollary 5.130 with the displayed λ_m and J_m. By construction $J_m \geq 2\nu_m$, and

$$\sum_m \exp(-c_{\mathrm{SP}} J_m) \leq \sum_m (e+m)^{-c_{\mathrm{SP}} Q} < \infty.$$

The exponential truncation term is also summable:

$$\sum_m \exp(-c_{\mathrm{ATI}} \lambda_m) = \sum_m (e+m)^{-c_{\mathrm{ATI}} A} < \infty.$$

The remaining term in that corollary is

$$\sum_m \frac{H_m \lambda_m^{s-1}}{\nu_m^s},$$

which is finite by the displayed hypothesis, up to the fixed factor A^{s-1}. Hence the overflow series is summable, and Corollary 5.123 gives the random

active-prefix certificate conclusion for any deterministic prefix certificate already verified up to J_m.

The four growth claims follow from the definition of J_m. In the all-integer and polynomial cases the logarithmic term is dominated by any fixed positive power once m is large. In the stretched-exponential polynomial-in-d case, $Q\log(e+m) \le Cd_m^{\Gamma}$ when $\Gamma > 0$. In the logarithmic-prefix case, $\log(e+m) \le C\ell_m^{\Gamma}$ for every fixed $\Gamma > 0$, since $\ell_m \ge cm^{\kappa}$. The cited active-prefix schedule corollary then supplies the deterministic-prefix sampled-certificate hypotheses. □

Corollary 5.132 (End-to-end automatic active-prefix sampled certificates)**.** *Work with a countable predictable catalog library indexed by $r \ge 1$, in the setting of Corollary 5.121, and let N_m be the random active count generated by activation indicators whose predictable total intensity V_m is dominated by*

$$V_m \le \sum_{b=1}^{B_m} X_{m,b}.$$

Fix $s_{\mathrm{act}}, s_{\mathrm{cat}} > 1$. Suppose that deterministic $\nu_m > 0$ and $G_m \ge 0$ satisfy

$$\sum_b \mathbb{E}[X_{m,b} \mid \mathcal{R}_{m,b-1}] \le \frac{\nu_m}{2}, \qquad \sum_b \mathbb{E}[X_{m,b}^{s_{\mathrm{act}}} \mid \mathcal{R}_{m,b-1}] \le G_m$$

almost surely. Choose $A, Q > 0$ so that

$$c_{\mathrm{ATI}}A > 1, \qquad c_{\mathrm{SP}}Q > 1,$$

and set

$$J_m := \lceil \max\{2\nu_m,\ Q\log(e+m)\} \rceil .$$

For every $1 \le r \le J_m$, assume the active-menu catalog variables

$$Z_{m,i}^{(r)} = \iota^{(p)}_{m,i,\Lambda_{m,i}^{(r)}}(a)$$

come from centered, variance-one, K-sub-Gaussian catalog laws and have deterministic conditional-moment envelopes

$$\mathbb{E}[Z_{m,i}^{(r)} \mid \mathcal{F}_{m,i-1}^{(r)}] \le b_{m,i}^{(r)}, \qquad \mathbb{E}[(Z_{m,i}^{(r)})^{s_{\mathrm{cat}}} \mid \mathcal{F}_{m,i-1}^{(r)}] \le h_{m,i}^{(r)},$$

and define

$$\bar{b}_m^{*,J} := \max_{1\le r\le J_m} \frac{1}{n_m}\sum_{i=1}^{n_m} b_{m,i}^{(r)}, \qquad H_m^{\mathrm{cat}} := \sum_{r=1}^{J_m}\frac{1}{n_m}\sum_{i=1}^{n_m} h_{m,i}^{(r)}.$$

Let $\alpha_m > 0$ satisfy one of the deterministic inverse-p schedules in Corollary 5.114, and assume

$$\bar{b}_m^{*,J} \le \frac{\alpha_m}{4} \qquad \textit{eventually.}$$

If

$$\sum_m \frac{G_m(\log(e+m))^{s_{\mathrm{act}}-1}}{\nu_m^{s_{\mathrm{act}}}} < \infty,$$

$$\sum_m \left[J_m \exp(-c_{\mathrm{E2E}} \lambda_m^{\mathrm{cat}}) + \frac{(\lambda_m^{\mathrm{cat}})^{s_{\mathrm{cat}}-1} H_m^{\mathrm{cat}}}{n_m^{s_{\mathrm{cat}}-1} \alpha_m^{s_{\mathrm{cat}}}} \right] < \infty$$

for some deterministic $\lambda_m^{\mathrm{cat}} > 0$ *and a sufficiently small absolute constant* $c_{\mathrm{E2E}} > 0$*, and*

$$\sum_m J_m d_m^{-A_0} < \infty, \qquad \sum_m J_m d_m^{-q} < \infty,$$

then, with $u_m = q\ell_m$*, the fixed-fraction, vanishing-fraction, and all-robust polynomial-batch Toeplitz/Hankel conclusions of Corollary 5.114 hold simultaneously for every randomly active stream* $1 \le r \le N_m$*, for all sufficiently large* m*, almost surely.*

In particular, with

$$\lambda_m^{\mathrm{cat}} = A_{\mathrm{cat}} \log(e + m + J_m)$$

and A_{cat} *large enough, the exponential catalog-menu series is automatic. The remaining hypotheses are exactly the active-intensity moment series, the catalog moment series, the active-menu matrix/sign tail series, and the deterministic inverse-p schedule for* α_m*. Thus the random proposal frontier and the predictable law-catalog exhaustion may be certified by two independent moment budgets and one deterministic prefix formula.*

Proof. The active-intensity assumptions are precisely the hypotheses of Corollary 5.131 with $H_m = G_m$, $s = s_{\mathrm{act}}$, and $\lambda_m = A \log(e + m)$. Hence

$$\sum_m \mathbb{P}\{N_m > J_m\} < \infty.$$

For this deterministic prefix J_m, the catalog hypotheses are those of Corollary 5.121, with $s = s_{\mathrm{cat}}$, $H_m = H_m^{\mathrm{cat}}$, $\lambda_m = \lambda_m^{\mathrm{cat}}$, and $\bar{b}_m^* = \bar{b}_m^{*,J}$. The displayed catalog summability and the two matrix/sign summability assumptions therefore give the simultaneous active-menu sampled Toeplitz/Hankel conclusions for every deterministic row $1 \le r \le J_m$, eventually almost surely.

Borel–Cantelli applied to the overflow series gives $N_m \le J_m$ eventually almost surely. Intersecting this event with the active-menu certificate event proves the asserted conclusions for all random active streams $r \le N_m$.

Finally, if $\lambda_m^{\mathrm{cat}} = A_{\mathrm{cat}} \log(e + m + J_m)$, then

$$J_m \exp(-c_{\mathrm{E2E}} \lambda_m^{\mathrm{cat}}) \le (e + m + J_m)^{1 - c_{\mathrm{E2E}} A_{\mathrm{cat}}},$$

which is summable after increasing A_{cat}. This leaves only the two moment budgets and the deterministic inverse-p, matrix, and sign schedules. □

Corollary 5.133 (Concrete end-to-end automatic active-prefix schedules)**.** *Use the notation, active-count setup, activation moment hypotheses, and catalog hypotheses of Corollary 5.132, and put*

$$\sigma_p := 1 + \frac{(2-p)_+}{2}.$$

Choose the constants A, Q, A_{cat} *in that corollary sufficiently large and take*

$$\lambda_m^{\mathrm{cat}} := A_{\mathrm{cat}} \log(e + m + J_m).$$

Assume $\bar{b}_m^{*,J} \le \alpha_m/4$ eventually. Then, with $u_m = q\ell_m$, the fixed-fraction, vanishing-fraction, and all-robust polynomial-batch Toeplitz/Hankel sampled-certificate conclusions of Corollary 5.132 hold for every randomly active stream $1 \le r \le N_m$, eventually almost surely, in each of the following regimes.

(i) *All integer dimensions. Suppose $d_m = m$ and, for some $\Gamma > 0$ and $\eta_{\mathrm{act}} \ge 0$,*

$$\nu_m \le C_\nu m^\Gamma, \qquad G_m \le C_G m^{\eta_{\mathrm{act}}}, \qquad s_{\mathrm{act}}\Gamma > \eta_{\mathrm{act}} + 1.$$

Choose

$$A_0 > \Gamma + 1, \qquad q > \Gamma + 1, \qquad \alpha_m = c_\alpha m^{-B} \quad \text{with } B > 1.$$

It is enough that

$$\sum_m \frac{m^{Bs_{\mathrm{cat}}}(\log(e+m))^{s_{\mathrm{cat}}-1} H_m^{\mathrm{cat}}}{n_m^{s_{\mathrm{cat}}-1}} < \infty.$$

(ii) *Polynomial dimension schedules. Suppose*

$$d_m \ge c_0 m^\kappa, \qquad \nu_m \le C_\nu d_m^\Gamma, \qquad G_m \le C_G d_m^{\eta_{\mathrm{act}}},$$

where $\Gamma > 0$, $\kappa > 0$, and

$$\kappa(s_{\mathrm{act}}\Gamma - \eta_{\mathrm{act}}) > 1.$$

Choose

$$A_0 > \Gamma + \frac{1}{\kappa}, \qquad q > \Gamma + \frac{1}{\kappa}, \qquad \alpha_m = c_\alpha d_m^{-B} \quad \text{with } B\kappa > 1.$$

It is enough that

$$\sum_m \frac{d_m^{Bs_{\mathrm{cat}}} \ell_m^{s_{\mathrm{cat}}-1} H_m^{\mathrm{cat}}}{n_m^{s_{\mathrm{cat}}-1}} < \infty.$$

(iii) *Stretched-exponential dimensions with polynomial-in-d active growth. Suppose*

$$d_m \ge \exp(c_0 m^\kappa), \qquad \nu_m \le C_\nu d_m^\Gamma, \qquad G_m \le C_G d_m^{\eta_{\mathrm{act}}},$$

where $\Gamma > 0$, $\kappa > 0$, and

$$s_{\mathrm{act}}\Gamma > \eta_{\mathrm{act}}.$$

Choose

$$A_0 > \Gamma, \qquad q > \Gamma, \qquad \alpha_m = c_\alpha \ell_m^{-B} \quad \text{with } B > \sigma_p + \frac{1}{\kappa}.$$

It is enough that

$$\sum_m \frac{\ell_m^{Bs_{\mathrm{cat}} + s_{\mathrm{cat}} - 1} H_m^{\mathrm{cat}}}{n_m^{s_{\mathrm{cat}}-1}} < \infty.$$

(iv) *Stretched-exponential dimensions with logarithmic active growth. Suppose*

$$d_m \geq \exp(c_0 m^{\kappa}), \qquad \nu_m \leq C_\nu \ell_m^{\Gamma}, \qquad G_m \leq C_G \ell_m^{\eta_{\mathrm{act}}},$$

where $\Gamma > 0$, $\kappa > 0$, *and*

$$\kappa(s_{\mathrm{act}}\Gamma - \eta_{\mathrm{act}}) > 1.$$

Choose

$$A_0 > 0, \qquad q > 0, \qquad \alpha_m = c_\alpha \ell_m^{-B} \quad \text{with } B > \sigma_p + \frac{1}{\kappa}.$$

It is enough that

$$\sum_m \frac{\ell_m^{B s_{\mathrm{cat}}} (\log(e+m))^{s_{\mathrm{cat}}-1} H_m^{\mathrm{cat}}}{n_m^{s_{\mathrm{cat}}-1}} < \infty.$$

Proof. The active-count truncation parameter in Corollary 5.132 is $A \log(e+m)$. With A large, its exponential term is summable; the displayed activation growth assumptions give the remaining active-intensity moment series. Indeed the four summands are bounded, respectively, by

$$C m^{\eta_{\mathrm{act}} - s_{\mathrm{act}}\Gamma} (\log(e+m))^{s_{\mathrm{act}}-1},$$

$$C m^{-\kappa(s_{\mathrm{act}}\Gamma - \eta_{\mathrm{act}})} (\log(e+m))^{s_{\mathrm{act}}-1},$$

a stretched-exponentially summable sequence, and

$$C m^{-\kappa(s_{\mathrm{act}}\Gamma - \eta_{\mathrm{act}})} (\log(e+m))^{s_{\mathrm{act}}-1}.$$

The strict inequalities in the statement make these series finite.

The automatic prefix satisfies the same growth as the displayed envelope:

$$J_m \leq C m^{\Gamma}, \quad J_m \leq C d_m^{\Gamma}, \quad J_m \leq C d_m^{\Gamma}, \quad J_m \leq C \ell_m^{\Gamma}$$

in the four cases. Hence $J_m d_m^{-A_0}$ and $J_m d_m^{-q}$ are summable under the stated bounds on A_0 and q: by p-series in the first two cases, by stretched-exponential decay with $A_0, q > \Gamma$ in the third, and by stretched-exponential decay against a logarithmic prefix in the fourth.

It remains to check the active-menu catalog moment series with $\lambda_m^{\mathrm{cat}} = A_{\mathrm{cat}} \log(e+m+J_m)$. In the all-integer case $\log(e+m+J_m) \leq C \log(e+m)$; in the polynomial and stretched-exponential polynomial-growth cases it is at most $C\ell_m$; in the logarithmic-growth case it is at most $C \log(e+m)$. Substituting the four displayed choices of α_m gives exactly the four catalog moment series in the statement. The same large choice of A_{cat} makes the exponential catalog-menu series summable by Corollary 5.132. The displayed choices of α_m are precisely the deterministic inverse-p schedules in Corollary 5.114. The cited end-to-end corollary now gives all claimed sampled-certificate conclusions. □

Corollary 5.134 (Sample-size forms of the end-to-end active schedules)**.** *Consider any one of the four regimes in Corollary 5.133, and keep all of its hypotheses except for the displayed catalog moment summability assumption. Fix $\epsilon > 0$. In the corresponding regime, that omitted catalog summability follows from the following simpler aggregate catalog-growth and sample-size conditions.*

(i) *All integer dimensions. If*

$$H_m^{\mathrm{cat}} \le C_H m^{\eta_{\mathrm{cat}}}$$

and

$$n_m \ge Q_n m^{(Bs_{\mathrm{cat}}+\eta_{\mathrm{cat}}+1+\epsilon)/(s_{\mathrm{cat}}-1)} \log(e+m),$$

eventually, then the all-integer catalog moment series is summable.

(ii) *Polynomial dimension schedules. If*

$$H_m^{\mathrm{cat}} \le C_H d_m^{\eta_{\mathrm{cat}}}$$

and

$$n_m \ge Q_n d_m^{(Bs_{\mathrm{cat}}+\eta_{\mathrm{cat}}+1/\kappa+\epsilon)/(s_{\mathrm{cat}}-1)} \ell_m,$$

eventually, then the polynomial-dimension catalog moment series is summable.

(iii) *Stretched-exponential dimensions with polynomial-in-d active growth. If*

$$H_m^{\mathrm{cat}} \le C_H d_m^{\eta_{\mathrm{cat}}}$$

and

$$n_m \ge Q_n d_m^{\eta_{\mathrm{cat}}/(s_{\mathrm{cat}}-1)} \ell_m^{(Bs_{\mathrm{cat}}+s_{\mathrm{cat}}-1+1/\kappa+\epsilon)/(s_{\mathrm{cat}}-1)},$$

eventually, then the stretched-exponential polynomial-active catalog moment series is summable.

(iv) *Stretched-exponential dimensions with logarithmic active growth. If*

$$H_m^{\mathrm{cat}} \le C_H \ell_m^{\eta_{\mathrm{cat}}}$$

and

$$n_m \ge Q_n \ell_m^{(Bs_{\mathrm{cat}}+\eta_{\mathrm{cat}}+1/\kappa+\epsilon)/(s_{\mathrm{cat}}-1)} \log(e+m),$$

eventually, then the logarithmic-active catalog moment series is summable.

Here $C_H, Q_n > 0$ and $\eta_{\mathrm{cat}} \ge 0$ are deterministic constants. Thus, whenever the corresponding lower bound is compatible with the ambient requirements $n_m \le d_m$, $d_m/n_m \to \infty$, and $\log d_m = o(n_m)$, these sample-size tests may replace the catalog moment series in Corollary 5.133. The same fixed-fraction, vanishing-fraction, and all-robust polynomial-batch Toeplitz/Hankel sampled certificates then hold for every random active stream $1 \le r \le N_m$, eventually almost surely.

Proof. In the all-integer case, the first two displays give

$$\frac{m^{Bs_{\mathrm{cat}}}(\log(e+m))^{s_{\mathrm{cat}}-1}H_m^{\mathrm{cat}}}{n_m^{s_{\mathrm{cat}}-1}} \le Cm^{-1-\epsilon},$$

which is summable.

In the polynomial-dimension case, the corresponding term is at most

$$Cd_m^{-1/\kappa-\epsilon} \le C'm^{-1-\kappa\epsilon},$$

because $d_m \ge c_0 m^\kappa$. This is summable.

In the stretched-exponential polynomial-active case, the displayed sample-size bound gives

$$\frac{\ell_m^{Bs_{\mathrm{cat}}+s_{\mathrm{cat}}-1}H_m^{\mathrm{cat}}}{n_m^{s_{\mathrm{cat}}-1}} \le C\ell_m^{-1/\kappa-\epsilon}.$$

Since $d_m \ge \exp(c_0 m^\kappa)$, we have $\ell_m \ge cm^\kappa$ eventually, so the last display is $O(m^{-1-\kappa\epsilon})$.

The logarithmic-active case is the same calculation:

$$\frac{\ell_m^{Bs_{\mathrm{cat}}}(\log(e+m))^{s_{\mathrm{cat}}-1}H_m^{\mathrm{cat}}}{n_m^{s_{\mathrm{cat}}-1}} \le C\ell_m^{-1/\kappa-\epsilon} \le C'm^{-1-\kappa\epsilon}.$$

Thus the omitted catalog moment series in Corollary 5.133 is finite in each regime. Applying that corollary gives the stated sampled certificate conclusions. □

Corollary 5.135 (Compatible sample windows for catalog-growth exponents)**.** *In the setting of Corollary 5.133, write $s = s_{\mathrm{cat}}$. The catalog moment hypotheses in that corollary are compatible with the ambient unbounded-aspect requirements*

$$n_m \le d_m, \qquad d_m/n_m \to \infty, \qquad \log d_m = o(n_m)$$

in the following explicit windows.

(i) *All integer dimensions. Suppose $d_m = m$ and*

$$H_m^{\mathrm{cat}} \le C_H m^\gamma$$

for a real exponent γ. If

$$\gamma < -s(B-1) - 2,$$

then choose

$$\max\left\{0, \frac{Bs+\gamma+1}{s-1}\right\} < \rho < \rho' < 1.$$

Every sample-size sequence satisfying

$$Q_n m^\rho \log(e+m) \le n_m \le m^{\rho'}$$

eventually has the ambient unbounded-aspect properties and makes the all-integer catalog moment series summable.

(ii) *Polynomial dimension schedules. Suppose* $d_m \geq c_0 m^\kappa$ *and*

$$H_m^{\mathrm{cat}} \leq C_H d_m^\gamma.$$

If

$$\gamma < s - 1 - Bs - \frac{1}{\kappa},$$

then choose

$$\max\left\{0, \frac{Bs + \gamma + 1/\kappa}{s-1}\right\} < \rho < \rho' < 1.$$

Every sample-size sequence satisfying

$$Q_n d_m^\rho \ell_m \leq n_m \leq d_m^{\rho'}$$

eventually has the ambient unbounded-aspect properties and makes the polynomial catalog moment series summable.

(iii) *Stretched-exponential dimensions with polynomial-in-d active growth. Suppose* $d_m \geq \exp(c_0 m^\kappa)$ *and*

$$H_m^{\mathrm{cat}} \leq C_H d_m^\gamma.$$

If $\gamma < s - 1$*, choose*

$$\max\left\{0, \frac{\gamma}{s-1}\right\} < \rho < \rho' < 1.$$

Every sample-size sequence satisfying

$$Q_n d_m^\rho \ell_m \leq n_m \leq d_m^{\rho'}$$

eventually has the ambient unbounded-aspect properties and makes the stretched-exponential polynomial-active catalog moment series summable.

(iv) *Stretched-exponential dimensions with logarithmic active growth. Suppose* $d_m \geq \exp(c_0 m^\kappa)$ *and*

$$H_m^{\mathrm{cat}} \leq C_H \ell_m^\gamma.$$

For every real γ*, choose*

$$r > \max\left\{1, \frac{Bs + \gamma + 1/\kappa}{s-1}\right\}$$

and any $0 < \rho' < 1$*. Every sample-size sequence satisfying*

$$Q_n \ell_m^r \log(e+m) \leq n_m \leq d_m^{\rho'}$$

eventually has the ambient unbounded-aspect properties and makes the logarithmic-active catalog moment series summable.

In each case the fixed-fraction, vanishing-fraction, and all-robust polynomial-batch Toeplitz/Hankel sampled certificates hold for every random active stream $1 \leq r \leq N_m$*, eventually almost surely.*

Proof. The displayed sample windows are nonempty for all sufficiently large m. In the first three cases this follows from $\rho < \rho'$. In the fourth case it uses the stretched-exponential lower bound on d_m, which makes every fixed power of ℓ_m negligible compared with $d_m^{\rho'}$. The ambient properties are immediate from the lower bounds, which dominate $\log d_m$, and from the upper bounds, which are $o(d_m)$.

It remains only to check summability of the catalog series from Corollary 5.133. In the all-integer case,

$$\frac{m^{Bs}(\log(e+m))^{s-1}H_m^{\mathrm{cat}}}{n_m^{s-1}} \le Cm^{Bs+\gamma-(s-1)\rho}.$$

The choice of ρ makes the exponent less than -1.

In the polynomial case,

$$\frac{d_m^{Bs}\ell_m^{s-1}H_m^{\mathrm{cat}}}{n_m^{s-1}} \le Cd_m^{Bs+\gamma-(s-1)\rho} \le C'm^{-\kappa((s-1)\rho-Bs-\gamma)}.$$

The choice of ρ makes $\kappa((s-1)\rho - Bs - \gamma) > 1$, so the series is finite.

In the stretched-exponential polynomial-active case,

$$\frac{\ell_m^{Bs+s-1}H_m^{\mathrm{cat}}}{n_m^{s-1}} \le Cd_m^{\gamma-(s-1)\rho},$$

up to a harmless power of ℓ_m. The exponent of d_m is negative, so the summand decays stretched-exponentially.

Finally, in the logarithmic-active case,

$$\frac{\ell_m^{Bs}(\log(e+m))^{s-1}H_m^{\mathrm{cat}}}{n_m^{s-1}} \le C\ell_m^{Bs+\gamma-(s-1)r}.$$

The choice of r makes the exponent less than $-1/\kappa$; since $\ell_m \ge cm^{\kappa}$ eventually, the resulting series is summable. Applying Corollary 5.133 proves the sampled-certificate conclusions. □

Corollary 5.136 (Per-stream catalog budgets for compatible active windows)**.** *Use the notation and hypotheses of Corollary 5.133, except for its catalog moment summability assumption, and write $s = s_{\mathrm{cat}}$. For the active prefix J_m, set*

$$\bar{h}_m^{*,J} := \max_{1\le r\le J_m} \frac{1}{n_m}\sum_{i=1}^{n_m} h_{m,i}^{(r)}.$$

Then $H_m^{\mathrm{cat}} \le J_m\bar{h}_m^{*,J}$. Consequently the compatible sample-window conclusions of Corollary 5.135 hold under the following per-stream row-catalog conditions.

(i) *All integer dimensions. Suppose $d_m = m$, $\nu_m \le C_\nu m^{\Gamma}$, and*

$$\bar{h}_m^{*,J} \le C_h m^{\theta}$$

for a real exponent θ. If

$$\Gamma + \theta < -s(B-1) - 2,$$

then choose

$$\max\left\{0, \frac{Bs+\Gamma+\theta+1}{s-1}\right\} < \rho < \rho' < 1.$$

Every sample-size sequence satisfying

$$Q_n m^{\rho}\log(e+m) \le n_m \le m^{\rho'}$$

eventually has the ambient unbounded-aspect properties and gives the sampled-certificate conclusions for every random active stream.

(ii) *Polynomial dimension schedules. Suppose* $d_m \ge c_0 m^{\kappa}$, $\nu_m \le C_\nu d_m^{\Gamma}$, *and*

$$\bar{h}_m^{*,J} \le C_h d_m^{\theta}.$$

If

$$\Gamma+\theta < s-1-Bs-\frac{1}{\kappa},$$

then choose

$$\max\left\{0, \frac{Bs+\Gamma+\theta+1/\kappa}{s-1}\right\} < \rho < \rho' < 1.$$

Every sample-size sequence satisfying

$$Q_n d_m^{\rho}\ell_m \le n_m \le d_m^{\rho'}$$

eventually has the ambient unbounded-aspect properties and gives the sampled-certificate conclusions for every random active stream.

(iii) *Stretched-exponential dimensions with polynomial-in-d active growth. Suppose* $d_m \ge \exp(c_0 m^{\kappa})$, $\nu_m \le C_\nu d_m^{\Gamma}$, *and*

$$\bar{h}_m^{*,J} \le C_h d_m^{\theta}.$$

If $\Gamma+\theta < s-1$, *choose*

$$\max\left\{0, \frac{\Gamma+\theta}{s-1}\right\} < \rho < \rho' < 1.$$

Every sample-size sequence satisfying

$$Q_n d_m^{\rho}\ell_m \le n_m \le d_m^{\rho'}$$

eventually has the ambient unbounded-aspect properties and gives the sampled-certificate conclusions for every random active stream.

(iv) *Stretched-exponential dimensions with logarithmic active growth. Suppose* $d_m \ge \exp(c_0 m^{\kappa})$, $\nu_m \le C_\nu \ell_m^{\Gamma}$, *and*

$$\bar{h}_m^{*,J} \le C_h \ell_m^{\theta}.$$

For every real θ, *choose*

$$r > \max\left\{1, \frac{Bs+\Gamma+\theta+1/\kappa}{s-1}\right\}$$

and any $0 < \rho' < 1$. *Every sample-size sequence satisfying*

$$Q_n \ell_m^{r}\log(e+m) \le n_m \le d_m^{\rho'}$$

eventually has the ambient unbounded-aspect properties and gives the sampled-certificate conclusions for every random active stream.

In each case, with the remaining activation, inverse-p, matrix/sign, and mean-budget hypotheses from Corollary 5.133, the fixed-fraction, vanishing-fraction, and all-robust polynomial-batch Toeplitz/Hankel certificates hold for every random active stream $1 \le r \le N_m$, eventually almost surely.

Proof. Writing

$$\bar{h}_{m,r} := \frac{1}{n_m} \sum_{i=1}^{n_m} h_{m,i}^{(r)},$$

the aggregate catalog moment budget satisfies

$$H_m^{\mathrm{cat}} = \sum_{r=1}^{J_m} \bar{h}_{m,r} \le J_m \bar{h}_m^{*,J}.$$

The automatic-prefix growth estimates in the proof of Corollary 5.133 give $J_m \le C m^{\Gamma}$ in the all-integer regime, $J_m \le C d_m^{\Gamma}$ in the polynomial and stretched-exponential polynomial-active regimes, and $J_m \le C \ell_m^{\Gamma}$ in the stretched-exponential logarithmic-active regime. Thus the aggregate hypotheses of Corollary 5.135 hold with catalog exponent $\gamma = \Gamma + \theta$ in the corresponding scale. Applying that corollary proves all four sample-window conclusions and the resulting sampled certificates. □

Corollary 5.137 (First-moment-tuned per-stream compatible active windows)**.** *Use the activation setup, catalog laws, and notation of Corollary 5.133, but replace its mean-budget and catalog summability assumptions by the per-stream assumptions below. Write $s = s_{\mathrm{cat}}$ and*

$$\sigma_p := 1 + \frac{(2-p)_+}{2}.$$

For the automatically chosen active prefix J_m, set

$$\bar{b}_m^{*,J} := \max_{1 \le r \le J_m} \frac{1}{n_m} \sum_{i=1}^{n_m} b_{m,i}^{(r)}, \qquad \bar{h}_m^{*,J} := \max_{1 \le r \le J_m} \frac{1}{n_m} \sum_{i=1}^{n_m} h_{m,i}^{(r)}.$$

Then the compatible active-window conclusions hold in the following first-moment-tuned regimes.

(i) *All integer dimensions. Suppose $d_m = m$,*

$$\nu_m \le C_\nu m^{\Gamma}, \qquad G_m \le C_G m^{\eta_{\mathrm{act}}}, \qquad s_{\mathrm{act}} \Gamma > \eta_{\mathrm{act}} + 1,$$

and, for real exponents β, θ,

$$\bar{b}_m^{*,J} \le C_b m^{-\beta}, \qquad \bar{h}_m^{*,J} \le C_h m^{\theta}.$$

If

$$\beta > 1, \qquad \Gamma + \theta < -2,$$

choose B so that

$$1 < B < \beta, \qquad \Gamma + \theta < -s(B-1) - 2,$$

take $A_0, q > \Gamma + 1$ *and* $\alpha_m = c_\alpha m^{-B}$, *and choose*

$$\max\left\{0, \frac{Bs + \Gamma + \theta + 1}{s - 1}\right\} < \rho < \rho' < 1.$$

Every sample-size sequence satisfying

$$Q_n m^\rho \log(e + m) \le n_m \le m^{\rho'}$$

eventually gives the sampled Toeplitz/Hankel certificate conclusions for every random active stream.

(ii) *Polynomial dimension schedules. Suppose* $d_m \ge c_0 m^\kappa$,

$$\nu_m \le C_\nu d_m^\Gamma, \qquad G_m \le C_G d_m^{\eta_{\mathrm{act}}}, \qquad \kappa(s_{\mathrm{act}}\Gamma - \eta_{\mathrm{act}}) > 1,$$

and

$$\bar{b}_m^{*,J} \le C_b d_m^{-\beta}, \qquad \bar{h}_m^{*,J} \le C_h d_m^\theta.$$

If

$$\beta > \frac{1}{\kappa}, \qquad \Gamma + \theta < s\left(1 - \frac{1}{\kappa}\right) - 1 - \frac{1}{\kappa},$$

choose B *so that*

$$\frac{1}{\kappa} < B < \beta, \qquad \Gamma + \theta < s - 1 - Bs - \frac{1}{\kappa},$$

take $A_0, q > \Gamma + 1/\kappa$ *and* $\alpha_m = c_\alpha d_m^{-B}$, *and choose*

$$\max\left\{0, \frac{Bs + \Gamma + \theta + 1/\kappa}{s - 1}\right\} < \rho < \rho' < 1.$$

Every sample-size sequence satisfying

$$Q_n d_m^\rho \ell_m \le n_m \le d_m^{\rho'}$$

eventually gives the sampled Toeplitz/Hankel certificate conclusions for every random active stream.

(iii) *Stretched-exponential dimensions with polynomial-in-d active growth. Suppose* $d_m \ge \exp(c_0 m^\kappa)$,

$$\nu_m \le C_\nu d_m^\Gamma, \qquad G_m \le C_G d_m^{\eta_{\mathrm{act}}}, \qquad s_{\mathrm{act}}\Gamma > \eta_{\mathrm{act}},$$

and

$$\bar{b}_m^{*,J} \le C_b \ell_m^{-\beta}, \qquad \bar{h}_m^{*,J} \le C_h d_m^\theta.$$

If

$$\beta > \sigma_p + \frac{1}{\kappa}, \qquad \Gamma + \theta < s - 1,$$

choose B *so that*

$$\sigma_p + \frac{1}{\kappa} < B < \beta,$$

take $A_0, q > \Gamma$ *and* $\alpha_m = c_\alpha \ell_m^{-B}$, *and choose*

$$\max\left\{0, \frac{\Gamma + \theta}{s - 1}\right\} < \rho < \rho' < 1.$$

Every sample-size sequence satisfying

$$Q_n d_m^\rho \ell_m \le n_m \le d_m^{\rho'}$$

eventually gives the sampled Toeplitz/Hankel certificate conclusions for every random active stream.

(iv) *Stretched-exponential dimensions with logarithmic active growth. Suppose* $d_m \ge \exp(c_0 m^\kappa)$,

$$\nu_m \le C_\nu \ell_m^\Gamma, \qquad G_m \le C_G \ell_m^{\eta_{\mathrm{act}}}, \qquad \kappa(s_{\mathrm{act}}\Gamma - \eta_{\mathrm{act}}) > 1,$$

and

$$\bar{b}_m^{*,J} \le C_b \ell_m^{-\beta}, \qquad \bar{h}_m^{*,J} \le C_h \ell_m^{\theta}.$$

If

$$\beta > \sigma_p + \frac{1}{\kappa},$$

choose B *so that*

$$\sigma_p + \frac{1}{\kappa} < B < \beta,$$

take arbitrary $A_0, q > 0$, $\alpha_m = c_\alpha \ell_m^{-B}$, *choose any* $0 < \rho' < 1$, *and choose*

$$r > \max\left\{1, \frac{Bs + \Gamma + \theta + 1/\kappa}{s-1}\right\}.$$

Every sample-size sequence satisfying

$$Q_n \ell_m^r \log(e+m) \le n_m \le d_m^{\rho'}$$

eventually gives the sampled Toeplitz/Hankel certificate conclusions for every random active stream.

In all four cases the conclusions are the fixed-fraction, vanishing-fraction, and all-robust polynomial-batch sampled certificates of Corollary 5.133, simultaneously for every randomly active stream $1 \le r \le N_m$, *eventually almost surely.*

Proof. The displayed strict inequalities ensure that the stated interval for B is nonempty in each case. Once such a B is fixed, the first-moment bounds imply

$$\bar{b}_m^{*,J} \le \alpha_m/4$$

eventually, because the assumed first-moment decay exponent is strictly larger than the chosen inverse-p exponent. The activation and matrix/sign summability hypotheses are exactly the corresponding hypotheses of Corollary 5.133 with this B and the displayed choices of A_0, q.

The s-moment bounds and the automatic prefix estimates give the per-stream catalog hypotheses of Corollary 5.136. Indeed the aggregate catalog exponent there is $\Gamma + \theta$ in the appropriate scale, and the displayed choices of B, ρ, ρ' or B, r are precisely its compatibility inequalities. Applying that corollary gives the sample windows, ambient unbounded-aspect properties, and all sampled certificate conclusions. □

Corollary 5.138 (Convex-concentrated sampled operator certificates)**.** *In the setting of Theorem 5.99, let* (d_m, n_m) *be any sequence satisfying*

$$n_m \le d_m, \qquad d_m/n_m \to \infty, \qquad \log d_m = o(n_m).$$

For each m, let $B_{m,i} = T_{d_m}(\xi^{(m,i)})$, $1 \le i \le n_m$, be independent full Toeplitz matrices from the law, put

$$A_{m,i} := \frac{B_{m,i}}{\|B_{m,i}\|_{\mathrm{op}}}, \qquad \widetilde{A}_{m,i} := J_{d_m} A_{m,i},$$

and sample independent uniform signs $\delta_m^{(1)}, \dots, \delta_m^{(M_m)}$, independent of the matrices, with $M_m \ge \lceil B \rceil$ for some fixed $B > 0$. Then, with probability tending to one jointly over the matrices and sampled signs, some sampled sign $\delta_m^{(s)}$ is the center of a Hamming ball of radius

$$\lfloor c \log(e + d_m) \rfloor$$

such that every ε' in that ball satisfies

$$\left\| \sum_{i=1}^{n_m} \varepsilon'_i A_{m,i} \right\|_{\mathrm{op}} \le C\sqrt{n_m}, \qquad \left\| \sum_{i=1}^{n_m} \varepsilon'_i \widetilde{A}_{m,i} \right\|_{\mathrm{op}} \le C\sqrt{n_m}.$$

Here $C < \infty$ and $c > 0$ depend only on the coefficient-law parameters and on the fixed confidence choices. More quantitatively, before the deterministic edge-to-operator transfer, for every $A_0 > 0$ the edge-certificate failure is at most

$$C_{A_0,K,C_{\mathrm{cc}}} d_m^{-A_0} + (\log(e + d_m))^2 (e + d_m)^{-\beta} + d_m^{-B},$$

for some $\beta = \beta(K, C_{\mathrm{cc}}) > 0$, once m is large enough. Thus the same conclusion applies to standard real or complex Gaussian coefficients, Rademacher coefficients, and bounded real or complex coefficient laws with mean zero and variance one.

Proof. The proof of Theorem 5.99 gives $a = 1/2$ and

$$p_d(a) = \mathbb{P}\{\|T_d(\xi)\|_{\mathrm{op}} < a\, \mathbb{E}\, \|T_d(\xi)\|_{\mathrm{op}}\} \le (e + d)^{-\beta}$$

for some $\beta = \beta(K, C_{\mathrm{cc}}) > 0$. Since $\log d_m = o(n_m)$, the size condition $n_m \ge C \log(e + d_m)$ in Corollary 5.31 holds eventually. Applying that corollary at (d_m, n_m), with the fixed sign-confidence parameter B, shows that with failure at most the displayed quantity, one sampled sign is the center of a Hamming ball of radius $c \log(e + d_m)$ on which every sign has realization-normalized edge maximum $O(\sqrt{n_m})$; here $\min\{n_m, \log^2(e + d_m)\} \le \log^2(e + d_m)$ was used.

On the edge-success event, apply Corollary 5.2 to that Hamming ball. The Toeplitz operator estimate follows from the deterministic circulant reduction, and the Hankel estimate follows from the reversal J_{d_m}, with only a constant enlargement of the $O(\sqrt{n_m})$ threshold. The displayed failure probability tends to zero, so the joint sampled-operator certificate holds with probability tending to one. The named examples are those already covered by Theorem 5.99. □

Corollary 5.139 (Convex-concentrated operator-certificate abundance)**.** *In the setting of Corollary 5.138, write $\ell_m = \log(e + d_m)$.*

First, for every $0 < \rho \le 1$ and $S > 0$, there is an integer $M_{\rho,S}$ and constants $C_{\rho,S} < \infty$, $c_{\rho,S} > 0$ such that, if $M_{\rho,S}$ independent uniform signs are sampled at level m, then with probability tending to one at least a $1-\rho$ fraction of the sampled signs are centers of Hamming balls of radius

$$\lfloor c_{\rho,S}\ell_m \rfloor$$

on which every sign vector ε' satisfies

$$\left\|\sum_{i=1}^{n_m} \varepsilon'_i A_{m,i}\right\|_{\mathrm{op}} \le C_{\rho,S}\sqrt{n_m}, \qquad \left\|\sum_{i=1}^{n_m} \varepsilon'_i \widetilde{A}_{m,i}\right\|_{\mathrm{op}} \le C_{\rho,S}\sqrt{n_m}.$$

More quantitatively, before the deterministic edge-to-operator transfer, for every $A_0 > 0$ the edge robust-fraction failure is at most

$$C_{\rho,S,A_0,K,C_{\mathrm{cc}}} d_m^{-A_0} + C_{\rho,S,A_0,K,C_{\mathrm{cc}}} \ell_m^2 (e+d_m)^{-\beta} + 2d_m^{-S},$$

for some $\beta = \beta(K, C_{\mathrm{cc}}) > 0$, once m is large enough.

Second, for every polynomial batch exponent $\theta \ge 0$ and confidence exponent $\sigma > 0$, there are constants $C_{\theta,\sigma} < \infty$, $c_{\theta,\sigma} > 0$ such that, whenever

$$1 \le M_m \le d_m^{\theta},$$

then with probability tending to one every sampled sign $\delta_m^{(s)}$, $1 \le s \le M_m$, is the center of a Hamming ball of radius

$$\lfloor c_{\theta,\sigma}\ell_m \rfloor$$

on which every ε' satisfies the two Toeplitz and Hankel operator-norm bounds above, with $C_{\theta,\sigma}$ in place of $C_{\rho,S}$. Before the deterministic transfer, the corresponding edge failure is at most

$$C_{\theta,\sigma,A_0,K,C_{\mathrm{cc}}} d_m^{-A_0} + C_{\theta,\sigma,A_0,K,C_{\mathrm{cc}}} \ell_m^2 (e+d_m)^{-\beta} + d_m^{-\sigma}.$$

Proof. As in the proof of Corollary 5.138, the convex-concentration theorem supplies $a = 1/2$ and

$$p_d(a) \le (e+d)^{-\beta}$$

for some $\beta = \beta(K, C_{\mathrm{cc}}) > 0$. Since $\ell_m = o(n_m)$, all size conditions of the normalizer-tail sampled-search corollaries hold for large m.

For the robust-fraction statement, choose B and $M_{\rho,S}$ as in Corollary 5.33, and apply that corollary with $u = S\ell_m$ and $a = 1/2$. At level m, outside the displayed edge failure, at least a $1-\rho$ fraction of the sampled signs are centers of radius $c\ell_m$ Hamming balls on which every sign has realization-normalized edge maximum $O(\sqrt{n_m})$.

For the polynomial-batch statement, apply Corollary 5.34 with $a = 1/2$, batch exponent θ, and confidence exponent σ. Outside the second displayed edge failure, every sampled sign in the batch is the center of such a logarithmic-radius edge-robust Hamming ball.

In either case, let $\mathcal{F}_m$ be the union of the Hamming balls supplied on the edge-success event. Corollary 5.2 applies to these sign families for all sufficiently large m, enlarging only the constant in the $O(\sqrt{n_m})$ bound and

giving both the Toeplitz and Hankel operator-norm estimates. The displayed edge failures tend to zero, so the operator-certificate abundance conclusions hold with probability tending to one. □

Corollary 5.140 (Convex-concentrated Borel operator batches)**.** *In the setting of Theorem 5.99, let* (d_m, n_m) *satisfy*

$$n_m \le d_m, \qquad d_m/n_m \to \infty, \qquad \log d_m = o(n_m),$$

and put $\ell_m = \log(e + d_m)$*. For each* m*, let* $B_{m,i} = T_{d_m}(\xi^{(m,i)})$*,* $1 \le i \le n_m$*, be independent full Toeplitz matrices from the law, and set*

$$A_{m,i} := \frac{B_{m,i}}{\|B_{m,i}\|_{\mathrm{op}}}, \qquad \widetilde{A}_{m,i} := J_{d_m} A_{m,i}.$$

Independently of the matrices, sample independent uniform signs

$$\delta_m^{(1)}, \dots, \delta_m^{(M_m)} \in \{-1, 1\}^{n_m}, \qquad M_m \ge 1,$$

and let $u_m \ge 0$*,* $1 \le r_m \le M_m$ *be deterministic. Let* $\beta = \beta(K, C_{\mathrm{cc}}) > 0$ *be the convex-concentration exponent from the proof of Theorem 5.99. If, for some* $A > 0$*,*

$$\sum_{m=1}^{\infty} \left[d_m^{-A} + \ell_m^2 (e + d_m)^{-\beta} + e^{-u_m} + \binom{M_m}{r_m} d_m^{-A r_m} \right] < \infty,$$

then, with probability one over the matrices and sampled signs, all but finitely many m *have at least* $M_m - r_m + 1$ *sampled signs that are centers of Hamming balls of radius*

$$\lfloor c \ell_m \rfloor$$

on which every signing ε' *satisfies*

$$\left\| \sum_{i=1}^{n_m} \varepsilon_i' A_{m,i} \right\|_{\mathrm{op}} \le C\sqrt{n_m}, \qquad \left\| \sum_{i=1}^{n_m} \varepsilon_i' \widetilde{A}_{m,i} \right\|_{\mathrm{op}} \le C\sqrt{n_m}.$$

On the same eventual event, the whole sampled list has the entropy-sensitive edge-grid maximum bound

$$\max_{1 \le s \le M_m} \max_{\omega^{d_m} = 1} \left| \sum_{i=1}^{n_m} \delta_{m,i}^{(s)} \frac{P_{m,i}(\omega)}{\|B_{m,i}\|_{\mathrm{op}}} \right| \le C \sqrt{n_m \left(1 + \frac{\log(e M_m) + u_m}{\ell_m} \right)}.$$

Consequently, on any stretched-exponential dimension schedule

$$d_m \ge \exp(c_0 m^{\kappa}) \qquad (c_0, \kappa > 0),$$

every sampled sign in every polynomial batch $M_m \le d_m^{\theta}$ *is eventually a Toeplitz/Hankel operator-norm robust center almost surely, after choosing any* $A > \theta$*, taking* $r_m = 1$*, and taking* $u_m = s \log(e + m)$ *with* $s > 1$*. On a polynomial schedule* $d_m \ge c_0 m^{\kappa}$*, the same conclusion holds whenever* $\beta\kappa > 1$*, after choosing* $A > \theta + 1/\kappa$*.*

Proof. Theorem 5.99 gives $a = 1/2$ and

$$p_d(a) \le (e+d)^{-\beta}.$$

Since $\ell_m = o(n_m)$, the size condition in Corollary 5.55 holds eventually. Applying that corollary from the first sufficiently large index onward, the displayed summability condition implies that, almost surely, all but finitely many sampled batches have the stated edge-grid list maximum and at least $M_m - r_m + 1$ logarithmic-radius robust centers with $O(\sqrt{n_m})$ edge maximum. Corollary 5.4 transfers exactly these same centers and radii to realization-normalized full Toeplitz and full Hankel operator norm, because $d_m/n_m \to \infty$ and $\log d_m = o(n_m)$.

For the stretched-exponential consequence, the matrix terms d_m^{-A} and $\ell_m^2(e+d_m)^{-\beta}$ are summable for every $A, \beta > 0$, and e^{-u_m} is summable because $s > 1$. With $r_m = 1$ and $M_m \le d_m^{\theta}$,

$$\binom{M_m}{1} d_m^{-A} \le d_m^{\theta - A},$$

which is summable on the stretched-exponential schedule when $A > \theta$. On the polynomial schedule, the lower-normalizer contribution is bounded by a constant times $m^{-\beta\kappa}\log^2(e+m)$, which is summable under $\beta\kappa > 1$, while $d_m^{\theta-A} \le Cm^{-(A-\theta)\kappa}$ is summable under $A > \theta + 1/\kappa$. These choices make every sampled sign a robust center eventually. □

Theorem 5.141 (Gaussian realization-normalized edge profile)**.** *Let $B_i = T_d(g^{(i)})$, $1 \le i \le n$, be independent full Toeplitz matrices whose lag variables are standard real Gaussians, and let P_i be their symbols. Assume $n \le d$ and $\log d = o(n)$. Then*

$$\max_{\omega^d=1} \sum_{i=1}^{n} \frac{|P_i(\omega)|^2}{\|B_i\|_{\mathrm{op}}^2} \le C\,\frac{n}{\log(e+d)}$$

with probability tending to one. Consequently, for the corresponding realization-normalized edge vectors $v_i(\omega) = P_i(\omega)/\|B_i\|_{\mathrm{op}}$, independent Rademacher signs satisfy, conditionally on the matrices and with probability at least $1 - d^{-A}$,

$$\max_{\omega^d=1} \left| \sum_i \varepsilon_i v_i(\omega) \right| \le C_A\sqrt{n}$$

for every fixed $A > 0$, on an event whose matrix probability tends to one.

Proof. Write $\ell = \log(e+d)$ and $N_d = \mathbb{E}\,\|T_d(g)\|_{\mathrm{op}}$. By Meckes' lower bound, $N_d \ge c\sqrt{d\ell}$. It is enough, by Theorem 5.14, to verify the one-matrix lower-tail hypothesis with $a = 1/2$. The map $g \mapsto \|T_d(g)\|_{\mathrm{op}}$ is $\sqrt{2d}$-Lipschitz for the Euclidean norm of the $2d-1$ lag variables, because

$$\|T_d(a)\|_{\mathrm{op}} \le \sup_{\theta} \left| \sum_{|k|<d} a_k e^{ik\theta} \right| \le \sqrt{2d-1}\;\|a\|_2\,.$$

Gaussian concentration therefore gives

$$\mathbb{P}\{\|T_d(g)\|_{\mathrm{op}} < N_d/2\} \le d^{-\beta}$$

for some absolute $\beta > 0$. Thus the hypothesis of Theorem 5.14 holds, and the theorem gives the displayed profile. The final signed edge bound is Proposition 5.5. □

Corollary 5.142 (Gaussian realization-normalized full Toeplitz discrepancy)**.** *Let $B_i = T_d(g^{(i)})$, $1 \le i \le n$, be independent full Toeplitz matrices whose lag variables are standard real Gaussians, and put $A_i = B_i/\|B_i\|_{\mathrm{op}}$. If*

$$n \le d \le e^{o(n)} \qquad \text{and} \qquad d/n \to \infty,$$

then

$$\mathrm{disc}_{\mathrm{op}}(A_1, \dots, A_n) \le C\sqrt{n}$$

with probability tending to one, with an absolute constant C. More quantitatively, on a matrix event whose probability tends to one, independent random signs achieve the bound $C_A\sqrt{n}$ with conditional probability at least $1 - d^{-A}$ for every fixed $A > 0$. The same conclusion holds for the corresponding full Hankel model.

Proof. The assumption $d \le e^{o(n)}$ is exactly $\log d = o(n)$, so Theorem 5.141 gives the variance profile required by Proposition 5.5 with probability tending to one over the matrices. On this event, Proposition 5.5 shows that independent signs satisfy

$$\max_{\omega^d=1} \left| \sum_i \varepsilon_i P_i(\omega)/\|B_i\|_{\mathrm{op}} \right| \le C_A\sqrt{n}$$

with conditional probability at least $1 - d^{-A}$. In particular such a sign vector exists on the event. Proposition 5.1 then transfers the edge bound from the circulant model back to full Toeplitz contractions, with only the $o(\sqrt{n})$ de la Vallée Poussin boundary error already accounted for there. The Hankel statement follows by applying the Toeplitz result to the reversed matrices $J_d B_i$, which have the same operator norms and the same discrepancy. □

5.3. **The obstruction: the multi-symbol spectral edge.** For the general sub-Gaussian realization-normalized model, the remaining obstacle is to control $\max_\omega |\sum_i \varepsilon_i v_i(\omega)|$ over the $O(d)$ edge nodes for a well-chosen sign pattern. For a fixed edge node ω, the Rademacher variance is $\sum_i |v_i(\omega)|^2$. Proposition 5.5 shows that the square-root bound follows as soon as this variance is $O(n/\log d)$ uniformly in ω, since the final maximum is taken over only d nodes. Theorem 5.14 shows that it is enough to control the low-normalizer tail of one random Toeplitz matrix at logarithmic density, and Theorem 5.99 verifies this for convex-concentrated laws, including real or complex Gaussian and bounded coefficients, and Rademacher coefficients, while Corollaries 5.100 and 5.101 add strongly log-concave laws and finite

independent linear combinations of convex-concentrated laws, and Corollary 5.102 adds bounded Gaussian scale mixtures. Theorem 5.103 also allows the individual matrices to be drawn from different laws in any uniform lower-normalizer library, Corollary 5.104 allows arbitrary positive matrix-level radial amplitudes after normalization, and Theorem 5.105 allows nonuniform lower-tail quality as long as the aggregate bad-normalizer budget is small, while Theorem 5.106 permits the corresponding aggregate dyadic-depth budget, with Corollary 5.107 giving the aggregate inverse-moment checkable form. Outside these concentration, finite-convolution, bounded scale-mixture, heterogeneous uniform-library, radial-scaling, average-budget, dyadic-budget, and inverse-moment-budget settings, the deterministic worst-case lower bound is exactly the failure mode one must rule out. Adversarial symbols can make many v_i's peak at the same common frequency, and for independent sub-Gaussian symbols without this lower-tail input one still has to prove that this common-frequency conspiracy does not persist through the random normalizers. This is the multi-symbol form of the near-circle edge-fluctuation problem of the companion paper [7].

Remark 5.143 (Meeting point of [7], [6], and [8]). The symbol P_i of a random Toeplitz matrix is, after the change of variable to $G_m(z) = z^w P_i(z)$, a Kac polynomial, and its operator-norm edge $\max_\omega |P_i(\omega)|$ is governed by the values of G_m on and near the unit circle, where its zeros concentrate. The one-symbol boundary theory has now split into two distinct inputs. For standard real Gaussian coefficients, [8] proves the sharp whole-circle boundary small-ball estimate

$$\mathbb{P}\{\min_{z\in\mathbb{T}} |G_m(z)| \le t\} \le C\sqrt{m}\, t,$$

while its finite-section conditioning theorem needs only a weaker one-threshold boundary-minimum event. Consequently every bounded-density variance-one law satisfying the same Konyagin–Schlag-type circle-clearance estimate has polynomial conditioning on the balanced stable subevent, with no tail assumption. This removes the single-symbol conditioning problem, and the Gaussian one-symbol boundary-small-ball hypothesis, from the list of open inputs.

What remains open from [7] is the disk-count fluctuation and local-limit problem: the current proved bound is $\mathrm{Var}(\nu) = O(m)$, while the matching $\Omega(m)$ lower bound, sharp constant, and CLT/LLT remain open. The discrepancy obstruction in Proposition 5.1 is stronger still for general laws: instead of one symbol one has the superposition $\sum_i \varepsilon_i P_i$, and instead of the boundary minimum of one Kac polynomial one needs the maximum of the signed edge over $O(d)$ nodes for an optimized sign pattern. Theorem 5.14 reduces this empirical edge-profile input, for general sub-Gaussian laws, to a one-matrix lower-tail estimate for $\|T_d(\xi)\|_{\mathrm{op}}$; Theorem 5.99 and Corollaries 5.101 and 5.102 settle that estimate for the standard convex-concentrated classes, their fixed finite independent linear combinations, and bounded Gaussian scale

mixtures, and Theorem 5.103 permits non-identically distributed samples from any such uniform library, and Corollary 5.104 permits arbitrary positive matrix-level radial amplitudes without moment assumptions, and Theorem 5.105 permits sparse bad-law contamination under a small aggregate lower-tail budget, while Theorem 5.106 allows the sharper aggregate dyadic-depth accounting and Corollary 5.107 gives the corresponding aggregate inverse-normalizer moment input. Thus Question 1.1 is now the point at which the deterministic discrepancy theory of [6], the disk-count edge theory of [7], and the boundary/conditioning theorems of [8] meet; the one-symbol estimates are useful anti-concentration and conditioning inputs, not yet a substitute for the remaining arbitrary-sub-Gaussian normalizer-tail control.

Neither [7] nor [8] resolves Question 1.1. The deterministic theory stops at $w \le n$; the parity paper leaves the disk-count matching variance lower bound, sharp constant, and CLT/LLT open; and the conditioning paper solves the one-symbol balanced finite-section problem under clearance, not the general sub-Gaussian low-normalizer tail isolated in Theorem 5.14. The convex-concentrated theorem resolves the realization-normalized full-bandwidth model for real or complex Gaussian and bounded coefficient laws, and for Rademacher coefficients; the strongly log-concave and finite-convolution corollaries add the unbounded concentration and convolution closures above, the bounded scale-mixture corollary adds conditional-Gaussian variance mixtures, and the heterogeneous-library theorem removes identical distribution inside those uniform classes. The matrix-level radial-scaling corollary shows that arbitrary positive heavy-tailed amplitudes can be multiplied into those matrices before normalization. The average-budget theorem further shows that the theorem does not require every law in the row to be good, only a small aggregate population of low-normalizer samples; the dyadic-budget theorem further charges that population by depth below the cutoff, and the heterogeneous inverse-moment corollary makes that dyadic control checkable through aggregate negative normalizer moments. The stated question remains open at the full generality of arbitrary fixed sub-Gaussian coefficient laws without such aggregate lower-normalizer, dyadic-depth, or inverse-moment control.

6. Exact moment identities

The fixed-frequency fourth moment is completely explicit. This is the first place where the sign average sees more than the variance profile used in Proposition 5.5.

Proposition 6.1 (Fixed-edge signed fourth moment)**.** *Let $a_1, \dots, a_n \in \mathbb{C}$, let $\varepsilon_1, \dots, \varepsilon_n$ be independent Rademacher signs, and put $S = \sum_i \varepsilon_i a_i$. Then*

$$\mathbb{E}_\varepsilon |S|^4 = 2\Big(\sum_i |a_i|^2\Big)^2 + \Big|\sum_i a_i^2\Big|^2 - 2\sum_i |a_i|^4.$$

Consequently

$$\mathbb{E}_\varepsilon |S|^4 \le 3\Big(\sum_i |a_i|^2\Big)^2,$$

and if the square phase sum $\sum_i a_i^2$ *is negligible then the fourth moment improves from the real Khintchine constant* 3 *toward the complex constant* 2.

Proof. Expand $S^2\overline{S}^2$:

$$\mathbb{E}_\varepsilon |S|^4 = \sum_{i,j,k,\ell} \mathbb{E}(\varepsilon_i\varepsilon_j\varepsilon_k\varepsilon_\ell)\, a_i a_j \overline{a_k}\,\overline{a_\ell}.$$

The expectation is zero unless every index occurs an even number of times. The three pairings give

$$\sum_{i,j} a_i^2 \overline{a_j}^{\,2} + \sum_{i,k} |a_i|^2|a_k|^2 + \sum_{i,\ell} |a_i|^2|a_\ell|^2,$$

but the completely diagonal terms $i=j=k=\ell$ have then been counted three times instead of once. Subtracting the two extra diagonal copies gives the displayed identity. The inequality follows from $|\sum_i a_i^2| \le \sum_i |a_i|^2$. □

The phase-square term is not a new obstruction in the expected-normalized model: away from the real edge nodes it cancels uniformly at a scale smaller than the variance profile.

Proposition 6.2 (Expected-normalized phase-square cancellation)**.** *Work in the expected-normalized model of Theorem 4.1, with*

$$v_i(\omega) = \frac{P_i(\omega)}{N_d}, \qquad \omega^d = 1, \qquad N_d = \mathbb{E}\,\|T_d(\xi)\|_{\mathrm{op}}.$$

For every $A > 0$*, if* $n \le d$ *and* $n \ge C_{A,K}\log(e+d)$*, then with probability at least* $1 - d^{-A}$*,*

$$\max_{\substack{\omega^d=1\\ \omega^2\neq 1}} \left|\sum_{i=1}^n v_i(\omega)^2\right| \le C_{A,K}\sqrt{\frac{n}{\log(e+d)}}.$$

Consequently, on the same event, for every non-real edge node ω*, writing* $V_\omega^2 = \sum_i |v_i(\omega)|^2$*,*

$$\mathbb{E}_\varepsilon \left|\sum_i \varepsilon_i v_i(\omega)\right|^4 \le 2V_\omega^4 + C_{A,K}\frac{n}{\log(e+d)}.$$

Proof. Put $\ell = \log(e+d)$. Fix a root $\omega^d = 1$ with $\omega^2 \neq 1$. Since

$$P_i(\omega) = \sum_{|k|<d} \xi_k^{(i)} \omega^k,$$

we have

$$\mathbb{E}P_i(\omega)^2 = \mathbb{E}(\xi^2) \sum_{|k|<d} \omega^{2k}.$$

The geometric sum equals -1: if $\zeta = \omega^2 \neq 1$, then

$$\sum_{|k|<d} \zeta^k = 1 + \sum_{k=1}^{d-1} (\zeta^k + \zeta^{-k}) = 1 - 1 - 1 = -1,$$

because $\zeta^d = 1$. Hence, using $N_d \geq c_K \sqrt{d\ell}$,

$$\left| \mathbb{E} v_i(\omega)^2 \right| \leq \frac{C_K}{d\ell}.$$

The same sub-Gaussian estimate used in Proposition 5.13 gives $\|P_i(\omega)\|_{\psi_2} \leq C_K \sqrt{d}$. Therefore

$$Y_i(\omega) := v_i(\omega)^2 - \mathbb{E} v_i(\omega)^2$$

is a centered complex sub-exponential variable with $\|Y_i(\omega)\|_{\psi_1} \leq C_K/\ell$. Bernstein's inequality, applied to real and imaginary parts, gives

$$\mathbb{P}\left\{ \left| \sum_{i=1}^{n} Y_i(\omega) \right| > t \right\} \leq 4 \exp\left[-c_K \min\left(\frac{t^2 \ell^2}{n}, t\ell \right) \right].$$

With $t = C_{A,K} \sqrt{n/\ell}$, and after increasing $C_{A,K}$ so that $n \geq C_{A,K} \ell$ puts us in the quadratic Bernstein regime, this probability is at most d^{-A-2}. A union bound over the d roots gives

$$\max_{\omega^d=1,\ \omega^2 \neq 1} \left| \sum_i Y_i(\omega) \right| \leq C_{A,K} \sqrt{\frac{n}{\ell}}$$

with probability at least $1 - d^{-A-1}$. The deterministic mean contribution satisfies

$$\max_{\omega^d=1,\ \omega^2 \neq 1} \left| \sum_i \mathbb{E} v_i(\omega)^2 \right| \leq \frac{C_K n}{d\ell} \leq \frac{C_K}{\ell} \leq C_K \sqrt{\frac{n}{\ell}},$$

since $n \leq d$ and $n \geq 1$. This proves the phase-square bound.

The fourth-moment consequence is Proposition 6.1 with $a_i = v_i(\omega)$; the diagonal correction is non-positive and the square of the phase-square term is at most $C_{A,K} n/\ell$. □

Lemma 6.3 (Expected-normalized edge-entry bound)**.** *Work in the expected-normalized model, with*

$$v_i(\omega) = \frac{P_i(\omega)}{N_d}, \qquad \omega^d = 1, \qquad N_d = \mathbb{E} \left\| T_d(\xi) \right\|_{\mathrm{op}}.$$

For every $A > 0$, if $n \leq d$, then with probability at least $1 - d^{-A}$,

$$\max_{\substack{1 \leq i \leq n \\ \omega^d = 1}} |v_i(\omega)| \leq C_{A,K}.$$

Proof. For each i and ω, the root value $P_i(\omega)$ has sub-Gaussian norm at most $C_K \sqrt{d}$. Meckes' lower bound gives $N_d \geq c_K \sqrt{d \log(e+d)}$. Hence, after increasing $C_{A,K}$,

$$\mathbb{P}\{ |P_i(\omega)| > C_{A,K} N_d \} \leq d^{-A-3}.$$

There are at most d^2 pairs (i, ω), so the union bound gives the claim. □

Corollary 6.4 (Expected-normalized fourth-moment profile)**.** *In the expected-normalized model, for every $A > 0$, if $n \le d$ and $n \ge C_{A,K} \log(e+d)$, then with probability at least $1 - d^{-A}$,*

$$\max_{\omega^d=1} \mathbb{E}_\varepsilon \left| \sum_i \varepsilon_i v_i(\omega) \right|^4 \le C_{A,K} \left(\frac{n}{\log(e+d)} \right)^2.$$

Moreover, away from the real roots,

$$\max_{\substack{\omega^d=1 \\ \omega^2 \neq 1}} \left[\mathbb{E}_\varepsilon \left| \sum_i \varepsilon_i v_i(\omega) \right|^4 - 2 \left(\sum_i |v_i(\omega)|^2 \right)^2 \right] \le C_{A,K} \frac{n}{\log(e+d)}.$$

Proof. Intersect the high-probability events of Propositions 5.13 and 6.2, with the constants enlarged so that the intersection has probability at least $1 - d^{-A}$. On this event,

$$V_\omega^2 := \sum_i |v_i(\omega)|^2 \le C_{A,K} \frac{n}{\ell}, \qquad \ell = \log(e+d),$$

for every edge node. Proposition 6.1 then gives the all-node fourth-moment bound

$$\mathbb{E}_\varepsilon \left| \sum_i \varepsilon_i v_i(\omega) \right|^4 \le 3 V_\omega^4 \le C_{A,K} \left(\frac{n}{\ell} \right)^2.$$

For $\omega^2 \neq 1$, Proposition 6.2 improves the same identity to

$$\mathbb{E}_\varepsilon \left| \sum_i \varepsilon_i v_i(\omega) \right|^4 \le 2 V_\omega^4 + C_{A,K} \frac{n}{\ell},$$

which is the second display. □

The sixth moment is also explicit. It is the first fixed-node layer where the phase-square statistic interacts with a weighted phase-square statistic.

Proposition 6.5 (Fixed-edge signed sixth moment)**.** *Let $a_1, \dots, a_n \in \mathbb{C}$, let $\varepsilon_1, \dots, \varepsilon_n$ be independent Rademacher signs, and put $S = \sum_i \varepsilon_i a_i$. Define*

$$\begin{aligned} A_2 &:= \sum_i |a_i|^2, & B_2 &:= \sum_i a_i^2, \\ A_4 &:= \sum_i |a_i|^4, & B_{2,1} &:= \sum_i a_i^2 |a_i|^2, \\ A_6 &:= \sum_i |a_i|^6. \end{aligned}$$

Then

$$\mathbb{E}_\varepsilon |S|^6 = 6A_2^3 + 9A_2 |B_2|^2 - 18 A_2 A_4 - 12 \operatorname{Re}(B_{2,1} \overline{B_2}) + 16 A_6.$$

Consequently, if for some $0 \le \eta \le 1$,

$$|B_2| \le \eta A_2, \qquad A_4 \le \eta A_2^2,$$

then

$$\left|\mathbb{E}_\varepsilon|S|^6 - 6A_2^3\right| \le (34\eta + 21\eta^2)A_2^3.$$

In all cases, $\mathbb{E}_\varepsilon|S|^6 \le 61A_2^3$.

Proof. Use the moment-cumulant expansion for three copies of S and three copies of $\overline{S}$. For one Rademacher sign the only nonzero cumulants up to order six are

$$\kappa_2 = 1, \qquad \kappa_4 = -2, \qquad \kappa_6 = 16.$$

Since the signs are independent, each block contributes the corresponding cumulant times the same-index coefficient product summed over i.

The pair partitions give two types. There are 6 fully mixed pairings, contributing $6A_2^3$, and 9 pairings with one SS-pair, one $\overline{S}\,\overline{S}$-pair, and one mixed pair, contributing $9A_2|B_2|^2$. For the $4+2$ partitions, the 9 blocks of type $SS\overline{S}\,\overline{S}$ paired with one mixed pair contribute $-18A_2A_4$. The 3 blocks of type $SSS\overline{S}$ paired with $\overline{S}\,\overline{S}$, together with their conjugates, contribute

$$-6B_{2,1}\overline{B_2} - 6\overline{B_{2,1}}B_2 = -12\,\mathrm{Re}(B_{2,1}\overline{B_2}).$$

Finally, the single six-block partition contributes $16A_6$. This proves the identity.

For the perturbative estimate, use

$$|B_{2,1}| \le A_4, \qquad A_6 \le A_2A_4,$$

and subtract $6A_2^3$ from the identity. The universal bound follows from $|B_2| \le A_2$, $A_4 \le A_2^2$, $|B_{2,1}| \le A_2^2$, and $A_6 \le A_2^3$. □

Corollary 6.6 (Expected-normalized sixth-moment profile)**.** *In the expected-normalized model, for every $A > 0$, if $n \le d$ and $n \ge C_{A,K}\log(e+d)$, then with probability at least $1 - d^{-A}$,*

$$\max_{\omega^d=1} \mathbb{E}_\varepsilon \left|\sum_i \varepsilon_i v_i(\omega)\right|^6 \le C_{A,K}\left(\frac{n}{\log(e+d)}\right)^3.$$

Moreover, away from the real edge nodes,

$$\max_{\substack{\omega^d=1\\ \omega^2\neq 1}} \left|\mathbb{E}_\varepsilon \left|\sum_i \varepsilon_i v_i(\omega)\right|^6 - 6\left(\sum_i |v_i(\omega)|^2\right)^3\right| \le C_{A,K}\left(\frac{n}{\log(e+d)}\right)^2.$$

Moreover, at any fixed edge node, if $0 \le \eta \le 1$ and

$$\left|\sum_i v_i(\omega)^2\right| \le \eta V_\omega^2, \qquad \sum_i |v_i(\omega)|^4 \le \eta V_\omega^4, \qquad V_\omega^2 := \sum_i |v_i(\omega)|^2,$$

then

$$\left|\mathbb{E}_\varepsilon \left|\sum_i \varepsilon_i v_i(\omega)\right|^6 - 6V_\omega^6\right| \le (34\eta + 21\eta^2)V_\omega^6.$$

Proof. Put $\ell = \log(e+d)$ and $x = n/\ell$. Intersect the events of Propositions 5.13 and 6.2, and Lemma 6.3, with constants enlarged so that the intersection has probability at least $1 - d^{-A}$. On this event,

$$V_\omega^2 \le C_{A,K} x, \qquad L_\omega := \max_i |v_i(\omega)|^2 \le C_{A,K}$$

for every edge node. Proposition 6.5 gives the universal bound

$$\mathbb{E}_\varepsilon \left| \sum_i \varepsilon_i v_i(\omega) \right|^6 \le 61 V_\omega^6,$$

and the displayed uniform estimate follows after enlarging $C_{A,K}$.

For a non-real edge node, Proposition 6.2 gives

$$|B_2| = \left| \sum_i v_i(\omega)^2 \right| \le C_{A,K} \sqrt{x}.$$

The entry bound also gives

$$A_4 = \sum_i |v_i(\omega)|^4 \le L_\omega V_\omega^2 \le C_{A,K} x,$$

$$|B_{2,1}| = \left| \sum_i v_i(\omega)^2 |v_i(\omega)|^2 \right| \le L_\omega V_\omega^2 \le C_{A,K} x, \qquad A_6 \le L_\omega^2 V_\omega^2 \le C_{A,K} x.$$

Since $x \ge 1$, the exact identity of Proposition 6.5 gives

$$\left| \mathbb{E}_\varepsilon \left| \sum_i \varepsilon_i v_i(\omega) \right|^6 - 6 V_\omega^6 \right| \le C_{A,K} x^2,$$

uniformly over the non-real edge nodes. The final assertion is exactly the perturbative part of Proposition 6.5. □

Corollary 6.7 (Expected-normalized scalar even-moment profile)**.** *In the expected-normalized model, for every $A > 0$, if $n \le d$ and $n \ge C_{A,K} \log(e+d)$, then with probability at least $1 - d^{-A}$, simultaneously for every integer $k \ge 1$,*

$$\max_{\omega^d = 1} \mathbb{E}_\varepsilon \left| \sum_i \varepsilon_i v_i(\omega) \right|^{2k} \le \left(C_{A,K} k \, \frac{n}{\log(e+d)} \right)^k.$$

Proof. On the event of Proposition 5.13,

$$V_\omega^2 := \sum_i |v_i(\omega)|^2 \le C_{A,K} \frac{n}{\log(e+d)}$$

for every edge node. Proposition 5.7 applies on this event and gives the claim, after enlarging $C_{A,K}$. □

Corollary 6.8 (Expected-normalized edge-maximum moment package)**.** *In the expected-normalized model, for every $A > 0$, if $n \le d$ and $n \ge$*

$C_{A,K}\log(e+d)$, then with probability at least $1-d^{-A}$, simultaneously for every integer $k\ge 1$,

$$\mathbb{E}_\varepsilon \max_{\omega^d=1}\left|\sum_i \varepsilon_i v_i(\omega)\right|^{2k} \le d\left(C_{A,K}k\,\frac{n}{\log(e+d)}\right)^k.$$

Consequently, on the same matrix event,

$$\left(\mathbb{E}_\varepsilon \max_{\omega^d=1}\left|\sum_i \varepsilon_i v_i(\omega)\right|^{2k}\right)^{1/(2k)} \le C_{A,K}\sqrt{n}\sqrt{\frac{k}{\log(e+d)}} \qquad (k\ge \log(e+d)).$$

In particular,

$$\mathbb{E}_\varepsilon \max_{\omega^d=1}\left|\sum_i \varepsilon_i v_i(\omega)\right| \le C_{A,K}\sqrt{n},$$

and for every $B>0$,

$$\mathbb{P}_\varepsilon\left\{\max_{\omega^d=1}\left|\sum_i \varepsilon_i v_i(\omega)\right| > C_{A,B,K}\sqrt{n}\right\} \le d^{-B}.$$

Thus, on the same high-probability matrix event, there exists a sign vector $\varepsilon\in\{\pm1\}^n$ with

$$\max_{\omega^d=1}\left|\sum_i \varepsilon_i v_i(\omega)\right| \le C_{A,K}\sqrt{n}.$$

Proof. On the event of Corollary 6.7, for every $k\ge 1$,

$$\max_{\omega^d=1}\mathbb{E}_\varepsilon\left|\sum_i \varepsilon_i v_i(\omega)\right|^{2k} \le \left(C_{A,K}k\,\frac{n}{\ell}\right)^k, \qquad \ell=\log(e+d).$$

Since the $2k$-th power of the edge maximum is bounded by the sum of the $2k$-th powers over the d edge nodes, the first display follows. Taking $2k$-th roots and using $d^{1/(2k)}\le e^{1/2}$ for $k\ge\ell$ gives the second display. The expectation bound follows with $k=\lceil\ell\rceil$ and Hölder's inequality. Markov's inequality with $k=\lceil(B+2)\ell\rceil$, after enlarging the threshold constant, gives the sign-tail display. The deterministic signing is bounded by the sign average. □

Corollary 6.9 (Expected-normalized sampled robust certificates)**.** *In the expected-normalized model, for every $A>0$, if $n\le d$ and $n\ge C_{A,K}\log(e+d)$, then with probability at least $1-d^{-A}$ over the matrices the following conditional sign statements hold. Let $\delta^{(1)},\dots,\delta^{(M)}$ be independent uniform sign vectors in $\{-1,1\}^n$, independent of the matrices. For every $M\ge 1$ and $u\ge 0$,*

$$\mathbb{P}_\delta\left\{\max_{1\le s\le M}\max_{\omega^d=1}\left|\sum_i \delta_i^{(s)} v_i(\omega)\right| > C_{A,K}\sqrt{n\left(1+\frac{\log(eM)+u}{\log(e+d)}\right)}\right\} \le e^{-u}.$$

Moreover, for every $B > 0$, if $M \geq \lceil B \rceil$, then with δ-probability at least $1 - d^{-B}$ at least one sampled sign vector is the center of a Hamming ball of radius

$$r_0 = \lfloor c_{A,K} \log(e + d) \rfloor$$

on which every sign vector ε' satisfies

$$\max_{\omega^d = 1} \left| \sum_i \varepsilon'_i v_i(\omega) \right| \leq C_{A,K} \sqrt{n}.$$

More generally, for every $S > 0$ and every integer $1 \leq r \leq M$, if H_M counts sampled signs that are centers of such good Hamming balls, with the constants allowed to depend on S, then

$$\mathbb{P}_\delta\{H_M \leq M - r\} \leq \binom{M}{r} d^{-Sr}.$$

Proof. On the event of Proposition 5.13,

$$V^2 := \max_{\omega^d = 1} \sum_i |v_i(\omega)|^2 \leq C_{A,K} \frac{n}{\log(e + d)}.$$

The sampled-list maximum is therefore exactly Corollary 5.6, with $C_0 = C_{A,K}$. The constant-size robust-certificate statement is Corollary 5.11 on the same variance-profile event. Finally, the binomial robust-center count follows from Corollary 5.10, with exponent S. □

Corollary 6.10 (Expected-normalized sampled operator certificates)**.** *Let (d_m, n_m) satisfy*

$$n_m \leq d_m, \qquad d_m / n_m \to \infty, \qquad \log d_m = o(n_m).$$

For each m, work in the expected-normalized full Toeplitz model:

$$B_{m,i} = T_{d_m}(\xi^{(m,i)}), \qquad A_{m,i} := \frac{B_{m,i}}{N_{d_m}}, \qquad N_{d_m} := \mathbb{E} \left\| T_{d_m}(\xi) \right\|_{\mathrm{op}},$$

and put $\widetilde{A}_{m,i} = J_{d_m} A_{m,i}$. If $\delta_m^{(1)}, \ldots, \delta_m^{(M_m)}$ are independent uniform signs, independent of the matrices, with $M_m \geq \lceil B \rceil$ for a fixed $B > 0$, then with probability tending to one jointly over the matrices and sampled signs, some sampled sign is the center of a Hamming ball of radius

$$\lfloor c \log(e + d_m) \rfloor$$

on which every sign vector ε' satisfies

$$\left\| \sum_{i=1}^{n_m} \varepsilon'_i A_{m,i} \right\|_{\mathrm{op}} \leq C \sqrt{n_m}, \qquad \left\| \sum_{i=1}^{n_m} \varepsilon'_i \widetilde{A}_{m,i} \right\|_{\mathrm{op}} \leq C \sqrt{n_m}.$$

More generally, for every $S > 0$ and every $1 \leq r_m \leq M_m$, if

$$\binom{M_m}{r_m} d_m^{-S r_m} \to 0,$$

then with probability tending to one at least $M_m - r_m + 1$ sampled signs are centers of such operator-norm robust Hamming balls.

Proof. Let

$$Q_{m,i}(\omega) := \frac{P_{m,i}(\omega)}{N_{d_m}}, \qquad \omega^{d_m} = 1,$$

be the expected-normalized edge symbols. Since $\log d_m = o(n_m)$, the size condition $n_m \geq C \log(e+d_m)$ in Corollary 6.9 holds eventually. That corollary gives a matrix event of probability tending to one on which the claimed edge-grid Hamming balls exist, with sign failure d_m^{-B} for the single-certificate statement and with binomial failure $\binom{M_m}{r_m} d_m^{-Sr_m}$ for the count statement.

It remains only to ensure that the expected-normalized summands are uniformly operator bounded. Apply Theorem 4.1 with $n = 1$ and a large fixed exponent $L > 2$. For one matrix,

$$\mathbb{P}\{\|B_{m,i}/N_{d_m}\|_{\mathrm{op}} > C_{L,K}\} \leq d_m^{-L}.$$

Since $n_m \leq d_m$, a union bound gives

$$\mathbb{P}\left\{\max_{1\leq i\leq n_m} \|A_{m,i}\|_{\mathrm{op}} > C_{L,K}\right\} \leq d_m^{1-L} = o(1).$$

On the intersection of this uniform-boundedness event with the edge-success event, apply Corollary 5.3 to the union of the supplied Hamming balls. This transfers the same centers and radii to the full Toeplitz operator-norm estimate, and the Hankel estimate follows from the same corollary. □

Applied to a fixed edge node $\omega^d = 1$, with $a_i = v_i(\omega)$, Proposition 6.1 says that the sign average is governed by the variance $V_\omega^2 = \sum_i |v_i(\omega)|^2$, the phase-square statistic $\sum_i v_i(\omega)^2$, and the diagonal correction $\sum_i |v_i(\omega)|^4$. Proposition 6.2 shows that, in the expected-normalized model, this phase-square term is already lower order uniformly away from the real roots, and Corollary 6.4 packages the resulting uniform fourth-moment profile. Proposition 6.5 gives the next fixed-node layer: besides V_ω^2 and $\sum_i v_i(\omega)^2$, the sixth moment only adds the weighted phase-square statistic $\sum_i v_i(\omega)^2|v_i(\omega)|^2$ and the diagonal sums $\sum_i |v_i(\omega)|^4, \sum_i |v_i(\omega)|^6$. Corollary 6.6 packages the resulting uniform sixth-moment scale in the expected-normalized model, and away from the real edge nodes it improves the leading constant to the complex value 6, with an $O((n/\log d)^2)$ error. Finally, Corollary 6.7 shows that all scalar edge-grid sign moments in the expected-normalized model have the sub-Gaussian size forced by the variance profile, while Corollary 6.8 packages the same moment control into an averaged edge maximum, sign-tail estimate, and deterministic signing on the expected-normalized high-probability matrix event, while Corollary 6.9 turns the same event into finite sampled lists and logarithmically Hamming-stable robust centers. Corollary 6.10 then transfers those robust centers to actual full Toeplitz/Hankel operator-norm certificates in the unbounded-aspect range. The remaining moment-method problem is therefore not a fixed-node fourth-, sixth-, or scalar edge-maximum calculation;

it is the all-frequency and all-order control of the higher pairing analogues in the realization-normalized model.

We close with the higher-moment template, stated as a route rather than a completed proof. The normalized signed superposition $M = \frac{1}{\sqrt{n}} \sum_i \varepsilon_i A_i$ has operator norm controlled by its even moments $\mathbb{E}\,\mathrm{tr}\, M^{2k}$, and for the circulant model these expand combinatorially over pair partitions, with the Toeplitz structure recorded by the volumes of associated transportation polytopes.

Remark 6.11 (Pair-partition expansion). For the periodic model with i.i.d. symbol-entry circulants A_i, averaging over independent Rademacher signs and over the entry law gives an exact expansion

$$\mathbb{E}\,\mathrm{tr}\Big(\frac{1}{\sqrt{n}} \sum_{i=1}^{n} \varepsilon_i A_i\Big)^{2k} = \sum_{\pi \in \mathcal{P}_2(2k)} \mathrm{vol}(\Pi_\pi)\, d^{\,1+|\pi|} \cdot n^{-k} \cdot (1 + o(1)),$$

where $\mathcal{P}_2(2k)$ is the set of pair partitions of $\{1, \dots, 2k\}$ and $\mathrm{vol}(\Pi_\pi)$ is the volume of the transportation polytope cut out by the lag constraints of π. The non-crossing/crossing dichotomy is visible in these volumes: a non-crossing pairing contributes the full volume $\mathrm{vol} = 1$, whereas a crossing pairing for the Toeplitz lag structure contributes a strictly smaller volume, at most $2/3$ at the first crossing. The cleanest expression of the resulting gain is at the fourth moment: the single crossing pairing has volume bounded by $2/3$, which gives a normalized fourth moment

$$m_4 := \mathbb{E}\,\mathrm{tr}\, M^4 \Big/ \big(\mathbb{E}\,\mathrm{tr}\, M^2\big)^2 < 3,$$

strictly below the Gaussian value 3, i.e. a sub-Gaussian fourth moment. A fourth moment strictly below the semicircular threshold is exactly the kind of input a moment method would convert into an edge bound at the $\sqrt{n}$ scale, and the strict deficit $m_4 < 3$ coming from the crossing-volume bound $2/3$ is the quantitative seed. Turning this into the high-probability operator-norm bound of Question 1.1 would require controlling all k and the maximum over edge nodes at the matrix-pairing level, beyond the scalar profile certificate above. The one-symbol boundary and conditioning theorems of [8] are useful inputs but not a substitute for this all-k, all-node matrix control. We record the fourth-moment gain only as evidence pointing that way, not as a proof.

Acknowledgements and disclosure

The author is responsible for all mathematical claims and bibliographic information herein. The pathwise bounds of Section 3, the expected-normalized full-bandwidth theorem of Section 4, the normalizer-tail criterion, the convex-concentrated realization-normalized theorem, the strongly log-concave, finite-convolution, and bounded Gaussian scale-mixture specializations, the heterogeneous lower-normalizer library theorem, the heterogeneous average-budget and dyadic-budget theorems, the heterogeneous inverse-moment

budget corollary and sampled-operator schedule, robust-fraction, batch-maximum, batch-schedule, inverse-p schedule, random-environment inverse-p, and expectation/random-catalog, bounded-catalog Bernstein/Freedman, and predictable-catalog and active-intensity Freedman corollaries, sampled operator certificates, the Gaussian specialization of Section 5, and the single-cutoff and dyadic empirical normalizer decompositions, edge-profile random search with success-count control, Hamming stability with robust centers, and fixed-edge fourth- and sixth-moment/phase-square/scalar even-moment estimates of Section 6 are proved; the arbitrary sub-Gaussian realization-normalized contraction question and the higher-moment all-edge route of Section 6 are stated as such and remain open.

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